

Local Control Network Planning

SW02-601

System Site Planning - 1

***Local Control Network
Planning***

SW02-601

Release 684

CE Compliant

03/2013

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Revision 02– March 2013

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About This Publication

This manual provides information and references that are useful during the planning, designing, and building phases of a project site.

There is a substantial amount of technical information that is unique to each individual project. This specific information is required to design, build, and install the Honeywell TPS system equipment and is not addressed in this manual.

Information on the system's requirements and layout must be made available to Honeywell during the initial design stages, and as the equipment design begins to take shape, the information will be transmitted back to the customer for site drawing preparation.

This publication supports **TotalPlant** Solution (TPS) Network Release 684 and earlier software releases. TPS is the evolution of TDC 3000^X.

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Standard Symbols

Scope

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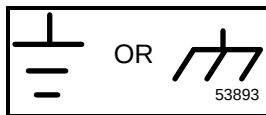
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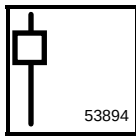
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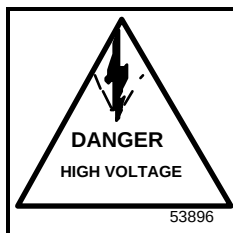
Ground connection to building safety ground



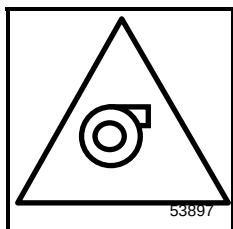
Ground stake for building safety ground



Electrical Shock Hazard—can be lethal



Electrical Shock Hazard—can be lethal



Rotating Fan—can cause personal injury

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Acronyms

AM.....	Application Module
APM.....	Advanced Process Manager
ARM.....	Archive Replay Module
AUI.....	Attachment Unit Interface
A ^X M.....	Application Module ^X (UNIX)
CG.....	Computer Gateway
CS/R.....	Clock Source/Repeater
DAT.....	Digital Audio Tape
DEC.....	Digital Equipment Corporation
DSAP.....	Destination Service Access Point
EMI.....	Electromagnetic Interference
FOC/RCVR.....	Fiber Optic Clock Receiver
FOC/XMTR.....	Fiber Optic Clock Transmitter
HG.....	Hiway Gateway
HM.....	History Module
LAT.....	Local Area Transport
LCNE.....	LCN Extender
LCNFL.....	LCN Fiber Link
LCN.....	Local Control Network
LED.....	Light Emitting Diode
LM.....	Logic Manager
MB.....	megabyte
NCF.....	Network Configuration File
NEC.....	National Electrical Code
NFPA.....	National Fire Protection Association
NG.....	Network Gateway
NIM.....	Network Interface Module
OTDR.....	Optical Time Domain Reflectometer
PCS/R.....	Precision Clock Source/Repeater
PC.....	Personal Computer
PG.....	Processor Gateway
PLCG.....	Programmable Logic Controller Gateway
PLNM.....	Plant Network Module
PM.....	Process Manager
RFI.....	Radio Frequency Interference
SAM.....	Scanner Application Module
SNAP.....	Sub-Network Access Protocol
SOE.....	Sequence Of Events
SSAP.....	Source Service Access Point
TCP/IP.....	Transmission Control Protocol/Internet Protocol
UCN.....	Universal Control Network
UCNE.....	Universal Control Network Extender
US.....	Universal Station
UWS.....	Universal Work Station
U ^X S.....	Universal Station ^X (UNIX)
XID.....	Exchange Identification

References

Publication Title	Binder Title	Binder Number
TPS System Site Planning	System Site Planning - 1	TPS 3020-1
LCN System Installation	LCN Installation	TPS 3025
LCN Guidelines - Implementation, Troubleshooting, and Service	LCN Installation	TPS 3025
LCN System Checkout	LCN Installation	TPS 3025
Deskside Global User Station Service	_____	_____
Console Global User Station Service	_____	_____
Classic Console Global User Station Service	_____	_____
Five/Ten-Slot Module Service	LCN Service - 1	TPS 3060-1
Dual Node Module Service	LCN Service - 1	TPS 3060-1
History Module Service	LCN Service - 1	TPS 3060-1
Engineer's Reference Manual	Implementation/Startup & Reconfiguration	TPS 3030-2

Section 1 – Introduction

1.1 Overview

Section contents The topics covered in this section are:

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1.2 Description	4

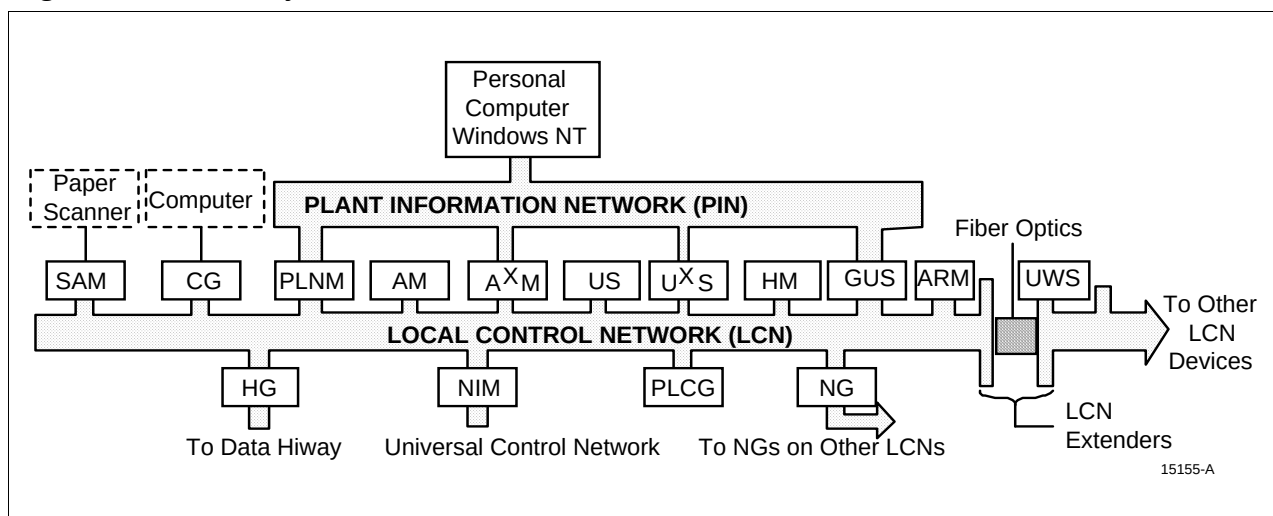
Introduction

This manual introduces you to the Local Control Network (LCN) which is a major component of the Honeywell **TotalPlant** Solution (TPS) System. It will assist you in understanding and selecting the functional components that can comprise the LCN. Based on your requirements, you will be able to select the components necessary to control and monitor your process system.

The LCN is a coaxial and, sometimes additionally, a fiber optic network that provides the data path for communication between the TPS system nodes.

Figure 1-1 is an illustrative overview of the TPS system, of which the Local Control Network is an integral part.

Figure 1-1 TPS System Overview



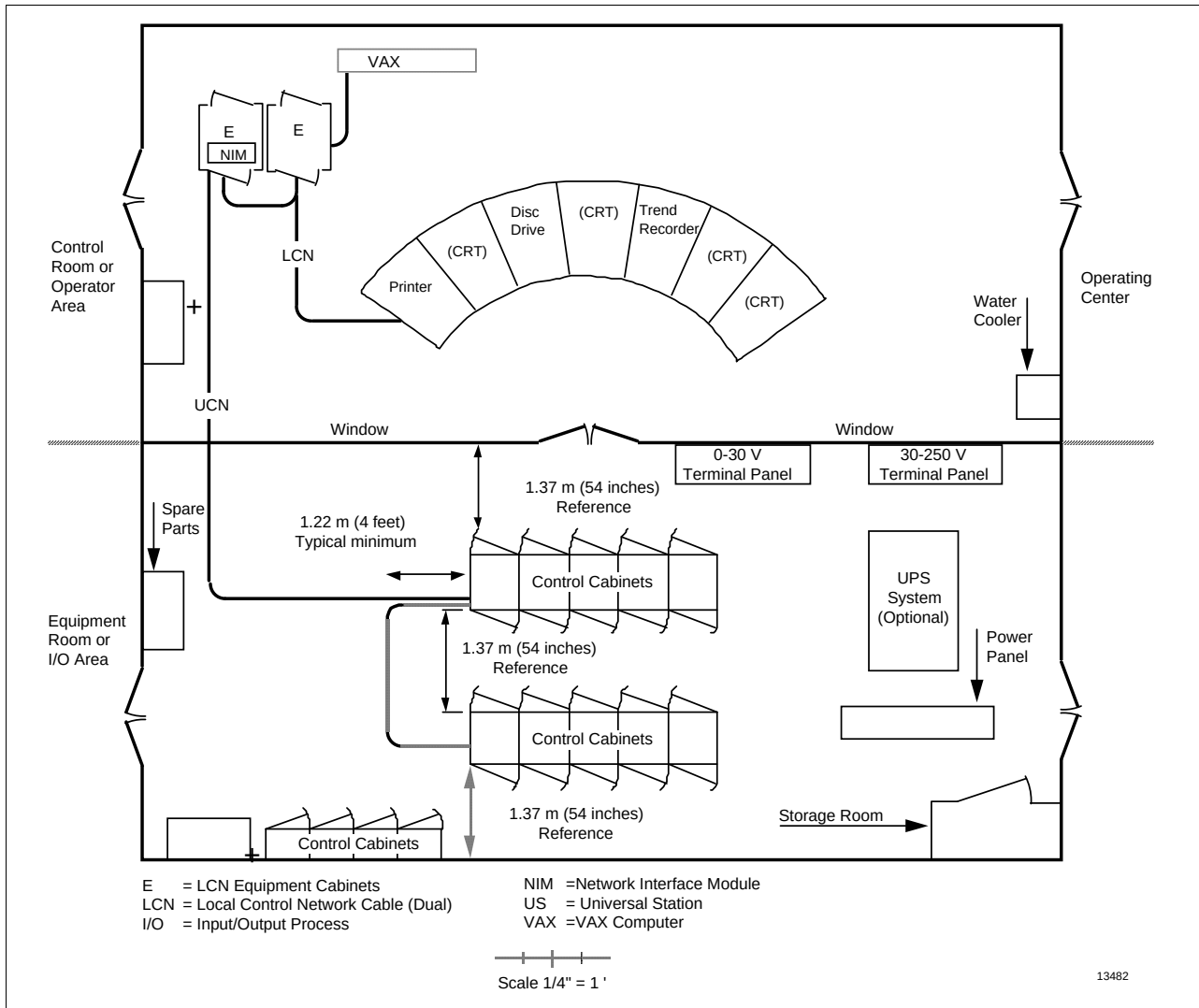
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1.1 Overview, Continued

Operating center

The Operating Center typically includes Operator Station(s), Universal Stations, and the equipment cabinets that support the Local Control Network (LCN) nodes and gateways. See Figure 1-2 for a layout example.

Figure 1-2 TPS System Operating Center Layout



Continued on next page

1.1 Overview, Continued

Supporting documentation

This manual, in combination with a companion manual, the *TPS System Site Planning* manual, provides information and references needed to prepare site facilities for the installation of your Honeywell TPS system's operating center equipment before its delivery. The *TPS System Site Planning* manual guides you through the planning and preparation phases necessary for proper selection of operating center system equipment, regardless of whether your engineering personal or outside consultants formalize the site preparation plans.

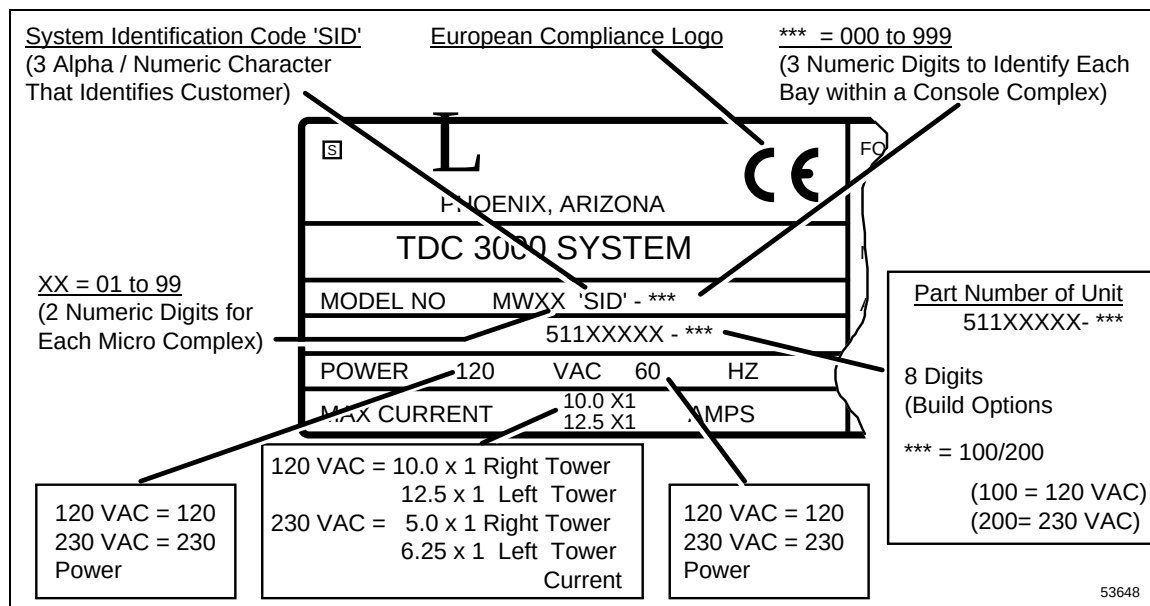
The operating center typically includes an operator station composed of Universal Stations and equipment cabinets that contain the LCN node equipment and the gateway equipment.

Product certification label

LCN equipment was redesigned for Electro Magnetic Compatibility (EMC) compliance. The European EMC directive (89/366/EEC) requires that an electronics product operate reliably in its intended EMC environment. It also requires that the product not detrimentally affect other products operating in their own environment.

The hardware will contain a product certification label to indicate the hardware is in compliance. This label is placed inside, at the bottom front of the cabinet or station. As shown in Figure 1-3, a "CE" logo on the product label indicates the product compliance with the European EMC directive along with other descriptive information about the product.

Figure 1-3 Product Certification Label



1.2 Description

Cable network	Physically, the Local Control Network is composed of modules that represent address nodes on the LCN. The modules are interconnected by dual coaxial and/or fiber optic cables for remote applications. The dual cables offer the reliability of redundancy for the network.
LCN modules types	Many different types of modules are available, such as History Module, Universal Station, Universal Station ^X , Application Module, Hiway Gateway, or Network Interface Module to identify a few.
Module board complements	The circuit board complement for a module varies but always contains a processor (kernel) board, a controller board or a board that provides an interface functionality, and optionally, one or more memory boards. These boards are installed in the front of the module chassis. Typically, an I/O or similar board is associated with the processor board or functional board. The memory boards have no associated I/O board. These boards are installed in the rear of the module chassis in the same slot as the board it is associated with. However, sometimes an I/O type board is not associated with any particular board.
Processor boards	There are four types of processor circuit boards. HMPU Consists of 68020 microprocessor with a coprocessor and 2 megabytes of memory. The HMPU requires an LCN or LLCN board to control the interface to the LCN. HPK2 Consists of 68020 microprocessor and 2 or 3 megabytes of memory. The HPK2 requires an LCN or LLCN (low power) board to control the interface to the LCN. K2LCN The K2LCN board is designated a kernel because it consists of 68020 microprocessor and 2, 3, 4, 6, or 8 megabytes of memory and LCN interface control circuitry. Its associated LCN I/O board provides the actual interface to the LCN cables. K4LCN The K4LCN board is designated a kernel because it consists of 68040 microprocessor and 4, 8, or 16 megabytes of memory and LCN interface control circuitry. Its associated LCN I/O board provides the actual interface to the LCN cables.
Memory boards	There are two types of memory circuit boards. • EMEM The EMEM board contains 1 megaword of memory. • QMEM The QMEM board contains 2, 3, or 4 megawords of memory.

Continued on next page

1.2 Description, Continued

Controller or function boards	The choice of the controller or function circuit board determines the functionality of the LCN module. Some boards require an associated I/O board, such as the PLCI I/O or SPC I/O boards to name a few. In some cases there are enhanced versions of the controller or function board, such as PLCG and EPLCG boards. Refer to the <i>Five/Ten-Slot Service</i> or <i>Dual Node Service</i> manuals for additional details.
Five/Ten-Slot Module chassis	An LCN Five-Slot Module is a chassis with five circuit board slots, a power supply, and fan assembly. Each circuit board slot at the front of the chassis has a complementary I/O circuit board slot that is located in a card cage at the rear of the chassis. An LCN Ten-Slot Module is a chassis with ten circuit board slots, a power supply, and fan assembly.

Continued on next page

1.2 Description, Continued

Five-Slot Module chassis

Figure 1-4 is an illustration of the front of a Five-Slot Module chassis. The Ten-Slot Module chassis is very similar, except it has ten circuit board slots instead of five. Figure 1-5 is an illustration of the rear of a Five-Slot Module.

Figure 1-4 Five-Slot Module Chassis (Front View)

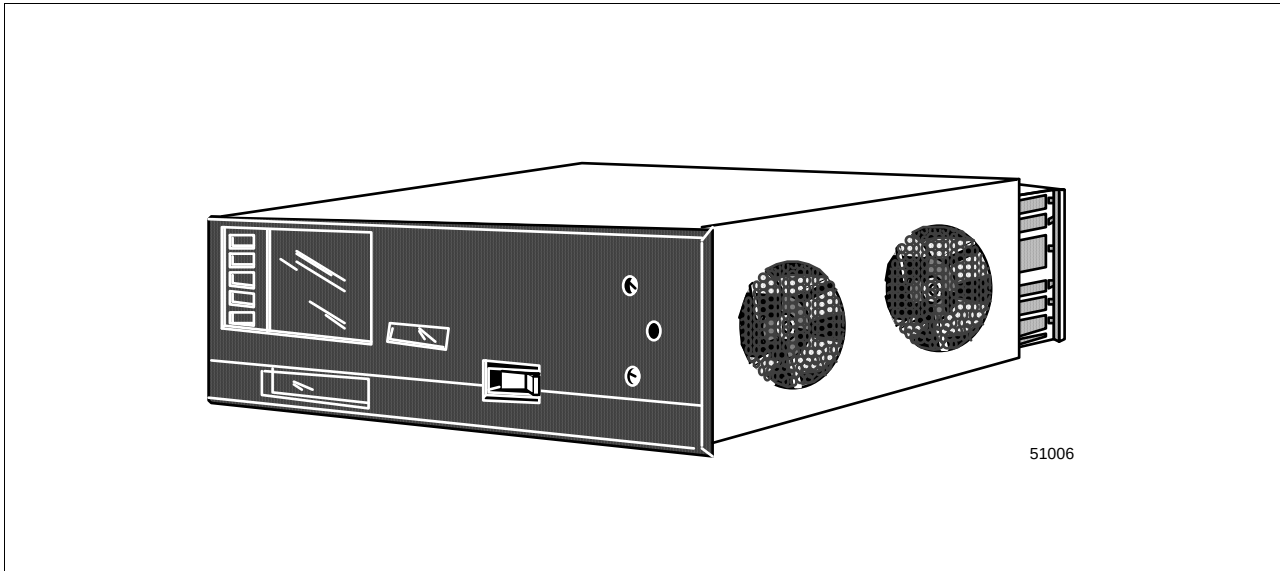
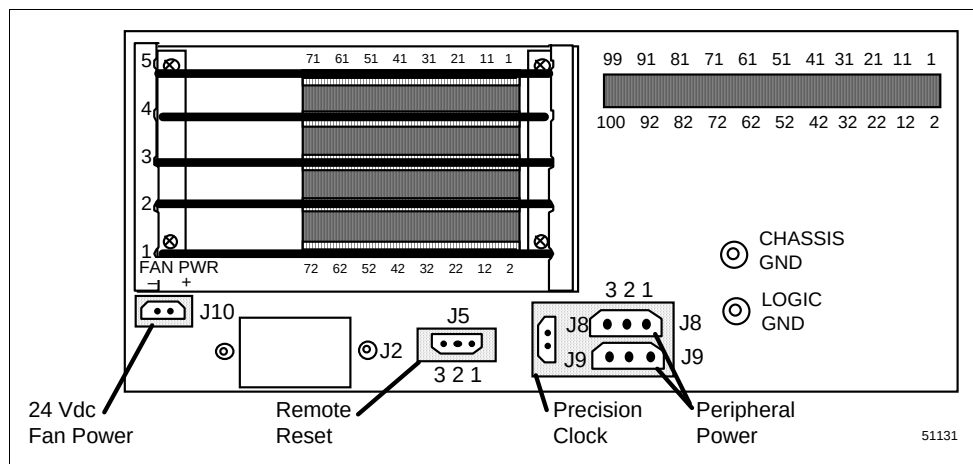


Figure 1-5 Five-Slot Module Chassis (Rear View)



Continued on next page

1.2 Description, Continued

Dual Node Module chassis

The third type of LCN module chassis is the Dual Node Module. This chassis houses the electronics for two LCN nodes. The Dual Node Module is partitioned into nodes that accommodate two and three circuit boards with associated I/O circuit board slots at the rear of the chassis. Each node has a dedicated power supply. A fan assembly is common to both nodes.

Figure 1-6 is an illustration of the front of a Dual Node Module chassis, while Figure 1-7 is an illustration of the rear of a Dual Node Module chassis.

Figure 1-6 Dual Node Module Chassis (Front View)

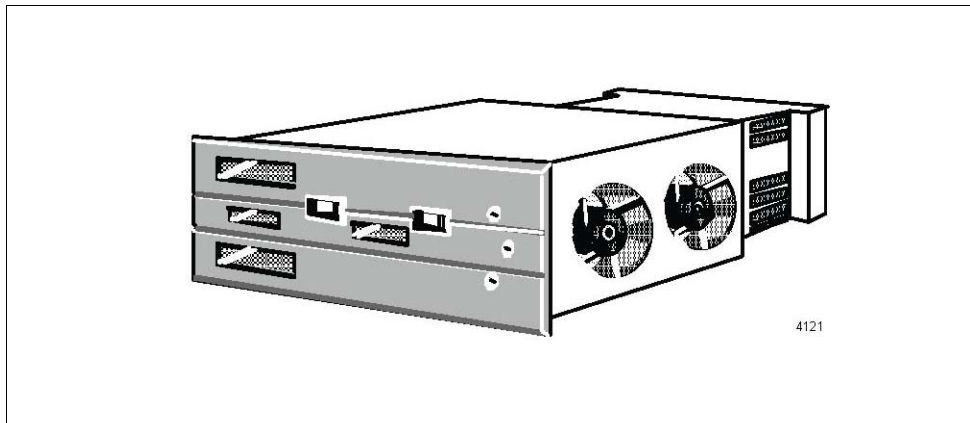
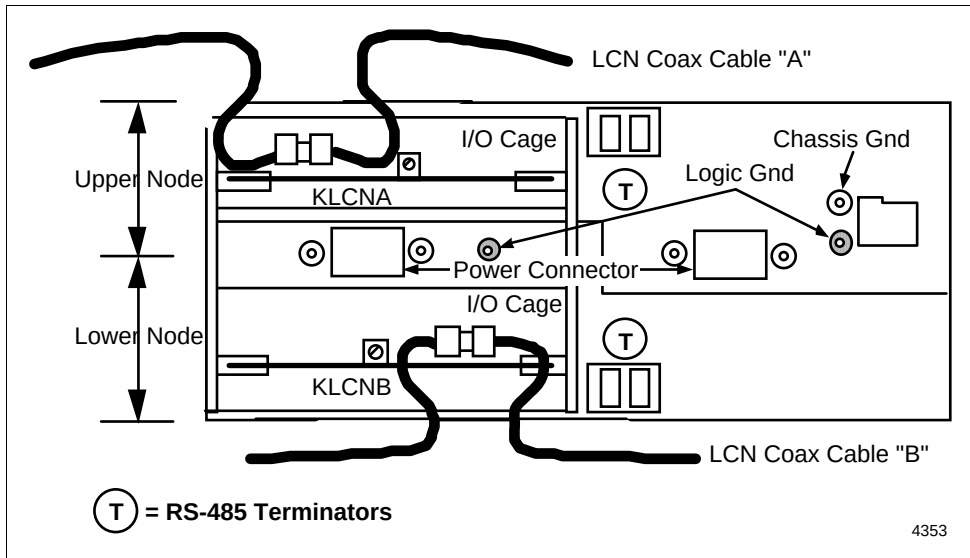


Figure 1-7 Dual Node Module Chassis (Rear View)



Continued on next page

1.2 Description, Continued

Module packaging

Depending upon the number of circuit boards required, economics, and convenience, the electronics that comprise an LCN module (node) can be packaged in either a Five-Slot, Ten-Slot, or Dual Node Module chassis. Many of the LCN module types, such as the History Module, are packaged in both a Five-Slot Module chassis and a Dual Node Module chassis.

Section 2 – LCN Modules

2.1 Overview

Section contents

The topics covered in this section are:

Topic	See Page
SECTION 2 – LCN MODULES	ERROR! BOOKMARK NOT DEFINED.
2.1 Overview	Error! Bookmark not defined.
2.1.1 Universal Station (US)	Error! Bookmark not defined.
2.1.2 Global User Station.....	Error! Bookmark not defined.
2.1.3 Universal Station ^X (U ^X S).....	Error! Bookmark not defined.
2.1.4 Universal Work Station (UWS)	Error! Bookmark not defined.
2.1.5 History Module (HM)	Error! Bookmark not defined.
2.1.6 Archive Replay Module (ARM)	Error! Bookmark not defined.
2.1.7 Application Module (AM).....	Error! Bookmark not defined.
2.1.8 Enhanced Redundant Applications Module.....	Error! Bookmark not defined.
2.1.9 Application Module ^X (A ^X M).....	Error! Bookmark not defined.
2.1.10 Plant Network Module (PLNM)	Error! Bookmark not defined.
2.1.11 Network Interface Module (NIM).....	Error! Bookmark not defined.
2.1.12 Programmable Logic Controller Gateway (PLCG).....	Error! Bookmark not defined.
2.1.13 Hiway Gateway (HG)	Error! Bookmark not defined.
2.1.14 Network Gateway (NG).....	Error! Bookmark not defined.
2.1.15 Computer Gateway (CG)	Error! Bookmark not defined.
2.1.16 Processor Gateway (PG).....	Error! Bookmark not defined.
2.1.17 Scanner Application Module (SAM).....	Error! Bookmark not defined.

Introduction

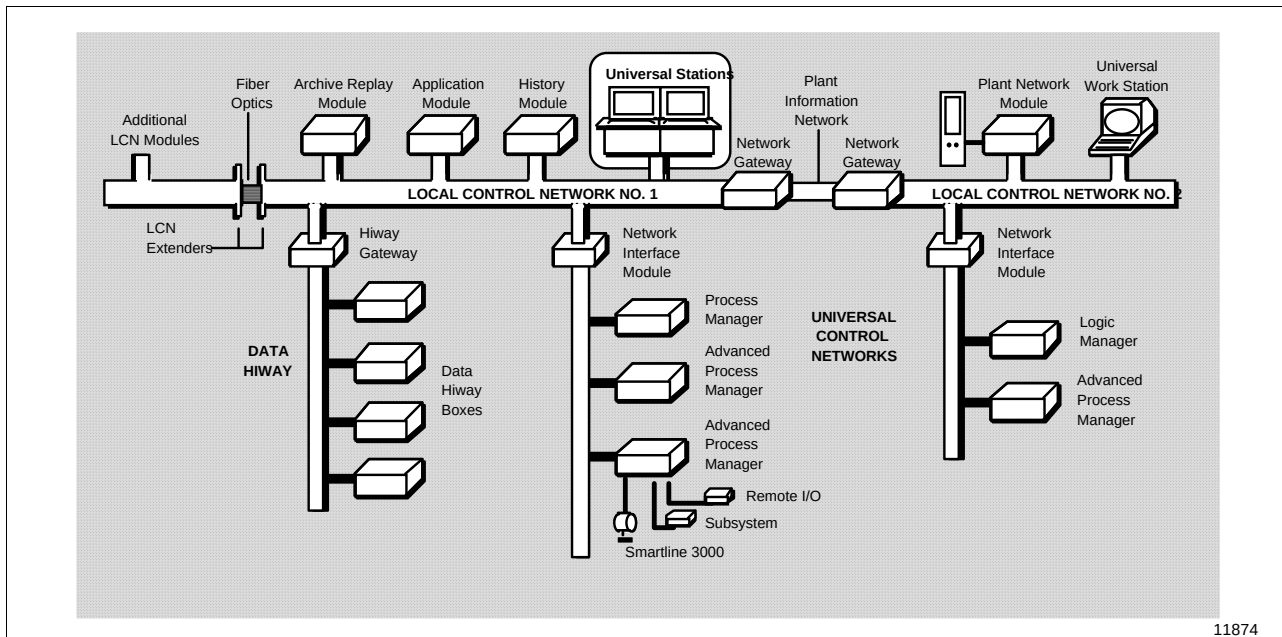
The Local Control Network offers a number of hardware modules with individual functionalities. Each module type is a node in the network. The modules are implemented in either Five-Slot, Ten-Slot, or Dual Node Module chassis. The Dual Node Module can accommodate hardware for two LCN nodes. The subsections that follow describe the types of LCN modules (nodes) that are available.

2.1.1 Universal Station (US)

Description

The Universal Station (US) module provides a window to the process or processes that the Local Control Network controls. As illustrated in Figure 2-3, the Universal Station communicates with other modules in the Local Control Network, modules in remote LCNs through the Network Gateway, process devices on Universal Control Networks through the Network Interface Module, and with process devices in Data Hiways through Hiway Gateways.

Figure 2-1 TPS System with Universal Station



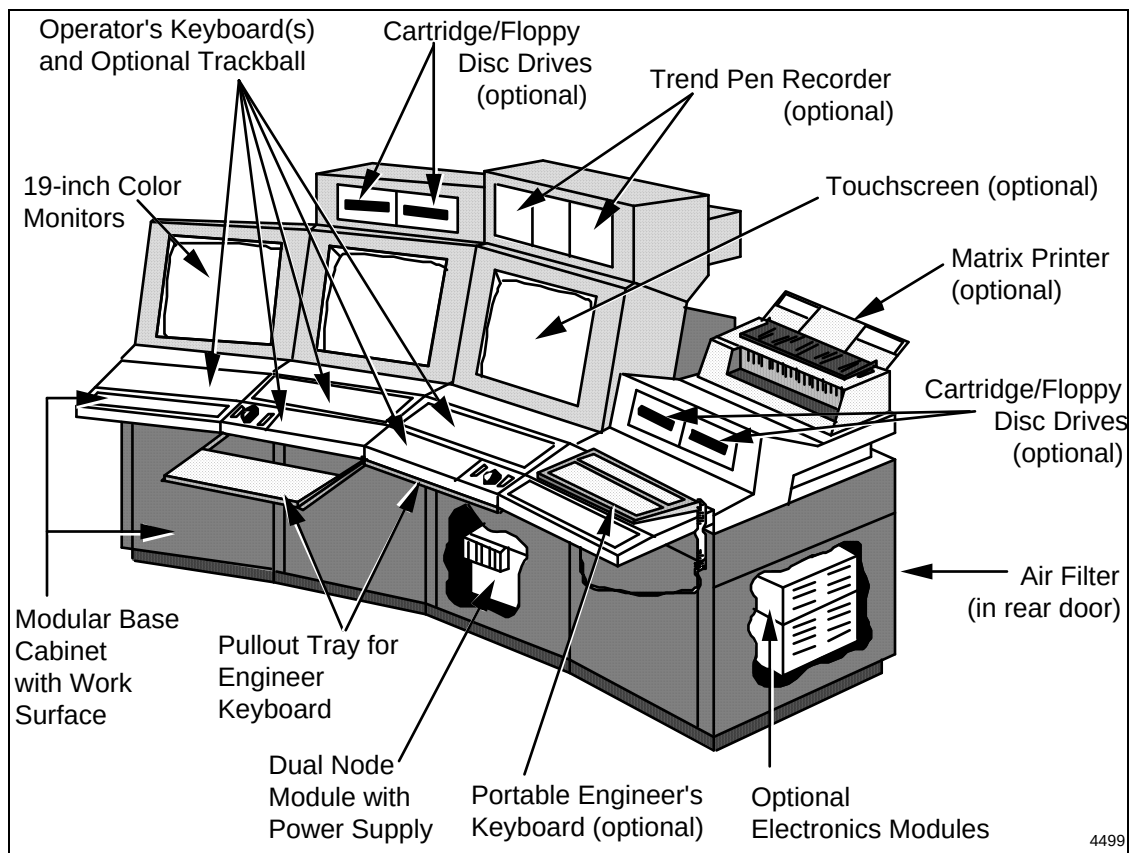
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2.1.1 Universal Station (US), Continued

US configurations

The Universal Station's hardware and optional peripheral equipment requirements are based on the the **TotalPlant** Solution (TPS) System's applications. Figure 2-2 is an illustration of a typical console complex of Universal Stations.

Figure 2-2 Universal Station Console Complex (Classic Furniture)



Continued on next page

2.1.1 Universal Station (US), Continued

US features

A Universal Station module supports the following features.

- A 19- or 21-inch color monitor with optional touchscreen capability to view displayed information (maximum of two units - one for each US)
 - Keyboards for input data
 - Operator's keyboard with optional trackball
 - U^XS Auxiliary keyboard
 - Portable Engineer's keyboard
 - Removable media drives for loading/unloading data
 - Cartridge drive (maximum of two) for Bernoulli cartridges, or
 - Zip drive (maximum of two) for Zip disks
 - Matrix printer
 - Module (node) electronics to control the devices listed above
 - Alarm annunciation
 - Trend Pen recorder – complex of 3 or 6 drives
 - Optional Remote User LCN Access (RULA)
-

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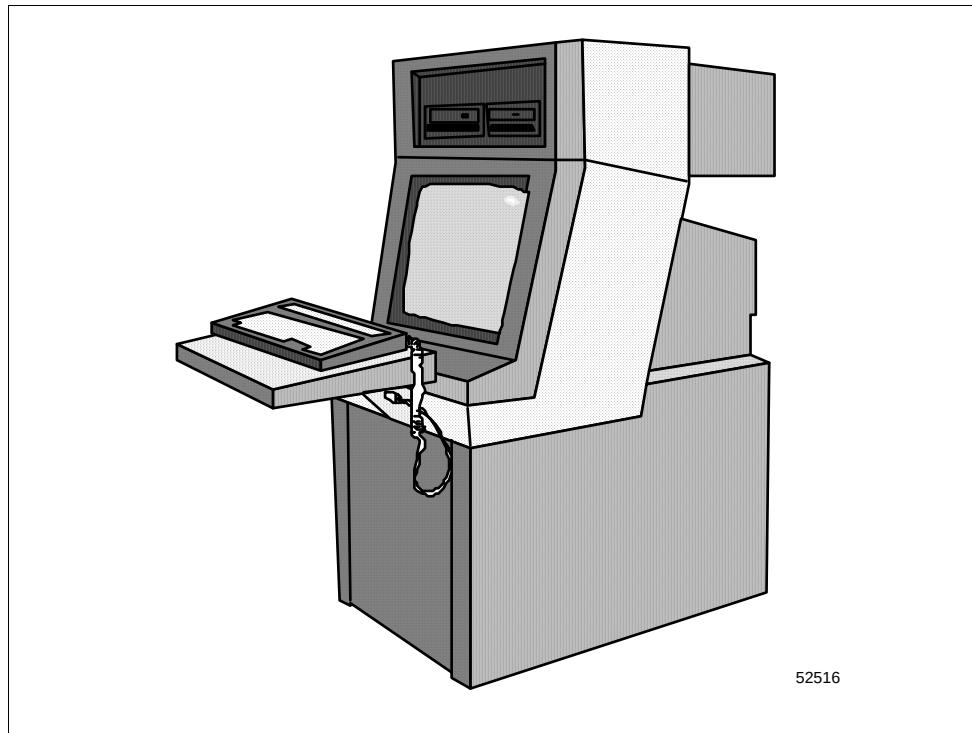
2.1.1 Universal Station (US), Continued

Furniture styles

The Universal Station module is installed in two styles of furniture. The Classic furniture style (1983 design) is illustrated in Figure 2-3 and the Ergonomic furniture style (1993 design) is illustrated in Figure 2-4.

US Classic furniture style

Figure 2-3 Universal Station – Classic Furniture Style

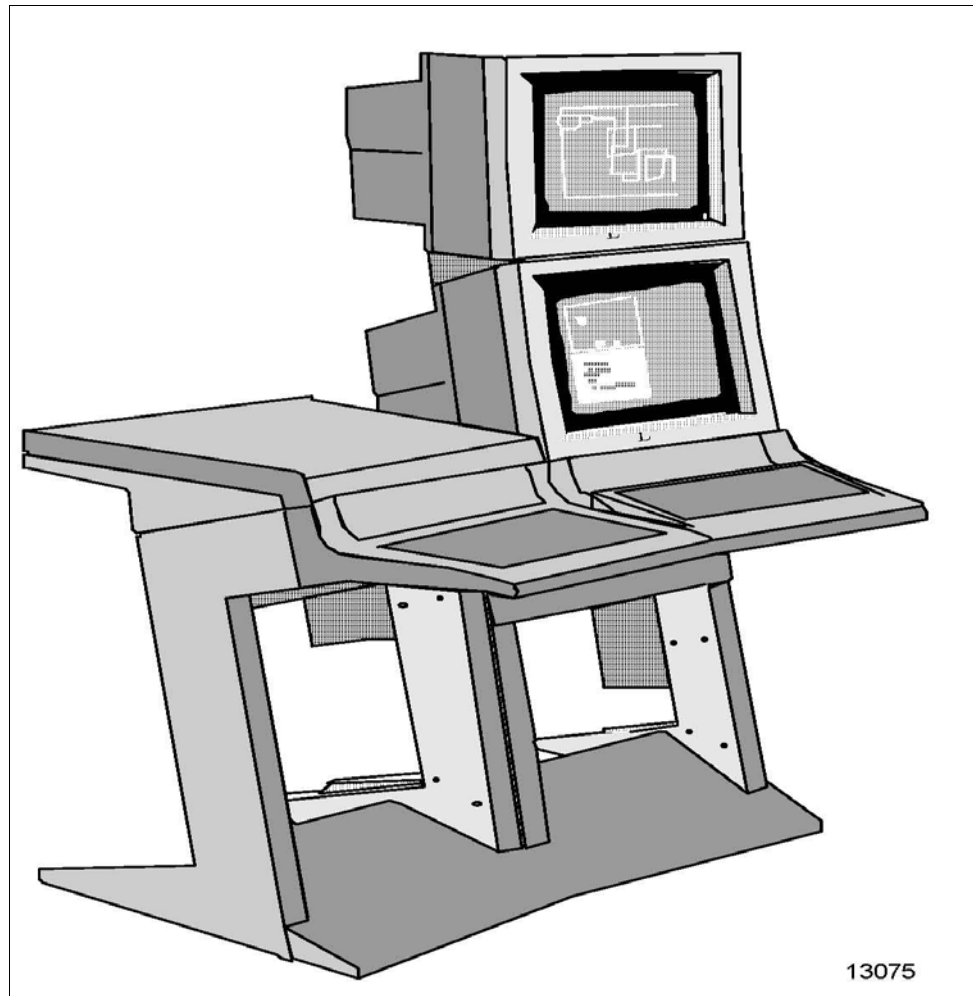


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2.1.1 Universal Station (US), Continued

US Ergonomic furniture style

Figure 2-4 Universal Station – Ergonomic Furniture Style



Module packaging

The Universal Station is implemented in either the Five-Slot and Dual Node Module chassis.

References

Additional information can be found in the *Universal Station Specification and Technical Data* manual and the *Universal Station Installation, Operation, and Service* manual.

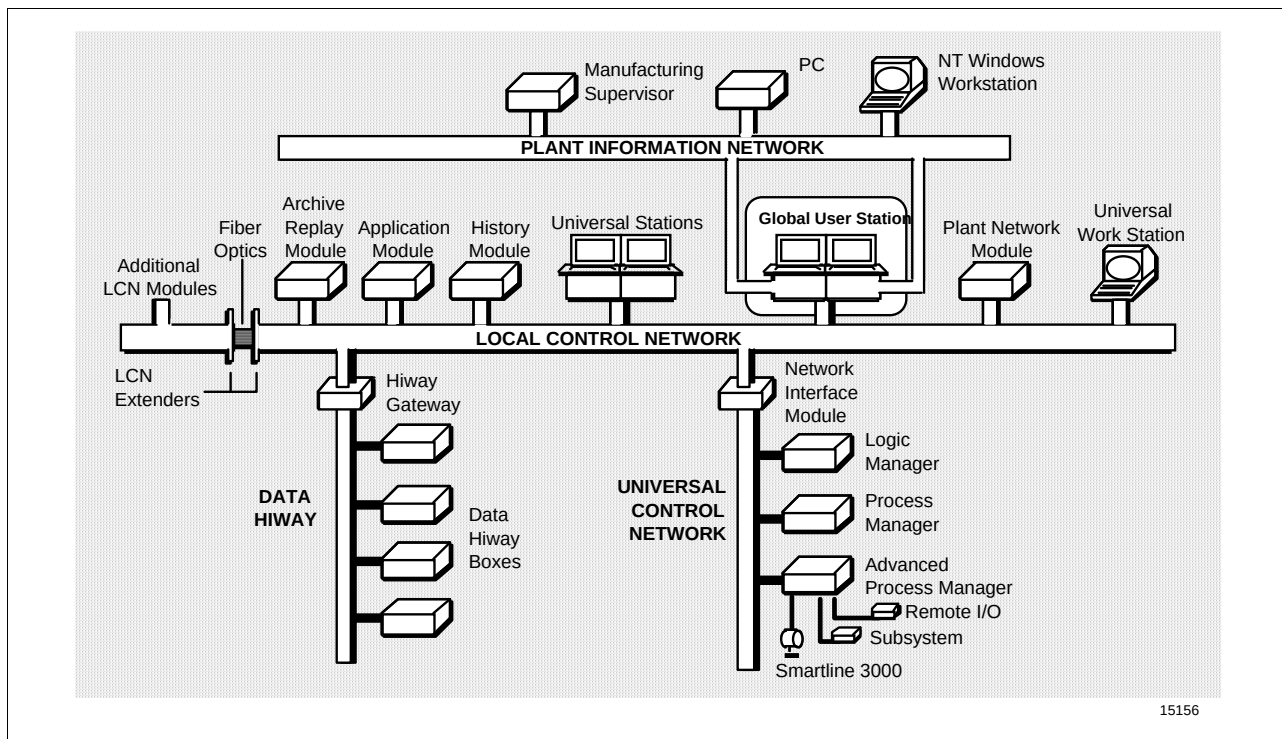
2.1.2 Global User Station (GUS)

Description

The Global User Station module provides a window to the process, or processes, that the Local Control Network controls. As illustrated in Figure 2-5, the Global User Station communicates with other modules in the Local Control Network:

- Modules in remote LCNs through the Network Gateway
- Process devices on Universal Control Networks through the Network Interface Module
- Process devices in Data Hiways through Hiway Gateways

Figure 2-5 TPS System with Global User Station



Description

The Global User Station comes in three styles:

- Deskside
- Console Furniture
- Classic Console Furniture

ATTENTION

See Section 9 for RFI classifications applicable to each Global User Station style.

Continued on next page

2.1.2 Global User Station, Continued

Deskside

A Deskside Global User Station consists of

- Computer tower
 - Hard Disk Drive(s)
 - CD-ROM Drive
 - Floppy Drive
- 17-inch or 21-inch Color Monitor
- Keyboard
 - Standard PC Keyboard
 - Integrated Keyboard (IKB)
- Zip Drive (optional)
- Tape Drive (optional)

Figure 2-6 Deskside Global User Station



Continued on next page

2.1.2 Global User Station (GUS), Continued

A Global User Station in new console furniture (XC3000 Z-Style) consists of

- Global User Station Module
 - 1 or 2 GB Hard Disk Drive (Second Drive Optional)
 - 3.5" - 1.44 MB Floppy Drive
 - CD-ROM Drive
 - 2 GB Tape Drive
 - 100 MB Zip Drive
- 21-inch Color Multisync Monitor (single or dual)
- Integrated Keyboard (IKB with trackball)
- Touchscreen (optional, lower monitor only)

Figure 2-7 Console Global User Station



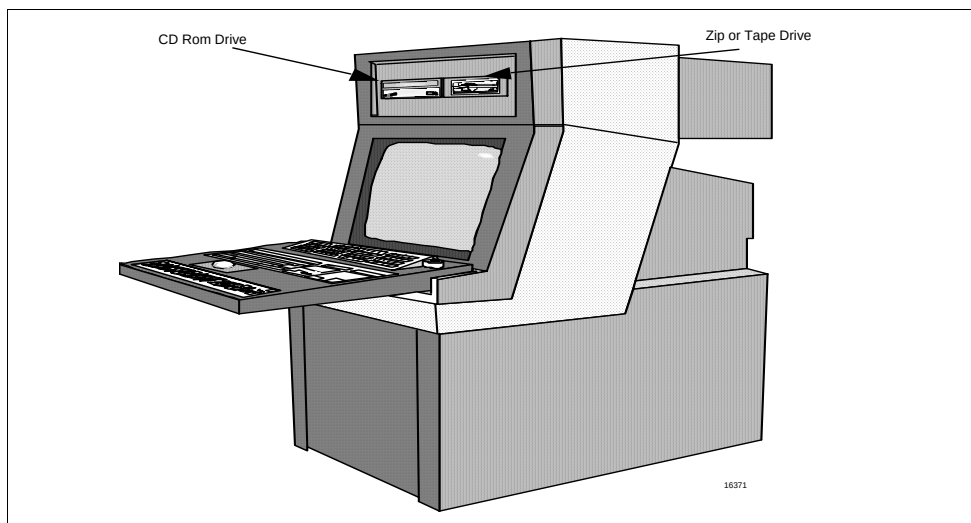
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2.1.2 Global User Station (GUS), Continued

A Global User Station in Classic Console furniture consists of

- Global User Station Module
 - 1 or 2 GB Hard Disk Drive (Second Drive Optional)
 - 3.5" - 1.44 MB Floppy Drive
 - CD-ROM Drive
 - 2 GB Tape Drive
 - 100 MB Zip Drive
- 19-inch Color Multisync Monitor (single or dual)
- Integrated Keyboard (IKB with trackball)
- Touchscreen (optional, lower monitor only)

Figure 2-8 Classic Console Global User Station



2.1.3 Universal Station^X (UXS)

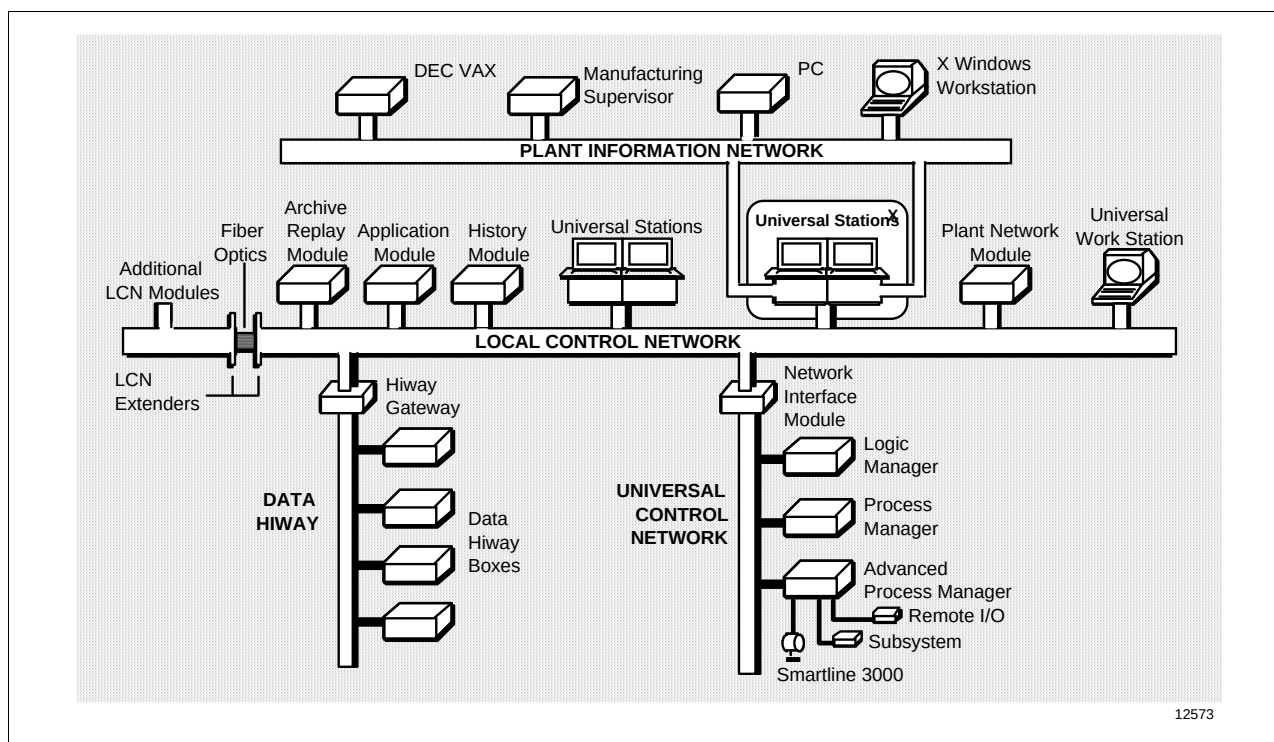
Description

The Universal Station^X (UXS) module provides a window to the process or processes that the Local Control Network controls. As illustrated in Figure 2-9, the Universal Station^X communicates with other modules in the Local Control Network, modules in remote LCNs through the Network Gateway, process devices on Universal Control Networks through the Network Interface Module, and with process devices in Data Hiways through Hiway Gateways.

The Universal Station^X (UXS) module supports:

- Open Systems applications
- Third party software
- Extended operators features
- Extended engineering functions

Figure 2-9 TPS System with Universal Station^X



Continued on next page

2.1.3 Universal Station^X (U^XS), Continued

U^XS features

A Universal Station^X supports the following features.

- A coprocessor for X Windows and the Unix operating system
 - A 19- or 21-inch high-resolution color monitor with optional touchscreen to view displayed information (maximum of two units)
 - Optional low resolution monitor
 - Keyboards for data input
 - QWERTY Operator's keyboard
 - Engineer's keyboard/U^XS Auxiliary keyboard
 - Optional mouse or trackball
 - Removable media drives for loading/unloading data
 - Cartridge drive (maximum of two) for Bernoulli cartridges
 - Digital Audio Tape (DAT) drive (maximum of two)
 - CD-ROM for UNIX Bookset
 - 400 megabyte (MB) hard drive to store X Windows and the UNIX operating system
 - Matrix Printer (optional)
 - Alarm annunciator
 - Optional Remote User LCN Access (RULA)
 - Module (node) electronics to control the devices listed above
-

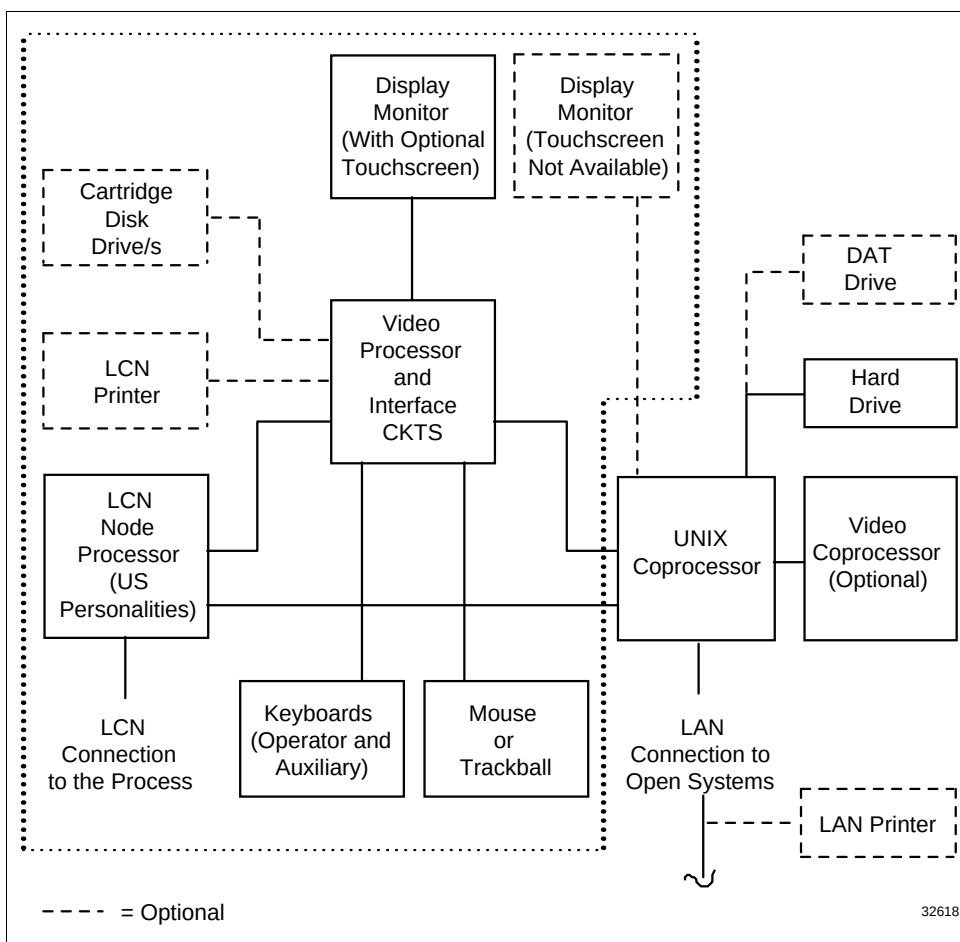
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2.1.3 Universal Station^X (UXS), Continued

Functionality

The Universal Station^X provides a simultaneous view of the process and LAN (Ethernet environments). An embedded UNIX coprocessor shares the monitor display and keyboard. However, functions are executed in their native processor. Figure 2-10 is a functional diagram of the Universal Station^X.

Figure 2-10 Universal Station^X Functional Diagram



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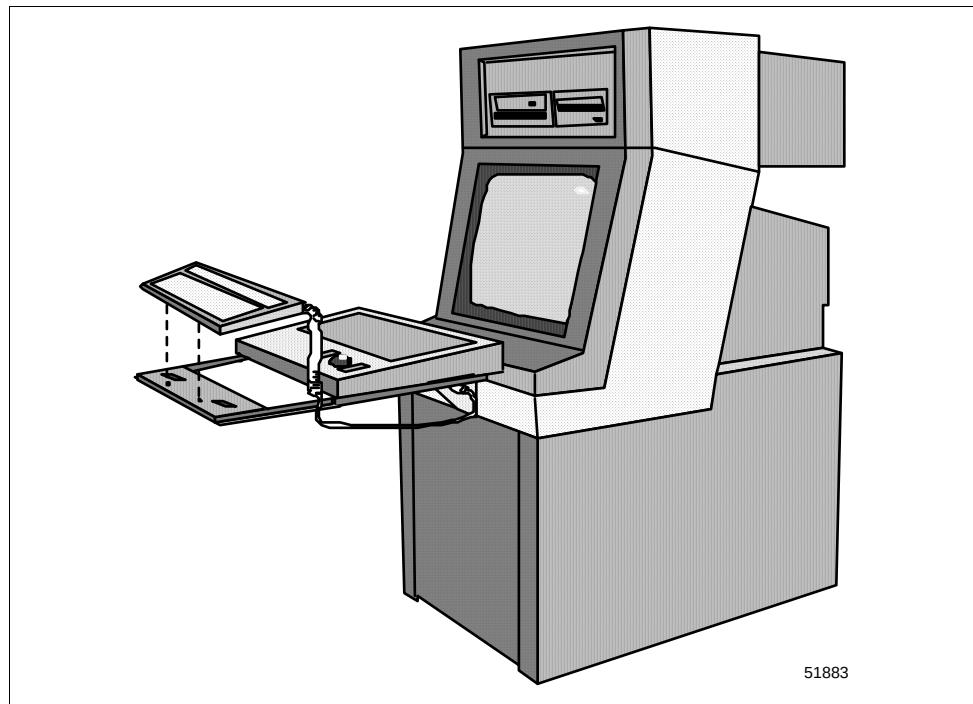
2.1.3 Universal Station^X (U^XS), Continued

Furniture styles

The Universal Station^X module is available in two furniture styles. The Classic furniture style (1983 design) is illustrated in Figure 2-11 and the Ergonomic furniture style (1993 design) is illustrated in Figure 2-12.

U^XS Classic furniture

Figure 2-11 Universal Station^X – Classic Furniture Style

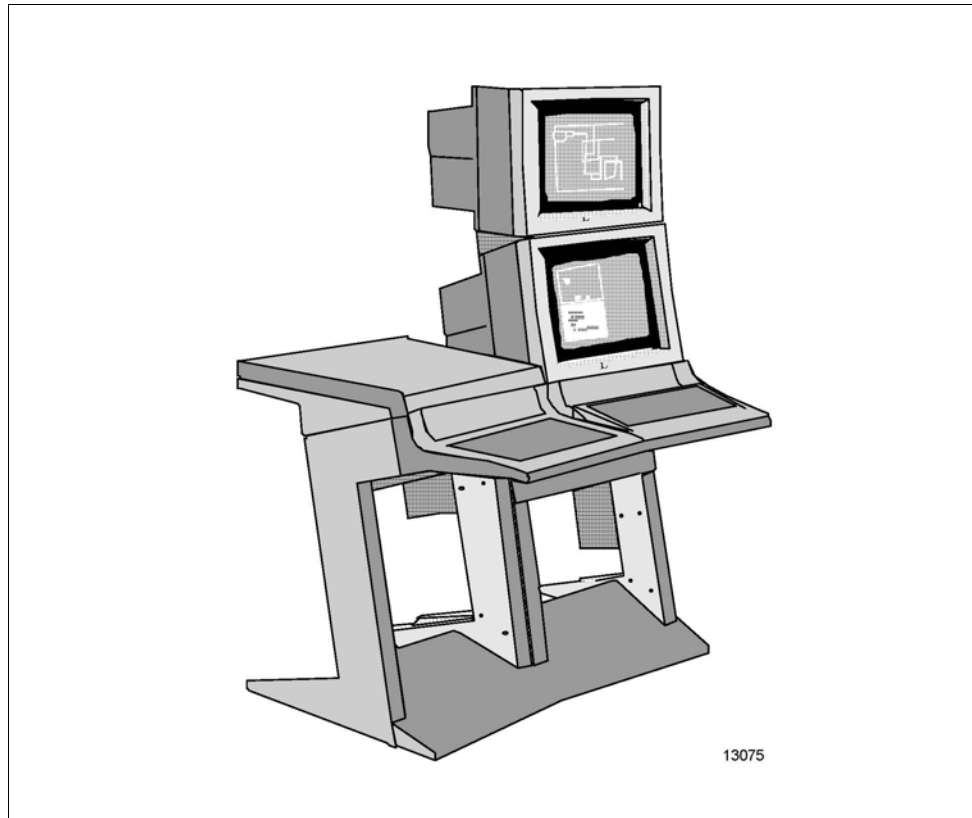


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2.1.3 Universal Station^X (UXS) , Continued

UXS Ergonomic furniture

Figure 2-12 Universal Station^X – Ergonomic Furniture Style



Module packaging

The Universal Station^X is implemented in either the Five-Slot and Dual Node Module chassis.

References

Additional information can be found in the *Universal Station^X Specification and Technical Data* manual, *Universal Station^X Service*, and the *Universal Station^X (Ergonomic) Service* manuals.

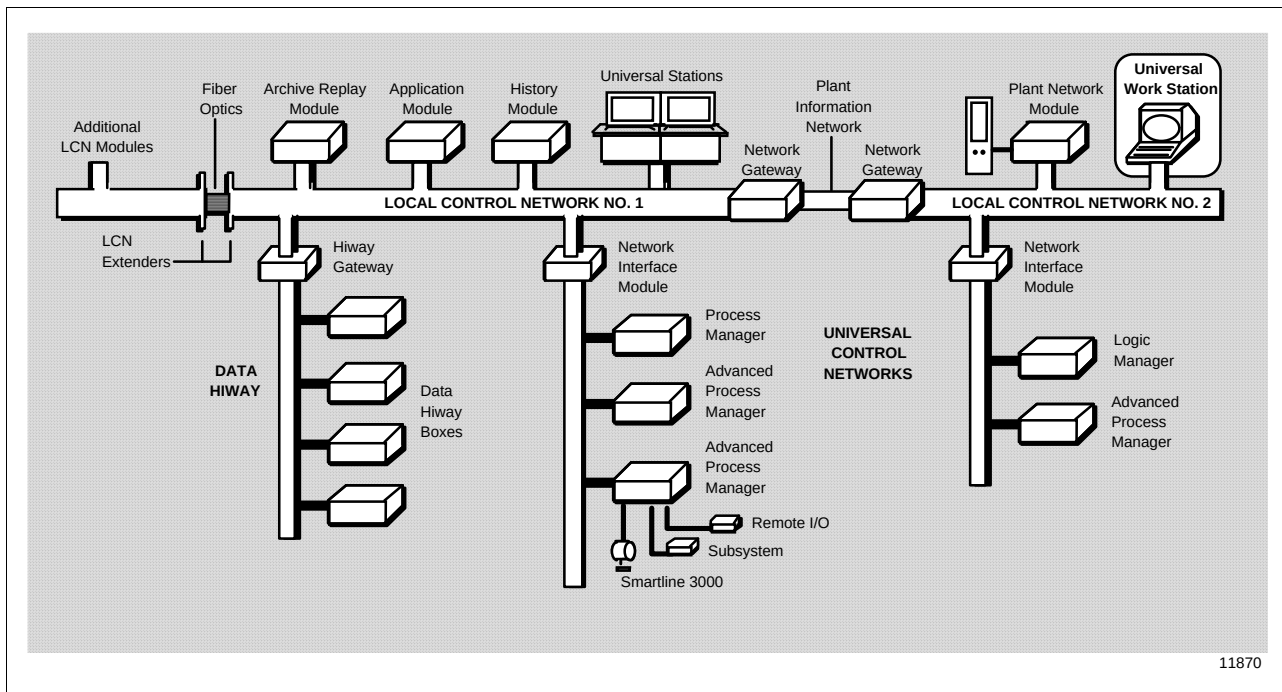
2.1.4 Universal Work Station (UWS)

Description

The Universal Work Station (UWS) module is an alternative to Universal Stations and is intended for use by engineers and supervisors. It can be located away from the control room in an office or some other site.

As illustrated in Figure 2-13, the Universal Work Station communicates with other modules in the Local Control Network, modules in remote LCNs through the Network Gateway, process devices on Universal Control Networks through the Network Interface Module, and with process devices in Data Hiways through Hiway Gateways.

Figure 2-13 TPS System with Universal Work Station



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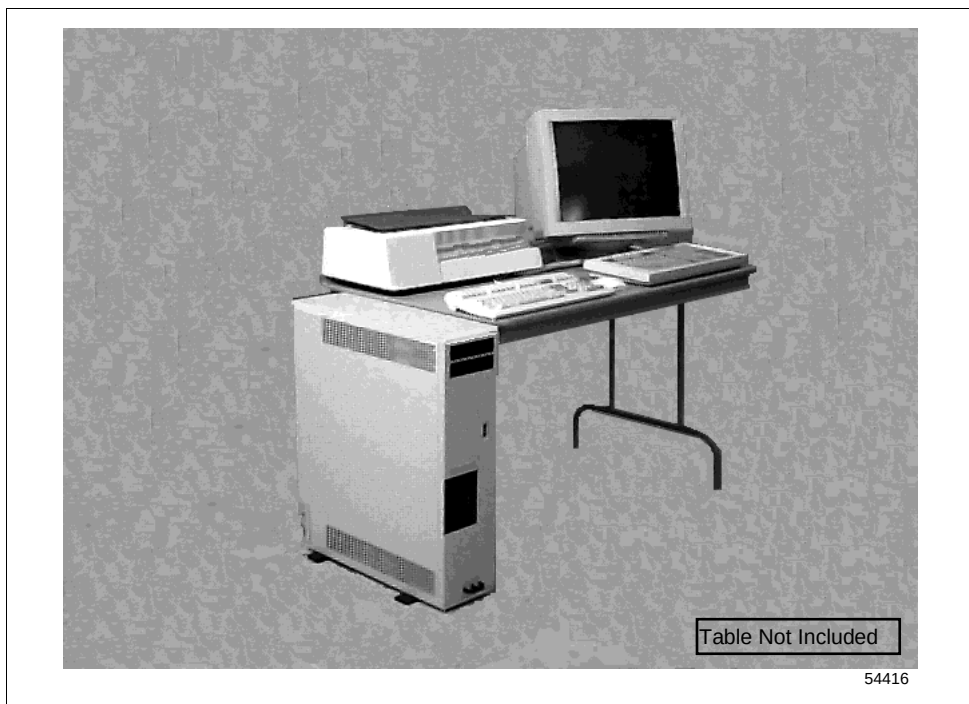
2.1.4 Universal Work Station (UWS), Continued

UWS features

The Universal Work Station has the following features. A typical Universal Work Station is illustrated in Figure 2-14.

- A color monitor
21-inch standard monitor
- Keyboards for data input
Supervisor's keyboard and/or
Engineer's keyboard
- Removable media drives for loading/unloading data
Cartridge drive (maximum of two) for Bernoulli cartridges
- An electronic tower that contains:
A Five-Slot Module
The removable media drives
Optional Remote User LCN Access (RULA)

Figure 2-14 Universal Work Station



Module packaging

The Universal Work Station is implemented in the Five-Slot Module chassis.

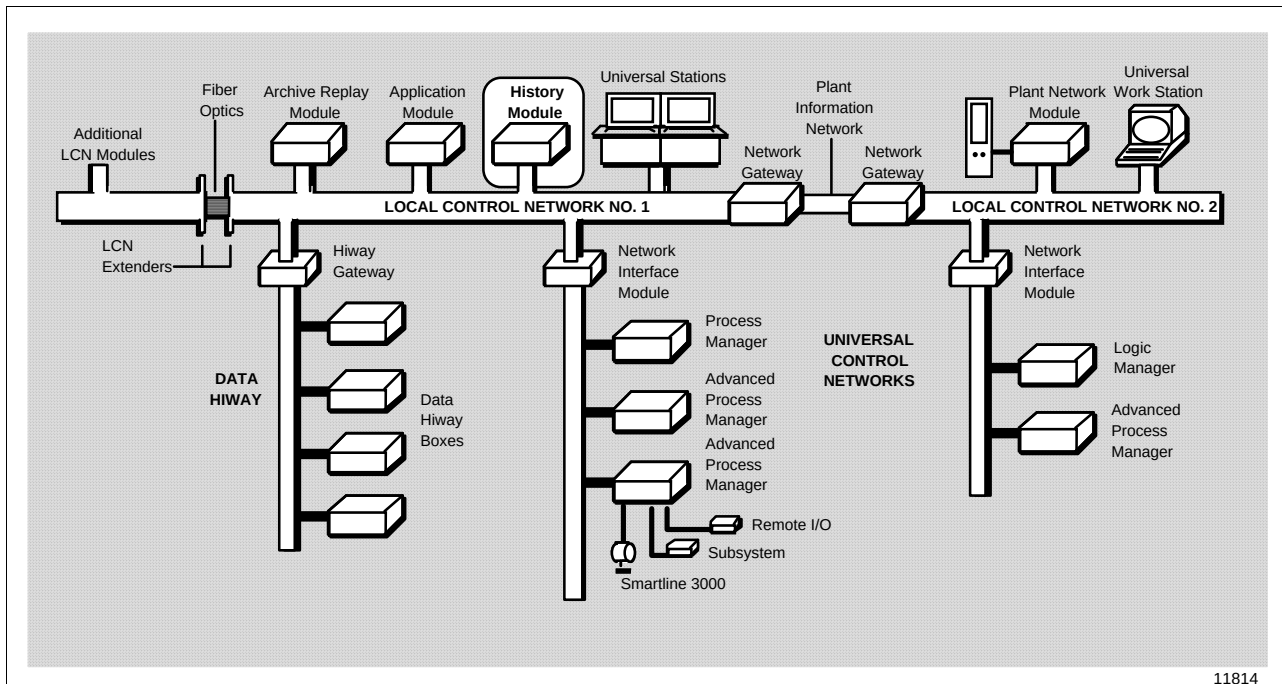
2.1.5 History Module (HM)

Overview

The History Module (HM) provides mass storage for history files, system software, and customer files for the TPS system. More than one History Module can reside in the Local Control Network.

As illustrated in Figure 2-15, the History Module can communicate with all modules (nodes) on the Local Control Network (LCN) and with process controllers on Universal Control Networks (UCNs) and Data Hiways.

Figure 2-15 TPS System with History Module



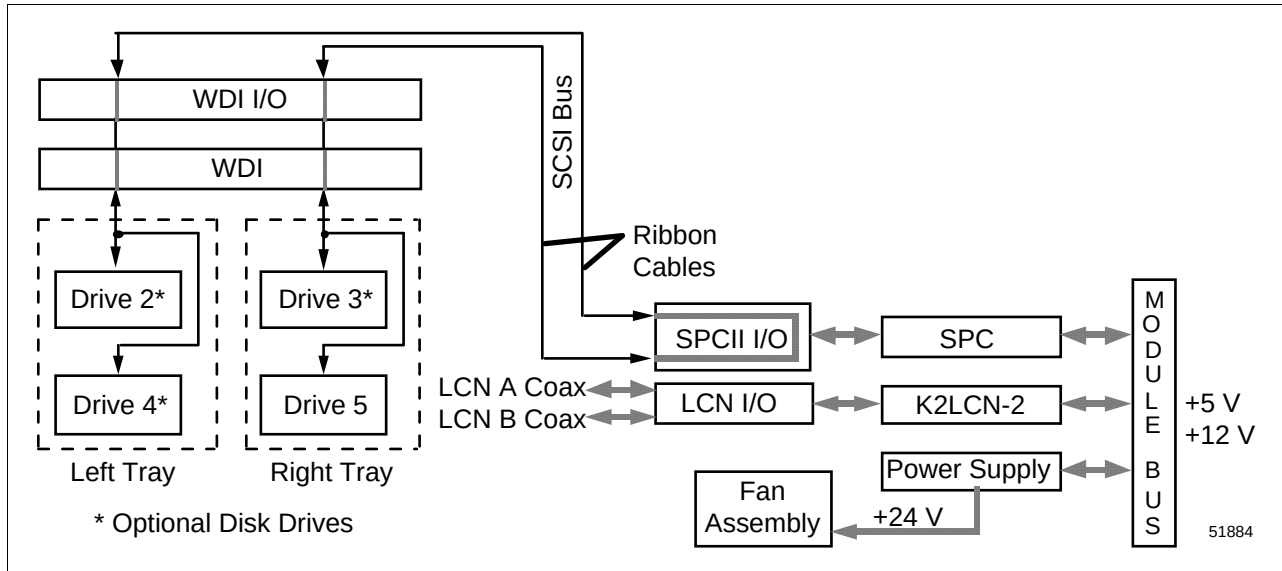
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2.1.5 History Module (HM), Continued

Block diagram

Figure 2-16 is a block diagram of the History Module.

Figure 2-16 History Module Block Diagram



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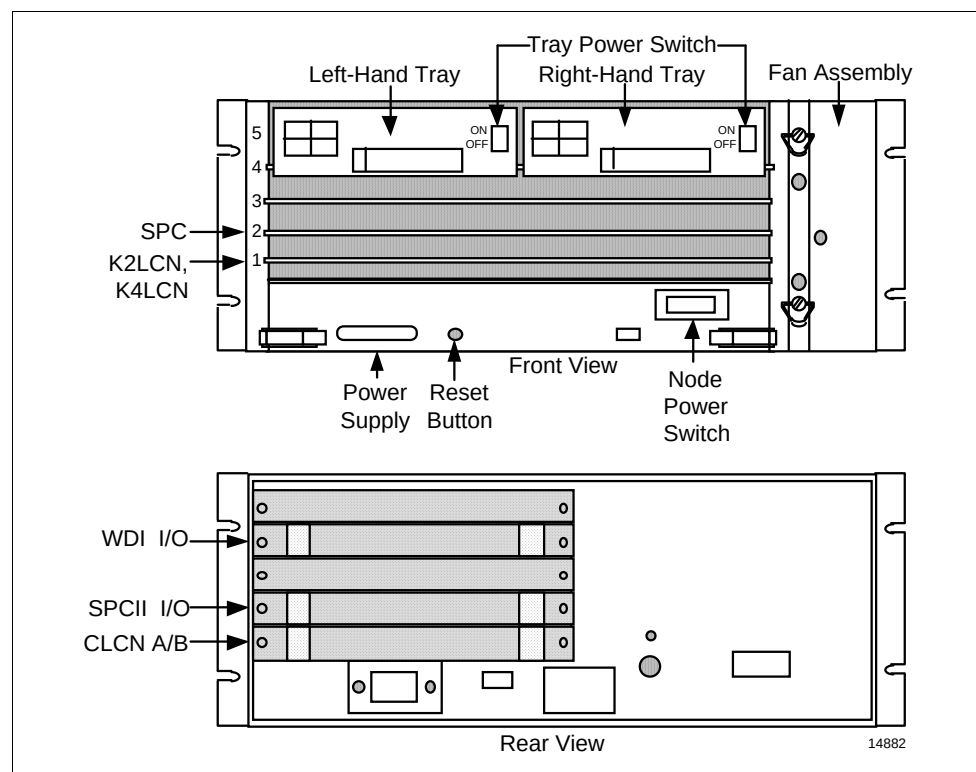
2.1.5 History Module (HM), Continued

Description

The History Module is presently implemented in a Five-Slot Module chassis. A Winchester Drive Assembly occupies the upper two board slots and accepts one or two Winchester Drive Trays. Depending upon the History Module's configuration, nonredundant or redundant single or dual 3.5-inch Winchester drives are mounted on the trays.

Figure 2-17 is an illustration of the History Module as implemented in a Five-Slot Module chassis. It is designated a Winchester Drive Assembly (WDA) History Module.

Figure 2-17 History Module (Front and Rear View)



Storage capacity

Each drive has a formatted storage capacity of approximately 875 megabytes. A dual drive configuration would have a formatted storage capacity of 1024 megabytes.

References

Additional information can be found in the *History Module Specification and Technical Data* manual and the *History Module Service* manual.

2.1.6 Archive Replay Module (ARM)

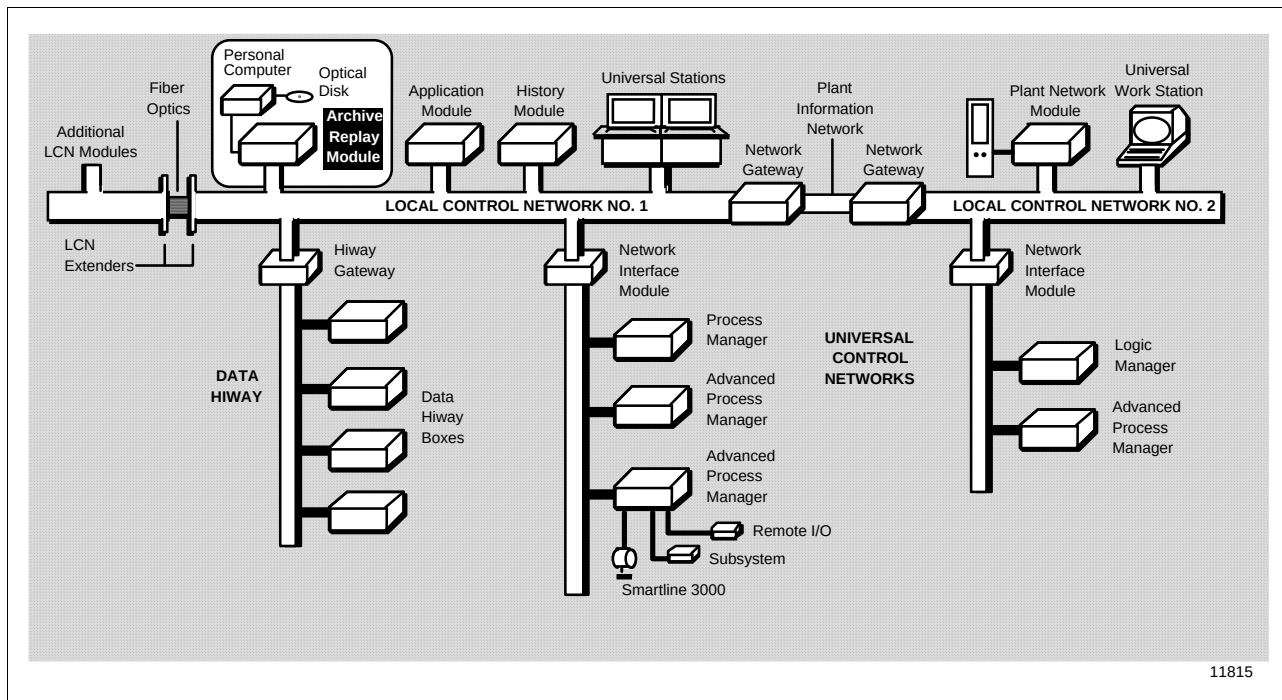
Overview

The Archive Replay Module (ARM) integrates a Personal Computer (PC) into the LCN for the specific purpose of gathering, archiving, analyzing, and replaying historical process data.

The basic components of the Archive Replay Module consist of LCN node hardware that is housed in a Dual Node Module, an IBM 486 PC, and an Optical Disk Drive that is capable of storing 940 megabytes of historical data.

As illustrated in Figure 2-18, the Archive Relay Module resides in the Local Control Network. Its two primary components are LCN node hardware, that is implemented as a Computer Gateway, and a UNIX-based IBM Personal Computer. Historical data is ultimately stored on an optical disk by the Personal Computer.

Figure 2-18 TPS System with Archive Replay Module



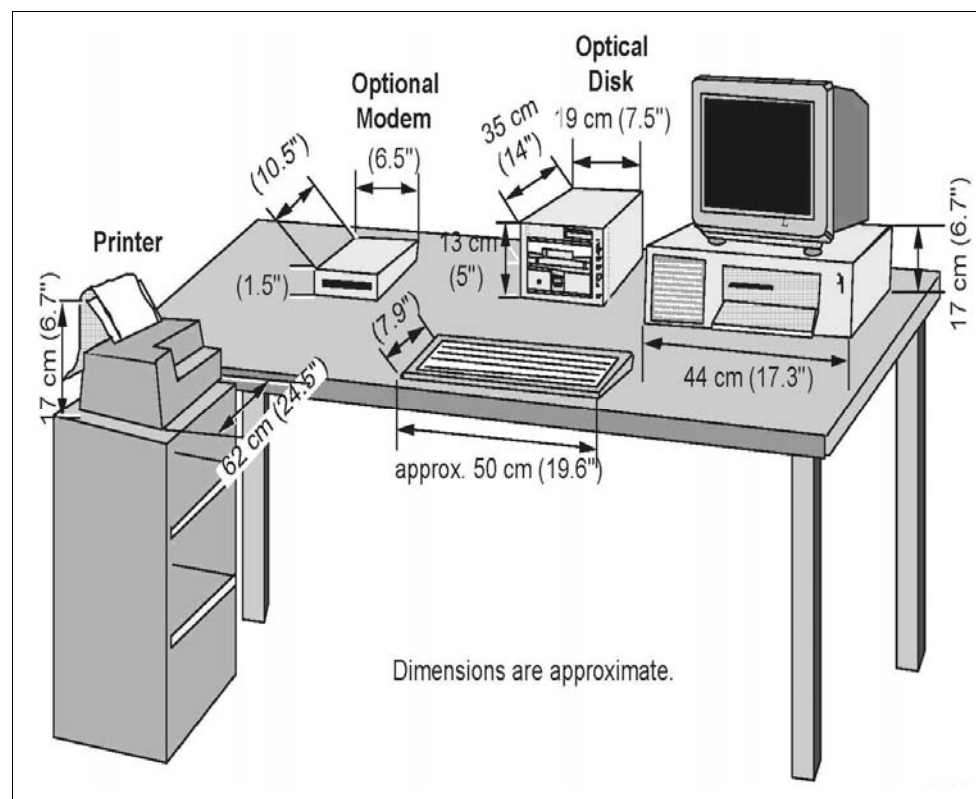
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2.1.6 Archive Replay Module (ARM), Continued

Functional description

Data that is archived using the Archive Replay Module can be retrieved for display and printout at both the TPS Universal Station and at the local Personal Computer monitor and printer. In addition, a remote IBM terminal can access this data and use available software packages, such as spreadsheet programs, to process and present the data to meet almost any specific user requirement. Figure 2-19 is an illustration of an Archive Replay Module application.

Figure 2-19 Archive Replay Module Application



Module packaging

The Archive Replay Module is implemented in a Five-Slot Module chassis.

References

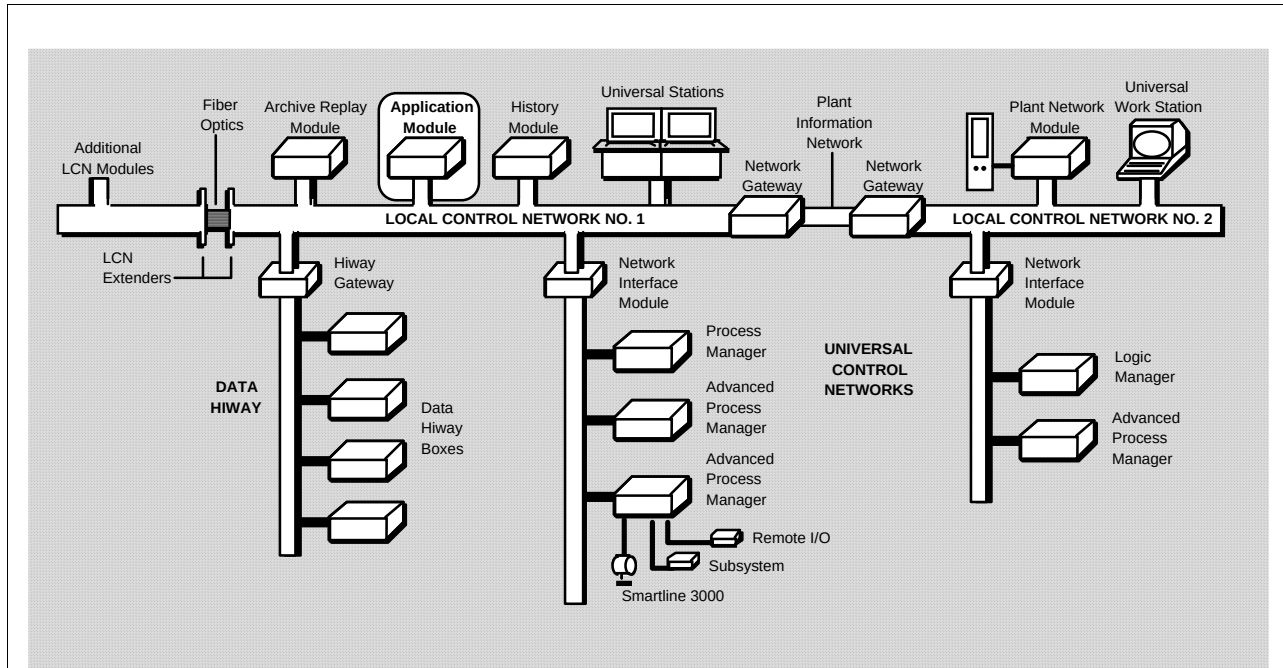
Additional information can be found in the *Archive Replay Module Specification and Technical Data* manual and the *Archive Replay Module Planning, Installation, and Service* manual.

2.1.7 Application Module (AM)

Overview

As illustrated in Figure 2-20, the Application Module (AM) communicates with other modules on the same or other Local Control Networks (LCNs), and process controllers on Universal Control Networks (UCNs) and Data Hiways.

Figure 2-20 TPS System with an Application Module



Redundant AMs

Two Applications Modules can be joined together to form a redundant pair. The joining is accomplished by adding an AMR board to each module with a private communications bus cable connecting the two AMR boards together. This private bus allows the secondary AM node to track the primary AM. This allows the secondary to continue the applications process should the primary fail or be taken out of service.

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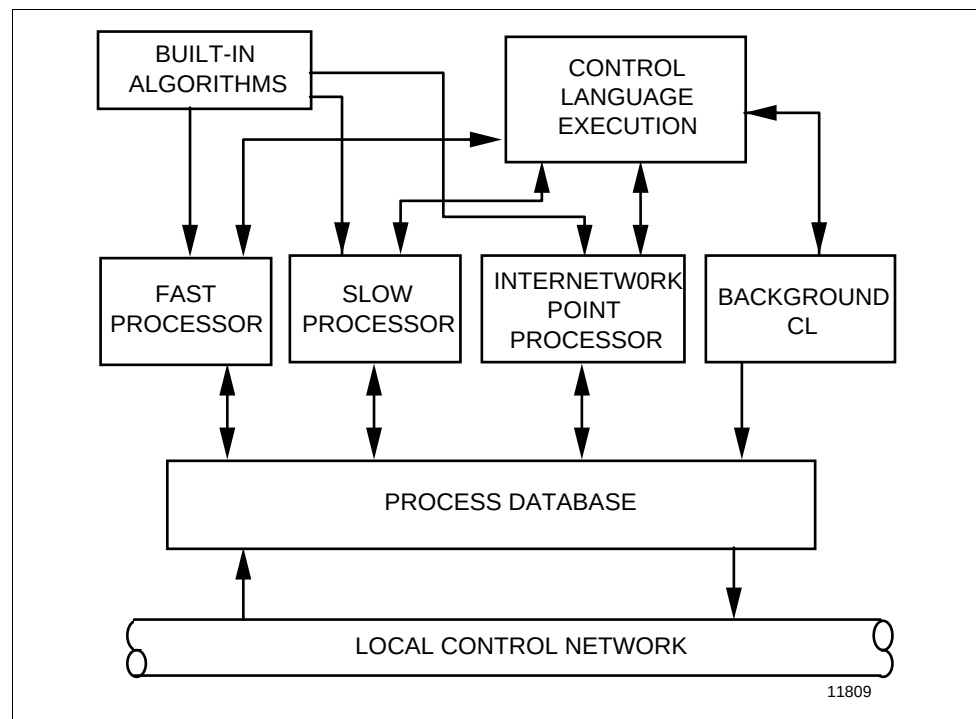
2.1.7 Application Module (AM), Continued

Functionality

Because of its position in the system architecture, the Application Module can accept inputs from multiple process controllers as well as other modules on different Local Control Networks.

The Application Module can also provide control outputs to control elements in the process or to other data points in process control modules on other Local Control Networks. Refer to Figure 2-21.

Figure 2-21 Application Module Functions



Module packaging

The Application Module is implemented in either the Dual Node or Five-Slot Module chassis.

Redundant AM nodes are implemented in separate Five-Slot Modules.

Reference

Additional information can be found in the *Application Module Specification and Technical Data* manual.

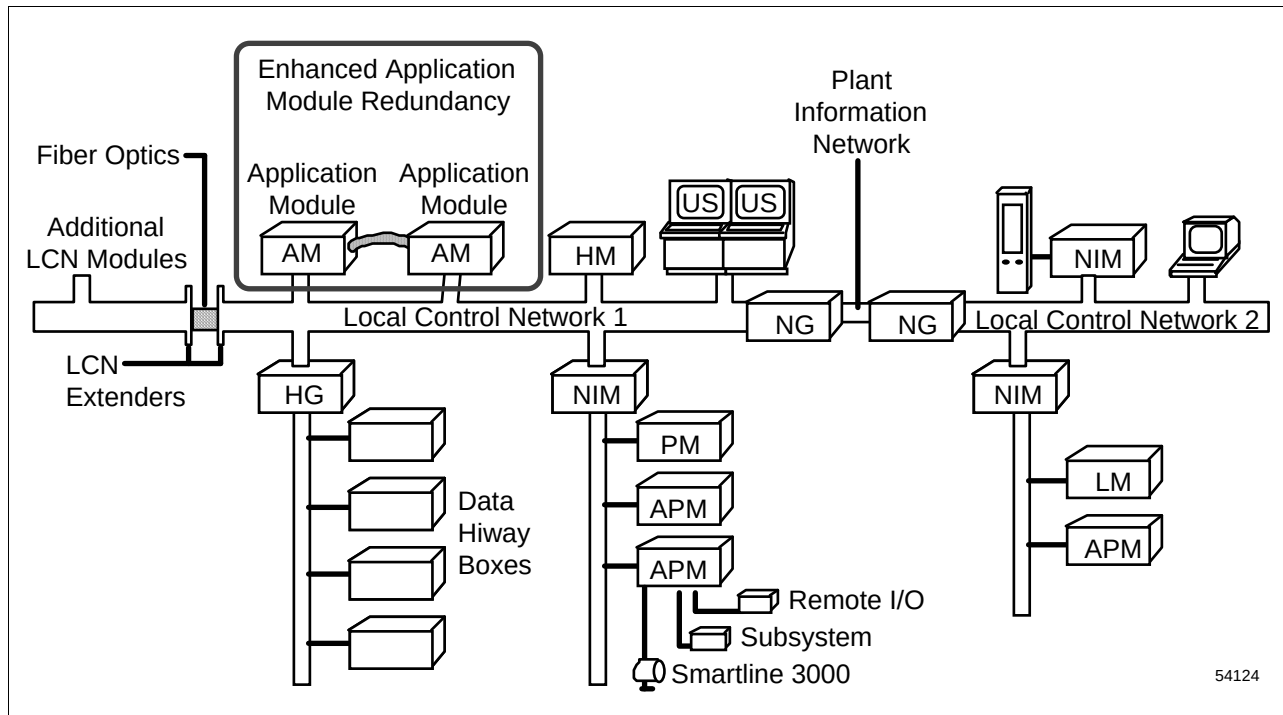
2.1.8 Enhanced Redundant Applications Module

Overview

The enhanced Redundant Applications Module couples 68040 performance with redundancy.

- Each node has a 68040 performance K4LCN with processing, LCN communications and node memory.
- Each K4LCN can have 4, 8 or 16 MB of memory.
- Each node has an Enhances Applications Module Redundancy (EAMR)board.
- Each module interfaces the LCN coax independently.
- Each node is located in a different module (Five-Slot or Dual Node)

Figure 2-22 TPS System with Enhanced Application Redundancy Module



Continued on next page

2.1.8 Enhanced Redundant Applications Module, Continued

The redundant AM nodes can be placed in any combination of node within two vertically adjacent Dual Node module. One example follows.

Figure 2-23 Example of EAMR Placement in Two Adjacent Modules (Front View)

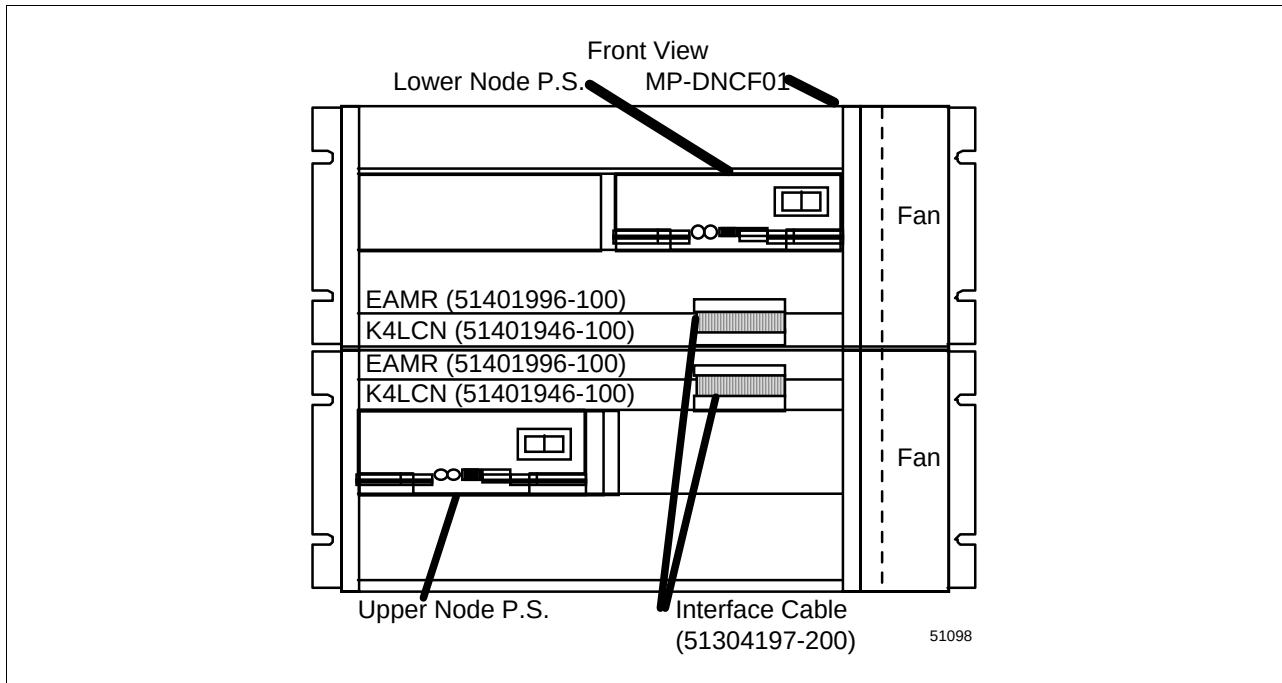
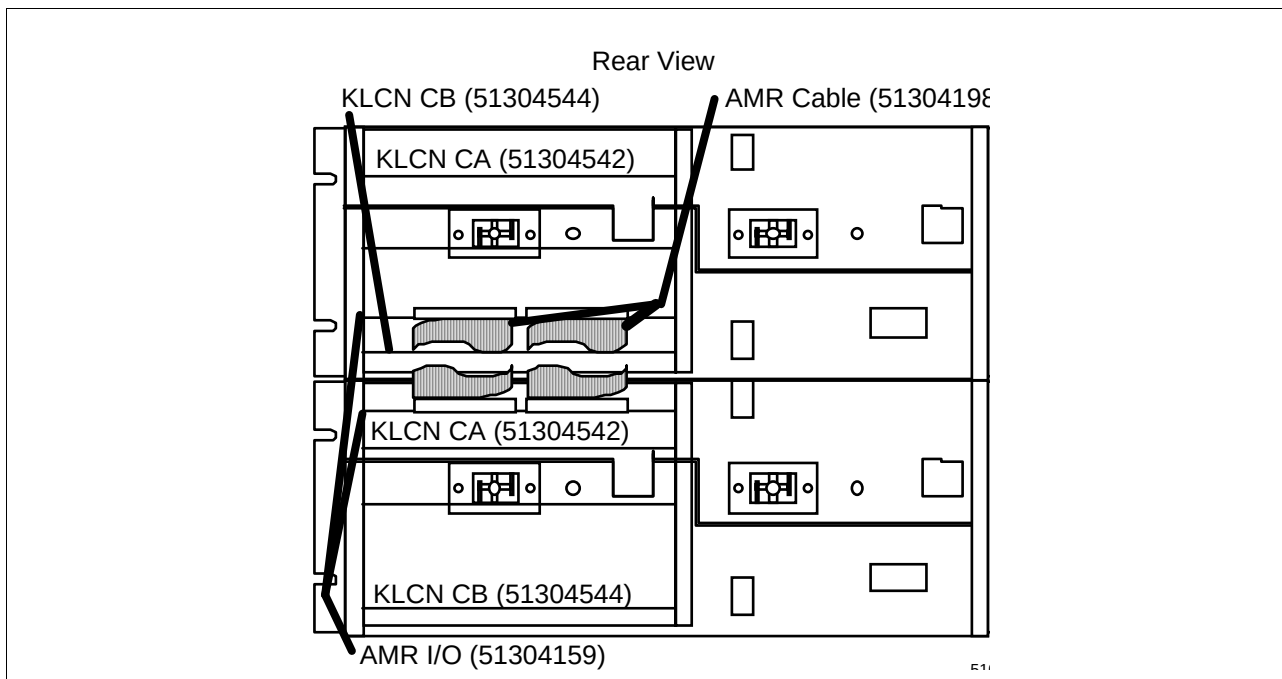


Figure 2-24 Example of EAMR Placement in Two Adjacent Modules (Rear View)



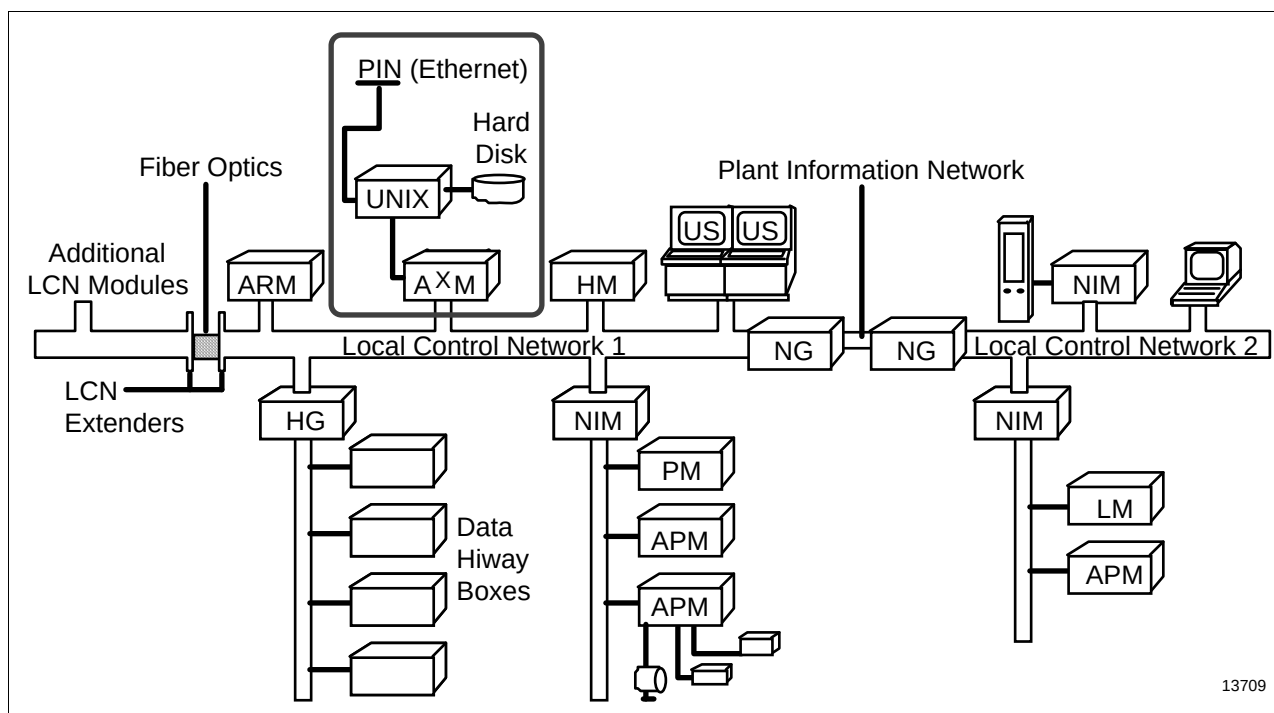
2.1.9 Application Module^X (A^XM)

Introduction

The Application Module^X (A^XM) provides an interface between the Local Control Network (LCN) and the Plant Information Network (PIN). It supports the X layer functionality. An operator within the PIN can access any LCN, UCN, or Data Hiway node through an X Window terminal.

As illustrated in Figure 2-25, the Application Module^X (A^XM) communicates with other modules on the same or other Local Control Networks (LCNs), and process controllers on Universal Control Networks (UCNs) and Data Hiways.

Figure 2-25 TPS System with an Application Module^X



Continued on next page

2.1.9 Application Module^X (A^XM), Continued

Functionality

The Application Module^X (A^XM) provides an interface between the Local Control Network (LCN) and an X Window terminal operating in the Plant Information Network (PIN). The heart of the A^XM is the PA-RISC coprocessor which allows the HP-UX UNIX operating system to function. It provides a window to the process from the PIN.

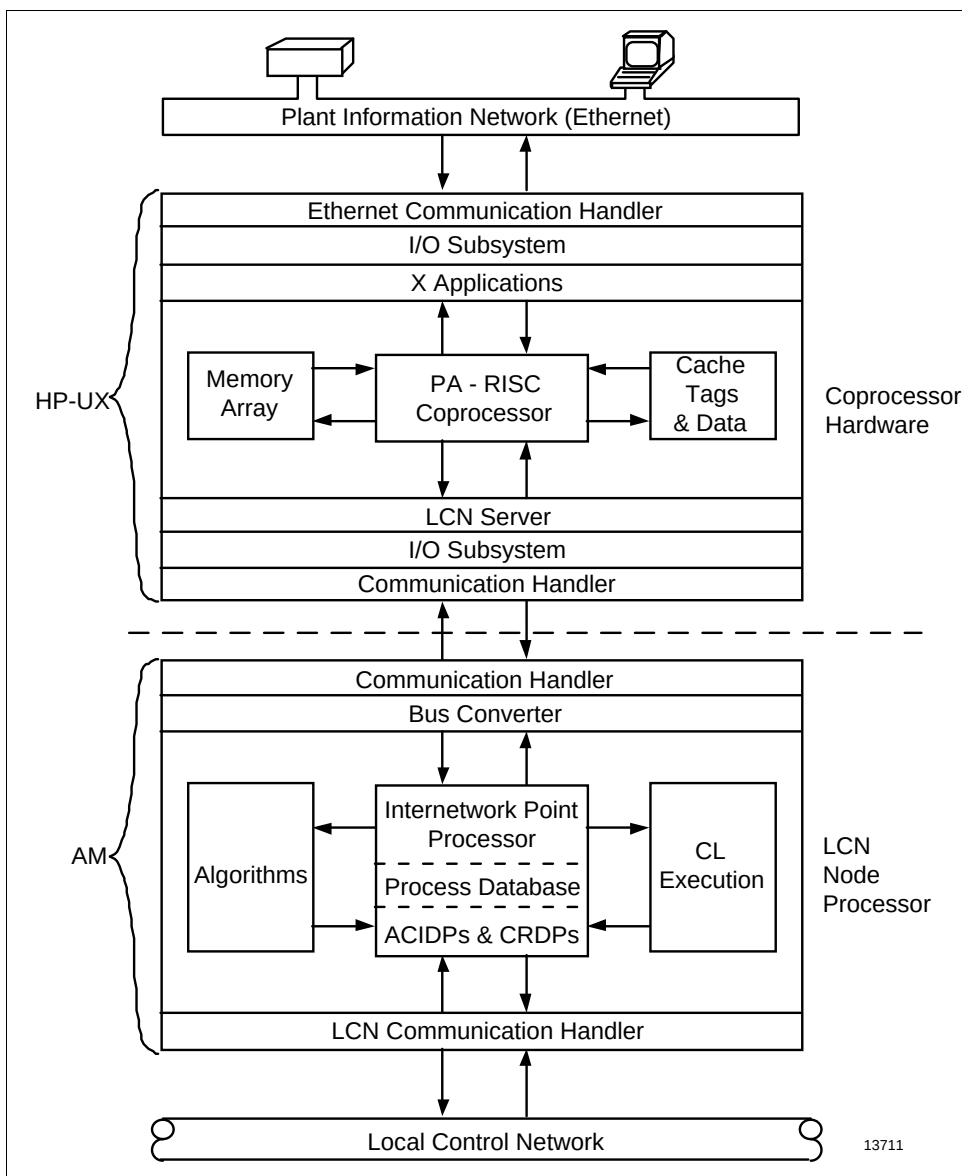
As shown in Figure 2-26, the A^XM provides the functionality of a standard Application Module (AM) coupled with an HP-UX-based coprocessor. The A^XM resides on the LCN and has an AM front end. The A^XM is a tightly coupled integration of workstation technology with current LCN technology. This integration provides an operating environment for a UNIX-compliant interface which dramatically increases the power, performance, and flexibility of the AM control node. It allows state-of-the-art commercial software applications to execute in the coprocessor while maintaining security for the control system.

Continued on next page

2.1.9 Application Module^X (A^XM), Continued

Functionality,
continued

Figure 2-26 A^XM Interface Relationships and Functionality



Continued on next page

2.1.9 Application Module^X (A^XM), Continued

Module packaging

The Application Module^X can be implemented in either the Five-Slot or Ten-Slot module chassis. Currently, redundancy is not supported in the A^XM.

References

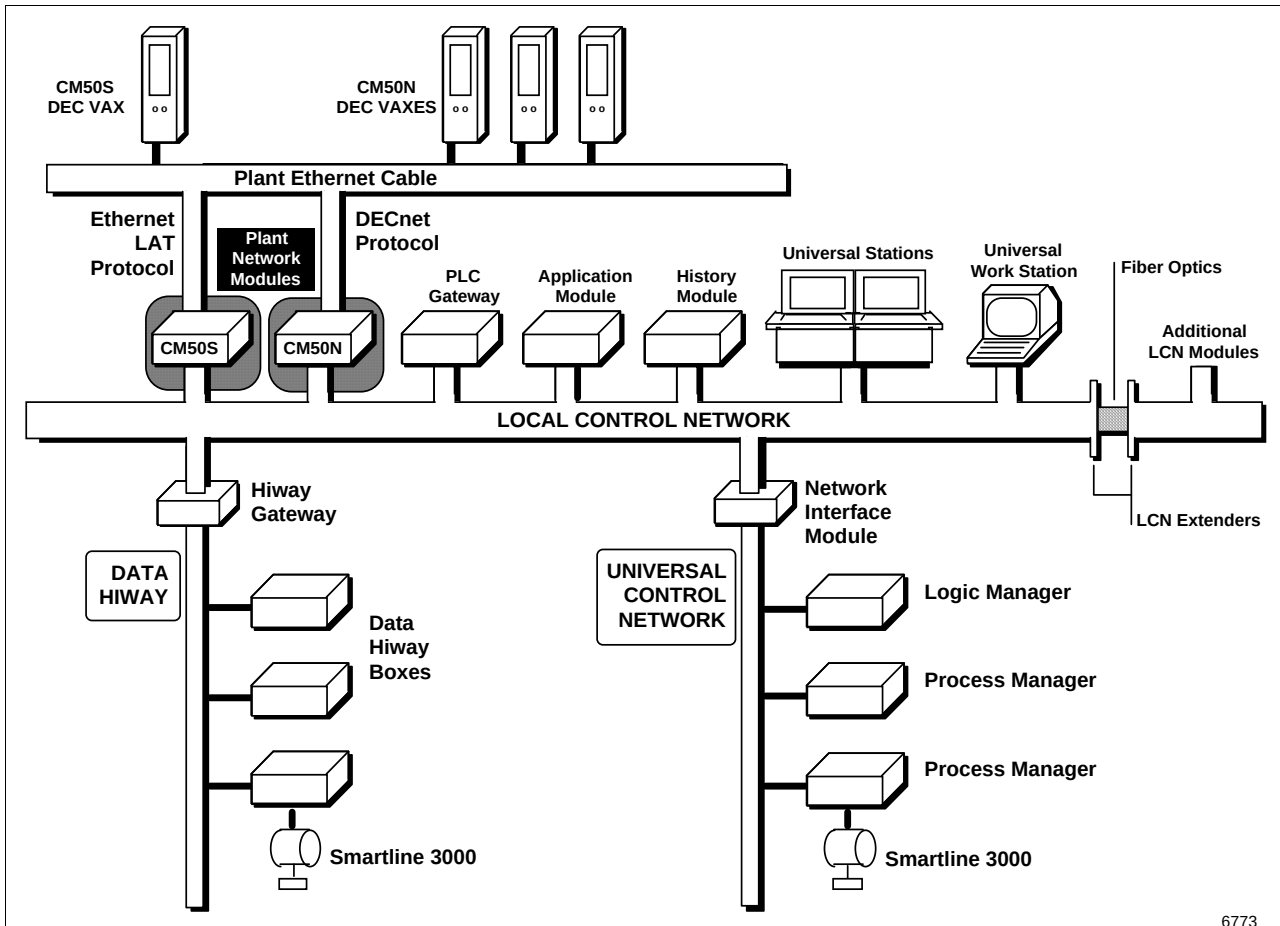
Additional information can be found in the *Application Module^X User Guide* manual and the *Application Module^X System Administration* manual.

2.1.10 Plant Network Module (PLNM)

Introduction

The Plant Network Module (PLNM) provides an interface from the Local Control Network (LCN) to DEC VAX computers. As illustrated in Figure 2-27, the Plant Network Module is a node on the Local Control Network that allows one or more DEC VAX computers to communicate with all modules (nodes) on the LCN, associated Universal Control Networks, and Data Hiways.

Figure 2-27 TPS System with Plant Network Modules



CM50S and CM50N software packages

Two software packages are available for use with the Plant Network Module. One package is the enhanced edition of CM50S which uses the Ethernet Local Area Transport (LAT) communication channel between a VAX computing module and one to four PLNMs. The other package, known as CM50N, communicates with multiple VAX systems over a DECnet network. Figures 2-28 and 2-29 illustrate the CM50S and CM50N network applications, respectively.

Continued on next page

2.1.10 Plant Network Module (PLNM), Continued

Typical Networks

Figure 2-28 Plant Network Module to VAX Interface with CM50S

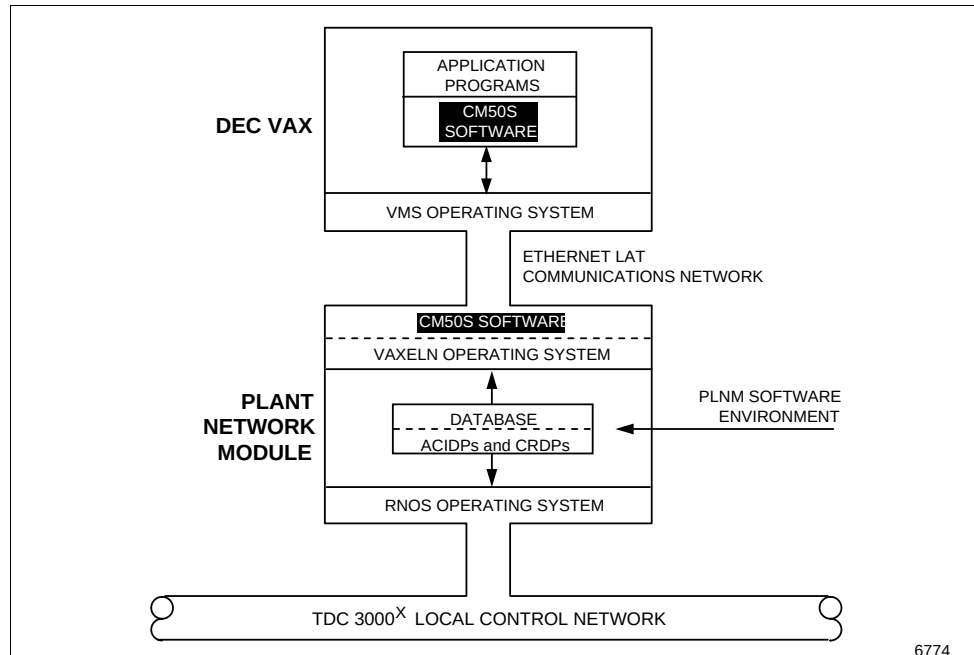
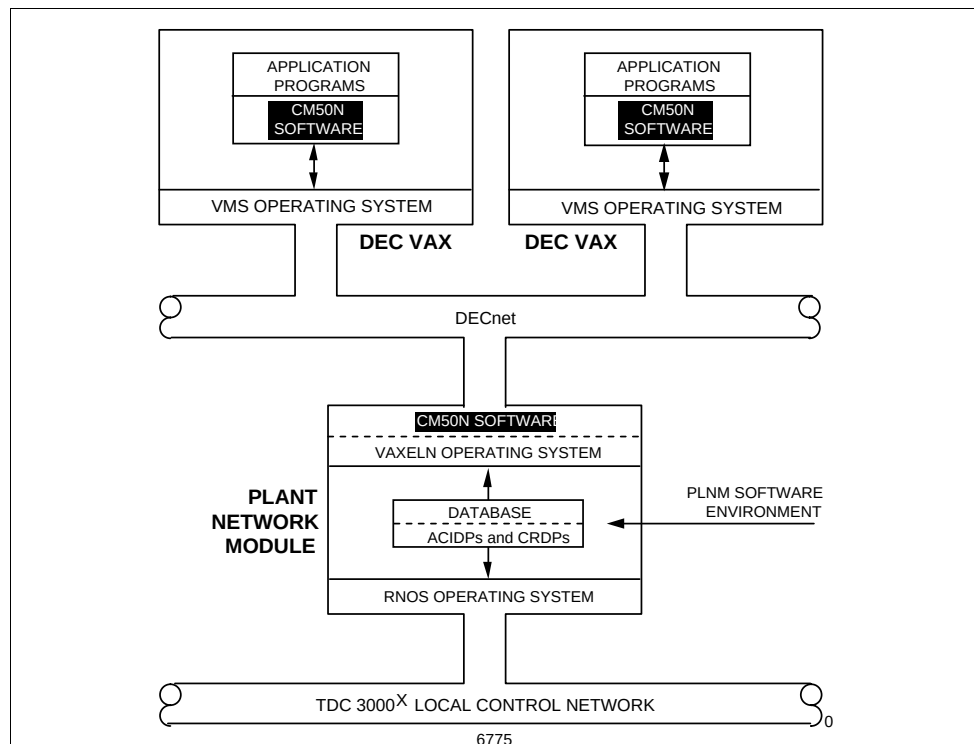


Figure 2-29 Plant Network Module to VAX Interface with CM50N



Continued on next page

2.1.10 Plant Network Module (PLNM), Continued

Module packaging

The Plant Network Module is implemented in either the Five-Slot or Dual Node Module chassis.

References

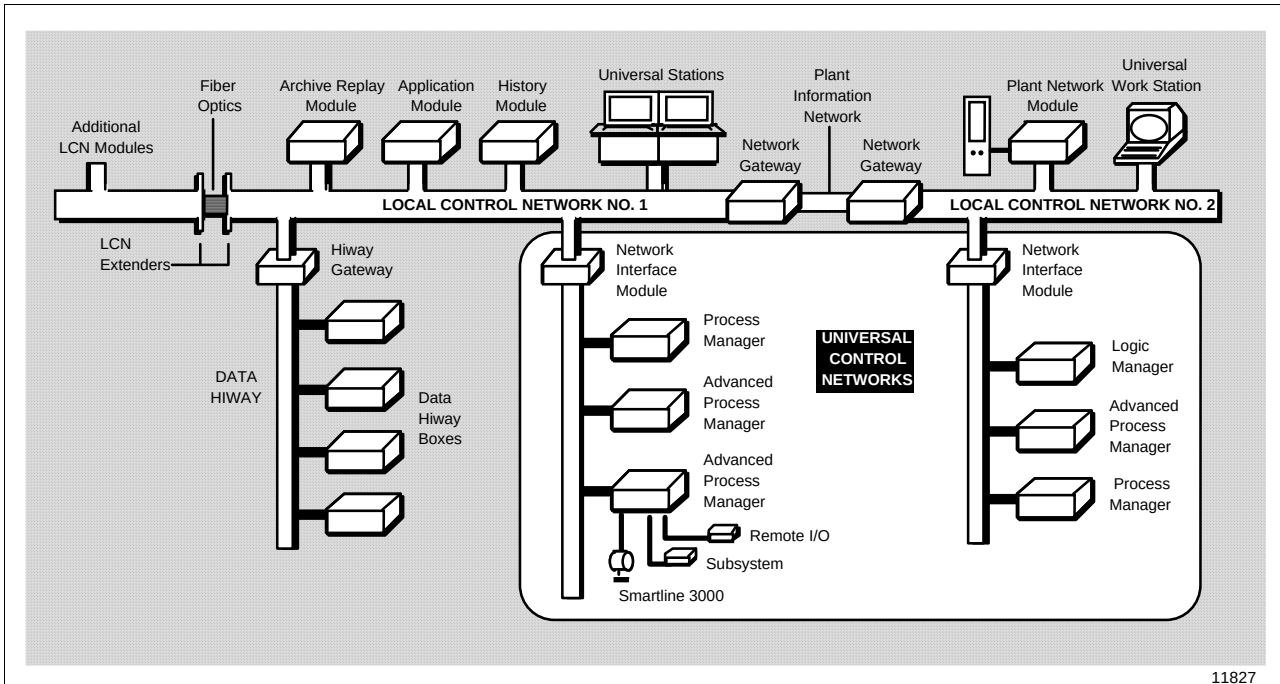
Additional information can be found in the *Plant Network Module Specification and Technical Data* manual and the *Plant Network Module Site Planning and Installation* manual.

2.1.11 Network Interface Module (NIM)

Introduction

As illustrated in Figure 2-30, the Network Interface Module (NIM) provides the interface between the Local Control Network (LCN) and the Universal Control Network (UCN). The UCN support process controllers such as the Process Manager (PM), Advanced Process Manager (APM), and the Logic Manager (LM).

Figure 2-30 TPS System with Network Interface Modules



11827

Redundancy

The Network Interface Module is capable of redundant applications.

Information exchange

Information concerning the process, such as status and configuration, is transferred through the Network Interface Module from the UCN to the LCN.

The Network Interface Module provides the protocol conversion and buffering necessary to efficiently exchange information between the UCN controllers and the LCN the LCN modules, such as Universal Stations, History Modules, Application Modules, and Plant Network Modules.

LCN time

The Network Interface Module broadcasts the LCN time to all UCN controllers, thereby synchronizing the UCN time with the LCN time.

Continued on next page

2.1.11 Network Interface Module (NIM), Continued

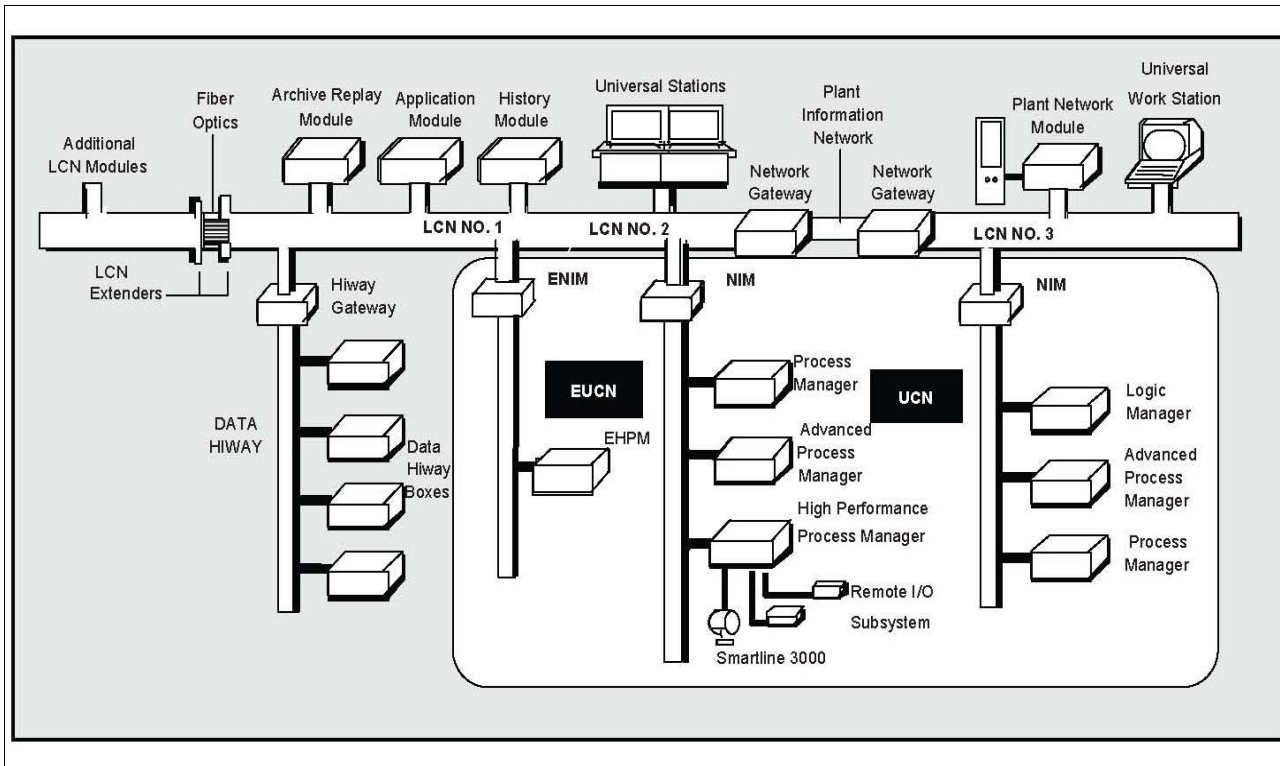
Module packaging	The Network Interface Module is implemented in either a Five-Slot or a Dual Node Module chassis.
Reference	Additional information can be found in the <i>Universal Control Network Specification and Technical Data</i> manual, <i>Universal Control Network Site Planning</i> manual, <i>Universal Control Network Guidelines</i> manual, and the <i>Universal Control Network Installation</i> manual.

2.1.12 Enhanced Network Interface Module (ENIM)

Introduction

As illustrated in Figure 2-31, the Enhanced Network Interface Module (ENIM) provides the interface between the Local Control Network (LCN) and the Enhanced High-Performance Process Manager (EHPM).

Figure 2-31 TPS System with Enhanced Network Interface Module



Redundancy

The Enhanced Network Interface Module is capable of redundant applications.

Information exchange

Information concerning the process, such as status and configuration, is transferred through ENIM from the EUCN to the LCN and EHPM controllers.

ENIM provides the protocol conversion and buffering necessary to efficiently exchange information between the EUCN controllers and the LCN modules, such as Universal Stations, History Modules, Application Modules, and Plant Network Modules.

LCN time

ENIM broadcasts the LCN time to all EUCN controllers, thereby synchronizing the EUCN time with the LCN time.

Continued on next page

2.1.12 Enhanced Network Interface Module (ENIM), Continued

Module packaging

You can implement ENIM either in a Dual Node or in a Five-Slot Module chassis.

Reference

Additional information can be found in the *Enhanced Universal Control Network Specification* manual, *Universal Control Network Site Planning* manual, *Universal Control Network Guidelines* manual, *Universal Control Network Installation* manual, *System Components Configuration Utility User's Guide*, and *TPS System Overview* manual.

2.1.13 Programmable Logic Controller Gateway (PLCG)

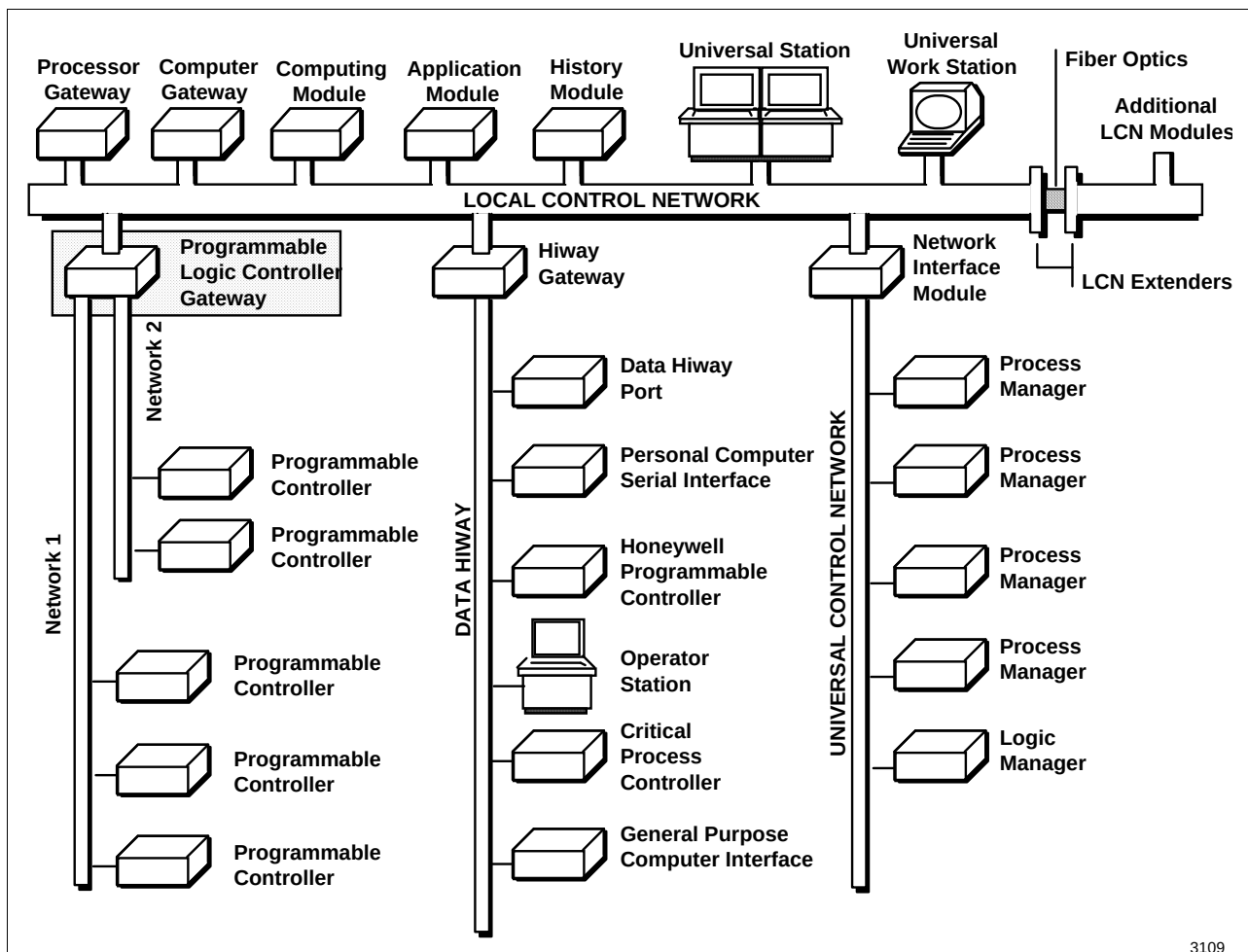
Introduction

The Programmable Logic Controller Gateway module (PLCG) provides a link between the Local Control Network (LCN) and programmable controllers that use Allen-Bradley or Modbus subsystem protocols.

Two ports

As illustrated in Figure 2-31, the Programmable Logic Controller Gateway module communicates with other modules (nodes) on the LCN and with programmable controllers, which are connected to one of the PLCG's two EIA-232 ports.

Figure 2-31 TPS System with PLC Gateway Module



3109

Allen-Bradley or Modbus protocol

The Programmable Logic Controller Gateway module provides the transition from the transmission technique and protocol of the LCN to the transmission techniques of the Allen-Bradley or Modbus protocols.

Continued on next page

2.1.13 Programmable Logic Controller Gateway (PLCG),

Continued

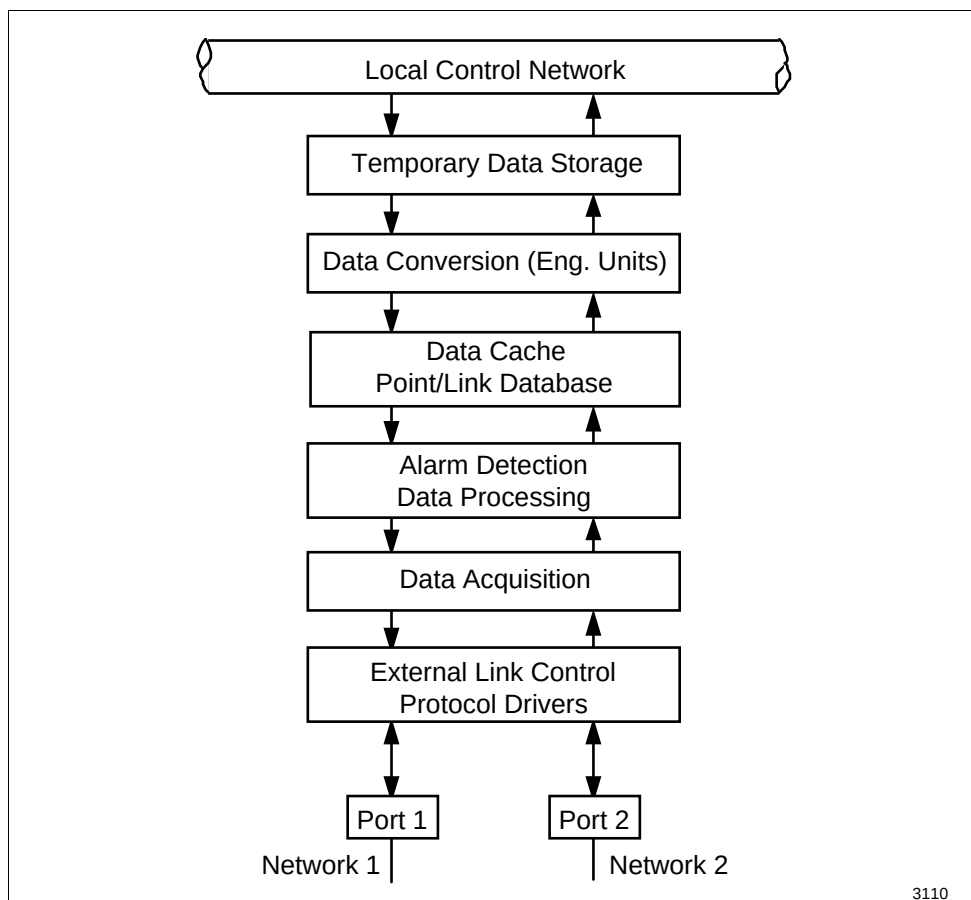
Port controllers

Each of the ports on the Programmable Logic Controller Gateway module serves as an independent programmable controller subsystem network. The two ports are guaranteed to support up to 16 programmable controllers up to 64 can be addressed.

Functionality

The Programmable Logic Controller Gateway's functionality is illustrated in Figure 2-32.

Figure 2-32 Programmable Logic Controller Gateway Functions



Redundancy support

The PLCG can operate as a single node in the Local Control Network or it can operate part of a node pair application. While one PLCG is operating, the other serves as the redundant partner with an exact up to date copy of the database. The redundant partner assumes operational control should the operating partner fail or be removed from operation.

Continued on next page

2.1.13 Programmable Logic Controller Gateway (PLCG),

Continued

Packaging

The Programmable Logic Controller Gateway module is implemented in either a Five-Slot or a Dual Node Module chassis.

References

Additional information can be found in the *PLC Gateway Specification and Technical Data* manual and the *PLC Gateway Planning, Installation, and Service* manual.

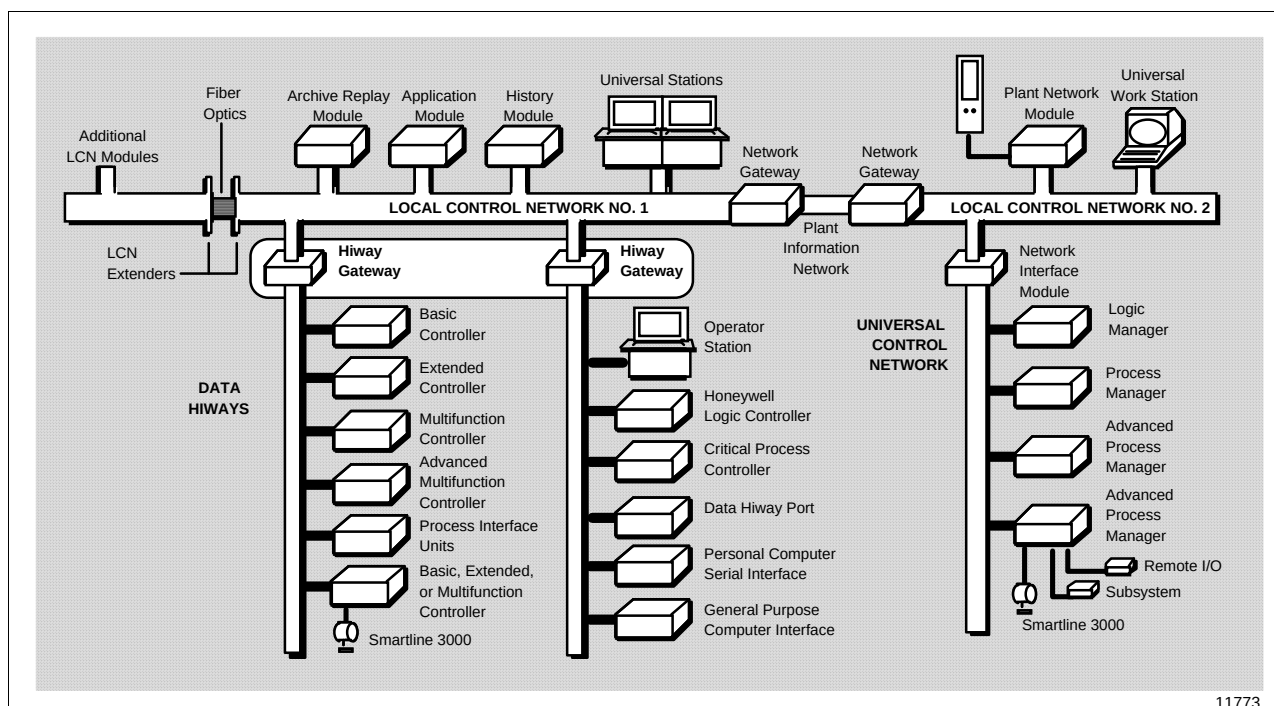
2.1.14 Hiway Gateway (HG)

Introduction

The Hiway Gateway module (HG) provides the interface between the Local Control Network (LCN) and a Data Hiway. As illustrated in Figure 2-33, the Hiway Gateway communicates with other modules (nodes) on the Local Control Network and with process controllers on the Data Hiway.

The Hiway Gateway module provides the transition from the transmission technique and protocol of the Local Control Network to the transmission technique and protocol of the Data Hiway.

Figure 2-33 TPS System with Hiway Gateway Modules



One Hiway Gateway for each Data Hiway

One Hiway Gateway module is required for each Data Hiway that is connected to a Local Control Network. Up to 20 Data Hiway modules can be connected to a Local Control Network.

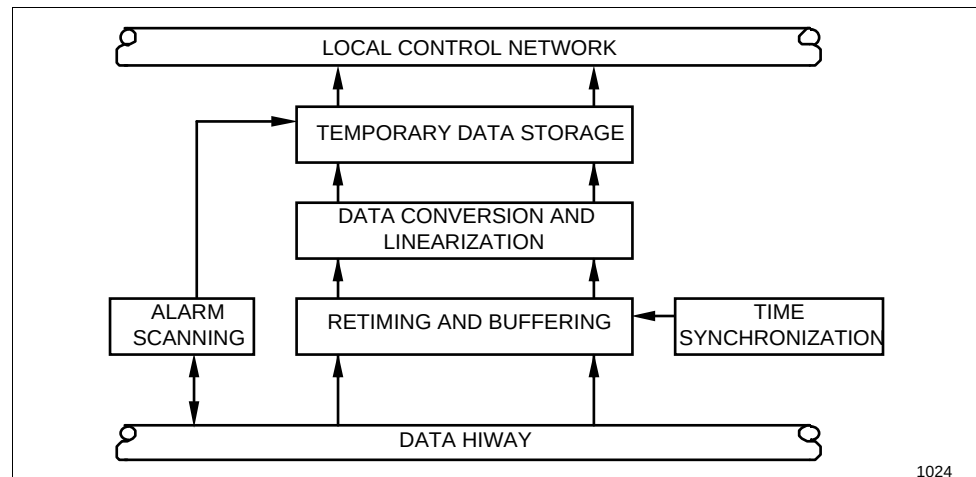
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2.1.14 Hiway Gateway (HG), Continued

Functionality

The Hiway Gateway module provides the data conversion, buffering, and sequencing necessary to provide an efficient interchange of information between the Local Control Network and Data Hiway. Although the Local Control Network and Data Hiway use similar dual coaxial cables and serial-bit communication, they employ different communication protocols because of the type of information they carry. This conversion of protocols and communication speeds is necessary to link the short-distance, high-speed communication that is typical of LCN-based modules with the widely dispersed, short-message, and less rapid communication that is characteristic of Data Hiway controllers. Refer to Figure 2-34.

Figure 2-34 Hiway Gateway Functions



Redundancy

The TPS full redundancy option is usually recommended for the Hiway Gateway, as the backup Hiway Gateway ensures the security of critical information and control in the event there is a primary failure. Redundant HG nodes should be placed in separate modules.

Module packaging

The Hiway Gateway is implemented in both Five-Slot and Dual Node Modules. Since there no longer is a requirement for logic ground to be connected to MRG, both node positions in a Dual Node Module can be used, but not for both nodes of a redundant pair.

References

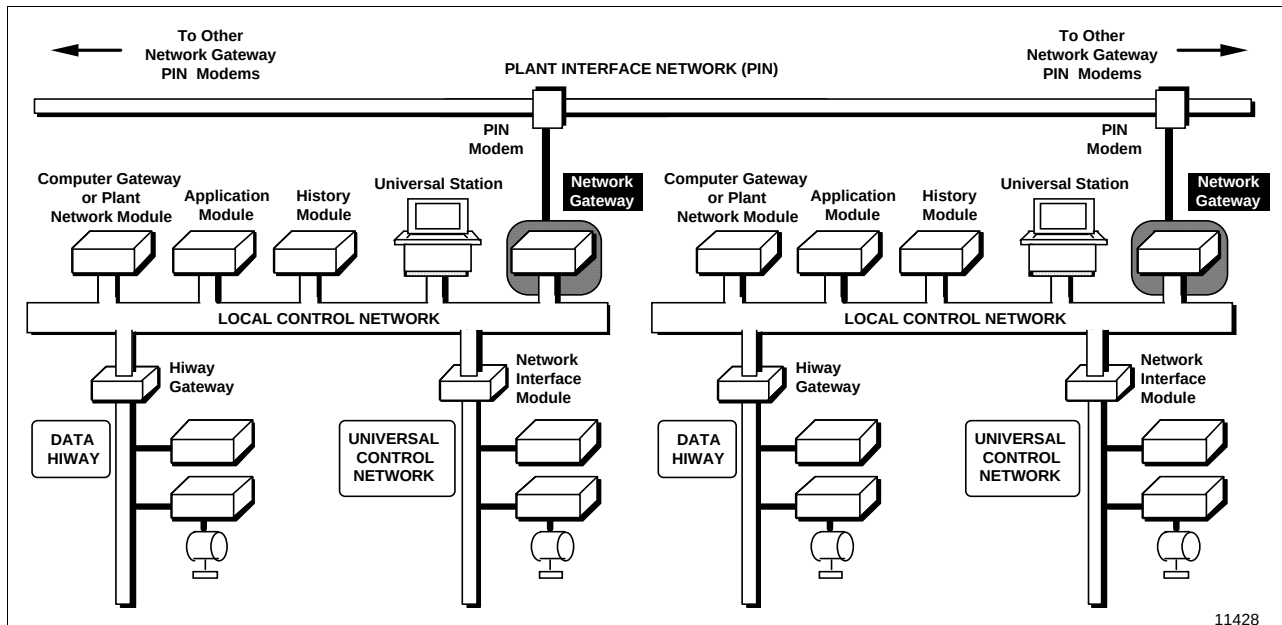
Additional information can be found in the *Hiway Gateway Specification and Technical Data* manual and the *Hiway Gateway Implementation Guidelines* manual.

2.1.15 Network Gateway (NG)

Introduction

The Network Gateway module (NG) provides communication between multiple geographically separated Local Control Networks (LCNs) through a Plant Information Network (PIN). Figure 2-35 illustrates an example of communication between two independent LCNs through their Network Gateway modules and a Plant Information Network.

Figure 2-35 TPS System with Network Gateway Modules



Plant Information Network

The Plant Information Network can be a customer's previously installed carrier band or fiber optic network, or a new installation of either type of network. The PIN is the same as a carrier band Universal Control Network (UCN).

A token-passing communications protocol that uses the IEEE 802.4 specification with IEEE 802.7 hardware is established on the Plant Information Network.

Module packaging

The Network Gateway is implemented in a Dual Node Module chassis.

References

Additional information can be found in the *Network Gateway Specification and Technical Data* manual and the *Network Gateway Site Planning and Installation* manual.

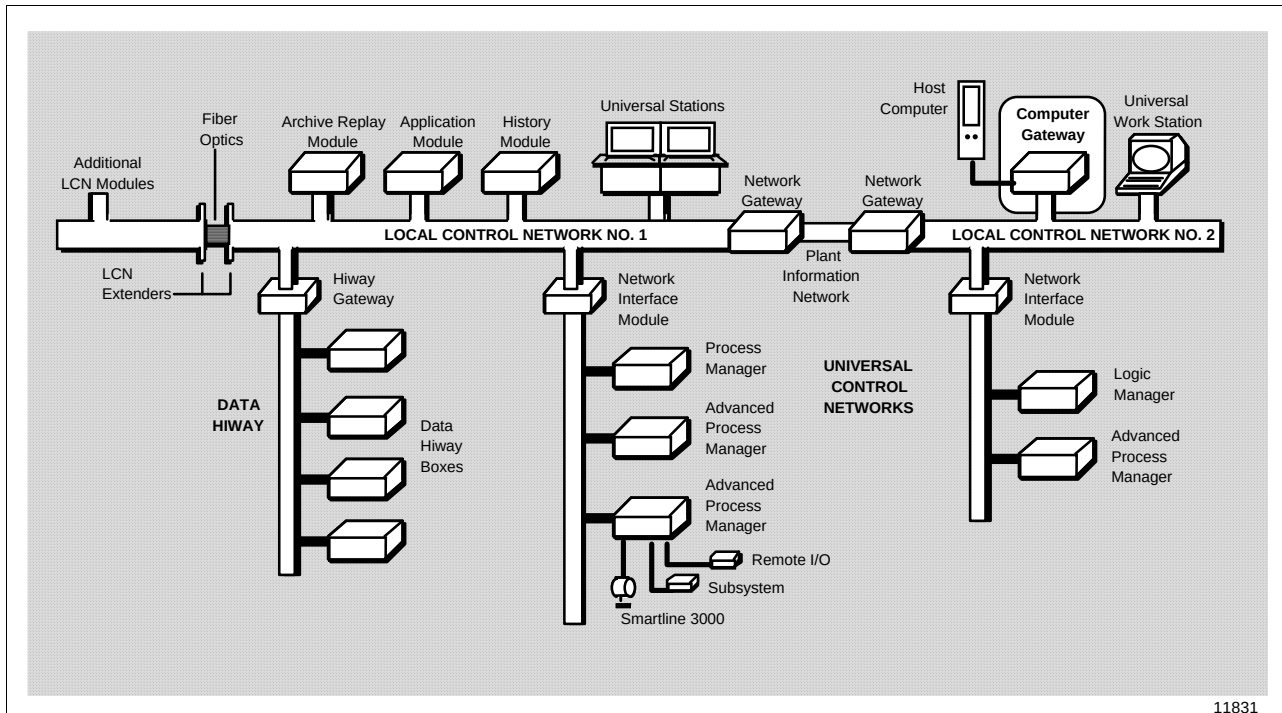
2.1.16 Computer Gateway (CG)

Introduction

The Computer Gateway module (CG) provides an interface between the Local Control Network (LCN) and a host computer. For information pertaining to interfacing a Digital Equipment Corporation (DEC) computer, refer to the *PLNM Specification and Technical Data* manual.

As illustrated in Figure 2-36, the Computer Gateway module allows a user-selected host computer to communicate with all modules (nodes) on a Local Control Network, modules on remote Local Control Networks through Network Gateways, Universal Control Network controllers, and Data Hiway controllers.

Figure 2-36 TPS System with Computer Gateway Modules



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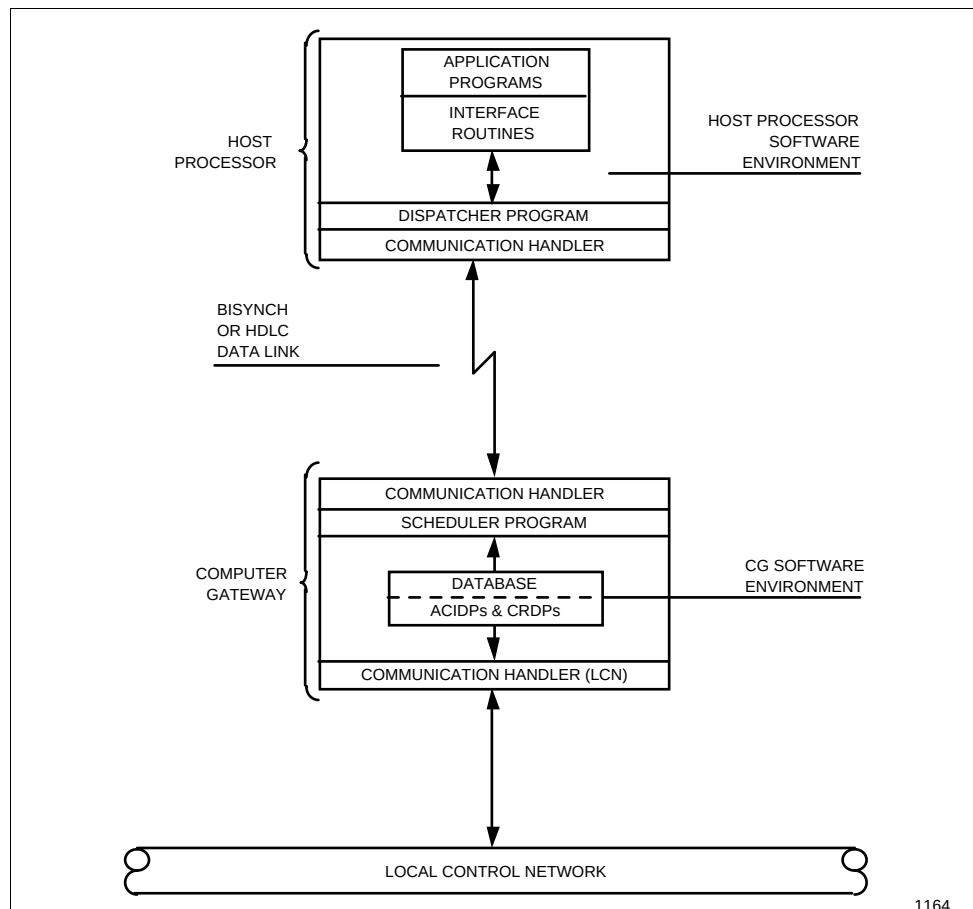
2.1.16 Computer Gateway (CG), Continued

Functionality

The relationships of the Computer Gateway functions are illustrated in Figure 2-37.

The Computer Gateway interfaces a user-selected host computer to the TPS system. The computer is expected to be used for data collection that is needed for a management information system or plant management requirements.

Figure 2-37 Computer Gateway Host Computer/LCN Relationship



1164

Bisynch or HDLC protocol

Information is exchanged between the computer and the Computer Gateway over a serial communications link using either Bisynch or HDLC protocol.

For a Bisynch system, either one or two half-duplex links are supported. For an HDLC system, one full-duplex link is supported.

Continued on next page

2.1.16 Computer Gateway (CG), Continued

Interface baud rates	The interface can be either EIA-232C with configurable speeds of up to 19.2 kilobaud, or EIA-422 with configurable speeds of up to 76.8 kilobaud for a Bisynch application and 56 kilobaud for an HDLC application.
Module packaging	The Computer Gateway is implemented in a Five-Slot or Dual Node Module chassis.
Reference	Additional information can be found in the <i>Computer Gateway Specification and Technical Data</i> manual.

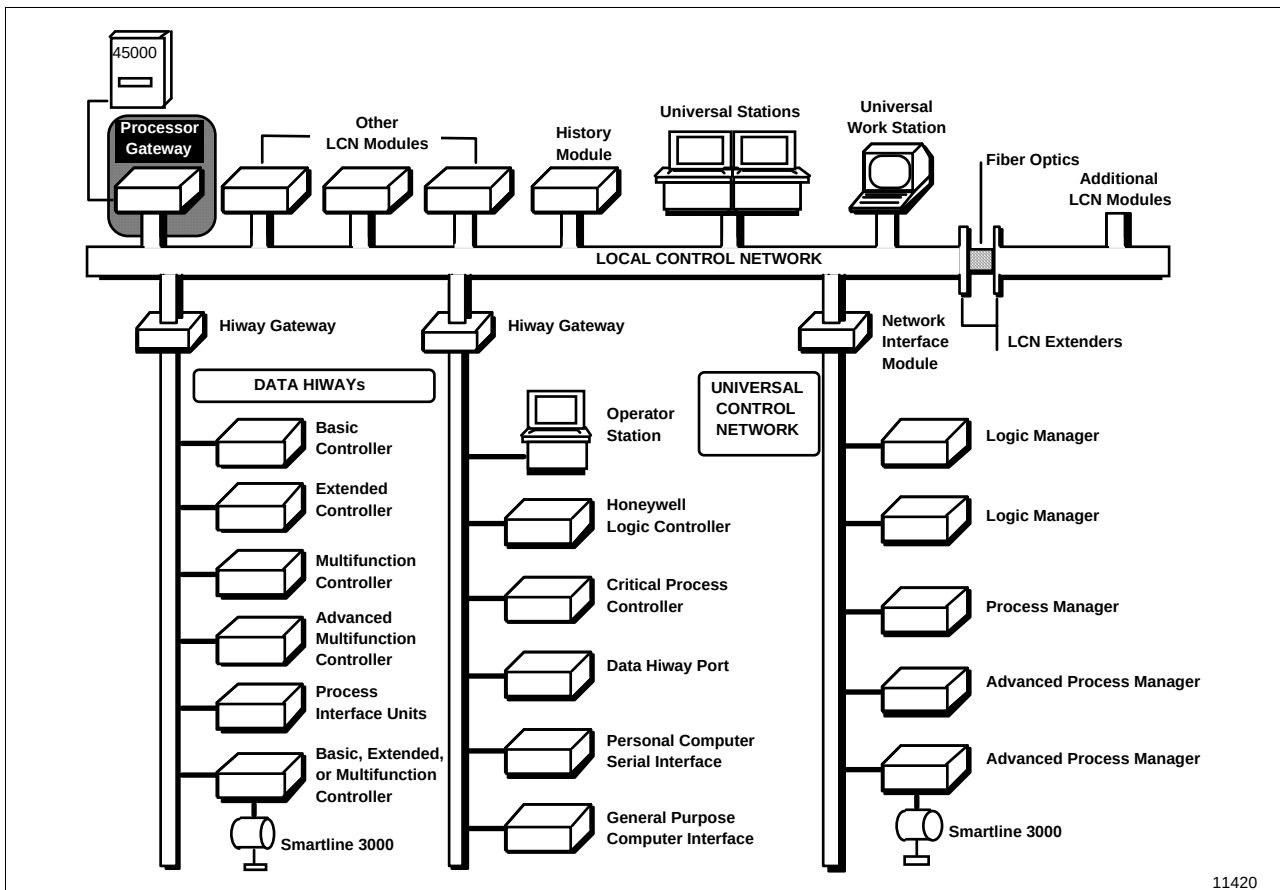
2.1.17 Processor Gateway (PG)

Introduction

The Processor Gateway module (PG) provides an interface between the Local Control Network (LCN) and a Honeywell model 45000 computer that contains PMC (450) or PMX software.

As illustrated in Figure 2-38, the Processor Gateway module provides a communication link between the PMC or PMX software and other LCN modules (nodes). See the *System Technical Data* manual for more information concerning the Processor Gateway's relationships with other modules and process controllers.

Figure 2-38 TPS System with a Processor Gateway Module



Bisynch protocol

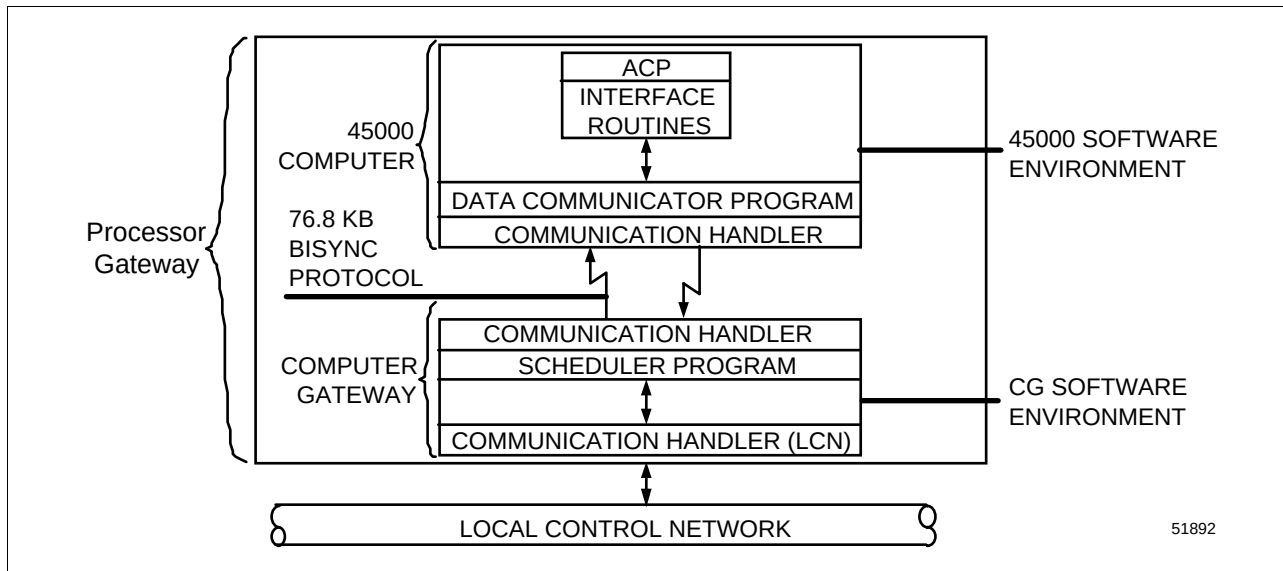
A serial communications link is provided between the Processor Gateway and the Honeywell 45000 computer using the IBM Binary Synchronous Communication protocol (Bisynch). Information is exchanged between the 45000 computer and the computer gateway, an inherent part of the Processor Gateway, by using an EIA-422 signal over an EIA-449 serial communication link that can operate up to 76.8 kilobaud using the Bisynch line protocol. The link can be connected directly or through modems.

Continued on next page

2.1.17 Processor Gateway (PG), Continued

Functionality The relationship of the Processor Gateway functions are illustrated in Figure 2-39.

Figure 2-39 Processor Gateway Functions



Module packaging The Processor Gateway is implemented in a Five-Slot or Dual Node Module chassis.

Reference Additional information can be found in the *Processor Gateway Technical Data* manual.

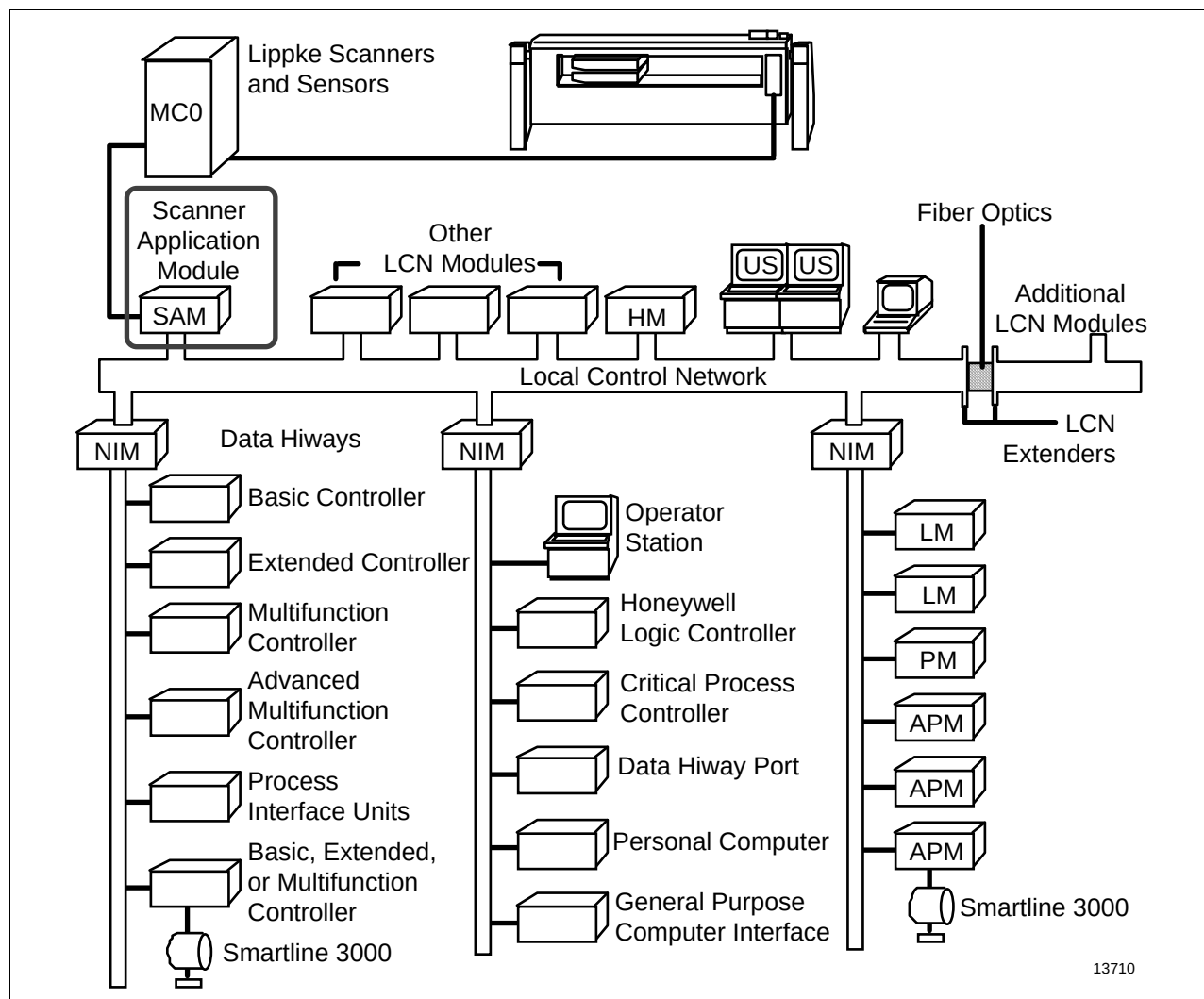
2.1.18 Scanner Application Module (SAM)

Introduction

As shown in Figure 2-40, the Scanner Application Module (SAM) provides the interface between the Local Control Network (LCN) and a Master Controller (MC0) cabinet. The MC0 cabinet then interfaces with paper sensing and scanning devices and provides measurement, service, and monitor modes of operation.

The SAM is part of the TPS Paper Machine Automation System and executes the scanning and sensing features used in paper-making process activities.

Figure 2-40 TPS System with a Scanner Application Module



Continued on next page

2.1.18 Scanner Application Module (SAM), Continued

Functionality

The Scanner Application Module (SAM) provides an interface between the MC0 and the Local Control Network (LCN). The SAM sends commands to the MC0 and receives data from the MC0. This interface to the TPS Paper Machine Automation System performs the following features:

- Encoding and decoding of the data link to the MC0 cabinet
- Measurement calculations
- Report calculations
- Interfacing with the AM database.

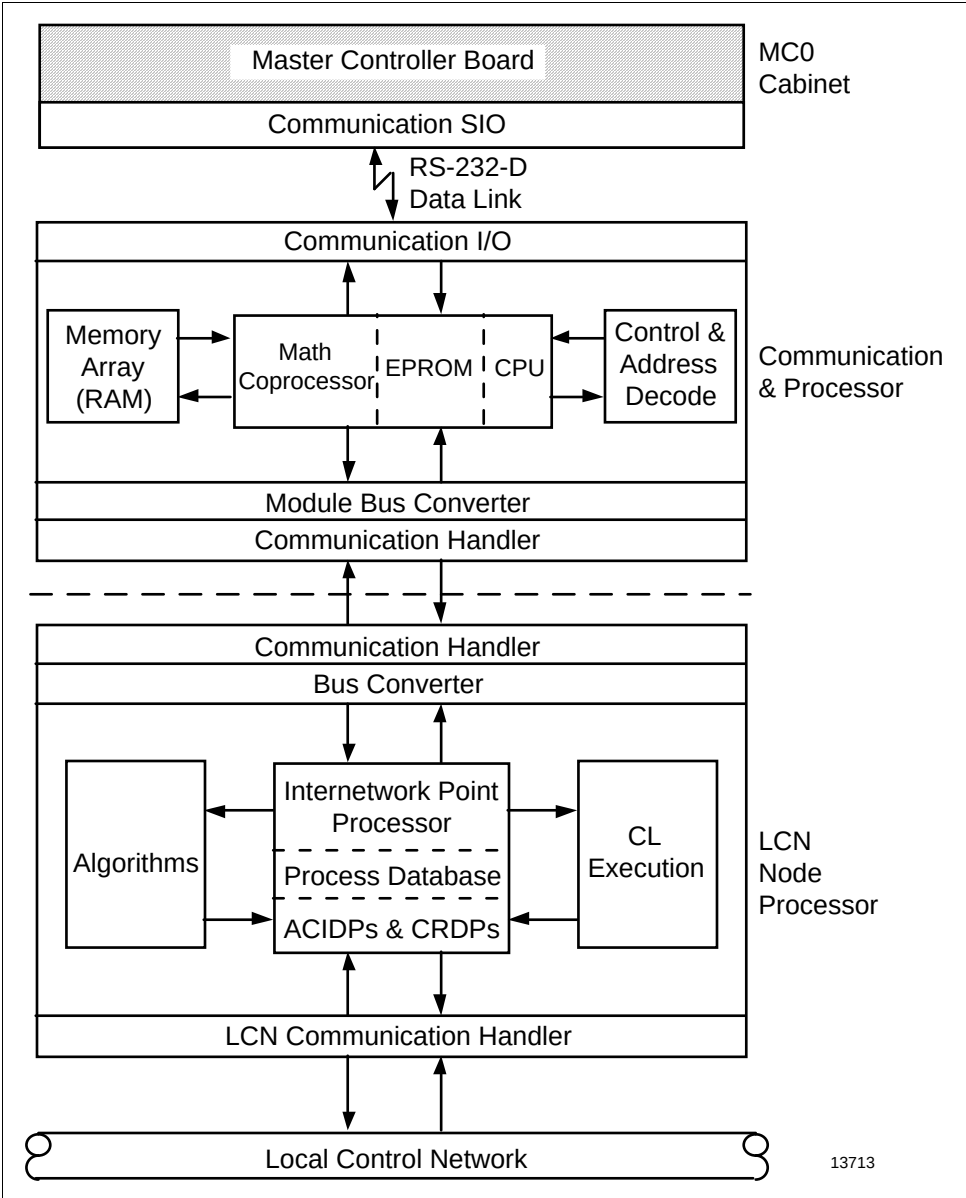
All these features are executed through the hardware in the SAM. As shown in Figure 2-41, the SAM essentially provides the same functionality of a standard Application Module (AM). However, the SAM contains an additional communication and processor board (CCP). The SAM may appear to the LCN as another AM but the additional loader modules and special software resident on the CCP make the unit a unique control node.

Continued on next page

2.1.19 Scanner Application Module (SAM), Continued

Functionality,
continued

Figure 2-41 SAM Interface Relationships and Functionality



Module packaging

The Scanner Application Module is implemented in a Five-Slot module chassis.

Section 3 – LCN Cable Hardware

3.1 Overview

Section contents

The topics covered in this section are:

Topic	See Page
SECTION 3 – LCN CABLE HARDWARE	61
3.1 Overview	61
3.2 Removable Media Requirements	63
3.3 LCN Hardware Limitations	64
3.4 Segment Planning Rules	65
3.5 Module Selection and Placement	67
3.6 LCN Node Address Selection Rules	68

Introduction

Local Control Network (LCN) modules (nodes) communicate with each other through the interconnections of coaxial and fiber optic cable. There are two cables for redundancy that are designated Cable A and Cable B. Each cable should be routed separately (to possibly prevent simultaneous damage to both cables) and must be marked to indicate that it is Cable A or Cable B. Cable A and Cable B should be routed separately. Cable A is marked with yellow and Cable B with green shrink-fit bands at the connectors. Only Honeywell specified cables must be used.

CAUTION

CAUTION—Local Control Network cables A and B must never be cross-connected.

Cable BNC tees

Each node in the LCN connects to both Cable A and Cable B through connectors called BNC tees. These connectors allow a node to be disconnected from the coaxial cable segment without disrupting communication between the other nodes. The LCN T-connectors have the Honeywell part number 51190728-105.

ATTENTION

ATTENTION—The Honeywell part numbers for the LCN and Data Hiway T-connectors are different. They cannot be substituted for each other.

Cable segments

An LCN can have up to seven coaxial cable segments, a central segment and up to six remote cable segments or remote nodes. The maximum length of each cable segment cannot exceed 300 meters (984 feet).

Continued on next page

3.1 Overview, Continued

ATTENTION

ATTENTION— The Remote Node option is not supported with the use of XLCEN2 boards.

Remote cable segments

The remote coaxial cable segments are connected to the central segment through LCN Extenders (LCNEs). LCN Extenders use fiber optic cables that provide ground isolation and can be routed through hazardous areas to interconnect LCN coaxial cable segments. The fiber optic cable length cannot exceed 2 kilometers (6562 feet). Four LCNE circuit boards are required to interconnect two LCN cable segments, one at each end of fiber optic Cable A and one at each end of fiber optic Cable B. Each fiber optic cable, A and B, contains at least two optic fiber filaments, one for transmit and one for receive.

Remote node

Two LCNE boards and one LCN Fiber Link (LCNFL) circuit board are required to connect a remote node to the central cable segment. An LCNE board is used for Cable A and another LCNE board is used for Cable B in the central cable segment. An LCNFL board is used in the remote node. It has fiber optic transmitters and receivers for both Cable A and Cable B. Like remote cable segments, remote nodes can only be connected to the central coaxial cable segment.

Number of LCN nodes

For each cable segment, the maximum number of nodes plus the number of LCNE board sets cannot exceed 40. A Dual Node Module is considered one or two nodes depending upon whether both nodes are populated. The maximum number of nodes allowed in a Local Control Network is 64.

3.2 Removable Media Requirements

Introduction	One or more removable media drives are required for the TotalPlant Solution (TPS) System. This subsection presents the requirements based upon the hardware configuration.
Minimum requirements	A removable media drive may be either a floppy disk drive, cartridge disk drive, or digital analog tape (DAT) drive on the coprocessor side of a Universal Station^X. The minimum removable media drive and associated equipment requirements for all systems are as follows. • At least one Universal Station with an Engineer's keyboard must have two removable media drives. • All other Universal Stations with an Engineer's keyboard must have at least one removable media drive.
No History Module	For systems without a History Module, the following rules also apply. • All Operator consoles must have two removable media drives. • All Universal Stations with an Engineer's keyboard must have two removable media drives.
With a History Module	For a system with a History Module, the following rule applies. • Each console must have two removable media drives. One of these drives must be in a Universal Station with an Engineer's keyboard.
Additional guidelines	Additional removable media disk drive guidelines are as follows. • Systems requiring fast reload should have the removable media cartridge disk drive option installed in each console. • A console in a remote cable segment (a coaxial cable segment separated from the main coaxial cable segment by a fiber optic link) of the LCN must have at least one Universal Station with at least one removable media drive.

3.3 LCN Hardware Limitations

Introduction

This subsection lists the hardware limitations for an Local Control Network. These maximum hardware numbers must not be exceeded.

Network maximums

- The maximum number of nodes in an LCN is 64 (software limitation).
 - For each cable segment, the number of nodes plus the number of LCN Extension sets must not exceed 40. An LCN Extension set is a pair of LCNE boards, one for Cable A and one for Cable B. A Dual Node Module counts as one node.
 - The maximum number of Application Modules (AMs) per LCN is not restricted.
 - The maximum number of Computer Gateways (CGs) per LCN is 10.
 - The maximum number of Process Networks per LCN is 20. This means a maximum number of 40 physical Process Network nodes per LCN (20 redundant pairs). A Process Network is defined as a Data Hiway, a Universal Control Network (UCN), or a Programmable Logic Controller Gateway (PLCG).
 - The maximum number of History Modules (HMs) per LCN is 20.
 - The maximum number of Universal Stations (USs) in a single console is 10.
 - The maximum number of Network Gateways (NGs) per LCN is 10.
 - The maximum number of consoles per LCN is 10.
-

3.4 Segment Planning Rules

Introduction	The following is a summary of rules that should be adhered to when planning your Local Control Network cable segments.
Two CS/R boards	A cable segment that has only Five-Slot Modules requires two Clock Source/Repeater (CS/R) boards, one for Cable A and one for Cable B. These boards can be located in any empty I/O slot, but not in the same module.
Maximum coaxial cable length	The maximum LCN coaxial cable length per segment is 300 meters. The segment cannot have stubs or branches. Use part number 51308112-xxx coaxial cable between consoles and between cabinets and consoles. The suffix “xxx” is the cable length in meters. DO NOT SUBSTITUTE another type of cable.
Minimum coaxial cable length	The minimum cable length between nodes in a cable segment is two meters. For adjacent nodes, use Honeywell cable set 51308111-002, which contains two cables marked “Cable A” and “Cable B.”
Difference in length between Cable A and B	The maximum acceptable difference in the total length of Cable A and Cable B between any two modules (nodes) in the network must not exceed 300 meters. The total length of each cable is the sum of all the coaxial cable and fiber optic cable sections between the two modules. The reason for this restriction is to limit the difference in communication delays between the A and B cables.
Maximum fiber optic cable length	Fiber optic cables can be up to 2000 meters (6562 feet) in length.
Node address selection	For each node, the node address is selected by switches on the CLCN A/B board, jumpers on the LCNFL transceiver board, or jumpers on the K2LCN/K4LCN board. For a remote node that has a K2LCN/K4LCN board and an LCNFL board, the pinning is done on the LCNFL board and all address jumpers must be removed from the K2LCN/K4LCN board. The address consists of 7 bits with odd parity. A jumper or a switch set to the “0” or “Off” position represents a “zero,” and the absence of a jumper or a switch set to the “1” or “On” position represents a “one.” Refer to subsection 3.6 for additional information concerning LCN node address selection.

Continued on next page

3.4 Segment Planning Rules, Continued

Node address assignment

The nodes' LCN addresses should be assigned in increasing numerical order. For communication efficiency, try not to leave gaps in the address numbers. Honeywell recommends that addresses be assigned in accordance with the physical placement of the modules in the control room. For example, start at one end of the LCN cable segment and assign consecutive addresses until the opposite end of the cable segment is reached. Repeat for each remote cable segment. Sometimes it is desirable to leave gaps in the node numbers when it is known that certain nodes will be added later. Refer to subsection 3.6 for additional information.

Cable termination

Both ends of the coaxial cable segment must be terminated. Use Honeywell terminator resistors, part number 30732052-001.

ATTENTION

ATTENTION—It is necessary that Cable A be connected only to the Cable A interface at each of the nodes, and that Cable B be connected only to the Cable B. **Cables must not be crossed. Do not substitute unapproved cables.**

Crossed cables will function until a communications error occurs. Then, some nodes or the entire system may isolate itself and/or fail.

Routing coaxial cable

LCN coaxial cable physical and electrical characteristics are suitable for internal building use only. **The LCN coax cables must not be routed outside the control room building.**

Locating XLCNE2, FOC/XMTR, and FOC/RCVR boards

Although XLCNE2, FOC/XMTR, and FOC/RCVR circuit boards can be installed in unused I/O slots in any node type, it is desirable to install them in nodes that are the least likely to have the power removed from them. It is recommended that the order listed in Table 4-4 be used for selecting module types to house these fiber optic cable extender boards.

Power for XLCNE2, FOC/XMTR, and FOC/RCVR boards

The XLCNE2, FOC/XMTR, and FOC/RCVR boards for Cable A must be located in a different node than those for Cable B, and should receive power from separate ac power entries.

If space is limited, an otherwise unused Dual Node Module with a power supply and KJMP load board (51401594-200) installed in the node should be used to house the extender board set. The LCNE board and associated FOC/XMTR or FOC/RCVR board for Cable A should be installed in the I/O section of the two-slot node and the corresponding boards for Cable B should be installed in the three-slot node. If FOC/XMTR/FOC/RCVR boards are not used, two XLCNE boards for Cable A should be installed in the two-slot node, and two XLCNE2 boards for Cable B should be installed in the three-slot node (the third slot could also be available for an extender board that is part of another fiber optic extender set).

3.5 Module Selection and Placement Rules

Introduction	This subsection lists rules for the proper selection of LCN modules and physical placement of modules in the Local Control Network.
Two Universal Stations per console desirable	There must be at least two USs per console, per area database, to ensure a window to the process in case of failure in the primary US.
Hiway Gateways	Because of the lower reliability of a fiber optic extender to remote cable segments, locate redundant Hiway Gateways (HGs) and added Hiway Gateways in the same coaxial cable segment. Also, locate redundant Network Interface Modules (NIMs) and redundant Programmable Logic Controller Gateways (PLCGs) on the same coaxial cable segment.
Process Network interfaces	Any coaxial cable segment with a Process Network interface (HG, NIM, or PLCG) must have at least one Universal Station in that cable segment (two are recommended).
Two node remote cable segment undesirable	If possible, avoid placing only two nodes in a remote cable segment because of error conditions that may occur when the fiber optic cable is defective or power is removed from an XLCNE2 board.

3.6 LCN Node Address Selection Rules

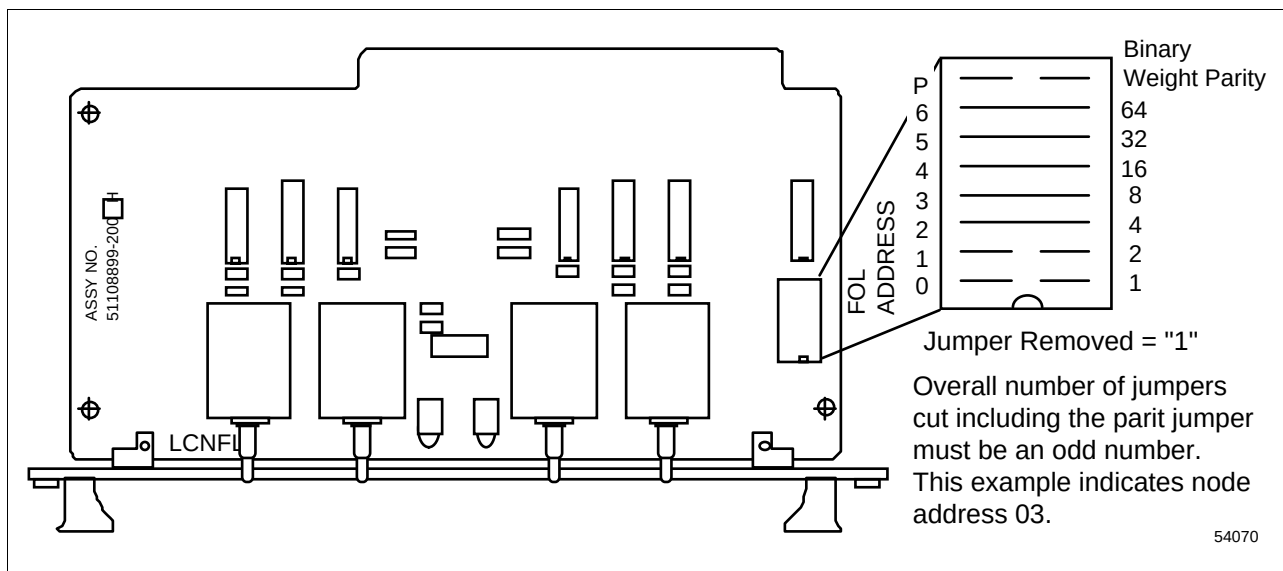
Introduction

This subsection defines the rules for node address assignment when planning a Local Control Network.

LCNFL board pinning procedure

The LCN address for each LCN node is in a range from 1 to 64. The address for each node is established by jumper pins on the LCN I/O board at the back of an LCN module chassis (LCN I/O board has the LCN T-connector connected to it), or by jumper pins on a LCNFL board at the back of an LCN module chassis for a single remote node at the end of a fiber optic link. The pinning method for an LCNFL board is illustrated in Figure 3-1. The method is the same for the LCN I/O board.

Figure 3-1 LCNFL Board Node Address Pinning



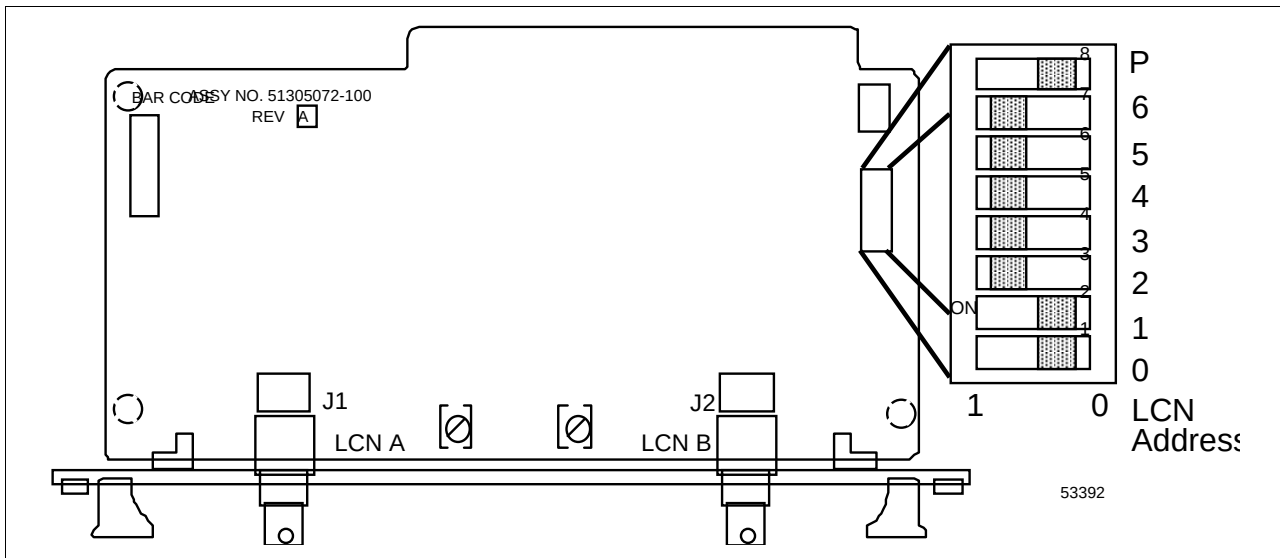
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3.6 LCN Node Address Selection Rules, Continued

CLCN A/B board LCN node address selection

Switches are used on the CLCN A/B board to select the LCN node address by setting each switch to the “0” or “1” position. The selection method for a CLCN A/B board is illustrated in Figure 3-2.

Figure 3-2 CLCN A/B Board Node Address Pinning



K2LCN/K4LCN board pinning procedure

Address pinning on the K2LCN/K4LCN board uses the same pinning method as the LCNFL and LCN I/O boards. When pinning is done on the K2LCN/K4LCN board, all pinning jumpers must be removed from the LCNFL or LCN I/O board. The opposite is true when address pinning is done on the K2LCN/K4LCN board.

If the CLCN A/B board is a substitute for the LCNFL and LCN I/O boards, and LCN node address selection is done on the K2LCN/K4LCN board, all the switches on the CLCN A/B board must be set to the “0” or “Off” position.

Pinning a remote node with a K2LCN/K4LCN board

For the special case of a single remote node that has both a K2LCN/K4LCN and LCNFL board, the LCNFL board is mounted in I/O slot one (the lowest) behind the K2LCN/K4LCN board. In this case, the address pinning should be done on the LCNFL board and all pinning jumpers should be removed from the K2LCN/K4LCN board.

Continued on next page

3.6 LCN Node Address Selection Rules, Continued

US and UWS node address assignments	Universal Stations and Universal Work Stations should be assigned the lowest LCN node addresses.
Redundant node address assignment	Redundant module pairs (HGs, NIMs, PLCGs, and AMs) should always be assigned consecutive LCN node addresses, one odd and one even.
History Module node pair number assignment	In addition to a node address, History Modules are assigned node pair numbers. For more information about node pair numbers, refer to subsection 7.2.1 in the <i>Engineer's Reference Manual</i> . In spite of the term “node pair number,” History Modules are not configured as redundant node pairs. History Modules can have redundant disk drives, but these History Modules still require only one LCN node address.

Section 4 – LCN Clock System

4.1 Overview

Section contents The topics covered in this section are:

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Introduction

The Local Control Network has a system clock that is used to time stamp events. The system clock can be implemented by two methods. Depending upon the hardware configuration, both methods can be present. LCN modules (nodes) that are hardware implemented with K2LCN/K4LCN circuit boards require a 5 Mbits/second digital clock. LCN modules (nodes) that are hardware implemented with non-K2LCN/K4LCN circuit boards (HMPU, or HPK2) require a 12.5 kHz clock.

12.5 kHz system clock

The original LCN system with only non-K2LCN/K4LCN nodes required a 12.5 kHz clock signal. The 12.5 kHz clock is embedded in the 5 MHz data transmissions in both LCN cables. The second configured system clock source that is loaded with operating software becomes the slave system clock source. It listens for the 12.5 kHz system clock and detects the master system clock signal. The slave system clock source synchronizes with the master system clock and transmits the system clock to its CS/R board which is connected to the second LCN cable.

The remaining non-K2LCN/K4LCN nodes listen for the 12.5 KHz system clock in both cables.

The master and slave source nodes must be located in the center (local) LCN cable segment, and be configured with a non-K2LCN/K4LCN processor (an HMPU, or HPK2 board) and a CS/R board.

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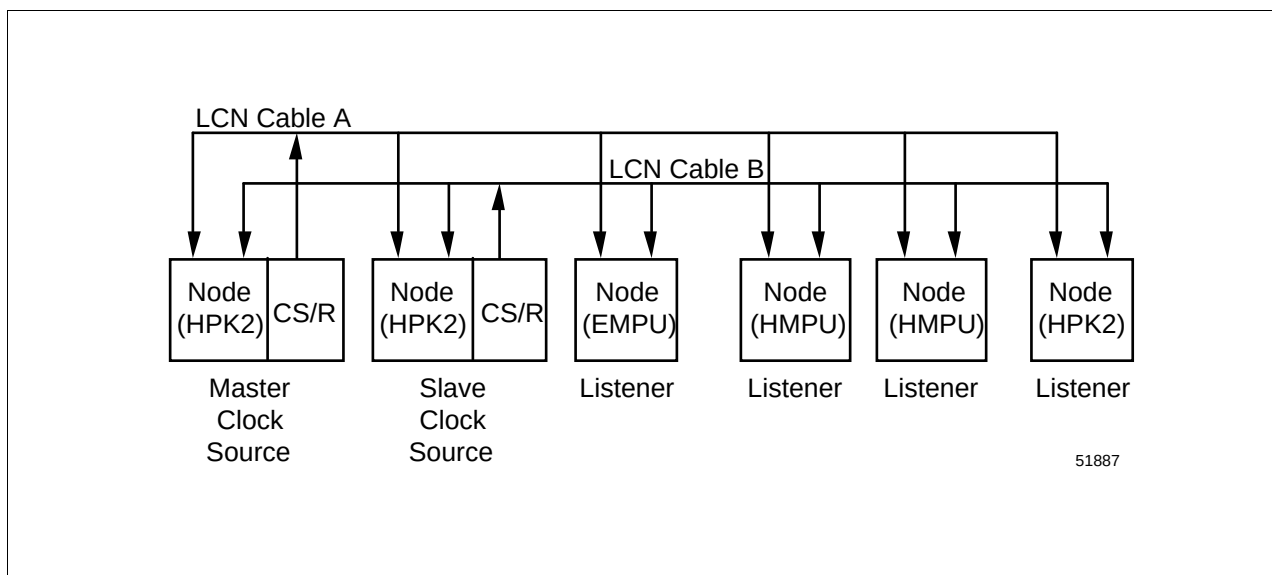
4.1 Overview, Continued

12.5 kHz system clock configuration

Figure 4-1 is an illustration of a 12.5 kHz system clock configuration with modules (nodes) that are implemented with HMPU, and HPK2 boards. There are no modules that contain a K2LCN/K4LCN board.

Table 4-1 contains a name and description of the basic nodes used in the 12.5 kHz system clock configuration.

Figure 4-1 12.5 kHz System Clock Configuration



Definitions

Table 4-1 12.5 kHz System Clock Configuration Nodes and Definitions

Node Name	Description
Master	Provides the clock data to the CS/R board for transmission at 12.5 kHz in LCN cable A. The clock is embedded on the 5 MHz LCN data signal in the cable. Once the master is established, it does not listen. The master assumes slave clock responsibility after recovering from a failure.
Slave	Once established as a slave, listens for clock messages on cable A. Synchronizes its own clock by using the messages from the master and provides clock data that is transmitted by the CS/R board in LCN cable B. If the master clock messages stop, it continues to transmit the clock. It then declares itself master and does not listen.
Listener	Listens for clock messages in both cables. The listener synchronizes itself with the master clock messages.

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4.1 Overview, Continued

5 Mbits/second system clock

The LCN modules (nodes) that are hardware implemented with K2LCN/K4LCN circuit boards use a 5 Mbits/second digital clock to time stamp events. Like the 12.5 kHz system clock, two separate nodes located in the central (local) cable segment in the LCN are identified in the Network Configuration File (NCF) and become the two clock sources for the digital clock in the LCN cables, A and B. The first of the two clock source nodes that is loaded with the operating software establishes itself as the master clock source by trying to detect a clock in both cables. Because it is the first clock source that is loaded with the operating software, it does not detect a clock and assumes the roll of master clock source. The master clock source transmits the 5 Mbits/second digital system information clock in both cables as a special diagnostic data frame.

When it is loaded with operating software, the second clock source node becomes the slave clock source because it will detect a clock in both cables. The slave clock source uses the synchronization data frame to synchronize its clock with the master clock source. However, it does not transmit a clock into the cable network unless the master clock source fails. If it detects a failure, the slave clock source becomes the master clock source.

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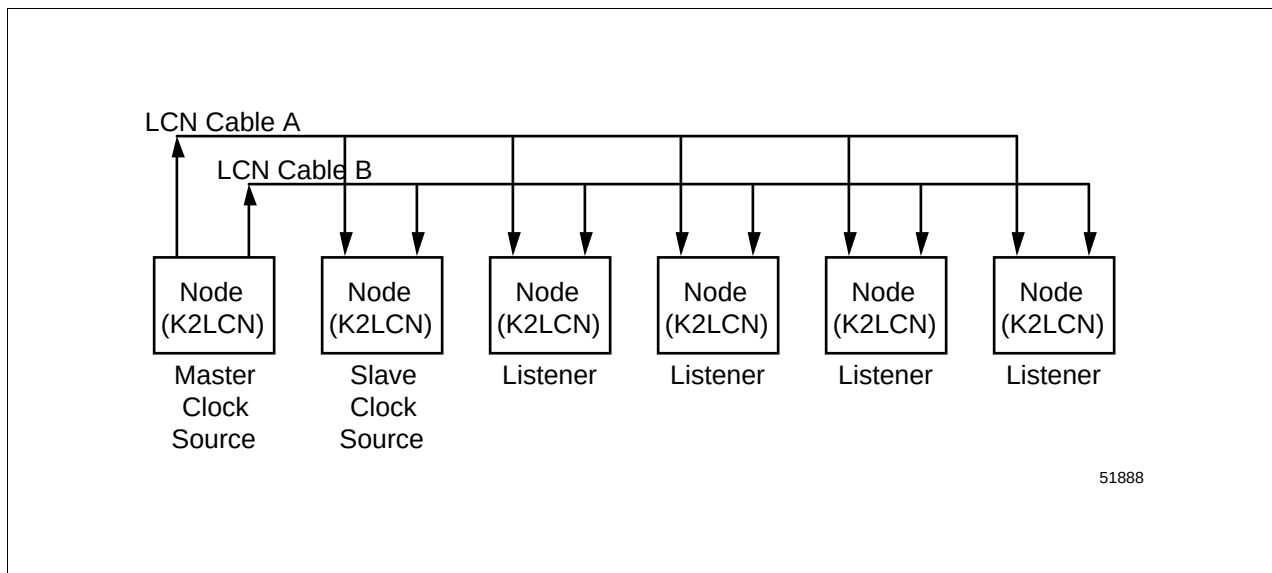
4.1 Overview, Continued

5 Mbits/second system clock configuration

Figure 4-2 is an illustration of a 5 Mbits/second system clock configuration with modules (nodes) that are implemented exclusively with K2LCN/K4LCN boards. There are no modules that contain a HMPU, or HPK2 boards.

Table 4-2 contains a name and description of the basic nodes used in the 5 Mbits/second system clock configuration.

Figure 4-2 5 Mbits/Second System Clock Configuration



Definitions

Table 4-2 5 Mbits/Second System Clock Configuration Nodes and Definitions

Node Name	Description
Master	After establishing itself as master, it does not listen. The master transmits clock synchronization messages on both LCN cables once every second.
Slave	After establishing itself as a slave, it listens for clock messages in both LCN cables. Synchronizes its own clock using the messages from the master. If the master clock messages stop, declares itself master and transmits clock messages on both LCN cables.
Listener	Listens for clock message on both LCN cables. The listener synchronizes itself with the master clock messages.

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4.1 Overview, Continued

Combined clock systems

Some Local Control Networks will have combined clock systems because both modules (nodes) with and without K2LCN/K4LCN boards will reside in the network. This can be the situation when modules that are implemented with K2LCN/K4LCN boards are added to an existing network that features only modules that contain non-K2LCN/K4LCN boards (HMPU, or HPK2). The non-K2LCN nodes require the 12.5 kHz system clock. The K2LCN nodes contain a subchannel clock mode that is compatible with the non-K2LCN/K4LCN node.

When a module that contains a K2LCN/K4LCN board operates in the subchannel clock mode, it listens for and detects the 12.5 kHz system clock synchronization signal that is superimposed on the 5 Mhz LCN data transmission generated by a non-K2LCN/K4LCN node. The K2LCN/K4LCN node can listen to the 12.5 KHz clock but cannot generate it. The 12.5 kHz system clock is generated in the non-K2LCN/K4LCN module with a CS/R as the source.

Two non-K2LCN/K4LCN nodes are configured as 12.5 kHz clock sources by the NCF. The first clock source that is loaded with operating software becomes the master clock source. It listens for a 12.5 kHz clock signal in the LCN cables and detects none. This master clock source then transmits the clock into the LCN cable that its CS/R interfaces.

A minimum of two LCN nodes must be in the central segment in order to transmit the digital clock when connecting it via LCNEs to an all digital clock segment. The two K2LCN/K4LCNs must be the lowest two physical addresses of the K2LCN/K4LCN nodes.

The first K2LCN/K4LCN node that is loaded with operating software will listen for 5 Mbits/second digital clock synchronization frames. It detects none and switches to the subchannel clock mode and checks for the 12.5 kHz system clock. Upon detecting the 12.5 kHz system clock, it synchronizes its clock to the master clock source.

If other K2LCN/K4LCN nodes are in the network, the first K2LCN/K4LCN node loaded with operating software becomes a digital clock translator for the other K2LCN/K4LCN nodes by transmitting the special clock synchronization message frames to any other K2LCN/K4LCN nodes. If there are no other K2LCN/K4LCN nodes present, the digital clock message is still transmitted but is unused.

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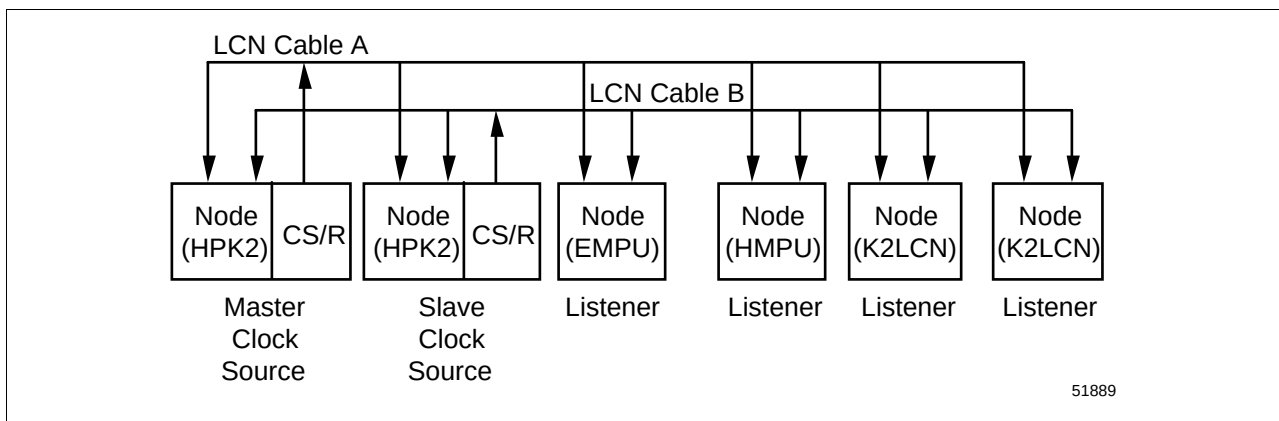
4.1 Overview, Continued

Combined clock systems configuration

Figure 4-3 is an illustration of a combined 12.5 kHz and 5 Mbits/second system clock configuration with modules (nodes) that are implemented in both non-K2LCN/K4LCN and K2LCN/K4LCN boards.

Table 4-3 contains a name and description of the basic nodes used in the combined 12.5 kHz and 5 Mbits/second system clock configuration.

Figure 4-3 Combined 12.5 kHz and 5 Mbits/second System Clock Configuration



Definitions

Table 4-3 Combined 12.5 kHz and 5 Mbits/second System Clock Configuration Nodes and Definitions

Node Name	Description
Master	Provides the clock data to the CS/R board for transmission at 12.5 kHz in LCN cable A. The clock is superimposed on the 5 MHz LCN data signal in the cable. Once the master is established, it does not listen. The master assumes slave clock responsibility after recovering from a failure.
Slave	After established as a slave, listens for clock messages on cable A. Synchronizes its own clock by using the messages from the master and provides clock data that is transmitted by the CS/R board in LCN cable B. If the master clock messages stop, it continues to transmit the clock. It then declares itself master and does not listen.
Non-K2LCN/K4LCN Listener	Listens for clock messages in both cables. The listener synchronizes itself with the master clock messages.
K2LCN/K4LCN Listener	When loaded with operating software, the K2LCN/K4LCN node listens to both cables for a digital clock synchronization message. If none is detected, it listens for the 12.5 kHz clock signal. When it is detected, it synchronizes its clock with the master and performs as a clock translator for other K2LCN/K4LCN nodes.

4.2 12.5 kHz Clock System

Introduction

Some Local Control Networks have only a 12.5 kHz clock system. These are LCNs that do not include K2LCN/K4LCN kernels (processor boards) circuit boards.

Two CS/R boards required

To implement the 12.5 kHz system clock, each coaxial cable segment must have two Clock Source/Repeater (CS/R) boards, one for Cable A and one for Cable B. The Clock Source/Repeater boards transmit the 12.5 kHz clock in each LCN cable and automatically provide a single point LCN shield connection to logic ground for each cable.

Model MP-MCSR02 includes two precision CS/R boards and associated cables. The two-meter cables connect the CS/Rs boards to the LCN coaxial cables (Cable A and Cable B).

Five-Slot Module installation

The CS/R board must be installed in slot one of the I/O card cage at the rear of a Five-Slot Module that includes an HMPU, or HPK2 board. For network integrity, it is desirable to locate them in modules that are least likely to have power removed at any time. The order of module preference type is shown in Table 4-4.

Table 4-4 LCN Clock Source Priority List

Priority	Node	Description
1	EAMR	Enhanced Redundant Application Module
2	HG	Hiway Gateway (preferably one of a redundant pair)
3	NIM	Network Interface Module (redundant)
4	PLCG	Programmable Logic Controller Gateway (redundant)
5	AM	Application Module
6	HM	History Module
7	PLNM	Plant Network Module
8	NG	Network gateway
9	CG	Computer Gateway
10	US	Universal Station
11	U ^X S	Universal Station ^X

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4.2 12.5 kHz Clock System, Continued

Transmit the clock through fiber optics

A CS/R board can be connected with twisted pair wires to a Fiber Optic Clock Transmitter (FOC/XMTR board) to propagate the 12.5 kHz clock to a remote cable segment through a fiber optic cable pair that has a complementing Fiber Optic Clock Receiver (FOC/RCVR board) and CS/R board set. The fiber optic cable's length should not exceed 2.2 kilometers. See subsection 5.7 for budget discussion.

Transmit the clock through a current loop

Alternatively, a current loop over twisted pair wires that does not exceed 300 meters in length can be used to directly interconnect CS/R boards to propagate the clock signal to a remote cable segment, although this is not recommended due to reduced reliability and possible noise susceptibility. Current loops cannot be used to transverse hazardous areas. The fiber optic clock approach is the preferred method of clock propagation to remote cable segments.

Clock source nodes must be configured in NCF

Two clock source nodes must be defined in the software Network Configuration File (NCF). These nodes must contain CS/R boards and they must be in the central cable segment. This allows the central cable segment to maintain synchronization with other remote cable segments if one remote link fails.

Node power supplies must be pinned

The power supply in each node of the LCN system must be configured by a pinning option for the internal (ac line frequency) clock unless the system is required to run Sequence of Events (SOE) functionality in the Network Interface Modules (NIMs). If SOE functionality is required, the clock master and the clock slave nodes must be pinned for external clock usage, which is normally provided by the Precision Clock board or the Precision Clock Source Repeater (PCS/R) board. All other nodes are pinned for the internal (ac line frequency) clock. For information on the clock pinning option, refer to the *LCN Guidelines - Implementation, Troubleshooting, and Service* manual. For Dual Node Module or Five-Slot Module, refer to *Dual Node Module Service* or *Five/Ten-Slot Module Service* manuals.

4.3 5 Mbits/Second Digital Clock System

Introduction

Some LCN systems use only K2LCN/K4LCN circuit boards as the kernel board (no HMPU, or HPK2 boards). These systems use only the digital mode of clock operation, in which clock synchronization signals are digital frames or messages incorporated in 5 Mbits/second LCN communications. Because remote cable segments and remote nodes receive the clock through LCN messages, fiber optic clock transmitters and receivers (FOC/XMTR and FOC/RCVR boards) are not required.

NCF configuration

Two clock source nodes must be configured in the software Network Configuration File (NCF). You must identify these two clock source nodes in the central coaxial cable segment. This allows the central cable segment to maintain synchronization with other remote cable segments if one remote link fails.

Note that systems containing only K2LCN/K4LCN boards do not use CS/R boards.

ATTENTION

ATTENTION—A remote node in an LCN that contains only K2LCN/K4LCN nodes will receive the clock through LCN communications messages. This allows the remote node to have the Sequence Of Events (SOE) function.

CLCNA and CLCNB board requirements

Each CE Compliant Dual Node Module requires a CLCNA board in I/O slot 1 of the upper two-slot node, and a CLCNB board in I/O slot 1 of the lower three-slot node, except when a remote node uses the LCNFL board. The KLCNA and KLCNB boards are used in pre-CE Compliant units. The CLCNA and CLCNB board are compatible replacements for KLCNA and KLCNB boards.

Dual Node Module grounding

If an LCN coaxial cable segment does not have CS/R boards (which is true for a cable segment that contains only K2LCN/K4LCN nodes), a grounding wire must be installed on one, and only one, CLCNA board and on one, and only one, CLCNB board. This grounds the shield of Cable A and the shield of Cable B at one point only to chassis ground. If possible, do not ground Cable A and Cable B in the same module. Refer to *Dual Node Module Service* manual for additional information on LCN grounding.

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4.3 5 Mbits/Second Digital Clock System, Continued

Non-Dual Node Module grounding

A non-dual node, using only K2LCN/K4LCN boards, provides several grounding methods. In a five or ten-slot LCN that is exclusively K2LCN/K4LCN equipped, the LCN cable(s) may be grounded using either of the following methods:

- If CS/Rs are available, one per cable may be plugged into any empty chassis slot to provide a ground path for that cable. No clock messages will be generated.
- In the absence of CS/Rs, the revision T and later LCN I/O board can be used to mechanically ground the LCN cables. These paddle boards are provided with holes at J1 and J2 that will accommodate spade lugs with wires to chassis ground.

Node power supplies must be pinned

The power supply in each node of the system must be configured by a pinning option to use the internal (ac line frequency) clock. If Sequence Of Events (SOE) functionality in the NIM is required, the module containing the clock master and the clock slave must be pinned for external clock. For these nodes, the Precision Clock Oscillator on the K2LCN/K4LCN board will then be used as the clock source instead of the ac line frequency. All other nodes are pinned for the internal (ac line frequency) clock. For information on the clock pinning option, refer to the *LCN Guidelines - Implementation, Troubleshooting, and Service* manual. For Dual Node Module or Five-Slot Module, refer to *Dual Node Module Service* or *Five/Ten-Slot Module Service* manuals.

4.4 Combined 12.5 kHz and Digital Clock System

Description

Combined clock systems are systems that include both the 12.5 kHz (subchannel) and 5 Mbits/second (digital) clocks. The 12.5 kHz clock is required for nodes that use HMPU, or HPK2 processor boards, while the 5 Mbits/second digital clock is used by nodes that have K2LCN/K4LCN processor boards.

Board requirements

Each combined clock cable segment and each 12.5 kHz clock only cable segment must have two Clock Source/Repeater (CS/R) boards, one for Cable A and one for Cable B. The Clock Source/Repeater boards transmit the 12.5 kHz clock in each LCN cable and automatically provide the single point LCN shield connection to logic ground at the host module for each cable. The CS/R boards must be installed in the non-K2LCN/K4LCN nodes. Digital clock only cable segments do not use CS/R boards. Grounding for the cable shields in digital clock only cable segments is accomplished by the connection of a ground wire at one CLCNA board and one CLCNB board as described in the subsection 4.3. The exception is for grounding of a non-dual node using K2LCN/K4LCN (see subsection 4.3).

CS/R board requirement

Each CS/R board must be installed in I/O slot 1 of a module that includes an HMPU, or HPK2 board. For network integrity, it is desirable to locate them in nodes that are least likely to have power removed from them. The order of module preference is listed in Table 4-1.

WARNING

WARNING—Do not install a CS/R board in I/O slot 1 of a K2LCN/K4LCN node. Damage could result causing the node to become inoperative.

Processor board requirements

In a combined clock system, the central cable segment must contain at least two nodes with an HMPU, or HPK2 processor board. These are used in conjunction with CS/R boards to source the 12.5 kHz clock. If one or more remote cable segments have only K2LCN/K4LCN boards, or could contain all K2LCN/K4LCN boards through upgrades or replacements, then the central cable segment must contain at least two nodes with K2LCN/K4LCN boards. These nodes must be the lowest addressed on the network. One K2LCN/K4LCN board will serve as a translator, receiving the 12.5 kHz clock and generating the 5 Mbits/second digital clock to be used by other K2LCN/K4LCN boards in the system. The second K2LCN/K4LCN board required in the central cable segment serves as the backup for the translator function, and will take over automatically if the primary translator fails.

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4.4 Combined 12.5 kHz and Digital Clock System, Continued

NCF configuration	Two clock source nodes must be defined in the software Network Configuration File (NCF). These nodes must contain CS/R boards and they must be in the central cable segment. This will allow the central cable segment to maintain synchronization with other remote segments if one clock source node fails.
LCN/K4LCN nodes	A minimum of two K2LCN/K4LCN nodes must be in a cable segment in order to connect it to an all-digital clock segment using XLCNE2 boards.
FOC/XMTR/FOC/RCVR or CS/Rs board requirements	FOC/XMTR/FOC/RCVR or CS/Rs boards with twisted pair wires must be used to attach an all-12.5 kHz clock cable segment with XLCNE2 boards to a combined clock cable segment, or to attach one combined clock cable segment to another combined clock cable segment.
Cable segment connection restrictions	An all-12.5 kHz cable segment cannot be connected with XLCNE2 boards to an all-digital clock cable segment. At least one of the cable segments must be a combined clock cable segment.
Software algorithm	A software algorithm automatically configures a K2LCN/K4LCN node to the digital clock source node that translates the 12.5 kHz clock into the digital clock. The algorithm also configures backups as needed.
Adding non-K2LCN/K4LCN board nodes	Adding non-K2LCN/K4LCN board nodes to a digital clock only system requires that at least two nodes be added to the central cable segment and that they contain the CS/R boards. This scenario might occur when adding EAMR Modules to a digital clock only system.
Remote single node requirements	Two XLCNE2 and one LCNFL boards are required to connect a single remote node to an LCN coaxial cable segment. The clock signal cannot be transmitted to this node unless it is a K2LCN/K4LCN board based node, in which case the digital clock is transmitted through the LCN connection.
Remote dual node module requirements	There can be only one K2LCN/K4LCN and one LCNFL board in one node of a Dual Node Module. The other node must be unused. The LCNFL board must be Revision F to be compatible with the K2LCN/K4LCN board. The LCNFL board is mounted in I/O slot 1 (the lowest), behind the K2LCN/K4LCN board.
ATTENTION	ATTENTION—Installing an LCNFL board in a module that contains a K2LCN/K4LCN board allows this remote node to have the digital clock. This allows for Sequence Of Events (SOE) functionality, if configured.

Continued on next page

4.4 Combined 12.5 kHz and Digital Clock System, Continued

CLCNA and CLCNB requirements

Each Dual Node Module requires a CLCNA board in the two-slot node and a CLCNB board in the three-slot node except for remote nodes using the LCNFL board.

Node power supplies must be pinned

The power supply in each node of the LCN system must be configured by a pinning option for the internal (ac line frequency) clock, unless the system is required to run Sequence Of Events (SOE) functionality in the NIMs. If SOE functionality is required, the clock master and the clock slave must be pinned for an external clock, which is normally provided by Precision Clock Source Repeater (PCS/R) boards. All other nodes are pinned for an internal (ac line frequency) clock. For additional clock pinning option information, refer to the *LCN Guidelines - Implementation, Troubleshooting, and Service* manual. For Dual Node Module or Five-Slot Module, refer to *Dual Node Module Service* or *Five/Ten-Slot Module Service* manuals.

4.5 Remote Segment Clock Requirements

Remote K2LCN/K4LCN node cable segment

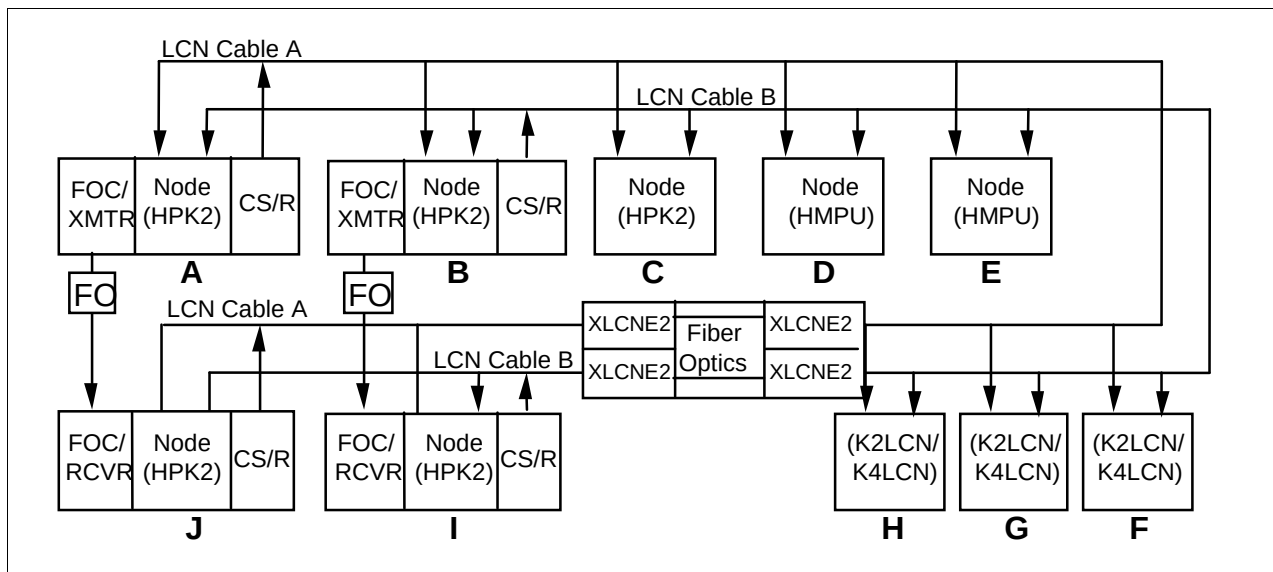
The need to transmit a system clock between LCN coaxial cable segments is mandatory to guarantee the accuracy of alarm events, specifically Sequence Of Events (SOE) alarms. If the remote cable segment requires only a 5 Mbits/second digital clock because K2LCN/K4LCN nodes are present, and the central cable segment can generate this clock through a K2LCN/K4LCN node, additional hardware is not required. Figure 4-4 is an illustration of an LCN configuration that does not require transmission of the 12.5 kHz system clock to a remote cable segment.

Table 4-5 contains a name and description of the basic nodes used in the remote K2LCN/K4LCN node segment combined clock configuration.

Continued on next page

4.5 Remote Segment Clock Requirements, Continued

Figure 4-4 Remote K2LCN/K4LCN Node Segment Combined Clock System Configuration



Continued on next page

4.5 Remote Segment Clock Requirements, Continued

Definitions

Table 4-5 Remote K2LCN/K4LCN Node Segment Combined Clock System Configuration Nodes and Definitions

Node Name	Description
Node A	Configured as the 12.5 kHz master clock source, the node is loaded with operating software first. The node provides a clock to its CS/R board for transmission of the 12.5 kHz system clock that is superimposed on the 5 MHz data signal in LCN cable A. After the node is established as the master clock source, it does not listen. The node assumes responsibility as the slave clock source when it has recovered from a failure.
Node B	The node is configured as a clock source and loaded with operating software after the master clock source node is loaded. After the operating software is loaded in the node, the node detects the master clock and synchronizes its clock. The node provides the clock data that is transmitted by its CS/R board in LCN cable B. If the master clock source fails, the node declares itself the master clock source and does not listen.
Nodes C, D, E, I	These nodes listen for the clock messages in both LCN cables. The nodes synchronize themselves with the master clock messages.
Node F	Assume this is the first K2LCN/K4LCN node to be loaded with operating software after nodes A and B are loaded. The node first listens to both LCN cables for a digital clock synchronization message. If none are detected, it listens for the 12.5 kHz system clock signal. When the 12.5 kHz system clock is detected, the node synchronizes its clock with the master clock and becomes a 5 Mbits/second digital clock translator for other K2LCN/K4LCN node listeners in the LCN.
Nodes G, H, J, K	Assume that these nodes are loaded with operating software after the node F becomes a clock translator. They listen for digital clock synchronization messages on both cables and synchronize their clocks using the messages from node F.

Continued on next page

4.5 Remote Segment Clock Requirements, Continued

Remote non-K2LCN/K4LCN node cable segment

If the remote cable segment requires a 12.5 kHz system clock because it has non-K2LCN/K4LCN nodes and the distance to the remote cable segment exceeds 300 meters, fiber optic clock transmitter and receiver boards (FOC/XMTR and FOC/RCVR) must be used in addition to the CS/R boards and LCN Extender (XLCNE2) boards. Should the distance to the remote cable segment be less than 300 meters, optionally the 12.5 kHz system clock can be transmitted directly over twisted pair current loops between CS/R boards.

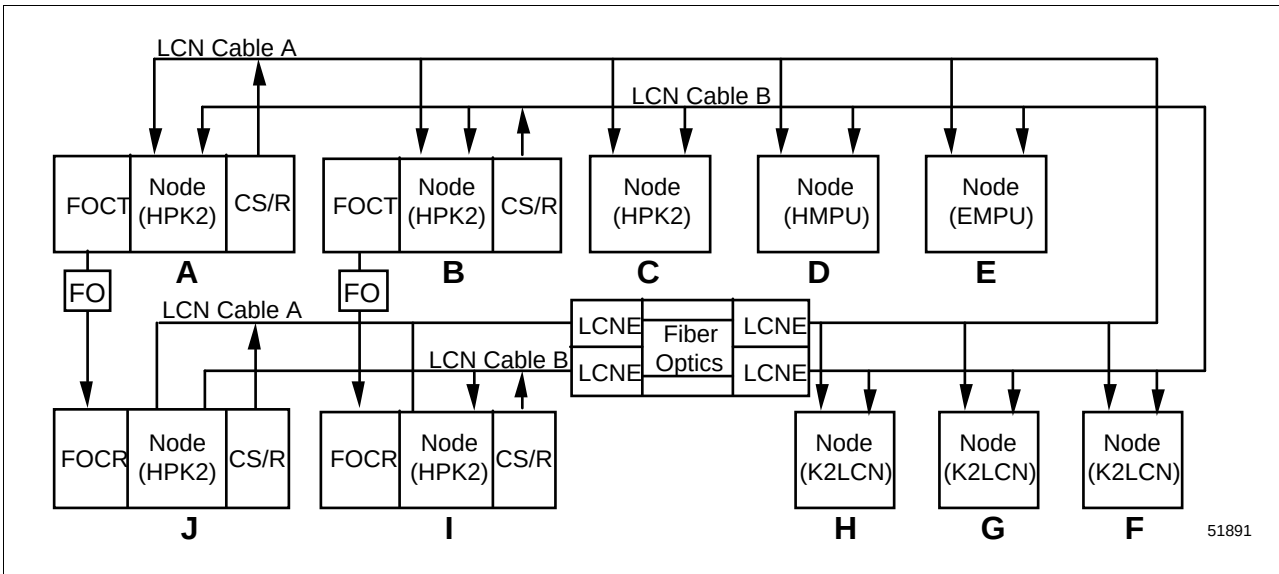
Refer to Section 5 for additional information on fiber optic links and twisted pair current loops.

Remote non-K2LCN/K4LCN node cable segment illustration

Figure 4-5 is an illustration of a typical LCN configuration that requires transmission of a 12.5 kHz system clock to a remote cable segment over a fiber optic link.

Table 4-6 contains a name and description of the basic nodes used in the remote non-K2LCN/K4LCN node segment combined clock configuration.

Figure 4-5 Remote Non-K2LCN/K4LCN Node Segment Combined Clock System



Continued on next page

4.5 Remote Segment Clock Requirements, Continued

Definitions

Table 4-6 Remote Non-K2LCN/K4LCN Node Segment Combined Clock System and Definitions

Node Name	Description
Node A	Configured as the 12.5 kHz master clock source, the node is loaded with operating software first. The node provides a clock to its CS/R board for transmission of the 12.5 kHz system clock that is superimposed on the 5 MHz data signal in LCN cable A. After the node is established as the master clock source, it does not listen. The node assumes responsibility as the slave clock source when it has recovered from a failure.
Node B	The node is configured as a clock source and loaded with operating software after the master clock source node is loaded. After the operating software is loaded in the node, the node detects the master clock and synchronizes its clock. The node provides the clock data that is transmitted by its CS/R board in LCN cable B. If the master clock source fails, the node declares itself the master clock source and does not listen.
Nodes C, D, E	These nodes listen for the clock messages in both LCN cables. The nodes synchronize themselves with the master clock messages.
Node F	Assume this is the first K2LCN/K4LCN node to be loaded with operating software after nodes A and B are loaded. The node first listens to both LCN cables for a digital clock synchronization message. If none are detected, it listens for the 12.5 kHz system clock signal. When the 12.5 kHz system clock is detected, the node synchronizes its clock with the master clock and becomes a 5 Mbits/second digital clock translator for other K2LCN/K4LCN listener nodes in the LCN.
Nodes G, H	Assume that these nodes are loaded with operating software after the node F becomes a clock translator. They listen for digital clock synchronization messages on both cables and synchronize their clocks using the messages from node F.
Nodes I, J	These non-K2LCN/K4LCN nodes in a remote cable segment receive and transmit the LCN 5 MHz data signal through the fiber optic XLCNE2 boards. The 12.5 kHz system clock for the remote cable segment is detected by the FOC/RCVR boards, also through fiber optic links. FOC/XMTR boards transmit the clock for the remote segment LCN A and B cables. The clock that is detected by the FOC/RCVR boards is synchronized and transmitted to the CS/R boards where it is retransmitted into the A and B cables.

Section 5 – LCN Fiber Optic Extenders

5.1 Overview

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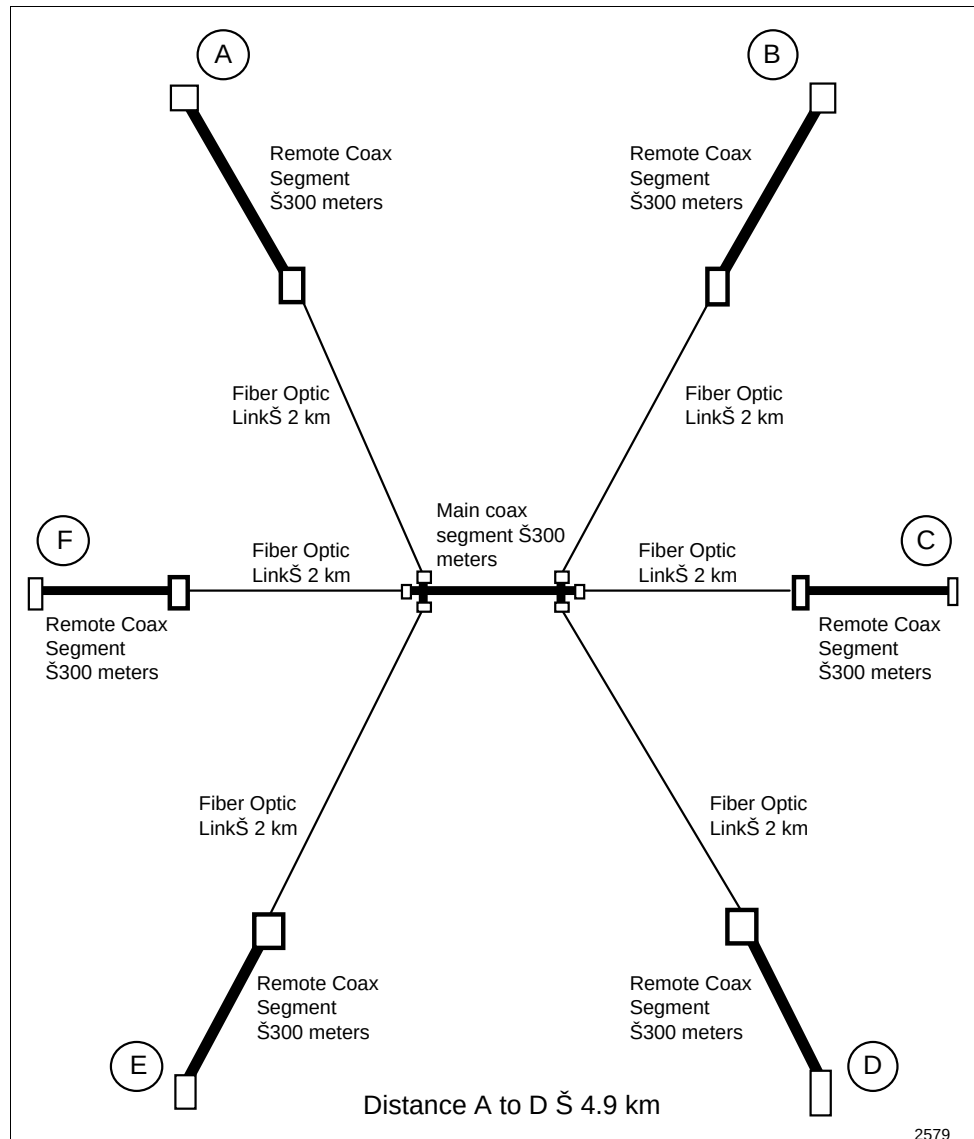
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5.1 Overview, Continued

Introduction

Figure 5-1 illustrates the maximum acceptable configuration of LCN coaxial cable segments. There can be no more than one central (main) and six remote coaxial cable segments in the network. The link between the central cable segment and the remote cable segments must be through fiber optic extender hardware. The hardware consists of LCN extender circuit boards and fiber optic cables.

Figure 5-1 LCN Segment Extender Configuration



Maximum link length

Each fiber optic link is limited to a maximum length of 2000 meters (6562 feet).

Continued on next page

5.1 Overview, Continued

Fiber optic properties	The fiber optic link is nonelectrical, so it can be routed safely through hazardous areas. Because fiber optic cables use light as the transmission media, they provide complete electrical isolation between the cable segments at each end of the link.
LED transmitters	The fiber optic transmitters use Light Emitting Diodes (LEDs) rather than lasers as the light source, so there is no danger of eye damage if you accidentally look directly at a transmitter output or into the end of a fiber cable. If you were to do this with a laser system, eye damage could result.

5.2 Description

LCN Extension set	An LCN Extension set is not an LCN module. It is a set of circuit boards the function of which is to link remote cable segments and/or remote nodes to the LCN using fiber optic cable transmission media. To connect a remote cable segment, an LCN Extension set is installed that includes two pairs of XLCNE2 boards, one pair for Cable A and one pair for Cable B. The next two subsections will cover these configurations in more detail.
Why use an LCN Extender?	There are several reasons why an LCN Extender may be necessary or desirable. • To provide control functions in an area that is too far away from the central area to be serviced by coaxial cable. • To traverse an area where hazardous or explosive conditions exist and electrical signals would be dangerous. • To traverse an area where environmental factors would prohibit coaxial cable, such as an outdoor or buried cable run. • To implement a system that requires more than 40 LCN nodes. Only 40 nodes are allowed for a cable segment.
LCN expansion rules	LCN Extension sets permit the extension (expansion) of the LCN as follows. • There must be a main or central coaxial segment and up to six remote cable segments, each of which must be located no greater than 2.2 kilometers (7218 feet) away from the main coaxial cable segment. • The maximum allowable length of each coaxial segment is 300 meters (984) feet). • For each cable segment, the number of nodes plus the number of XLCNE2 board pairs must not exceed 40. A Dual Node Module counts as one or two nodes, depending on whether one or two nodes are installed. • The maximum number of nodes in an LCN must not exceed 64.

Continued on next page

5.2 Description, Continued

Additional fiber optic link advantages

A fiber optic link has these additional advantages:

- A fiber optic cable is an intrinsically safe transmission medium that isolates the LCN from any hazardous area of a plant it passes through.
 - Fiber optic cables are nonelectric, and therefore offer extremely high immunity to electromagnetic interference (EMI) or radio frequency interference (RFI).
 - Fiber optic cables allow the extension of an LCN for a long distance without having to worry about the ground potential difference between the connected sites.
 - Using fiber optics, an LCN can be implemented in a star configuration. Such a topology can facilitate rapid diagnosis and improve survivability if a communications problem develops. First, the cable segments can be isolated by disabling the fiber optic links. Next, the faulty cable segment can be determined by noting communication statistics on the separate segments. Finally, the “good” cable segments can be reconnected.
 - An LCN coaxial cable segment is limited to 40 nodes (including each fiber optic link which counts as the equivalent of one node). Because 64 nodes are permissible in an LCN, that total can be achieved by implementing two cable segments with fiber optic links.
-

5.3 LCN Extension Set Components

Introduction

An LCN Extension set consists of optional circuit boards that are used to extend the LCN data cables, Cable A and Cable B, to a remote location. Also, when required, additional boards are used to transmit the 12.5 kHz system clock to the remote cable segment.

LCN Extension set description

A minimum LCN Extension set consists of four LCN Extender (XLCNE2) circuit boards, one for each end of fiber optic Cable A and one for each end of fiber optic Cable B. Each fiber optic cable, A and B, contains at least two fiber filaments, including one for transmission of data from the central cable segment to the remote cable segment and one for receiving data from the remote cable segment. This configuration extends the LCN data signals and the 5 Mbits/second digital clock, if present, to the remote cable segment, but does not extend the 12.5 kHz clock to the remote cable segment. If the 12.5 kHz clock is required at the remote cable segment, a Fiber Optic Clock Transmitter (FOC/XMTR) board is required in the main (central) cable segment, and a Fiber Optic Clock Receiver (FOC/RCVR) board is required in the remote cable segment. Also, two Clock Source/Repeater (CS/R) boards are required in the remote cable segment to transmit the 12.5 kHz clock into the coaxial cables (A and B).

Continued on next page

5.3 LCN Extension Set Components, Continued

LCN extension illustration

Figures 5-2 and Figure 5-3 illustrate an LCN Extension configuration that includes both data and 12.5 kHz clock extensions. Both figures must be used together.

Figure 5-2 LCN Extension Set Connections for One of a Pair of XLCNE2s

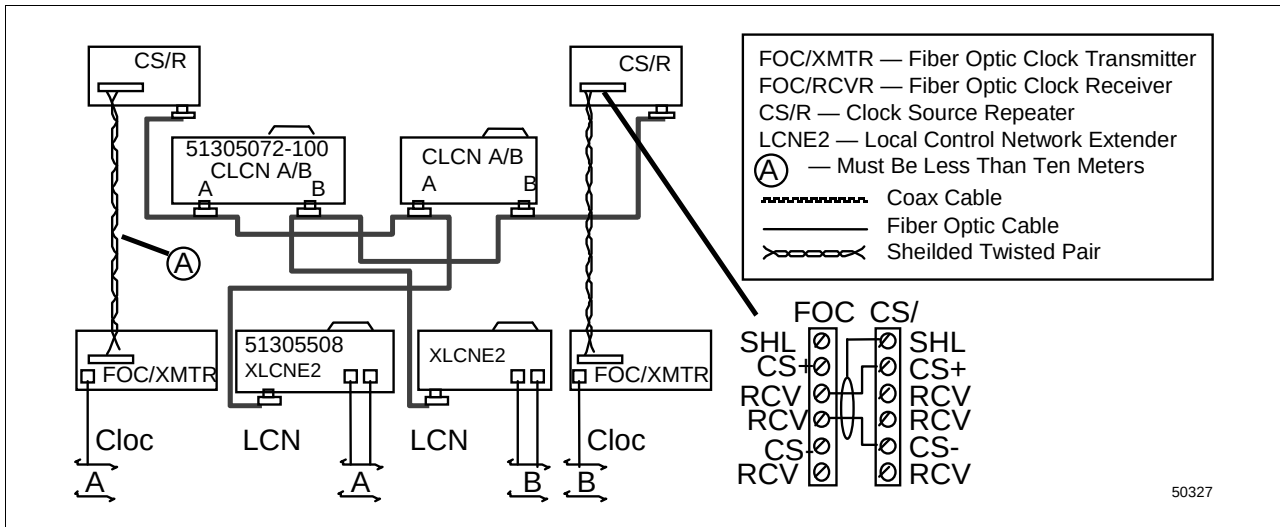
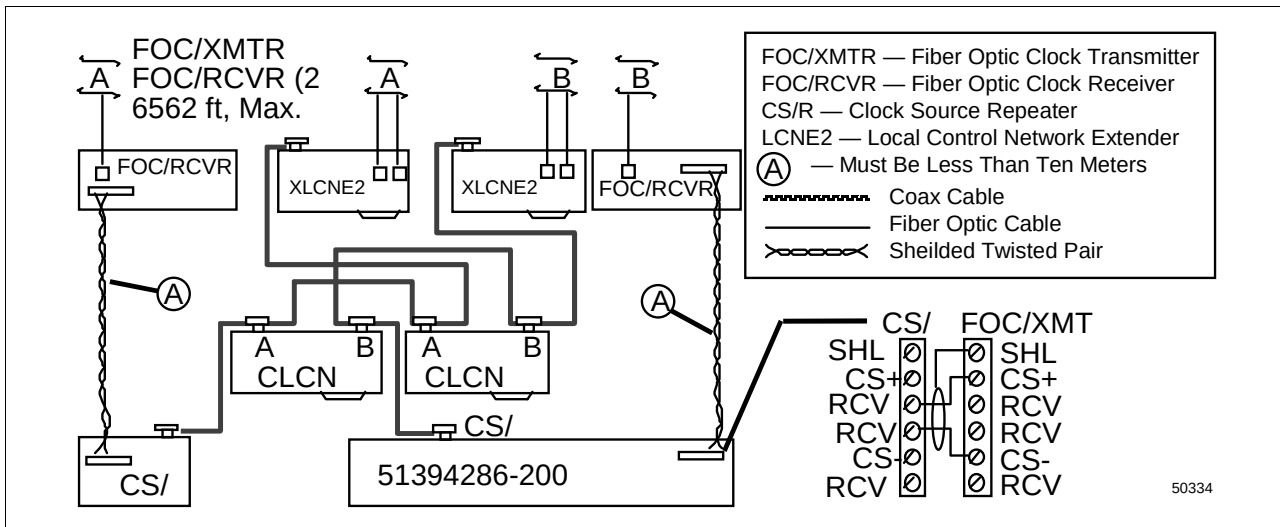


Figure 5-3 LCN Extension Set Connections for One of a Pair of XLCNE2s



Continued on next page

5.3 LCN Extension Set Components, Continued

LCN Extension boards

The circuit boards that comprise the LCN Extension family are described below.

- **LCN Extender (XLCNE2):** Provides an interface to one end of a fiber optic link that passes data. One XLCNE2 board is required for each end of fiber optic cables, A and B. Refer to subsection 5.3.1 for more detailed information.
Fiber Optic Clock Transmitter (FOC/XMTR): Receives the 12.5 kHz system clock from the CS/R board through a current loop interface and transmits it into a designated cable (A or B) over a fiber optic link to a remote cable segment. Refer to subsection 5.3.2 for more detailed information.
- **Fiber Optic Clock Receiver (FOC/RCVR):** Receives the 12.5 kHz system clock transmitted by an FOC/XMTR board and converts it to current loop format to send to a CS/R board. See subsection 5.3.3 for more detailed information.
- **Clock Source/Repeater (CS/R):** Transmits the 12.5 kHz clock into an LCN coaxial cable segment (cable A or cable B). It has an optional current loop input and output so that it can receive the clock from an FOC/RCVR board, another CS/R board, or transmit the clock to FOC/XMTR boards and/or another CS/R board. Refer to subsection 5.3.4 for more detailed information.

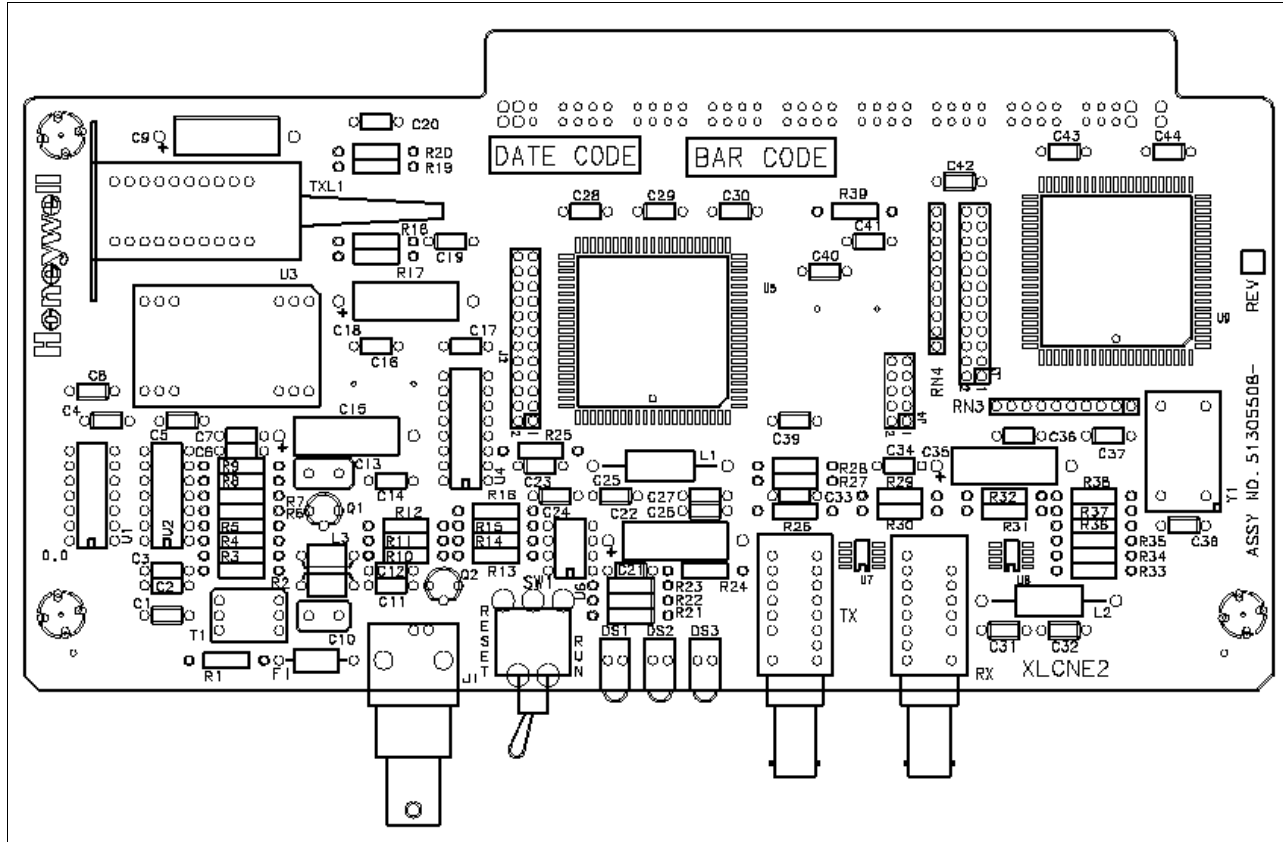
Description

The XLCNE2 board is used at both ends of a fiber optic link to extend the LCN signals to and from a remote cable segment. The XLCNE2 board includes both a fiber optic transmitter and receiver. This allows data to be sent in both directions between the main coaxial cable segment and the remote coaxial cable segment. The XLCNE2-to-XLCNE2 board fiber optic link cannot transmit the 12.5 kHz system clock to the remote cable segment. Refer to Section 4 for system clock system details. If the clock is required in the remote cable segment, such as for the Sequence Of Events function, a twisted pair loop that is no longer than 300 meters or a separate fiber link that is no longer than 2000 meters will be required for each LCN cable (A and B). FOC/XMTR and FOC/RCVR boards are used in the implementation of the fiber clock link. See the next two subsections for additional information.

5.3.1 LCN Extender (XLCNE2)

XLCNE2 board illustrations

Figure 5-4 XLCNE2 Circuit Board Layout (51305508)



ATTENTION

ATTENTION— The XLCNE2 board is a replacement for the LCNE2 board. The two boards perform the same function although they use different connectors. The allowable fiber optic cable lengths allowed are also different for each board. An XLCNE2 can only talk to another XLCNE2, and the LCNE2 and XLCNE2 cannot be mixed on the same fiber.

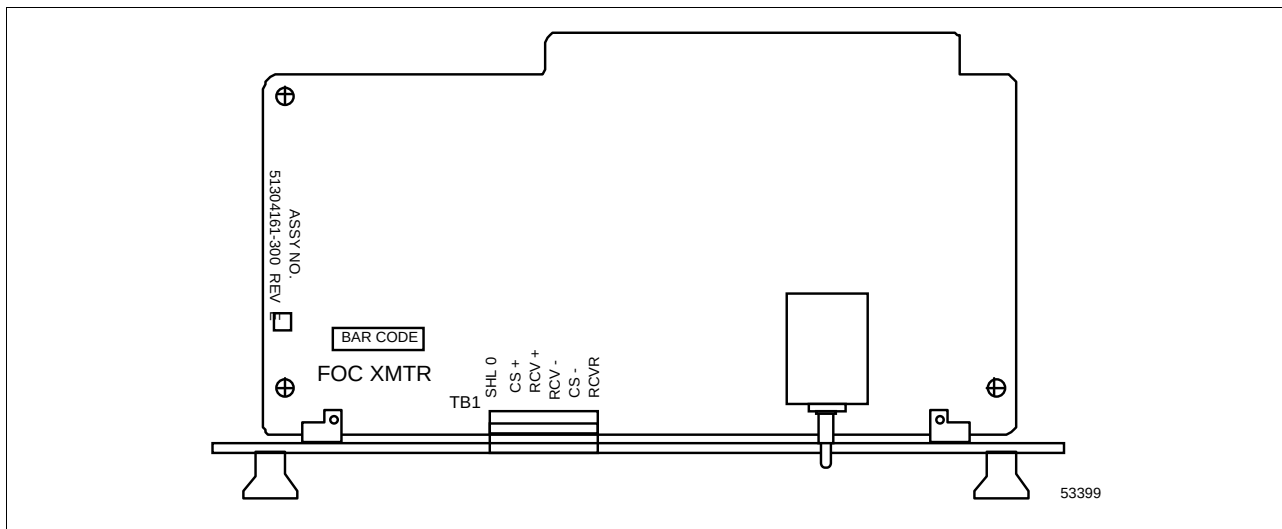
5.3.2 Fiber Optic Clock Transmitter (FOC/XMTR)

Description

The FOC/XMTR circuit board receives 12.5 kHz system clock information through a current loop from a CS/R board and transmits it to a remote coaxial cable segment through a single fiber that is up to 2 kilometers long to a FOC/RCVR board. If the remote cable segment is less than 300 meters away from the central cable segment, the clock can be transmitted directly with a shielded twisted pair current loop from the central CS/R board to the remote CS/R board, eliminating the need for the FOC/XMTR board, FOC/RCVR board, and fiber optic cable components.

FOC/XMTR board illustration

Figure 5-5 FOC/XMTR Circuit Board Layout (51304161-300)



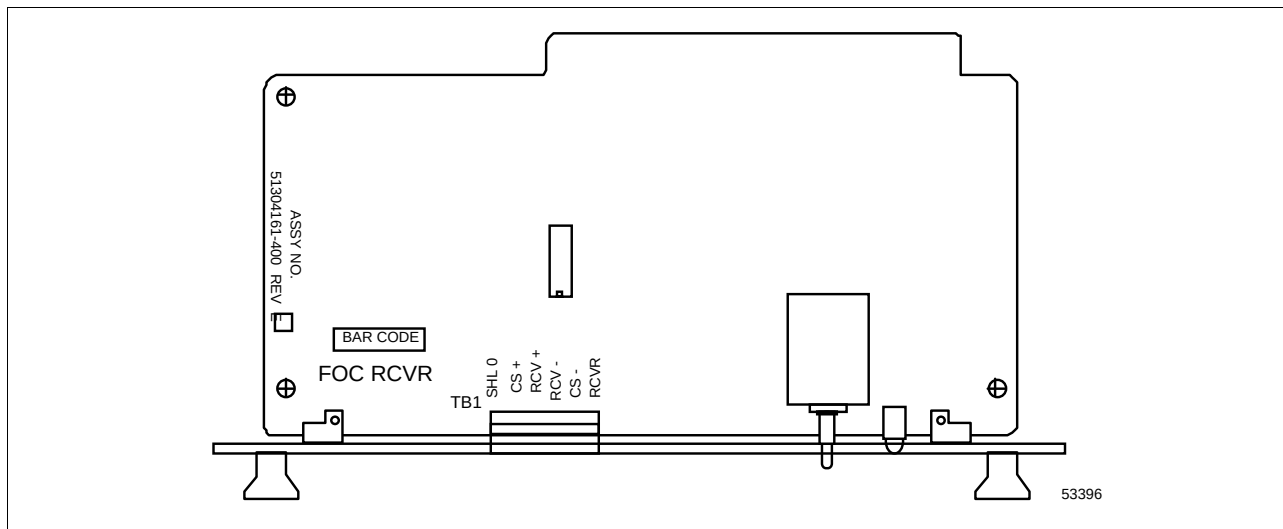
5.3.3 Fiber Optic Clock Receiver (FOC/RCVR)

Description

The FOC/RCVR circuit board receives 12.5 kHz system clock information that is transmitted by a FOC/XMTR board through a fiber optic cable and retransmits it through a current loop to a CS/R board, which in turn retransmits the clock into an LCN coaxial cable. The current loop may be up to 10 meters long. The FOC/RCVR board is used in applications that require transmission of the 12.5 kHz system clock data from a main cable segment to a remote cable segment, over a distance of up to 2000 meters. If the remote cable segment is less than 300 meters away from the central cable segment, the clock can be transmitted directly with a shielded twisted pair current loop from the central CS/R board to the remote CS/R board, eliminating the need for the FOC/XMTR board, FOC/RCVR board, and fiber optic cable components.

FOC/RCVR board illustration

Figure 5-6 FOC/RCVR Circuit Board Layout (51304161-400)



5.3.4 Clock Source/Repeater (CS/R)

Module slot restriction

The CS/R circuit board must be installed in slot 1 of the I/O card cage at the rear of a 5-Slot or 10-Slot Module, directly behind an HMPU, or HPK2 processor board. This is the only location where the CS/R board can obtain the ± 12 Vdc power that it requires.

ATTENTION

If a K2LCN/K4LCN board is installed in slot 1, the CS/R board must be moved to slot 1 of a module that does not have a K2LCN/K4LCN board in slot 1.

Description

In the central (main) cable segment, two CS/R boards (one for LCN Cable A and one for Cable B) receive the 12.5 kHz clock signal from the processor board. Each CS/R board transmits the clock into the LCN central coaxial cable segment. It also provides provisions for a maximum 300 meter length current loop output that can be used to drive up to four loads. The loads may be FOC/XMTR boards (used for transmission of the clock over fiber optic cables to remote segments) and/or other CS/R boards. If there are more than four remote cable segments (up to six are allowed), the first CS/R board can include in its loop a second CS/R board, which is used as a repeater and can drive the additional required FOC/XMTR boards or remote CS/R boards. These configurations are covered in more detail in subsection 5.4.

Twisted pair wiring current loop

If the remote cable segment is less than 300 meters away from the central cable segment, the clock can be transmitted directly with a shielded, twisted pair current loop from the central CS/R board to the remote CS/R board, eliminating the need for the FOC/XMTR board, FOC/RCVR board, and fiber optic cable components.

5.4 XLCNE2 Configuration Rules

Introduction

Section 3 in this publication contains rules and guidelines for the implementation of an Local Control Network from the standpoint of coaxial cable segments, while previous subsections introduced similar aspects of fiber optic extender links. This subsection includes some rules and guidelines concerning fiber optic extension links. It is intended to be supplemental to the material in Section 3. In some cases it may overlap material presented in Section 3 and elsewhere. Nevertheless, it is not intended to be a stand-alone reference for extending LCN cable segments. When planning or expanding a system, it is essential to consider all aspects, including rules and guidelines for the LCN, coaxial cable segments, and fiber optic extensions, because all the elements are interrelated.

Considerations

Consider the following general information, rules, and guidelines when you configure an LCN system that will contain one or more fiber optic links.

XLCNE2 description

LCN Extension sets with fiber optic cables up to two kilometers (6562 feet) in length can be used to interconnect LCN coaxial cable segments. Four LCN Extender (XLCNE2) boards with four fibers are required to extend both Cable A and Cable B to a remote cable segment. Two optic fibers, one for transmit and one for receive, are provided for the Cable A extension and, similarly, two fibers are provided for the Cable B extension. An XLCNE2 board is used at each end of the Cable A fiber optic pair and at each end of the Cable B fiber optic pair. For examples, see Figures 5-9 and 5-10.

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5.4 XLCNE2 Configuration Rules, Continued

Cabling overview

Figure 5-7 LCN Extension Set Connections for One of a Pair of XLCNE2s

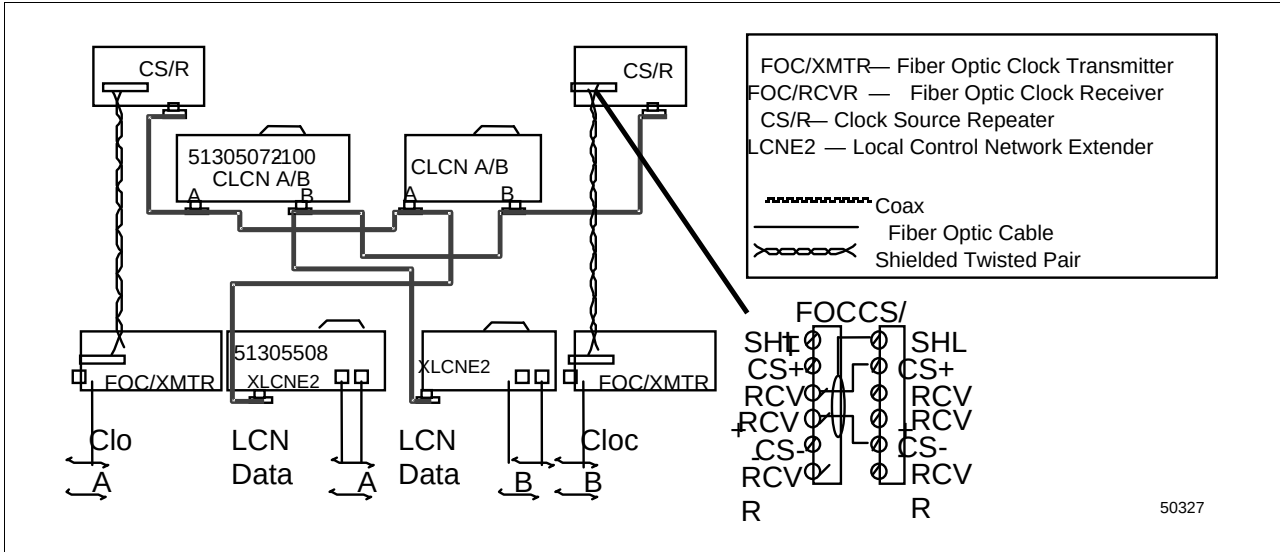
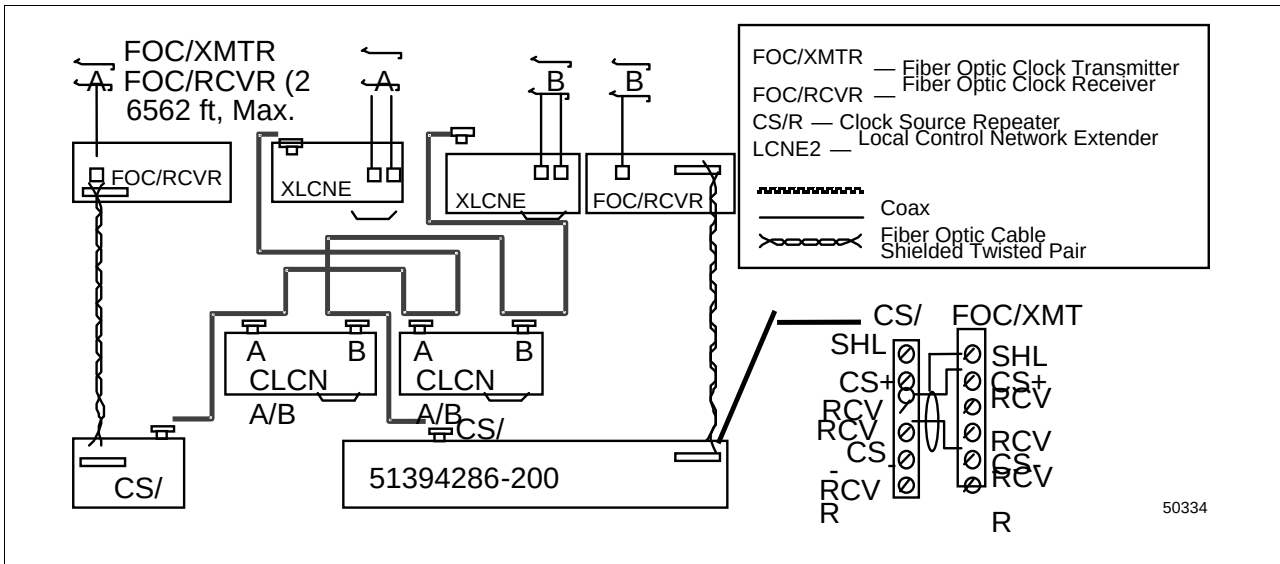


Figure 5-8 LCN Extension Set Connections for One of a Pair of XLCNE2s

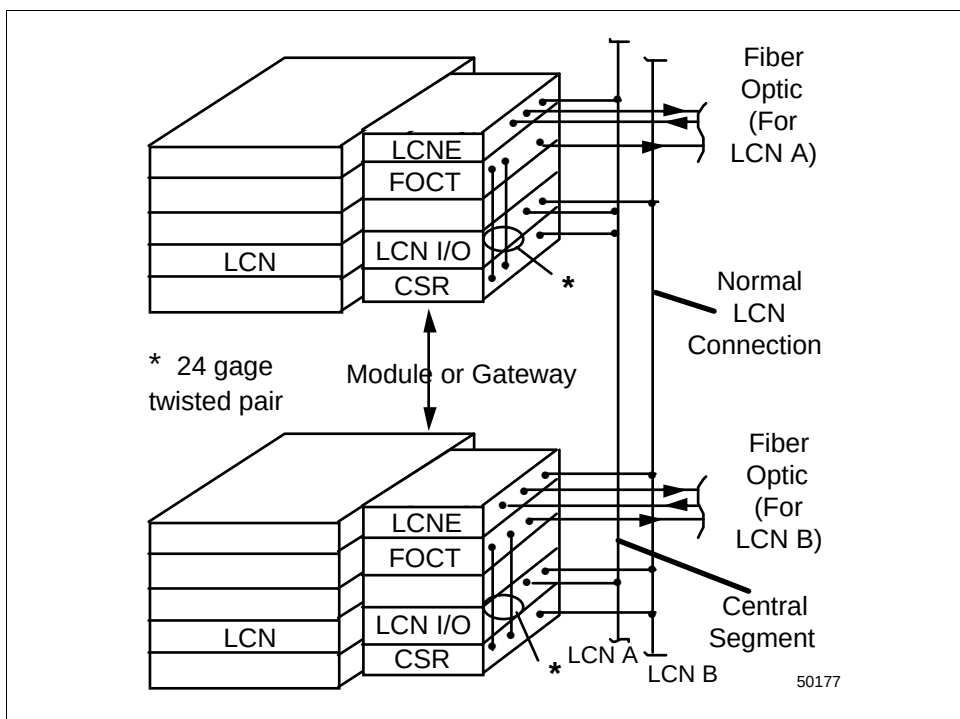


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5.4 XLCNE2 Configuration Rules, Continued

Interconnection diagram

Figure 5-9 LCN Extender Interconnection Diagram for Cable A



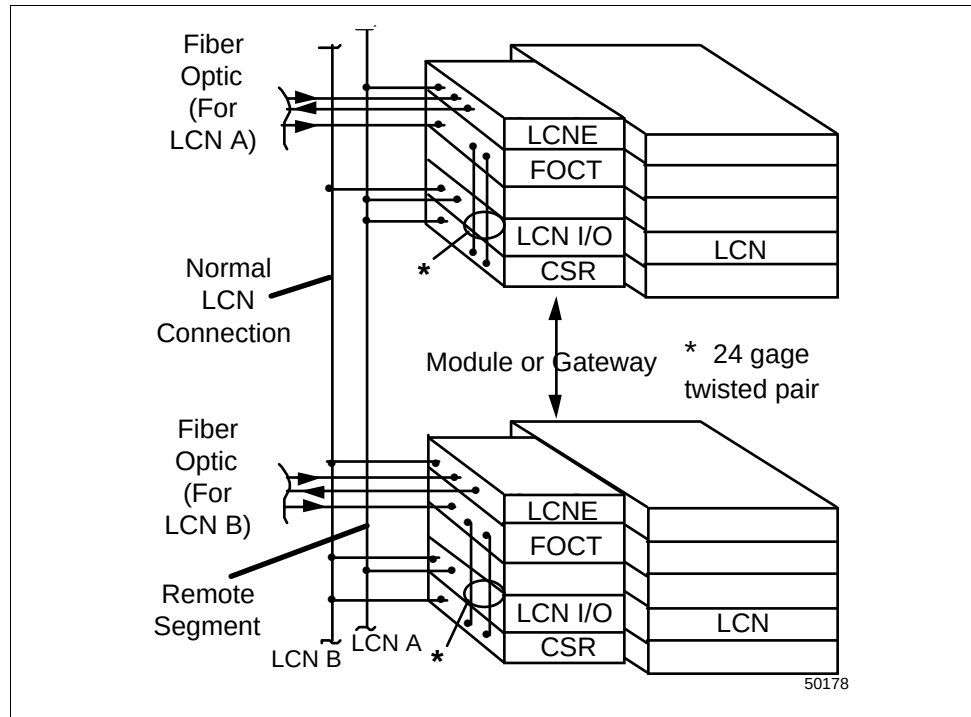
The LCN Extender for Cable A is in a different node from the LCN Extender for Cable B to ensure that both boards are not simultaneously made inoperative by turning the power off at one module.

Continued on next page

5.4 XLCNE2 Configuration Rules, Continued

Interconnection diagram

Figure 5-10 LCN Extender Interconnection Diagram for Cable B



Continued on next page

5.4 XLCNE2 Configuration Rules, Continued

12.5 kHz clock extension

An LCN extension that uses XLCNE2 boards with optic fibers provides extension of the Cable A and B data paths (which can include the 5 Mbits/second digital clock) to the remote cable segment. However, it does not provide transmission of the 12.5 kHz system clock to the remote cable segment. If the remote cable segment requires the system clock and cannot use the digital clock exclusively (for example, it contains HMPU, or HPK2 processor boards), then a separate means must be provided to transmit the 12.5 kHz system clock to the remote cable segment. Section 4 has additional information about the 5 Mbits/second and 12.5 kHz system clocks.

12.5 kHz clock fiber optic transmission

If a remote cable segment requires the 12.5 kHz system clock and it is located more than 300 meters from the central cable segment, then two additional optic fibers must be used to transmit the 12.5 kHz system clock from the central cable segment to the remote cable segment. One fiber carries the clock for Cable A and the other fiber carries the clock for Cable B. Two Fiber Optic Clock Transmitter (FOC/XMTR) boards are used in the central cable segment. They receive the clock over twisted pair current loops from the Clock Source/Repeater (CS/R) boards, also located in the central cable segment. The FOC/XMTR boards transmit the clock over fiber optic links to Fiber Optic Clock Receiver (FOC/RCVR) boards that are located in the remote cable segment. The FOC/RCVR boards transmit the clock over twisted pair current loops that must not exceed 10 meters in length to CS/R boards in the remote cable segment, which in turn, transmit the clock into the LCN segment's coaxial cables. This configuration is illustrated in Figures 5-9 through 5-12. The CS/R boards in the central cable segment are installed in nodes that have been configured as clock sources in the software Network Configuration File (NCF).

12.5 kHz clock current loop transmission

If a remote cable segment requires the 12.5 kHz system clock and it is located less than 300 meters from the central cable segment, optionally the clock can be transmitted directly from the CS/R boards in the central cable segment to the CS/R boards in the remote cable segment over shielded, twisted pair current loops that do not exceed 300 meters. This eliminates the need for the FOC/XMTR boards, FOC/RCVR boards, and clock optic fibers that were described in the previous paragraph. However, with this approach you lose the advantages of fiber optics that are described in subsection 5.1.

Continued on next page

5.4 XLCNE2 Configuration Rules, Continued

Avoid node power removal

Although XLCNE2, FOC/XMTR, and FOC/RCVR boards can be installed in unused I/O slots for any module type, it is desirable to locate them in modules that are the least likely to have their power removed. Table 4-1 lists the recommended order that should be used in selecting module types for housing these fiber optic extender boards.

Separate power source for cable A and B

The XLCNE2, FOC/XMTR, and FOC/RCVR boards for Cable A must be in a different node than those for Cable B and must be powered by separate ac power entries. If space is limited, an otherwise empty Dual Node Module with a power supply and KJMP board (51401594-200) installed in each node can be used to house the extender board sets. The XLCNE2 board and associated FOC/XMTR or FOC/RCVR board for Cable A are installed in the I/O card cage for the lower three-slot node and the corresponding boards for Cable B are installed in the upper two-slot node. If FOC/XMTR/FOC/RCVR boards are not used, two XLCNE2 boards for Cable A can be installed in the three-slot node and two XLCNE2 boards for Cable B can be installed in the two-slot node.

ATTENTION

ATTENTION— The LCNFL remote node is no longer supported and does not work with XLCNE2.

Rules and guidelines

When LCN extension hardware is used, the configuration of the LCN has increased importance from the standpoint of reliability and failure analysis. This hardware represents points in the system where failure or power loss of a relatively small percentage of the system can potentially isolate and thereby affect a whole segment or group of modules due to the disruption of communications. The extension hardware is transparent to the system communications software, and therefore during normal operation, the topology of the system may seem unimportant. However, when failure scenarios are considered, it is evident that system layout can dramatically affect diagnosis, recovery, and visibility of the process(es).

Continued on next page

5.4 XLCNE2 Configuration Rules, Continued

Rules and guidelines summary

The following rules and guidelines are essential to maximize system reliability in the event there is an LCN Extender failure.

- The configuration of removable media disk drives and Engineer's keyboards is very important. For details, refer to subsection 3.2.
- Optic fibers for Cable A and B should be in separate cables and routed along separate paths from source to destination. If one cable is damaged, the other will still be operable. However, for any two nodes in the network, the maximum difference in Cable A and Cable B lengths, including the sum of all coaxial and fiber optic segments, should not exceed 300 meters or communication problems may result. Refer to subsection 6.1 for additional details.
- The number of coaxial cable segments connected in serial, by way of fiber optic cables and XLCNE2 boards, must not exceed three.
- The XLCNE2 board for LCN Cable A should be installed in a different module than the XLCNE2 board for LCN Cable B. This is also true for FOC/XMTR, FOC/RCVR, and CS/R board pairs. For example, the XLCNE2 board for Cable A should be installed in one NIM of a redundant pair and the XLCNE2 board for cable B should be installed in its NIM partner. The same principle applies for redundant HG pairs.
- All nodes containing an XLCNE2 board should be served by an Uninterruptible Power Supply (UPS), if available, to ensure that the LCN coaxial cable segments will not be disconnected because of power outages and thus cause reconnection problems.
- If possible, avoid configuring only two nodes in a remote cable segment due to error situations that may occur if the fiber optic cable is broken or if XLCNE2 boards have their power removed.
- Because of the lower reliability of the fiber optic extension to remote cable segments, locate redundant HGs and added HGs in the same coaxial cable segment. Also, locate redundant NIMs and redundant PLCGs in the same coaxial cable segment.
- To prevent loss of view to the process, two or more Universal Stations (US) or Universal Work Stations (UWS) in a console should be in the same coaxial cable segment. In addition, for plant integrity, Honeywell recommends that the NIMs and HGs most often associated with a Universal Station be in the same coaxial cable segment.
- The two clock sources configured in the Network Configuration File must be located in the main coaxial cable segment.

Continued on next page

5.4 XLCNE2 Configuration Rules, Continued

- Unless the Precision Clock Source option is installed, the power supply in each node of the system must be set up by a pinning option for the internal (ac line frequency) clock. If the Precision Clock Source option is installed, the power supplies in the nodes that have the Precision Clock Source installed in them must be pinned for the external clock option and all other nodes pinned for internal clock.
-

5.5 Typical LCN Extender Installations

Overview

This subsection includes examples of LCN Extension hardware with XLCNE2 boards used to extend the LCN data path, an LCNFL board used for a single remote node, and FOC/XMTR, FOC/RCVR, and CS/R boards used to extend the 12.5 kHz system clock to a remote cable segment. The examples illustrate the optional use of a shielded, twisted pair current loop in place of the FOC/XMTR boards, FOC/RCVR boards, and clock optic fibers when the distance between the cable segments is 300 meters or less. Also shown in detail are the fiber and twisted pair interconnections between the various extension components for various configurations.

Cable segment interconnections

Figures 5-9 and 5-10 illustrates a central or main cable segment (in the upper half of the figure) with a remote cable segment (lower left) and a remote node (lower right). It illustrates the interconnections of the two coaxial cables, Cable A and Cable B, in both the central and the remote cable segments. The example includes a fiber optic link using FOC/XMTR and FOC/RCVR boards for transmission of the 12.5 kHz system clock from the central cable segment to the remote cable segment. In the central cable segment, note the twisted pair cables that carry the clock from the CS/R boards to the FOC/XMTR boards, and in the remote cable segment, note the twisted pair cables that carry the clock from the FOC/RCVR boards to the remote CS/R boards. The CS/R boards in the remote segment then transmit the clock into the A and B coaxial cables.

Continued on next page

5.5 Typical LCN Extender Installations, Continued

Cable A and B interconnections

Figures 5-11 and 5-12 illustrates the use of LCN Extender components to connect a remote cable segment to a central cable segment. This example also includes the FOC/XMTR boards, FOC/RCVR boards, and optic fibers required to transmit the 12.5 kHz system clock to the remote cable segment. In this illustration, the extension components (XLCNE2, FOC/XMTR, FOC/RCVR, and CS/R boards) associated with Cable A are located in a separate module from the components associated with Cable B. This separation occurs in both the central cable segment and the remote cable segment. As indicated in the figure, this placement of the Cable A components in a module that is separate from the Cable B components minimizes the possibility of total communications loss in the event of ac power loss or power supply failure in one module. Although this example has the CS/R, FOC/XMTR or FOC/RCVR boards, and the XLCNE2 board for each coaxial cable segment located in the same module, this is not required. It is desirable for reliability purposes that all of the components for at least one cable (A or B), including fiber interface components, be serviced by a UPS. It is also desirable that the optic fibers associated with each cable be routed separately, and that spare fibers be included in the cable, as is covered in more detail in subsection 6.1.

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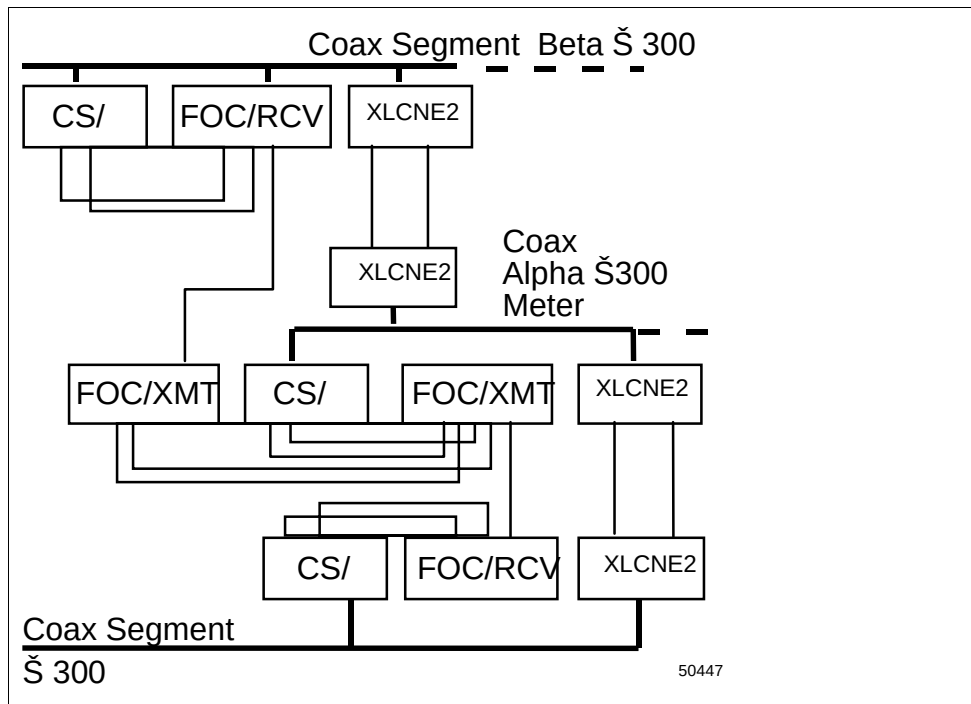
5.5 Typical LCN Extender Installations, Continued

System clock current loop interconnections example 1

Figure 5-11 illustrates a central segment “Alpha” with two remote cable segments, “Beta” and “Gamma.” Both remote cable segments receive the 12.5 kHz system clock through fiber optic links.

Figures 5-12, 5-13, and 5-14 show, in detail, the current loop interconnections between the central cable segment’s CS/R and FOC/XMTR boards and between the remote cable segment’s FOC/RCVR and CS/R boards. The figures also illustrate how the twisted pair shields must be grounded at one end of the loop only. Note that the maximum length of the loop between an FOC/RCVR and CS/R board is 10 meters, whereas the maximum length of the loop between the CS/R board and the two FOC/XMTR boards is 300 meters.

Figure 5-11 LCN Extender Fiber Optic Cabling Current Loop Interconnections – Example 1, Overview



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5.5 Typical LCN Extender Installations, Continued

Wiring details

Figure 5-12 Detail of Segment Beta

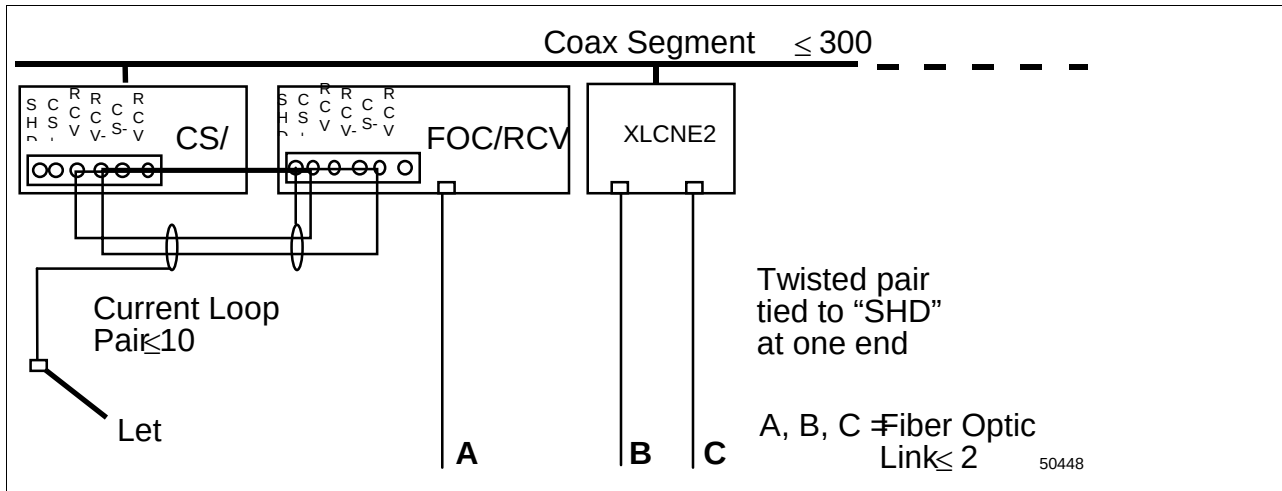
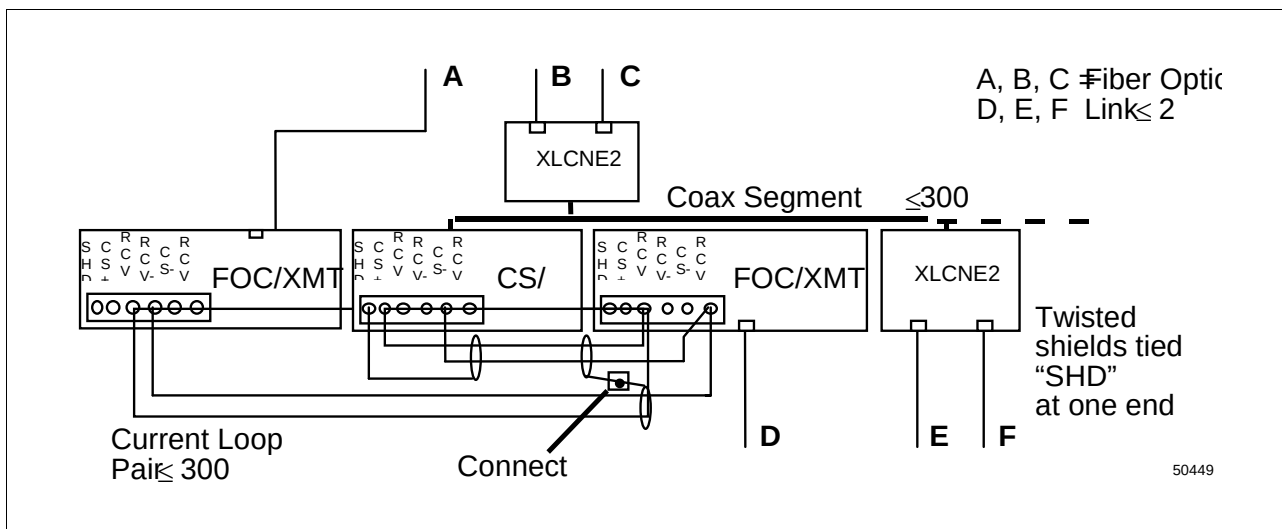


Figure 5-13 Detail of Segment Alpha

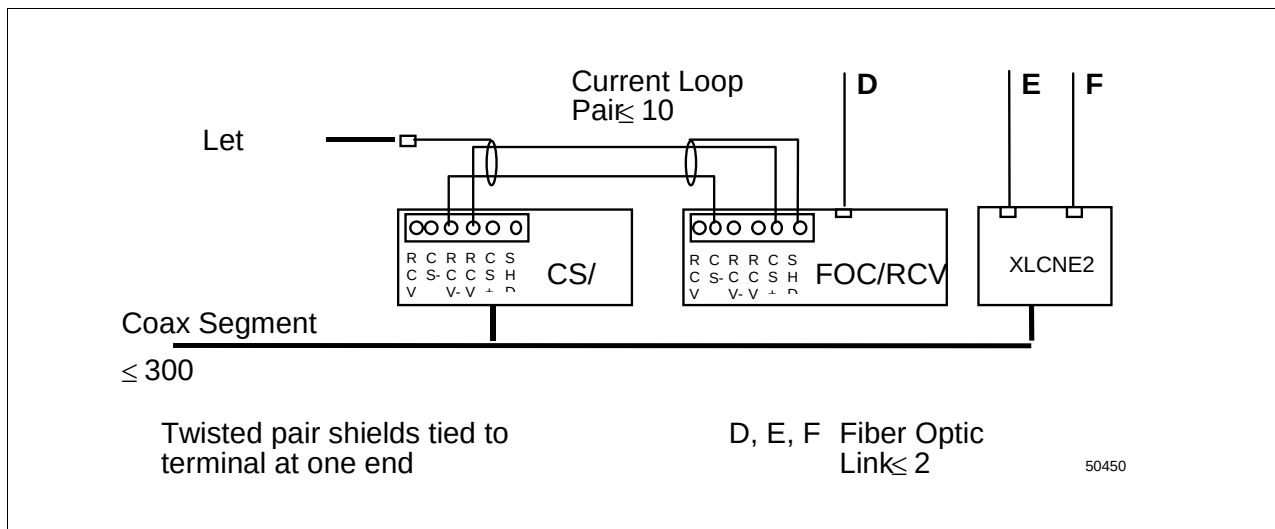


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5.5 Typical LCN Extender Installations, Continued

Wiring details,
continued

Figure 5-14 Detail of Segment Gamma



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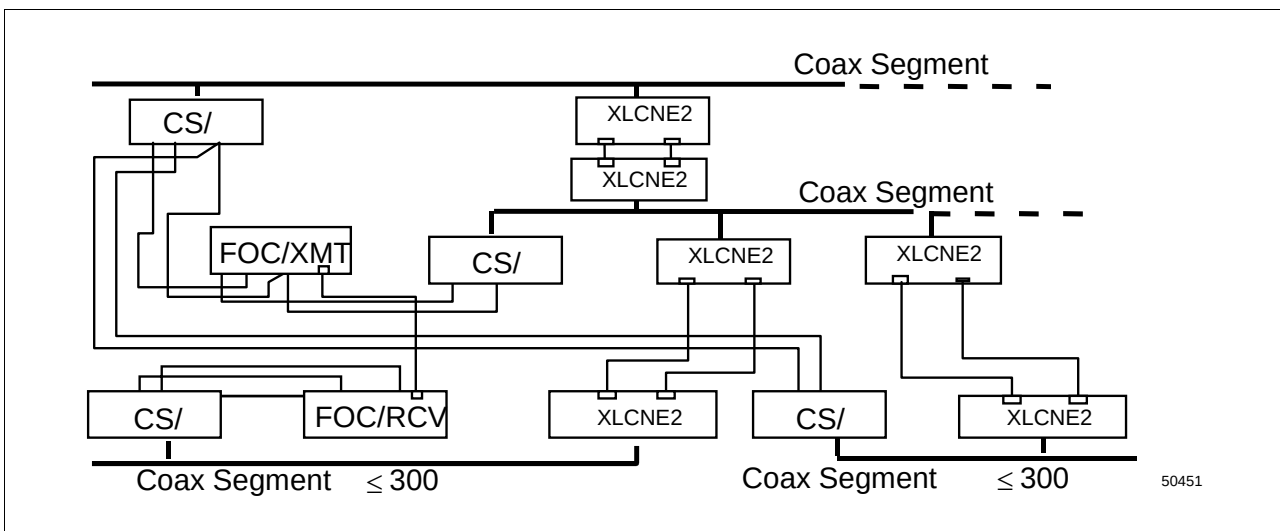
5.5 Typical LCN Extender Installations, Continued

System clock current loop mixed with fiber optic clock transmitters

Figure 5-15 illustrates a central cable segment “Alpha” that has three remote segments, “Beta,” “Delta,” and “Gamma.” Remote cable segment “Delta” has the 12.5 kHz system clock transmitted to it with fiber optic components, as in the previous example. However, remote cable segments “Beta” and “Gamma” have the clock transmitted to them by current loop.

Figures 5-20 and 5-21 show, in detail, the connections for the current loop twisted pairs and shields. Note that the current loop from the CS/R board in the central cable segment includes as loads the FOC/XMTR board (for transmitting the clock to segment “Delta”) as well as the CS/R boards for segments “Beta” and “Gamma.” The total length of this current loop, which includes the sections marked “X,” “Y,” and “Z” in the figure, must be no greater than 300 meters.

Figure 5-15 LCN Extender Fiber Optic Cabling Current Loop Interconnections – Example 2, Overview



Continued on next page

5.5 Typical LCN Extender Installations, Continued

Wiring details

Figure 5-16 Detail of Main Current Loop

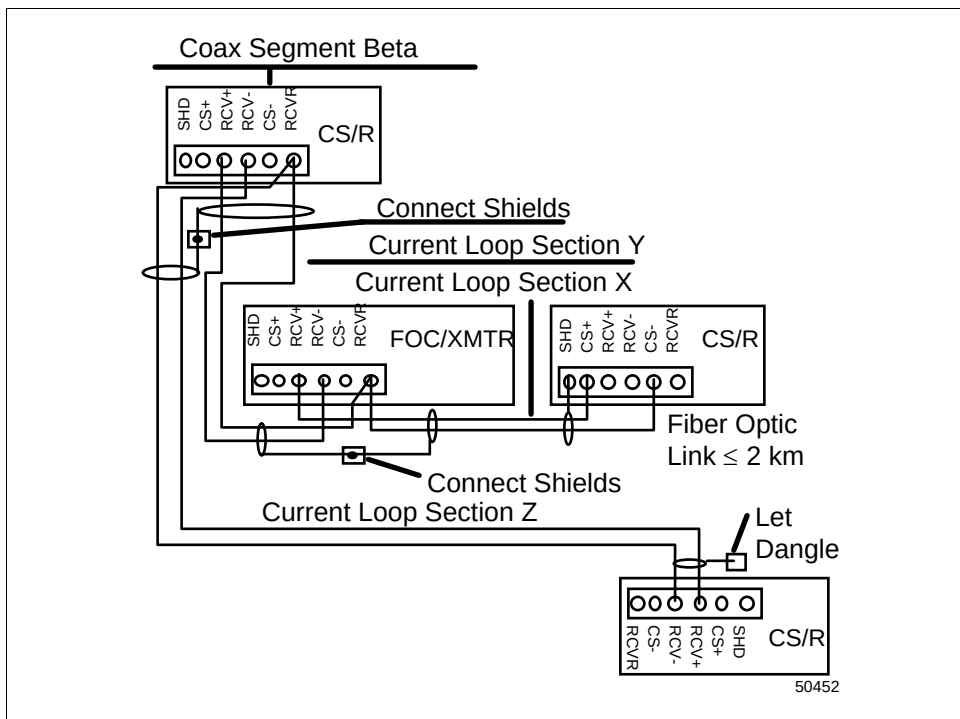
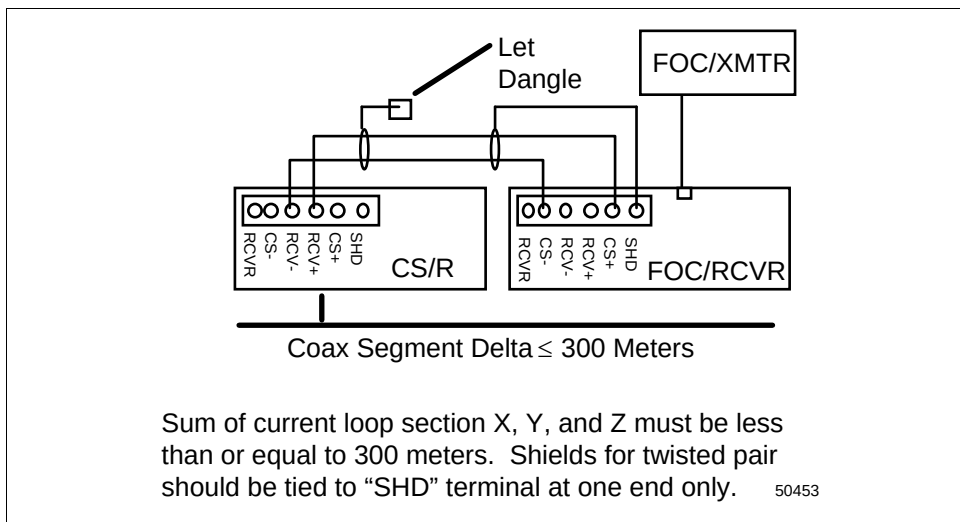


Figure 5-17 Detail of Remote Current Loop



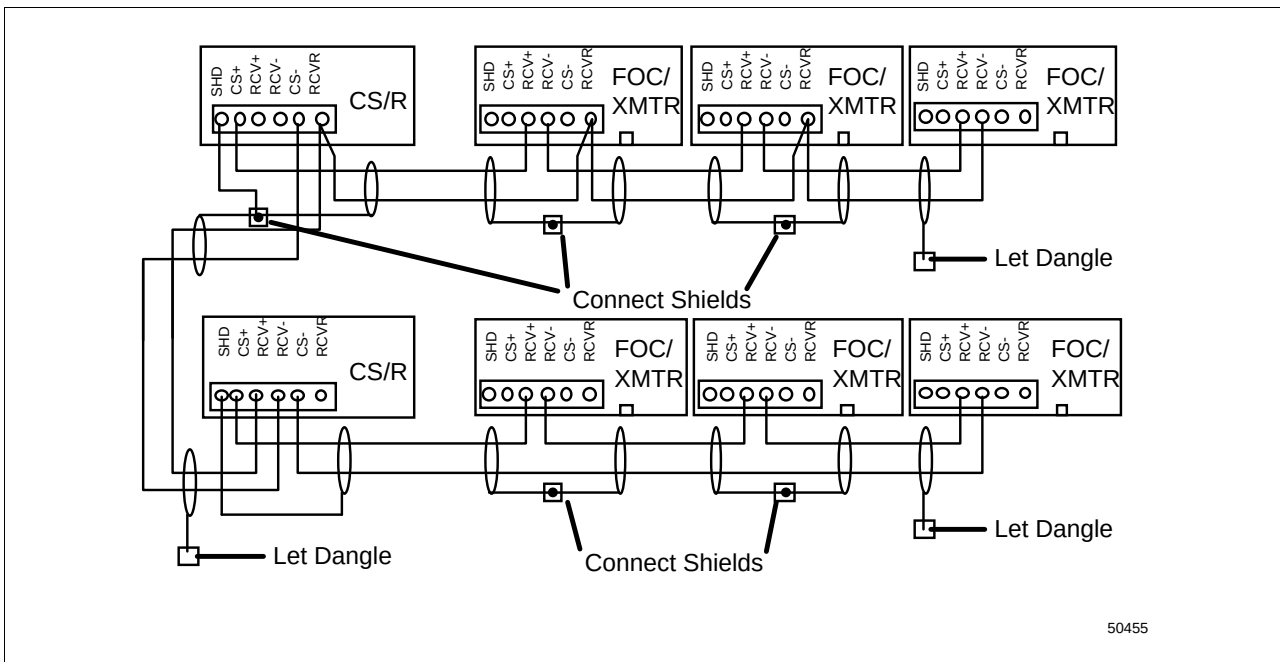
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5.5 Typical LCN Extender Installations, Continued

Retransmitting the 12.5 kHz clock

The current loop output of a CS/R board can drive up to four loads, that can be either FOC/XMTR or other CS/R board inputs. If more than four loads are required (for example, there are more than four remote segments which require the 12.5 kHz system clock), a CS/R board can be used to repeat, or retransmit, the clock. Figure 5-18 illustrates this function, showing a CS/R board driving three FOC/XMTR boards and another CS/R board, which in turn drives three additional FOC/XMTR boards. The system clock can be retransmitted only twice in this manner, or excessive distortion of the signal may result. The maximum length of each current loop is 300 meters, providing a total length of 900 meters, assuming the loop is retransmitted twice.

Figure 5-18 Multiple Current Loop Connections with Retransmitted Clock



ATTENTION

- CSR current loop can be repeated only twice.
- This set of boards and connections is for Cable A and must be repeated for Cable B.

5.6 Fiber Optic Cable Specifications

Overview

There are four Honeywell raw fiber optic cable specifications. They describe both indoor and outdoor versions of fiber optic cable in both the 100 micron and 62.5 micron optic fiber core sizes. In addition, there are three Honeywell fiber optic cable assembly drawings, each with associated marketing model numbers.

Terminology

“Micron” is another term for “micrometer,” a unit of measure, one thousandth of a millimeter or one millionth of a meter.

The symbol “ μm ” is often used to denote micron or micrometer.

The symbol “nm” means nanometer (one billionth of a meter) and is used as a unit of measurement for the wavelength of light.

5.6.1 100 and 62.5 Micron Optic Fiber

Overview

100 micron was the optic fiber core size originally specified for the TDC 3000^X system. To date, most TDC 3000^X systems that have fiber optic cable installed use 100 micron optic fiber. However, the industry has steadily moved toward 62.5 micron optic fiber as a de facto standard for multimode optic fiber. Hastening this process is the preference for 62.5 micron optic fiber in the recently completed ANSI Fiber Distributed Data Interface (FDDI) specification. As a result of this shift, 62.5 micron optic fiber is now more readily available at a lower cost than 100 micron optic fiber. Recognizing this industry trend, Honeywell has added 62.5 micron optic fiber as an option in implementing **TotalPlant** Solution (TPS) System LCN fiber optic installations. As a matter of fact, Honeywell now recommends that all new installations use 62.5 micron optic fiber.

Do not mix optic fiber sizes

Now that two different sizes of fiber are allowed in the TPS System, there exists the possibility of inadvertently mixing fiber sizes in a fiber optic link. This is generally undesirable. There is no additional loss when light travels from the smaller optic fiber to the larger one, but a 4.1 dB loss is experienced when going from a larger to smaller size. Because fiber optic cables are routed as a pair, one for transmit and one for receive, light will be traveling in opposite directions in the individual optic fibers of the pair. Adding a different optic fiber size to both cables at one end of a link, such as when transitioning from outdoor cable to indoor cable, means that at least one cable will suffer at least a 4.1 dB loss. Therefore, to be safe, DO NOT mix optic fiber sizes in a fiber optic link.

5.6.2 Cable Procurement Policy

Overview

In general, Honeywell does not wish to supply or install outdoor fiber optic cables. Honeywell relies on the cable installation expertise of cable vendors and installation contractors to perform the cable installation. Honeywell also does not wish to restrict the purchase of outdoor fiber optic cable to a particular vendor. Honeywell recognizes that vendors may be able to supply better service in some parts of the world than in others, thus making it desirable to have a choice worldwide. Also, the installation conditions at various project sites may call for widely differing types of cable construction. For these reasons, the outdoor cable specifications were written rather loosely with respect to physical construction details and mechanical parameters. The actual glass fiber itself is completely specified to insure proper operation of the fiber optic link.

Outdoor cable vendor and installer selection

If the customer desires, Honeywell will contract with cable vendors and installation contractors for the customer to purchase cable, oversee, and guarantee a proper installation. However, if the customer procures his fiber optic cable directly from the supplier and arranges his own installation, the cable supplier and/or installing contractor must certify to the customer that his cable fully meets or exceeds the applicable Honeywell cable specification. Honeywell will freely supply our outdoor cable specifications to our customers for this purpose.

Indoor cable procurement

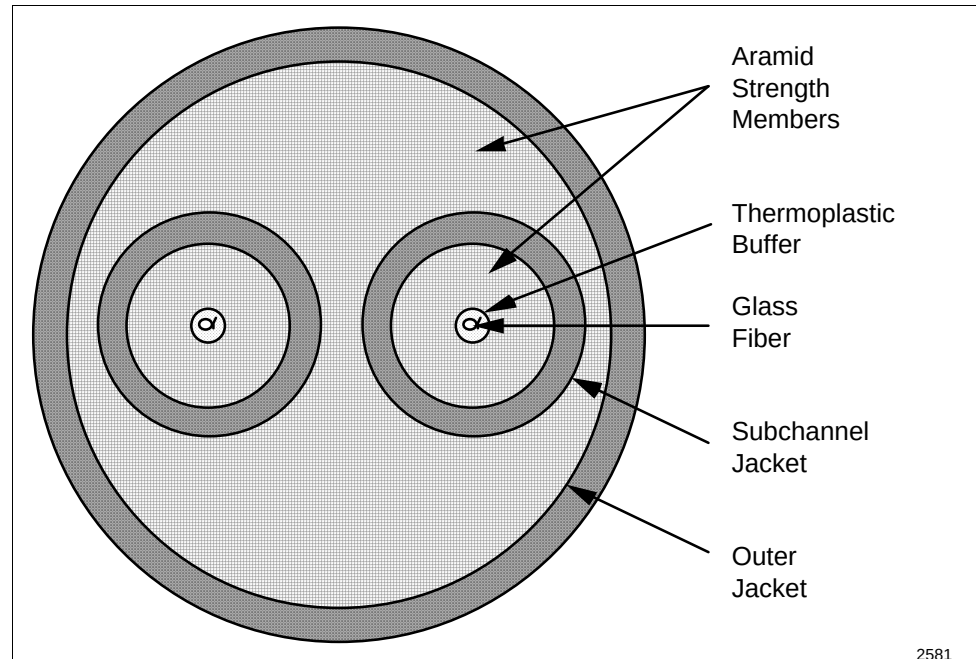
In contrast, the indoor cable specifications completely specify both the mechanical construction details of the cable, and the important parameters of the glass fiber. The recommended method of procurement is direct purchase from Honeywell as a finished cable assembly, including connectors, under the applicable Honeywell model number. See subsection 5.6.5 for a description of the cable assemblies that are available and the corresponding Honeywell model numbers.

5.6.3 Indoor Grade Cable Specifications

Cable construction illustration

Figure 5-19 illustrates the construction of indoor fiber optic cable as specified by Honeywell. Both 100 micron and 62.5 micron indoor cables share this basic construction, differing only in the optic fiber size.

Figure 5-19 Indoor Tight-Buffered Fiber Optic Cable



Tight-buffered cable description

This type of cable is known as tight-buffered cable because the optical fibers are tightly held by the cable fillers. Differing coefficients of expansion between the cable materials and the glass fiber can subject the fibers to significant microbending losses if the cable is exposed to temperature extremes. For this reason, tight-buffered cable is limited to indoor use. The advantage indoors is good physical protection for the fiber while maintaining the cable flexibility required for routing the cable inside electronic cabinets and under floors as examples.

Cable subunits

The cable contains two, four, or six subunits. Each subunit protects a single optic fiber and can have independent connectors. The standard indoor cable assemblies are duplex (two optic fibers). Finished cable assemblies with four or six optic fibers are available by special order.

Continued on next page

5.6.3 Indoor Grade Cable Specifications, Continued

Cable jacket options	There are two cable jacket options available for the indoor cable. The National Fire Protection Association (NFPA) publishes the National Electrical Code (NEC) to establish fire safety standards for premises wiring. Honeywell specifies jacketing material conforming to either NEC optical cable rating OFNR (Optical Fiber, Nonconducting, for Riser applications), or OFNP (Optical Fiber, Nonconducting, for Plenum applications). The standard cable assemblies which can be purchased from the Honeywell price book by model number are OFNR rated cables. Should OFNP rated cables be required, they can be special ordered through Honeywell Purchasing. OFNP rated cables are required only when routing indoor optic fiber runs through air handling chambers.
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62.5 micron cable parameters	The important parameters of the 62.5 micron indoor grade glass fiber specified by Honeywell are as follows. <table><tr><td>Core diameter:</td><td>62.5 ± 3 microns</td></tr><tr><td>Cladding diameter:</td><td>125 ± 2 microns</td></tr><tr><td>Bandwidth @ 1300 nm:</td><td>500 MHz-km minimum</td></tr><tr><td>Attenuation @ 1300 nm:</td><td>.9 dB/km maximum</td></tr></table>	Core diameter:	62.5 ± 3 microns	Cladding diameter:	125 ± 2 microns	Bandwidth @ 1300 nm:	500 MHz-km minimum	Attenuation @ 1300 nm:	.9 dB/km maximum
Core diameter:	62.5 ± 3 microns								
Cladding diameter:	125 ± 2 microns								
Bandwidth @ 1300 nm:	500 MHz-km minimum								
Attenuation @ 1300 nm:	.9 dB/km maximum								

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5.6.3 Indoor Grade Cable Specifications, Continued

Mechanical properties Table 5-1 lists the mechanical properties of indoor grade cable supplied by Honeywell. “Subchannel” refers to an individual jacketed optic fiber which is part of a multifiber cable (refer to Figure 5-19). “N” stands for “Newton,” a metric unit of force equal to approximately 0.225 lb.

Table 5-1 Mechanical Properties of 62.5 Indoor Grade Cable

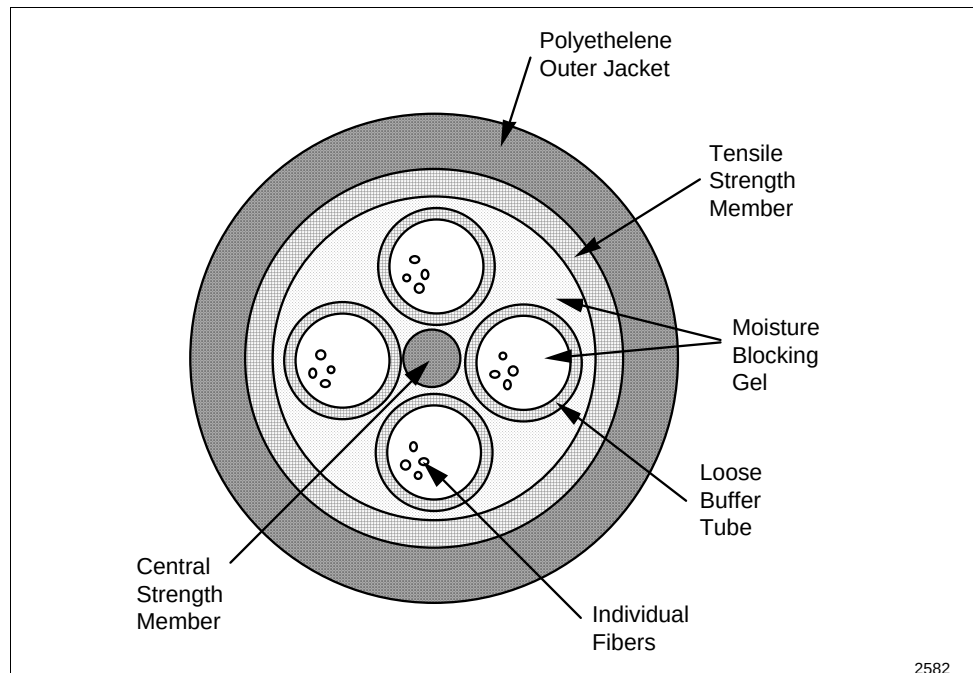
Characteristic	Two Fiber Cable	Four Fiber Cable	Six Fiber Cable
Tensile Load, Short Term, N (LBF)	900 (200)	1000 (225)	1000 (225)
Tensile Load, Log Term, N (LBF)	300 (68)	400 (90)	350 (79)
Minimum Bend Radius, Loaded, CM (IN)	9.0 (3.5)	10.2 (4.0)	11.9 (4.7)
Minimum Bend Radius, Installed, CM (IN)	5.9 (2.3)	6.8 (2.7)	7.9 (3.1)
Nominal Outer Diameter, MM (IN)	5.9 (0.3)	6.8 (0.3)	7.9 (0.3)
Nominal Weight KG/KB (LB/KFT)	34 (23)	46 (31)	65 (44)

5.6.4 Outdoor Grade Cable Specifications

Cable construction illustration

Honeywell specifies a cable construction known as loose-tube for its outdoor grade fiber optic cable. An illustration of this construction is shown in Figure 5-20.

Figure 5-20 Outdoor Loose-Tube Fiber Optic Cable



Cable construction characteristic

This construction is characterized by loose fitting, gel filled tubes into which the optic fibers are placed. The optic fibers are actually longer than the tubes so that when thermal expansion lengthens the buffer tubes, the glass fibers which have a lower coefficient of expansion are never subject to tensile stress.

Cable strength

The cable is given buckle resistance typically by a glass reinforced plastic (GRP) rod through the center of the cable. An aramid wrap around the buffer tubes provides tensile strength. This type of cable is significantly stiffer than tight-buffered cable and the jacketing material does not meet NEC requirements. These factors make it unsuitable for indoor use. The loose-tube design's advantage is being able to take environmental extremes without suffering any significant optical performance degradation.

Continued on next page

5.6.4 Outdoor Grade Cable specifications, Continued

Direct burial cable

Figure 5-20 shows a typical construction for an aerial/duct cable. A direct burial cable is similar, except that it would probably have two additional layers, a steel armor tape layer for rodent protection, covered by an additional polyethylene jacket. Vendors also may have double armoring options available. Again, exact construction details will vary from vendor-to-vendor, but the basic loose-tube concept remains the same.

62.5 micron cable parameters

The important parameters of the 62.5 micron outdoor grade glass fiber specified by Honeywell are identical to the parameters specified for indoor cable.

Core diameter:	62.5 ± 3 microns
Cladding diameter:	125 ± 2 microns
Bandwidth @ 850 nm:	500 MHz-km minimum
Attenuation @ 850 nm:	.9 dB/km maximum

Cable recommendation

As discussed in subsection 5.6.2, Honeywell does not recommend specific cable for outdoor use. When a cable is selected, refer to the vendor's specifications for that cable to obtain applicable mechanical properties.

5.6.5 Fiber Optic Cable Assemblies

Available duplex indoor cable assemblies

This subsection discusses the duplex indoor fiber optic cable assemblies with preinstalled connectors that are available from Honeywell. Table 5-2 lists the all of the standard assemblies and the information that follow the table discuss each type.

Table 5-2 Standard Duplex Indoor Fiber Optic Cable Assemblies

Model Number	Length (Meters)	Core Diameter (μm)	Description
P-KFB01 P-KFB02 P-KFB05 P-KFB10 P-KFB20 P-KFB50	1 2 5 10 20 50	62.5	ST to ST

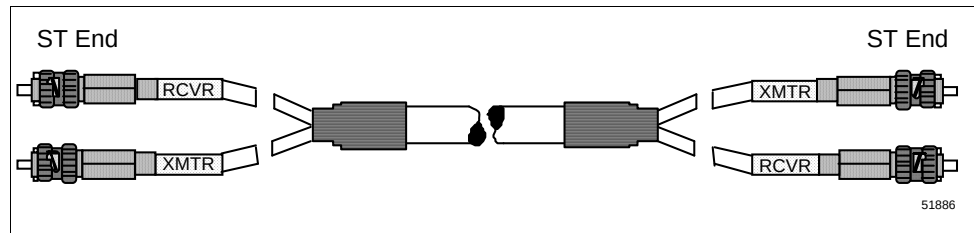
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5.6.5 Fiber Optic Cable Assemblies, Continued

Model P-KFBxx 62.5 micron ST/ST cable

This fiber optic cable assembly contains ST connectors at both ends. Standard LCN equipment does not use ST connectors but special long distance versions of the XLCNE2 have been developed which do. Also, the PM I/O Link Extender uses ST connectors. These cable assemblies may also be used in interconnect panels as patch cables. Refer to subsection 6.4 for a discussion of interconnect panels. Figure 5-21 is an illustration of the model P-KFBxx cable.

Figure 5-21 Model P-KFBxx Fiber Optic Cable Assembly (ST to ST)



5.6.6 Fiber Optic Cable Connectors

Recommended cable connectors

Honeywell recommends the fiber optic connectors listed in Table 5-3 for use on indoor tight-buffered fiber optic cable only because outdoor cable does not permit the small bend radii needed in a cabinet or equipment rack and is always spliced to a short length of indoor cable for connection purposes.

Table 5-3 Fiber Optic Cable Connectors

Supplier	Supplier's Part Number	Type	Power Loss (dB)	Fiber (μm)
Amphenol Fiber Optic Products 1925 Ohio Street Lisle, Ill, 60532 (708)-810-5800	953-101-5010	ST Ceramic Ferrule	<0.13 dB Typical	62.5

Continued on next page

5.6.6 Fiber Optic Cable Connectors, Continued

Connector tools

Special tools and procedures are required for installing connectors. The cable must be stripped, cut, finished, and attached to the connector. Typical tools and materials for performing these operations are listed in Table 5-4.

Table 5-4 Fiber Optic Cable Connector Tools

Supplier	Supplier's Part Number	Type	Use
Amphenol Fiber Optic Products 1925 A. Ohio Street Lisle, Ill 60532 Phone: 630-960-1010 Fax: 630-810-5640 sales@amphenol-fiberoptics.com http://www.amphenol-fiberoptics.com	927-100-2053	Add-on Kit	953 Series ST Add-on Kit
Amphenol Fiber Optic Products 1925 A. Ohio Street Lisle, Ill 60532 Phone: 630-960-1010 Fax: 630-810-5640 sales@amphenol-fiberoptics.com http://www.amphenol-fiberoptics.com	927-100-5000	Basic Termination Kit	Basic Termination Kit with multiport over (3 ST/FC ports and 3 SC/SMA ports)
Amphenol Fiber Optic Products 1925 A. Ohio Street Lisle, Ill 60532 Phone: 630-960-1010 Fax: 630-810-5640 sales@amphenol-fiberoptics.com http://www.amphenol-fiberoptics.com	927-100-5041	Designer Kit	ST Designer Kit includes Basic Termination Kit 927-100-5000, 953 Series ST Add-on Kit 927-100-2053 and samples for 6 practice terminations.

5.7 Power Budget Calculation

Overview

In planning the installation of a fiber optic link, one of the most important tasks is the calculation of the power budget. The power budget determines the achievable link distance and indicates how much power margin exists. It depends on a number of factors, such as the optic fiber attenuation, the number of connector joints, and the number and type of splices in the link.

XLCNE2 operating parameters

The transmitter module is guaranteed to couple 10 μW (-20 dBm) of optical power into 62.5 micron fiber when new. The receiver modules are guaranteed to operate properly with a bit-error-rate better than 1 in 10^9 with only .8 μW (-31.0 dBm) of received power.

5.7.1 Micron Fiber Optic Cable

Typical power budget calculation

An example of the calculation of a typical link's power budget using 62.5 micron optic fiber is as follows.

Minimum power coupled into the fiber	10 μ W =	-20.0 dBm
Minus the minimum power the receiver requires	(-) 0.8 μ W =	(-) -31.0 dBm
Total power budget	+	11.0 dB

Losses:

Connector loss (2 x 1.0 dB)	=	1.0 dB
Splice loss (2 x 0.25 dB)	=	0.5 dB
Safety Margin	=	<u>2.3 dB</u>
		7.2 dB

Divide by the specified attenuation per kilometer	÷	<u>0.9 dB</u>
Achievable fiber optic link distance		8.0 km

Total power budget calculation

Starting with the minimum power guaranteed to be coupled into 62.5 micron optic fiber, 10 μ W or -20 dBm, subtract from that figure the minimum power level guaranteed to be properly decoded by the receiver, 0.8 μ W or -31.0 dBm. This calculation provides a total power budget of 11.0 dB.

Connector losses

From this 11.0 dB, certain losses must be subtracted. The first shown is the loss of two connector-to-connector butt splices. Typically, two such connections are made in the link at interconnect boxes at the interface between the outdoor and the indoor cable, one at each end of the link.

Splice losses and safety margin

Similarly, each indoor-to-outdoor cable interface may require a fiber-to-fiber splice, so two splice losses are subtracted from the budget. Lastly, a 2.3 dB safety factor is subtracted. A safety factor of 2.3 dB is minimal, and it is strongly recommended that if a larger safety factor is achievable, either by shortening the required link distance, or by using fiber with a better attenuation figure, by all means do so. The larger the safety margin, the more reliable the link will be.

5.7.1 Micron Fiber Optic Cable, Continued

Fiber attenuation

After all the above loss factors have been subtracted, the 7.2 dB that remains is the power left to be expended in cable loss. It is this number, when divided by the attenuation figure of the optic fiber, that determines the achievable link distance.

ATTENTION

It is important that the fiber optic cable be positioned and supported such that no stress is placed on the fiber optic cable's mechanical connection at the board. Any stress on the fiber optic connections will increase the signal loss at the connection reducing the available operating distance.

Section 6 – LCN Fiber Optic Cabling Considerations

6.1 Cable Routing Rules

Section contents

The topics covered in this section are:

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6.3 Cable Installation	136
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Introduction

When extending the **TotalPlant** Solution (TPS) System LCN, two optic fibers are needed for each coaxial cable channel, a transmit and a receive fiber. Therefore, to extend both A and B channels of the redundant LCN, four fibers are required. In addition, if 12.5 kHz clock synchronization is required across the fiber optic link, two more optic fibers are required to carry the clock signal to the remote cable segment (one for cable A and one for cable B) for a total of six optic fibers.

Cable routing recommendations

Honeywell does not recommend that the data and the system clock for both A and B channels be carried in the same cable or cable group. To maintain the inherent reliability of redundant cables, Honeywell recommends that the optic fibers for cables A and B be carried in separate cables on separate routes. For example, the A cable can be buried on one side of a service road and the B cable on the other side; or bury one cable and hang the other aurally. Only one basic rule needs to be observed when routing the cables along separate paths. The maximum distance between any two nodes in the system as measured on the A cable must not differ from the distance measured along the B cable by more than 300 meters (984 feet). Note that the distance referred to includes the sum of all fiber and coaxial sections of cable.

non-LCN data routing and communications

Frequently, customers have a need to send data, other than LCN, over the same route as the fiber LCN extension. There are no limitations which would restrict LCN data to a dedicated fiber cable. That is, it is entirely permissible to use a multifiber cable where LCN fibers share the same cable sheath with other non-Honeywell data communications optical fibers.

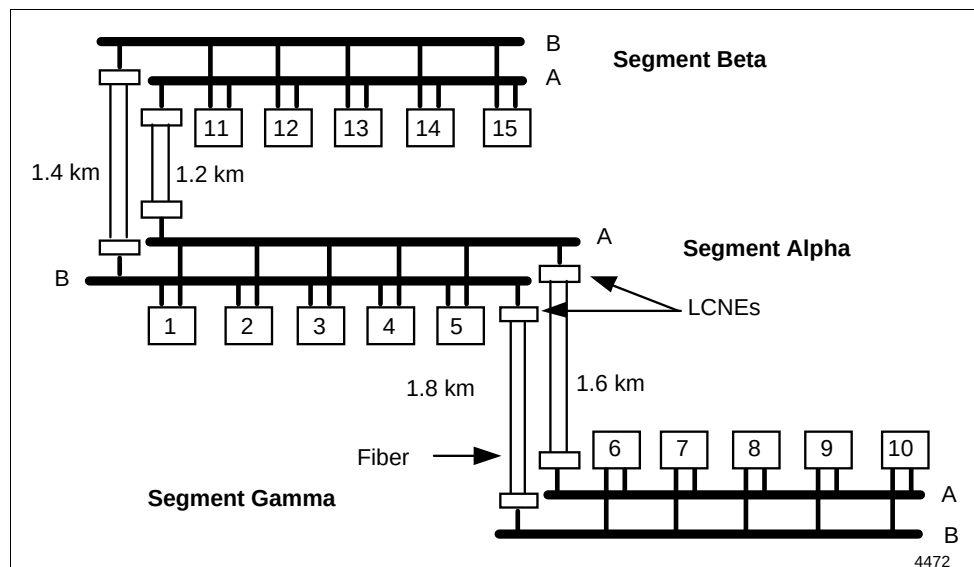
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6.1 Cable Routing Rules, Continued

Poor cabling configuration

Figure 6-1 is an example of a poor cable configuration. Segment “Alpha” is a central coaxial cable segment with two remote coaxial cable segments, “Beta” and “Gamma.” There is no problem with communications between the “Alpha” segment and either of the two remote cable segments. Cable A is 200 meters (656 feet) shorter than Cable B between cable segments “Alpha” and “Beta.” Cable A is also 200 meters shorter than Cable B between cable segment “Alpha” and “Gamma.” The 200 meters is within the 300-meter (984 feet) differential limit; however, Cable A is a total 400 meters (1312 feet) shorter than cable B between cable segments “Beta” and “Gamma.” Unreliable communications could be a result of the cable length difference.

Figure 6-1 Poor Redundant Fiber Optic Cable Routing Example



6.2 Cable Fiber Count

Optic fiber count recommendation

Due to the expense of installing outdoor cable runs, it is generally recommended that outdoor cable runs contain 33% to 100% more fibers than is presently required. The incremental cost of providing extra fibers in the cable is small compared with the cost of routing another cable if additional data communications capacity is required in the future.

For example, consider a single remote LCN extension requiring remote clock synchronization. If the recommendation for routing the A and B cables on separate paths is adhered to, two cables of at least three optic fibers each would be required. To provide the extra capacity for future contingencies, the recommended optic fiber count per cable would be four to six optic fibers. Cables are generally available only in an even number of optic fibers.

6.3 Cable Installation Methods

Overview

Aerial, underground duct, and direct cable burial are the three most popular ways to install outdoor cable runs. The method chosen for a particular site depends on any number of factors. Each has its own pluses and minuses.

Aerial cable installation

Of the major methods of outdoor cable installation, an aerial cable installation is subject to the greatest environmental stresses. An aerial cable is subject to wind loading, ice loading, extremes of temperature variation, and even gunshots.

There are two basic kinds of aerial cable available. One is designed to be lashed to a messenger wire because it cannot support itself. The other is self-supporting. Of the self-supporting aerial cables, there are two main subclasses. One type contains a tensile stress bearing member within the normal cable sheath. The other type, which is capable of longer unsupported runs, is the "figure-8" type of cable, where the cable contains its own built-in messenger wire from which hangs the capsule that contains the optic fiber.

Underground duct installation

The underground duct type of installation involves the burying of a plastic tube or tubes into which a cable or several cables can be routed. The recommended fill ratio is 50% by cross sectional area. Usually, the cable is of virtually the same construction as a lashed aerial cable. Under the ground it is well protected from environmental stress. It is recommended that the duct be buried as deeply as possible, definitely below the frost line. Frost heaving can crack the duct. The relatively large diameter of the duct provides rodent protection. Rodents cannot chew on it if they cannot get their mouths around it. If empty inner ducts are provided within a larger duct, then new cables could be routed in the future without too much trouble. Pulling new cable through an already occupied inner duct is not recommended.

Direct burial installation

The direct burial type of installation involves burying the cable directly in the earth. As with the underground duct method, bury the cable as deeply as possible. Make sure to document and mark where the cable is buried for future reference. An armor sheath is required on this cable for rodent protection. Keep in mind that the metal content of the cable can make it subject to lightning strikes. Also, the armor must be grounded at each end where it enters the building.

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6.3 Cable Installation Methods, Continued

Preinstallation of cable connectors

In certain situations, it may be desirable to preinstall connectors on the cable at the factory. This may be the case where the skilled labor required to perform splicing and installing connectors will not be available at the installation site. However, preinstallation of connectors makes pulling the cable through duct or conduit very difficult, so in duct situations it may not be feasible. If handled carefully, it should not be as much of a problem as aerial or direct burial installation methods. Still, the risk of damaging the connectors during installation is substantial; therefore, the general preference is to splice or attach connectors to the cable in the field, once the cable is installed.

6.4 Outdoor-to-Indoor Cable Transitioning

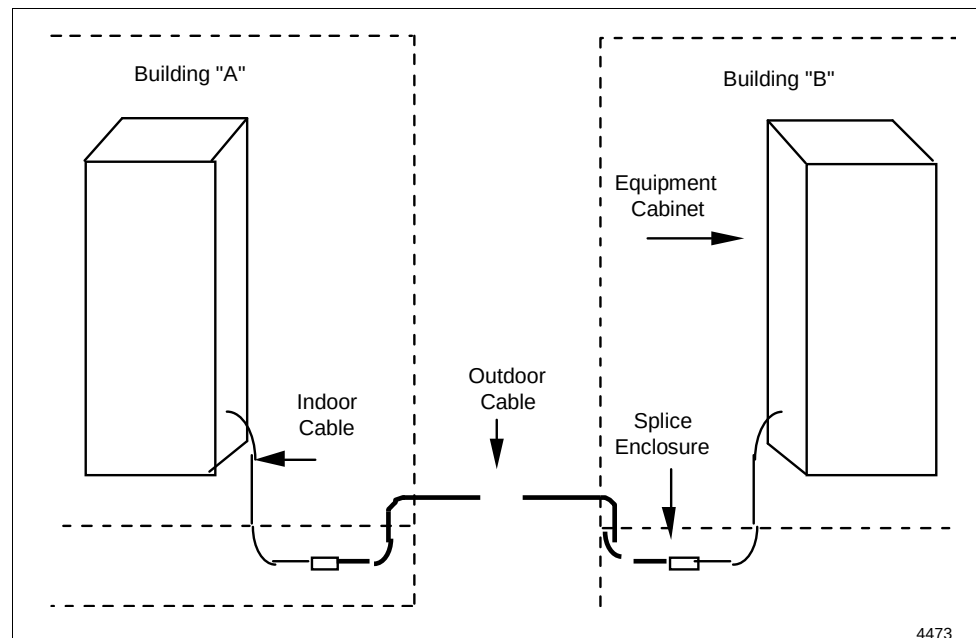
Overview

It is necessary to use both indoor and outdoor type fiber optic cable when installing a fiber optic link between two buildings. The indoor cable provides the flexibility necessary for routing under floors and into equipment cabinets, while the outdoor grade provides superior performance over temperature and environmental extremes.

Splicing

Perhaps the most straight forward method of transitioning from outdoor-to-indoor cable types is to use a splice that is housed in an in-line splice enclosure. Figure 6-2 illustrates an example of using in-line splices and in-line enclosures.

Figure 6-2 Outdoor-to-Indoor Cable Transition Using In-Line Splices



Advantages and disadvantages of splices

Since this method does not involve any connector-to-connector couplings, there is the least amount of power budget penalty. The drawback is the lack of flexibility. The in-line splice enclosures have provision for only one cable in and one cable out.

If unused fibers are present in the outdoor cable, and it becomes necessary to use them in the future, the in-line splice enclosure should probably be replaced with a wall-mounted interconnect panel. Multiple indoor cable runs cannot be fanned out from the simple in-line splice enclosure. Of course, if it is considered highly unlikely that any future expansion needs to be accommodated, this is a very cost-effective method for cable transitioning.

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6.4 Outdoor-to-Indoor Cable Transitioning, Continued

Two splicing methods	As far as the optic fiber splice itself is concerned, there are two basic methods used to join the fibers, fusion splicing and mechanical splicing.
Fusion splicing	Fusion splicing is accomplished by bringing the two fiber ends together in a precision micropositioning fixture, then fusing the joint with a precisely timed electric arc that melts the fiber ends together. Fusion splicing is the most reliable method and when done properly results in less than .2 dB of power loss in the link. The biggest drawback to fusion splicing is expense. The machine is quite expensive, as is the technical talent. The cost to acquire a trained individual with his fusion splicer to a project site will not be inexpensive.
Mechanical splicing	In recent years, fiber optics vendors have begun to produce mechanical splices with excellent insertion loss performance. The low insertion loss of fusion splicing can be achieved. The newest mechanical splices can be mated and remated if the first try is not acceptable. The mechanical splice aligns the two optic fibers to be joined in a V-groove or a precision capillary. The joint between the two fibers is filled with an index-matching gel to minimize back reflection and maximize forward coupling. Mechanical splices typically exhibit insertion losses of less than .5 dB or better if tuning is employed. Since these splices can be made with a minimum of training and the alignment jig is relatively inexpensive, the mechanical splice is quite cost-effective in most cases.

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6.4 Outdoor-to-Indoor Cable Transitioning, Continued

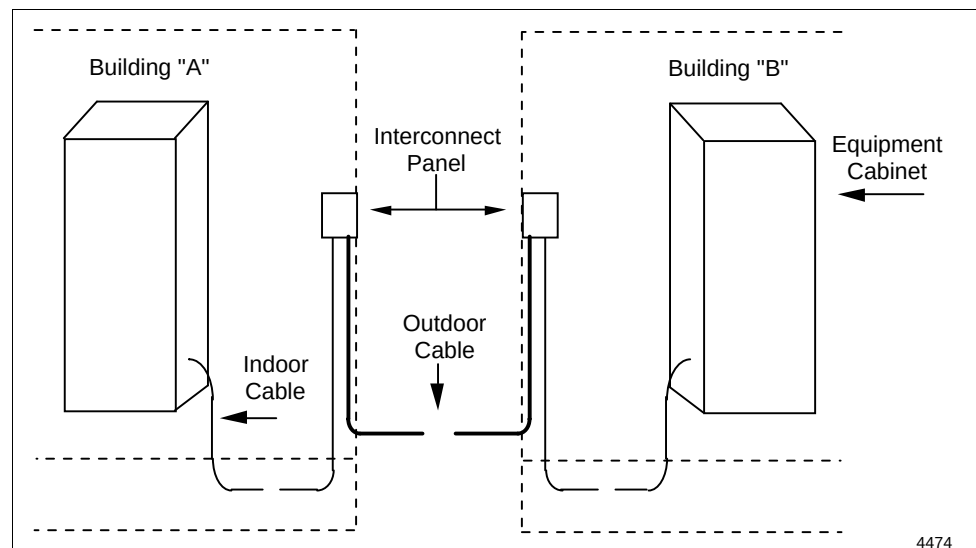
Mechanical splice hardware

Representative splicing kits and splice enclosures are available from several suppliers, one of which is Corning Cable Systems (formerly Siecor Corporation): 1-800-743-2671.

Interconnect panels

The use of interconnect (or patch) panels for a fiber optic cable complex is strongly encouraged. They allow the quick and easy future use of unused optic fibers. Also, cables from several remote buildings can be brought to a single interconnect panel to provide a central point of fiber cable management for plant-wide data communications. Figure 6-3 illustrates a typical indoor-to-outdoor cable transition.

Figure 6-3 Interconnect Panels for Indoor-to-Outdoor Cable Transition



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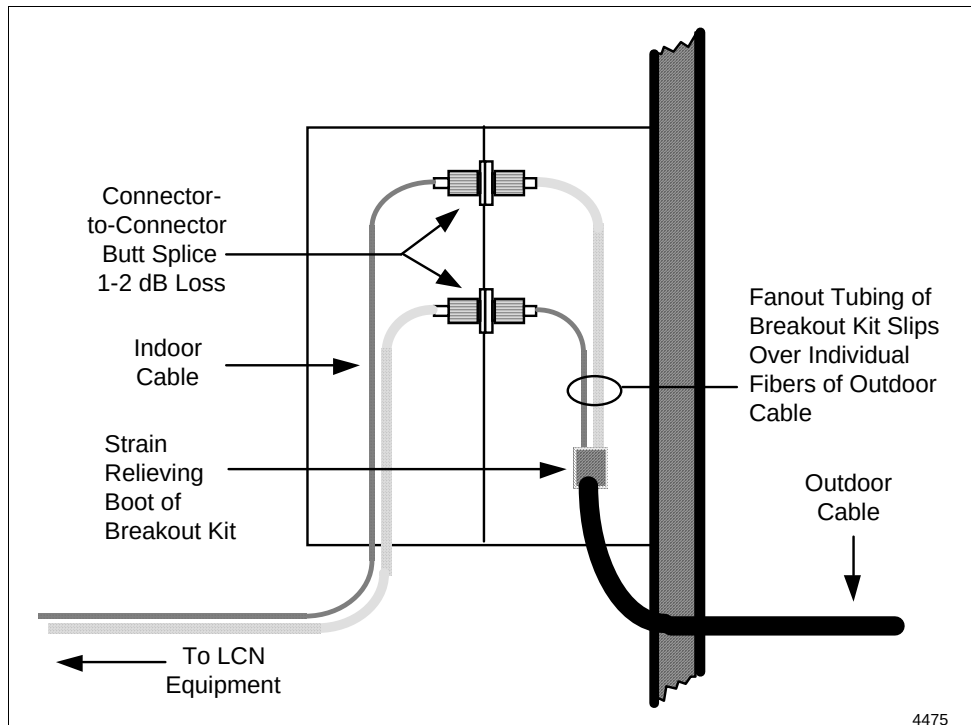
6.4 Outdoor-to-Indoor Cable Transitioning, Continued

Fanout tubing overview

Fanout tubing eliminates the need to perform any splicing. Fanout tubing is essentially a tight-buffer subunit without any fiber. A fanout tube is slipped over each of the individual fibers in a loose-tube cable after the sheath and buffer tubes have been stripped off. Cable vendors supply "break-out" or "fan-out" kits which consist of fanout tubing and a strain relieving boot. These kits allow the installation of connectors to the individual fibers. Figure 6-4 illustrates the use of fanout tubing.

This technique can jacket only about 3 to 15 feet of bare fiber, so it is really limited to use inside an interconnect panel.

Figure 6-4 Fanout Tubing Cable Transitioning



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6.4 Outdoor-to-Indoor Cable Transitioning, Continued

Fanout tubing method Each fiber in the outdoor cable (now jacketed by the breakout kit) is terminated and plugged into the back of a bulkhead-mounted barrel connector. These connectors will generally be the ST type of connector. On the other side, the connection to LCN equipment is made by means of the hybrid, duplex, cable assembly discussed in subsection 5.6.6. Unused receptacles should be capped off to prevent the ingress of contaminants.

Interconnect panels Representative interconnect panels are available from several suppliers, one of which is Corning Cable Systems (formerly Siecor Corporation): 1-800-743-2671.

Continued on next page

6.5 Fiber Optic Link Qualification

Overview

Once the cable complex is installed, a final fiber optic check should be made. For multimode fiber optic installations, an Optical Time Domain Reflectometer (OTDR) is not necessary. An end-to-end attenuation check plus a single-ended minimum receive power check is sufficient. Assuming that an XLCNE2 board is available as an optical power source, only a hand-held optical power meter and a short length of patch cable is required.

The tests are very easy to perform. First, the end-to-end attenuation check.

Power budget determination

When the fiber optic link is planned, the link's power budget should be calculated (refer to subsection 5.7) in order to determine where the optical losses will occur and how much safety margin will remain. The end-to-end attenuation check compares the planned link attenuation with the actual installation.

Measuring attenuation

Measure the output of an XLCNE2 board's fiber optic transmitter by connecting a short length of patch cable to a hand-held optical power meter. Use a patch cable of the same fiber diameter as the optic fiber used in the fiber optic link to be tested. Record the transmitter's power output in dBm. Now, connect the XLCNE2 board's transmitter to one end of the fiber optic link under test. At the other end, measure the optical power output in dBm. The difference between the power output recorded from the patch cable measurement and the power output measured over the link under test is the end-to-end attenuation of the link. The error introduced by the patch cable not being used in the second measurement is negligible because the length of the patch cable is so short. The end-to-end attenuation measured should substantially agree with the planned end-to-end attenuation.

Measuring the minimum receive power

To measure the minimum receive power, pay attention to the actual value of the power measurement at the far end of the link under test, as opposed to the difference from a reference. Recall from the discussion on power budget (subsection 5.7) that the XLCNE2 board's fiber optic receiver requires a minimum of .8 μ W (-31.0 dBm) of power to operate. Recall also that it is desirable to leave at least 3 dB as a general safety margin. Therefore, if the XLCNE2 board is new, the power measured at the far end of the link should be a minimum of -28.7 dBm (-31.0 dBm + 2.3 dB = -28.7 dBm).

Continued on next page

6.5 Fiber Optic Link Qualification, Continued

Link qualification

If both of these tests return satisfactory results, the link can be considered qualified.

ATTENTION

It is important that the fiber optic cable be positioned and supported such that no stress is placed on the fiber optic cable's mechanical connection at the board. Any stress on the fiber optic connections will increase the signal loss at the connection reducing the available operating distance.

Section 7 – Universal Control Network Extender Planning

7.1 Overview

Section contents

The topics covered in this section are:

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7.2 Description

Purpose

The Universal Control Network two-kilometer fiber optic Extender (UCNE) is used to:

- increase the end-to-end distance of a single UCN without resorting to special, wide-diameter coaxial trunk cables.
- increase the UCN signal quality at the ends of the UCN that would otherwise be connected by long coaxial trunk cables.
- implement certain UCN topologies, such as a “star configuration,” that is not possible by using coaxial trunk cables.
- provide ground isolation, when necessary, between UCN trunk segments.

Each UCNE provides coax-to-fiber optic and fiber optic-to coax cable conversion for the UCN signals.

Implementation

UCNEs are installed in pairs at each end of a redundant fiber optic link; that is, a UCNE-A and UCNE-B at one end of the fiber optic link and a UCNE-A and UCNE-B at the other end of the fiber optic link.

Therefore, a total of four UCNEs are required to implement one redundant fiber optic link.

The UCNE, at each end of the fiber optic link, never connects directly to the UCN coaxial trunk, but instead connects to the drop port of a UCN tap on that coaxial segment.

Standard coaxial UCN drop cables (RG6-type) are used to connect the UCNEs to the UCN tap drop ports.

7.3 Universal Control Network Topologies

Introduction

There are two classes of UCN topology when using UCN fiber optic Extenders (UCNEs) to link coaxial UCN trunk segments.

- Serial
- Star

Serial UCN topologies

The simplest “serial” UCN topology consists of two UCN coaxial trunk segments that are interconnected by a single fiber optic link. Figure 7-1 illustrates the “one series link” configuration.

Another serial UCN topology consists of three UCN coaxial trunk segments that are interconnected by two fiber optic links. Figure 7-2 illustrates the “two series link” configuration.

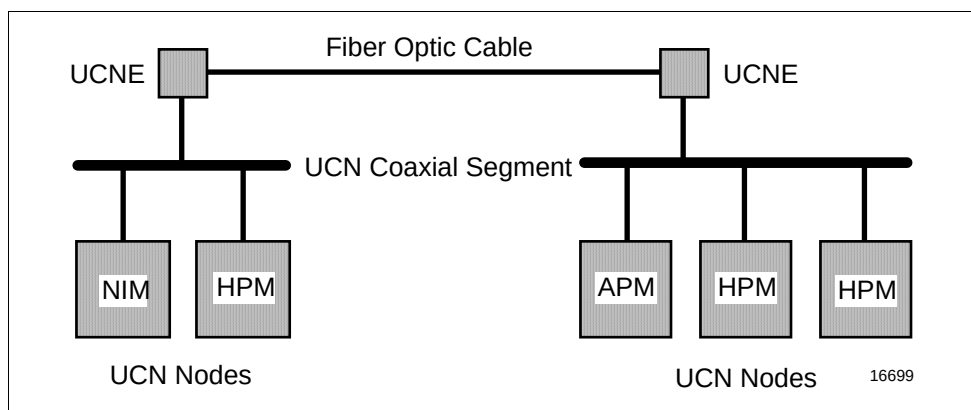
Each UCN redundant fiber optic link requires two UCNEs (UCNE-A and UCNE-B) at each end of the link, or a total of four UCNEs. Two fiber optic links, as illustrated in Figure 7-2, would require eight UCNEs.

The maximum number of series-connected fiber optic links allowed in a physical UCN is two. The use of three or more fiber optic links in a serial topology is prohibited, because the delay imposed by the additional UCNE(s) and fiber optic cable could violate the UCN’s slot time setting and cause UCN communication errors.

Serial UCN topology 2 coaxial segments

Figure 7-1 illustrates a typical serial UCN topology with two UCN coaxial segments and one fiber optic link. Only network “A” is shown. Network “B” is not shown.

Figure 7-1 Typical Serial UCN Topology – Two Coaxial Segments



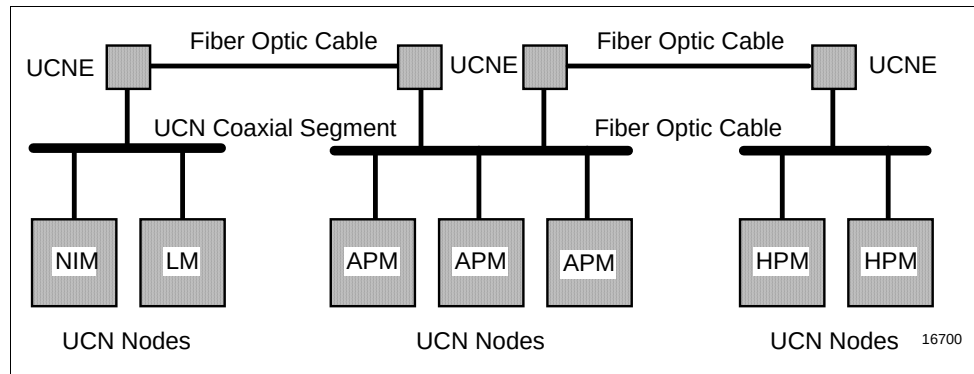
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7.3 Universal Control Network Topologies, Continued

Serial UCN topology 3 coaxial segments

Figure 7-2 illustrates a typical serial UCN topology with three UCN coaxial segments and two fiber optic links. Only network “A” is shown. Network “B” is not shown.

Figure 7-2 Typical Serial UCN Topology – Three Coaxial Segments



Star UCN topology

A “star” UCN topology consists of a central UCN coaxial cable segment with peripheral UCN coaxial cable segments connected to the central UCN coaxial cable segment through UCNEs. Figure 7-3 illustrates a typical star UCN “star” configuration example.

UCNEs must not be used with passive star couplers because passive couplers have excessive optical power loss. Instead, star topology is implemented by using multiple drop ports on a central UCN coaxial segment to branch out through multiple UCN fiber optic links to other UCN coaxial segments. An example of their use is a central control room that contains the NIM(s) with multiple radial UCN fiber optic links (using UCNEs) to remote UCN coaxial segments that accommodate other UCN nodes. Figure 7-3 illustrates four radial fiber optic links to four remote sites.

Only one of the redundant UCN cables, A or B, is shown in the figure for simplification.

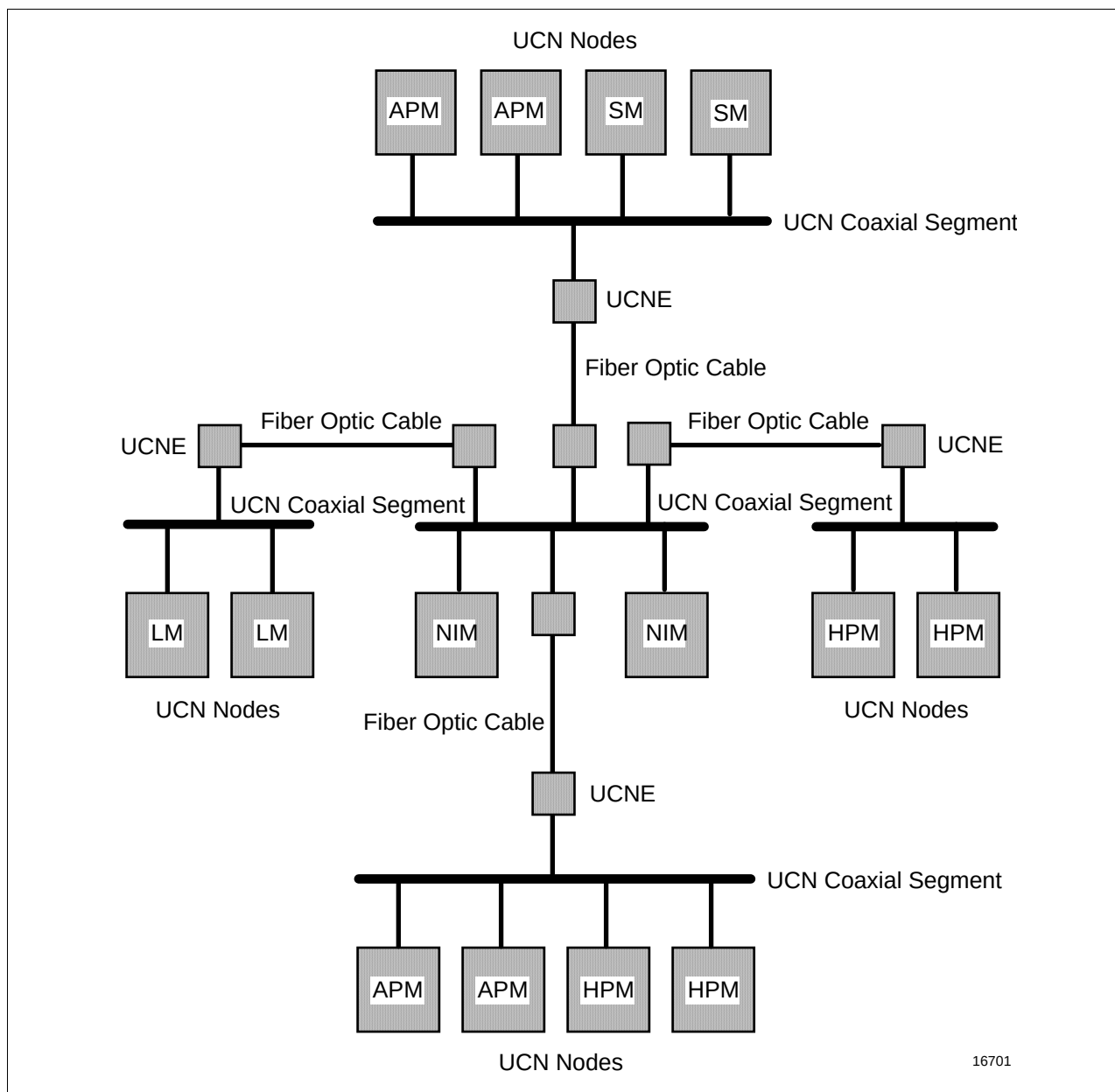
It is preferable that the NIM(s) is at the center of the “star,” but it is not a requirement, just a recommendation, to minimize the “loss of view” in the unlikely event that a redundant fiber optic link fails completely.

Continued on next page

7.3 Universal Control Network Topologies, Continued

Serial UCN topology Figure 7-3 illustrates a typical star UCN topology with five UCN coaxial segments. Only network “A” is shown. Network “B” is not shown.

Figure 7-3 Typical Star UCN Topology



7.4 UCNE Configuration Rules

Configuration rules

Regardless of the UCN topology, the following configuration requirements must be met.

1. The total length of the UCN (network A or B) between any two nodes, including all series-connected coaxial trunk cable segments and fiber optic links, but not including drop cable lengths, must not exceed 2 kilometers. Since any two UCN nodes can potentially communicate with each other, the restriction is determined by the two nodes that are farthest apart from each other along either UCN cable A or UCN cable B and separated by a combination of UCN coaxial trunk cable and UCN fiber optic cable. Violation of this rule could result in UCN errors that are due to communication delays that exceed the UCN slot time. UCN slot time is the maximum time that any UCN node waits for a response from any other UCN node on the network and is affected by cable and UCNE delays.
 2. Related to rule 1 above, the difference between the total length of UCN network A and UCN network B must be no greater than 500 meters. That is, the UCN A cable and UCN B cable communication paths between the two farthest apart UCN nodes must not be different by more than 500 meters. Violation of this rule could result in UCN errors that are due to communication delays that exceed the UCN slot time.
 3. Related to rule 1 above, the length of fiber optic cable that connects any two UCNEs must not be longer than 2 kilometers, but there is no minimum length requirement. Violation of this rule could result in UCN communications errors because the optical signal loss budget would be exceeded for the interconnected UCNEs (see the following rule).
 4. Related to rule 3 above, the optical signal loss budget for a fiber optic cable connection between two interconnected UCNEs is 10 dB maximum and must not be exceeded. This includes up to 2 kilometers of fiber optic cable that is specified in Honeywell drawing 51190918 that has a maximum loss of 4 dB/km, two ST-type connectors (one at each end of the fiber optic cable) with a maximum loss of 1 dB per connector, and two cable splices with a maximum loss of 0.5 dB per splice. Violation of this rule could result in UCN communications errors because the optical signal loss budget could be exceeded for the interconnected UCNEs.
 5. The UCN coaxial cable configuration rules found in other sections of the manual also apply for each separate coaxial trunk segment of the network.
-

7.5 UCNE Model Numbers

Model numbers	The UCNE model numbers that support the maximum 2 kilometer fiber optic cable distance are: • MU-FOED02 A pair of 18-36 Vdc powered UCNEs • MU-FOEA02 A pair of 85-264 Vac, 47-63 Hz powered UCNEs
DC-powered UCNE	The dc-powered UCNE is used where a PM, APM, or HPM Power System's 24 Vdc is available. Figure 7-4 is an illustration of a dc-powered UCNE.
AC-powered UCNE	The ac-powered UCNE is typically used at UCN locations where a PM, APM, or HPM Power System's 24 Vdc is not available, such as in cabinets or consoles that house a NIM, LM, or SM. However, the ac-powered UCNE should be powered by UPS-backed ac power if it is expected to continue supporting UCN communications during an ac line power outage. Figure 7-5 is an illustration of an ac-powered UCNE.
UCNE independence	UCNEs must be installed independently from each other such that a failed UCNE in a redundant pair (UCNE-A and UCNE-B) can be replaced while the other UCNE continues operating. This is accomplished using the standard mounting kits that are described in subsection 7.6.

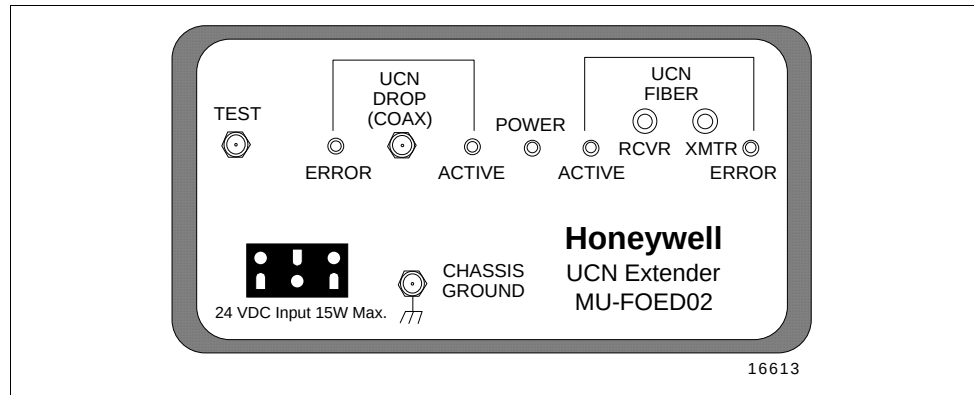
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7.5 UCNE Model Numbers, Continued

DC-powered UCNE

Figure 7-4 illustrates the front panel of a dc-powered UCNE.

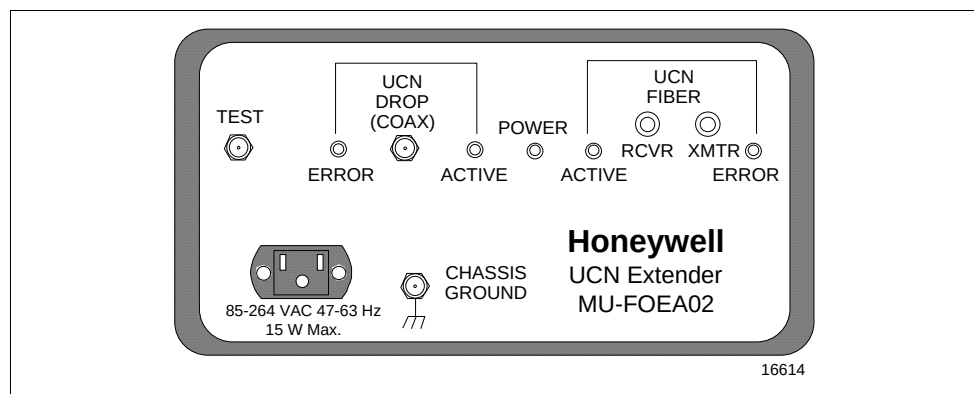
Figure 7-4 DC-Powered UCNE



AC-powered UCNE

Figure 7-5 illustrates the front panel of an ac-powered UCNE.

Figure 7-5 AC-Powered UCNE



7.6 UCNE Mounting Kits

7.6.1 PM, APM, and HPM Cabinets



SHOCK HAZARD

Troubleshooting, opening this equipment and removing any panels or covers will expose the **user** to the risk of a shock hazard. There are no **user** serviceable parts inside this equipment. Refer all servicing only to **qualified service personnel**.

ATTENTION

The information that follows is supplemental to the LCN cabinet and console information. Detailed PM, APM, or HPM cabinet information can be found in the *Universal Control Network Planning* and *Universal Control Network Installation* manuals.

Mounting location

The standard mounting location for a redundant pair of UCNEs is in model MU-FOMK01 mounting kit hardware. The kit's enclosure is installed in a standard Left or Right 7-slot card file position in the card file side of a PM, APM, or HPM cabinet. Typically, the kit is installed after first mounting all the other PM, APM, or HPM card files that connect to the UCN, and in a cabinet that contains UCN taps because UCNEs connect to the UCN through drop cables to the UCN taps.

7.6.1 PM, APM, and HPM Cabinets, Continued

Mounting kits MU-FOMK01 and MU-FOMK11

The model MU-FOMK01 mounting kit provides for vertical stacking of up to four UCNEs as illustrated by the layout in Figure 7-6. A redundant pair of dc-powered UCNEs (model MU-FOED02) will occupy vertical slots 1 and 3 as illustrated in Figure 7-6.

The model MU-FOMK11 mounting kit provides the hardware to accommodate a second redundant pair of dc-powered UCNEs for use in vertical slots 2 and 4 for an optional second UCN fiber optic link. The second pair of UCNEs must be ordered separately by ordering another model MU-FOED02.

Each of the four vertical mounting slots includes a slide-out mounting tray that attaches to the bottom of a UCNE by using four screws that are provided in the mounting kit.

Mounting positions 1 and 2 are identified by orange-colored “UCN-A” labels, and mounting positions 3 and 4 are identified by brown-colored “UCN-B” labels.

Mounting kits model MU-FOMK01 and MU-FOMK11 include two 5-meter power cables (51201397-002) to provide power to the dc-powered UCNEs. If longer power cables are needed, they must be ordered separately.

UCN tap drop cables

A pair of UCN drop cables of appropriate length (model MU-NKDxxx where xxx = length in meters) is required to connect a pair of UCNEs to a pair of UCN taps. The drop cables are not supplied with the UCNE mounting kits and must be ordered separately.

Continued on next page

7.6.1 PM, APM, and HPM Cabinets, Continued

UCN taps

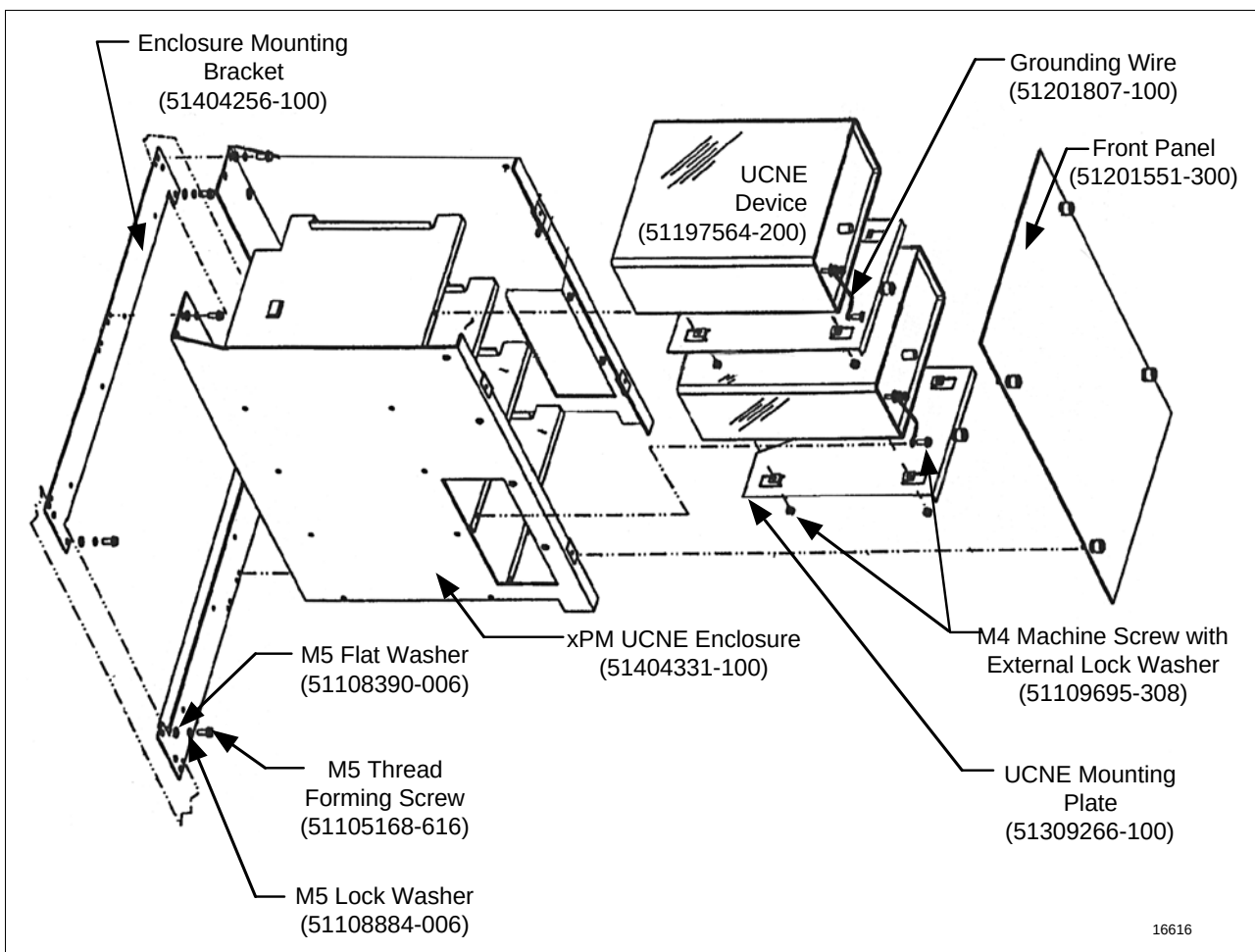
If sufficient UCN tap drop ports are not available when expanding a UCN with UCNEs, then either the number of ports per tap must be increased, or the number of taps in the existing UCN coaxial segments (A and B) must be increased. UCN taps can be ordered by model number as follows.

MU-NTAP02	Two 2-drop taps with mounting/grounding hardware
MU-NTAP04	Two 4-drop taps with mounting/grounding hardware
MU-NTAP08	Two 8-drop taps with mounting/grounding hardware

Pictorial view of model MU-FOMK01 kit

Figure 7-6 is a pictorial view of the model MU-FOMK01 mounting kit. The UCNEs are not included in the kit.

Figure 7-6 Pictorial View of Model MU-FOMK01 Mounting Kit



7.6.2 LCN Cabinets and Consoles



SHOCK HAZARD

Troubleshooting, opening this equipment and removing any panels or covers will expose the **user** to the risk of a shock hazard. There are no **user** serviceable parts inside this equipment. Refer all servicing only to **qualified service personnel**.

Mounting location

The standard mounting location for a redundant pair of UCNEs in LCN cabinets or consoles is in the UCNE enclosures that are included in the following mounting kits.

MP-FOMK02	LCN Cabinet 120 Vac UCNE Mounting Kit
MP-FOMK03	LCN Cabinet 240 Vac UCNE Mounting Kit
MP-FOMK04	Classic Console 120 Vac UCNE Mounting Kit
MP-FOMK05	Classic Console 240 Vac UCNE Mounting Kit
MP-FOMK06-100	Z-Console Upper Module UCNE Mounting Kit
MP-FOMK06-200	Z-Console Lower Module UCNE Mounting Kit

The LCN UCNE enclosures can also be installed in LM and SM cabinets. The model MP-FOMK02 and MP-FOMK03 mounting kits are recommended for that purpose.

Model MP-FOMK06 mounting kit

The model MP-FOMK06 mounting kits do not include a pair of replacement leg covers which are required when adding a UCNE enclosure to Z-console furniture. They must be ordered separately. Two factors must be considered before ordering, the height of the console furniture (high, medium, or low) and whether the UCNE enclosure's equipment pod will be the only (upper) pod or a second (lower) pod.

Continued on next page

7.6.2 LCN Cabinets and Consoles, Continued

Kit descriptions

For each of the kits, a redundant pair of ac-powered UCNEs (model MU-FOEA02), which must be ordered separately, are mounted side by side in a metal enclosure (module). Viewing the rear of the enclosure, the UCNE on the right side of the enclosure is connected to the UCN-A cable and the UCNE on the left side of the enclosure is connected to the UCN-B cable.

Labels on the enclosure identify the UCN-A and UCN-B mounting locations. The labels are color-coded orange and brown, respectively, to agree with the standard UCN redundant cable color-coding scheme.

Each mounting kit includes two ac power cables to power the pair of ac-powered UCNEs that must be installed in the enclosure. If redundant ac power sources are available, it is recommended that each UCNE in a redundant pair be connected to a different ac power source to maintain operation of at least one UCNE if a power source should fail. An Uninterruptible Power System (UPS) is recommended to maintain ac power to the UCNEs during ac line outages.

Continued on next page

7.6.3 LCN Cabinets and Consoles, Continued

UCN tap drop cables

A pair of UCN drop cables of appropriate length (model MU-NKDxxx where xxx = length in meters) is required to connect the pair of UCNE devices to a pair of UCN taps. The drop cables are not supplied with the LCN UCNE mounting kit and must be ordered separately.

UCN taps

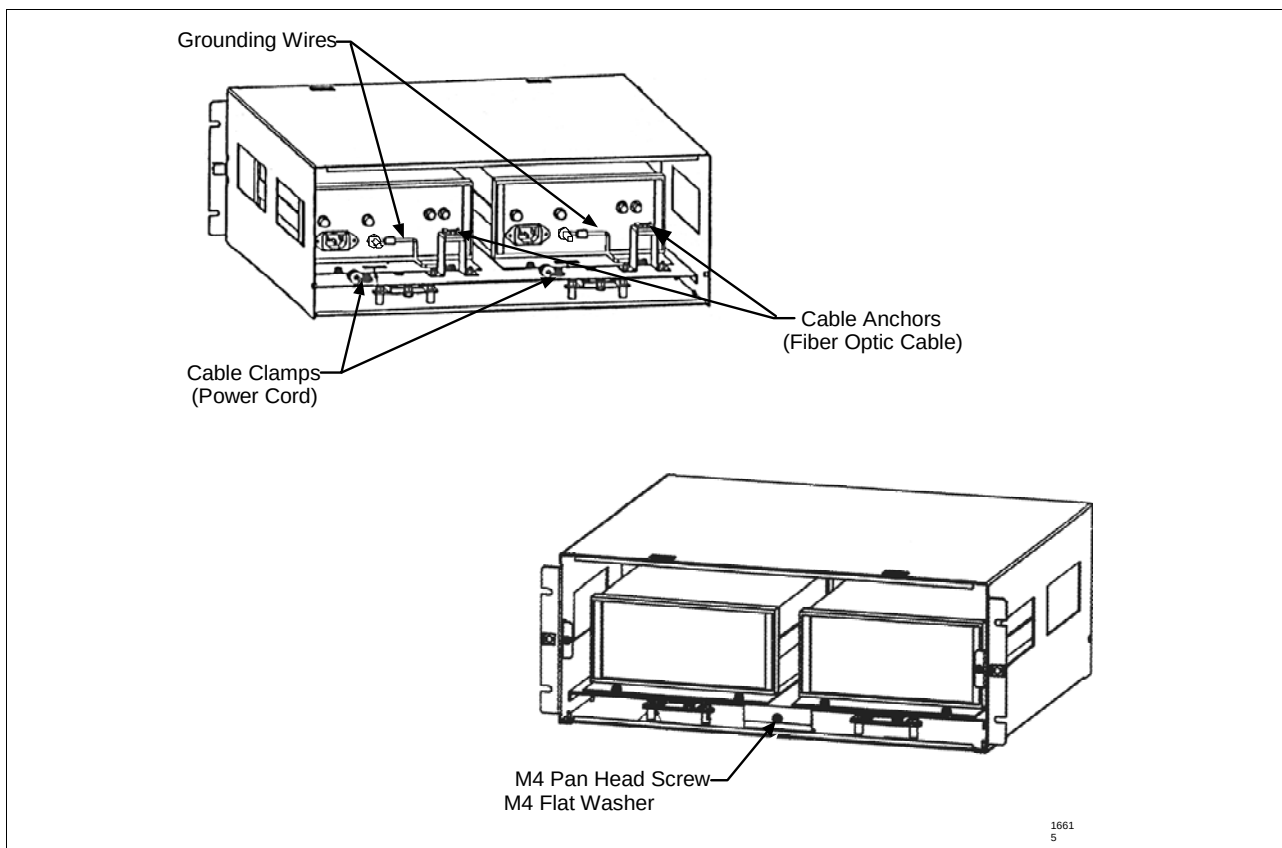
If sufficient UCN tap drop ports are not available when expanding a UCN with UCNEs, then either the number of ports per tap must be increased, or the number of taps in the existing UCN coaxial segments (A and B) must be increased. UCN taps can be ordered by model number as follows.

MU-NTAP02	Two 2-drop taps with mounting/grounding hardware
MU-NTAP04	Two 4-drop taps with mounting/grounding hardware
MU-NTAP08	Two 8-drop taps with mounting/grounding hardware

LCN UCNE mounting kit enclosure

Figure 7-7 illustrates pictorial views of the enclosure that is included in the mounting kits that mount in an LCN cabinet or console.

Figure 7-7 UCNE Enclosure Pictorial Views



Continued on next page

7.6.3 LCN Cabinets and Consoles, Continued

Model MP-FOMK02 mounting kit parts

Table 7-1 is the list of parts in the model MP-FOMK02 mounting kit. The two UCNEs that are mounted in the LCN UCNE enclosure must be ordered separately by model MU-FOEA02.

Table 7-1 Model MP-FOMK02 Parts List

Description	Part Number	Quantity
UCNE Enclosure	51403223-100	1
AC Power Cable, 120 Vac	51305409-500	2
UCNE Enclosure Front Panel	51305413-100	1
M5 x 16 mm Hex-Head Thread-Forming Screw	51195168-616	4
M5 x 16 mm Hex-Head Machine Screw	51108888-616	1
M5 Grounding Serrated Washer	51109931-100	2
M5 x 25 mm Hex-Head Machine Screw	51108888-625	1
Spacer	51108572-100	1
M5 Serrated Washer	51109933-003	4

Continued on next page

7.6.3 LCN Cabinets and Consoles, Continued

Model MP-FOMK03 mounting kit parts

Table 7-2 is the list of parts in the model MP-FOMK03 mounting kit. The two UCNEs that are mounted in the LCN UCNE enclosure must be ordered separately by model MU-FOEA02.

Table 7-2 Model MP-FOMK03 Parts List

Description	Part Number	Quantity
UCNE Enclosure	51403223-100	1
AC Power Cable, 240 Vac	51305410-500	2
UCNE Enclosure Front Panel	51305413-100	1
M5 x 16 mm Hex-Head Thread-Forming Screw	51195168-616	4
M5 x 16 mm Hex-Head Machine Screw	51108888-616	1
M5 Grounding Serrated Washer	51109931-100	2
M5 x 25 mm Hex-Head Machine Screw	51108888-625	1
Spacer	51108572-100	1
M5 Serrated Washer	51109933-003	4

Continued on next page

7.6.3 LCN Cabinets and Consoles, Continued

Model MP-FOMK04 mounting kit parts

Table 7-3 is the list of parts in the model MP-FOMK04 mounting kit. The two UCNEs that are mounted in the LCN UCNE enclosure must be ordered separately by model MU-FOEA02.

Table 7-3 Model MP-FOMK04 Parts List

Description	Part Number	Quantity
LCN UCNE Enclosure	51403223-100	1
AC Power Cable, 120 Vac	51305409-500	2
UCNE Enclosure Front Panel	51305413-100	1
M5 x 16 mm Hex-Head Thread-Forming Screw	51195168-616	4
M5 Serrated Washer	51109931-100	4

Model MP-FOMK05 mounting kit parts

Table 7-4 is the list of parts in the model MP-FOMK05 mounting kit. The two UCNEs that are mounted in the LCN UCNE enclosure must be ordered separately by model MU-FOEA02.

Table 7-4 Model MP-FOMK05 Parts List

Description	Part Number	Quantity
LCN UCNE Enclosure	51403223-100	1
AC Power Cable, 240 Vac	51305410-500	2
LCN UCNE Enclosure Front Panel	51305413-100	1
M5 x 16 mm Hex-Head Thread-Forming Screw	51195168-616	4
M5 Serrated Washer	51109931-100	4

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7.6.3 LCN Cabinets and Consoles, Continued

MP-FOMK06-100 mounting kit parts

Table 7-5 is the list of parts in the model MP-FOMK06-100 upper module mounting kit. The two UCNEs that are mounted in the LCN UCNE enclosure must be ordered separately by model MU-FOEA02.

Table 7-5 Model MP-FOMK06-100 Parts List

Description	Part Number	Quantity
UCNE Enclosure	51403223-100	1
AC Power Cable	51305450-500	2
Module Enclosure Upper Front Cover	51403089-100	1
Module Enclosure Assembly	51403117-100	1
Module Enclosure Mounting Bracket	51402745-100	1
Module Enclosure Mounting Bracket	51402745-200	1
Module Enclosure Front Cover Bracket	51308039-100	2
Module Enclosure Rear Cover Assembly	51403152-100	1
M5 x 10 mm Pan-Head Machine Screw	51108385-610	4
M5 x 16 mm Hex- Head Thread-Forming Screw	51195168-616	4
M6 x 16 mm Flat-Head Machine Screw	51198881-716	4
Air Filter	51308003-100	1
Foam Rubber Gasket	51197024-100	1

Continued on next page

7.6.3 LCN Cabinets and Consoles, Continued

MP-FOMK06-200 mounting kit parts

Table 7-6 is the list of parts in the model MP-FOMK06-200 lower module mounting kit. The two UCNEs that are mounted in the LCN UCNE enclosure must be ordered separately by model MU-FOEA02.

Table 7-6 Model MP-FOMK06-200 Parts List

Description	Part Number	Quantity
UCNE Enclosure	51403223-100	1
AC Power Cable	51305450-500	2
Module Enclosure Lower Front Cover	51403090-100	1
Module Enclosure Assembly	51403117-100	1
Module Enclosure Mounting Bracket	51402745-100	1
Module Enclosure Mounting Bracket	51402745-200	1
Module Enclosure Front Cover Bracket	51308039-100	2
Module Enclosure Rear Cover Assembly	51403152-100	1
M5 x 10 mm Pan-Head Machine Screw	51108385-610	4
M5 x 16 mm Hex- Head Thread-Forming Screw	51195168-616	4
M6 x 16 mm Flat-Head Machine Screw	51198881-716	4
Air Filter	51308003-100	1
Foam Rubber Gasket	51197024-100	1

7.6.3 LM and SM Cabinets



SHOCK HAZARD

Troubleshooting, opening this equipment and removing any panels or covers will expose the **user** to the risk of a shock hazard. There are no **user** serviceable parts inside this equipment. Refer all servicing only to **qualified service personnel**.

Mounting location

The standard mounting locations for a redundant pair of UCNEs in LM and SM cabinets is in the enclosures that are included in the model MP-FOMK02 or MP-FOMK03 mounting kit. The model MP-FOMK02 mounting kit is installed when the power source is 120 Vac, and the model MP-FOMK03 mounting kit is installed when the power source is 240 Vac.

Parts lists for the mounting kits are found in Tables 7-1 and 7-2.

Kit descriptions

For each of the kits, a redundant pair of ac-powered UCNEs (model MU-FOEA02), which must be ordered separately, are mounted side by side in a metal enclosure (module). Viewing the rear of the enclosure, the UCNE on the right side of the enclosure is connected to the UCN-A cable and the UCNE on the left side of the enclosure is connected to the UCN-B cable.

Labels on the enclosure identify the UCN-A and UCN-B mounting locations. The labels are color-coded orange and brown, respectively, to agree with the standard UCN redundant cable color-coding scheme.

Each mounting kit includes two ac power cables to power the pair of ac-powered UCNEs that must be installed in the enclosure. If redundant ac power sources are available, it is recommended that each UCNE in a redundant pair be connected to a different ac power source to maintain operation of at least one UCNE if a power source should fail. An Uninterruptible Power System (UPS) is recommended to maintain ac power to the UCNEs during ac line outages.

UCN tap drop cables

A pair of UCN drop cables of appropriate length (model MU-NKDxxx where xxx = length in meters) is required to connect the pair of UCNE devices to a pair of UCN taps. The drop cables are not supplied with the LCN UCNE mounting kit and must be ordered separately.

Continued on next page

7.6.3 LM and SM Cabinets, Continued

UCN taps

If sufficient UCN tap drop ports are not available when expanding a UCN with UCNEs, then either the number of ports per tap must be increased, or the number of taps in the existing UCN coaxial segments (A and B) must be increased. UCN taps can be ordered by model number as follows.

MU-NTAP02	Two 2-drop taps with mounting/grounding hardware
MU-NTAP04	Two 4-drop taps with mounting/grounding hardware
MU-NTAP08	Two 8-drop taps with mounting/grounding hardware

7.7 Grounding

Proper grounding required

The chassis ground stud at the front of the UCNE must be connected to Safety Ground, similar to the grounding of a UCN tap.

A #10 AWG ground wire is provided in all the UCNE mounting kits to connect the UCNE's front panel ground stud to ground on the metal enclosure in which the UCNEs are installed.

The eventual connection to Safety Ground depends on proper grounding of the metal enclosure. As required when installing a UCN tap, the installer must confirm that there is electrical continuity between the UCNE's front panel grounding stud and the cabinet Safety Ground stud after the installation is complete. The Safety Ground stud is typically located at the bottom of the cabinet.

7.8 Fiber Optic Cable

Recommended cable

The multi-node fiber optic cables recommended for use with a UCNE are the same as the 62.5-micron fiber optic cables that are recommended for the Honeywell I/O Link Extender in the PM, APM, or HPM. Honeywell drawings 51190918 (indoor) and 51190919 (outdoor) define the cables. The cable has 4 dB/km maximum optical attenuation according to the signal loss budget requirements of subsection 7.4.

Other fiber optic cable types, such as the 100-micron fiber optic cable, have not been tested with the UCNE by Honeywell and are not recommended. The UCNE is not compatible with single-mode fiber optic cable.

Proper cable construction

UCN indoor fiber optic cable assemblies should be fabricated using the information in Honeywell drawing 51195498. The RCVR and XMTR fibers at each end of the cable should be labeled as such to agree with the connector labeling on the UCNEs.

Cable composition

A UCN fiber optic cable link consists of two transmit-receive pairs of optic fibers (UCN-A and UCN-B). If possible, cables that have high installation costs, such as cables routed in conduit or buried in the ground should have spare optic fibers.

7.9 Fiber Optic Cable Routing

Routing methods	Fiber optic cable can be routed underground by using direct burial cable, through the air by using outdoor aerial cable, or in cable or electrical wiring trays. Fiber optic cable is immune to interference from electromagnetic fields or transmissions. Fiber optic cable is safe to route through Intrinsic Safety (IS) areas with no danger of explosion.
Cable A and B separation	Subject to the configuration rules in subsection 4.4, the A and B fiber optic links should take different routes to a common destination to avoid simultaneous damage to both cables.

7.9.1 Direct Burial

Under ground burial hazards	Heavy duty cables can be buried directly in the ground. Care should be taken to minimize the effects of freezing water, the heaving of rocks caused by the ground freezing, ground disruption because of construction, and rodents.
Type of cable to use	The most durable jacket material for environmental hazards, such as crushing and abrasion, is polyethylene. An inner layer of steel armor provides additional protection from rodents. Use of heavy duty cables can sometimes be avoided by protecting the cable with conduit or polyethylene pipe.

7.9.2 Aerial Lashing

Aerial lashing methods	Aerial Lashing methods are similar to those used for electrical cables. Most cables are compatible with helical lashing, clamping, or tied mounting.
-------------------------------	--

7.9.3 Vertical Installations

Cable clamping

Cable in vertical trays, raceways, shafts, etc., should be secured at frequent intervals so that cable weight is supported evenly. Clamping intervals can be as short as 1 meter (3.3 feet) outdoors to prevent wind slapping and minimize ice loading, or as long as 15 meters (49.2 feet) indoors. When securing the fiber optic cable, care must be taken not to compress the optic fibers or exceed the minimum bend radius specifications.

Fiber migration and cable loops

Optic fiber in vertical installations does not break because of its own weight. However, for vertical runs of 15 meters (50 feet) or greater, excess optic fiber can migrate downward. The crowding of excess optic fiber at the bottom can cause an undesirable increase in signal attenuation.

This downward migration of optic fiber in vertical runs can be reduced by placing loops in the cable approximately 0.3 to 0.5 meter (1 to 1.5 feet) in diameter at the top, bottom, and at 15-meter (50-foot) intervals.

7.9.4 Indoor Requirements

Cable jacket building code restrictions

Building code requirements frequently do not allow cables with polyethylene jackets. Jackets of polyvinyl chloride are frequently restricted to conduits, while fluoropolymer or other approved jacket material is required for use in cable trays and air plenums. Cable with suitable jacket material must be selected for the application.

7.9.5 Loose Buffered Cable

Ambient temperature variations

Where ambient temperature variations are 20°C or greater on a daily basis, the life of the optic fibers can be significantly reduced. In applications where the ambient temperature is not controlled, loose buffered cable must be specified.

Fire code requirements

Loose buffered cable is available in polyethylene jacket material only. Fire codes may dictate that the indoor portion of the cable installation be in metal conduit, or if the temperature is controlled, a splice may be required to convert to a tight-buffered cable with a PVC or fluoride copolymer jacket material.

7.9.6 Number of Optic Fibers

Multi-optic fiber cable required	Two optic fibers are required for each UCN-A or UCN-B link, one for transmit and one for receive.
Cable damage possible	As insurance against future damage, such as optic fiber breakage, or encountering excessive loss in any one optic fiber, the inclusion of spare optic fibers should be considered. This is especially important for cables that have a high installation cost.

7.9.7 Cable Installation

Use professional installers	It is recommended that the fiber optic cable be installed by professional fiber optic cable installers. The installers will provide the cable, install the cable, attach ST-type connectors, and test the connection with an Optical Time Domain Reflectometer (OTDR) and a power meter.
62.5 micron cables	62.5 micron cable usage is controlled by Honeywell drawing 51190918 for indoor cable and Honeywell drawing 51190919 for outdoor cable. Indoor cables are available with 2, 4, or 6 optic fibers. Outdoor aerial and direct burial cables are available with 4, 6, or 8 optic fibers.
100 micron cables	100 micron cable usage cannot be recommended at this time for the UCNE because it has not been tested with the UCNE.

7.10 Indoor Cable Bend Radius

Cable bend requirements

Fiber optic cable is easily damaged by excessive bending or kinking of the cable, so a minimum bend radius is established for each different cable size and cable construction. Indoor cable is the most flexible and outdoor direct burial cable is the least flexible. Table 7-7 specifies the minimum bend radius for indoor cable.

In the table, short term minimum bend radius refers to the minimum bend radius that is safe during installation. Long term minimum bend radius refers to the minimum bend radius that is safe after the cable installation is completed and settled.

Table 7-7 Minimum Bend Radius for Indoor Cable

Fibers Per Cable	Minimum Bend Radius (Short Term)	Minimum Bend Radius (Long Term)
2	17.0 cm (6.7 in.)	13.0 cm (5.1 in.)
4	22.0 cm (8.7 in.)	16.5 cm (6.5 in.)
6	26.0 cm (10.2 in.)	20.0 cm (7.9 in.)
8	30.0 cm (11.8 in.)	23.0 cm (9.1 in.)

7.11 Cable Construction

Fiber optic cable selection

The selection of fiber optic cable is dependent upon satisfying installation and environmental requirements without exceeding the maximum optical signal losses. The factors to be considered when selecting the cable are:

- the total fiber optic cable losses.
- the cable requirements caused by the desired routing. Routing requirements can include direct burial, conduits, trays, raceways, plenums, etc.
- construction code requirements.

Installation precautions

The installation procedures for placement of fiber optic cables are the same as for electrical wires. Care should be taken to avoid yanking, flipping, or warping that can result in the unnecessary tightening of the optic fibers. Fiber optic cables should not be subject to foot traffic or crushing forces. Sharp bending and scraping at entrances and covers should be avoided.

7.12 Cable Splices and Connections

Planning the cabling	When planning a system installation, design the system to have the minimum number of splices. For example, convert outdoor cable to indoor cable when entering a cabinet where the bend radius of outdoor cable will not fit in the cabinet. Where this splicing must be made, sufficient cable length must be provided for a splice loop. Thirty to 45 centimeters (12 to 18 inches) on each cable end is the usual allowance for a service loop. Also, when entering the equipment cabinet, sufficient cable length must be allowed for breakout (stripping and fanning out) and termination of the individual optic fibers.
Cable splice protection	Splices cannot withstand tensile forces and must be housed in a strain relief assembly. Moisture entry into the splice can cause degradation of performance. Therefore, the splice enclosure must be sealed, and if necessary, the splice encapsulated to minimize moisture entry.
Cable breakout	Breakout eliminates the outer sheath, leaving the more flexible individual optic fibers for routing within the cabinet. Tight buffered indoor cables (discussed in subsection 4.9.5) provide strength members with each optic fiber which eliminates the need to use breakout kits to add strength members for each optic fiber.
Use of a breakout kit	In outdoor loose buffered, gel-filled cable designs, there is no strength member or protective jacket for the individual optic fibers. Termination of this type of cable requires the use of a breakout or fan-out kit. The kit provides strength members with flexible jackets that are placed over the optic fibers after the outer jacket is removed. These kits can be installed over optic fibers as much as 10 meters (33 feet) in length. The kit provides optic fiber protection for in-cabinet routing and termination.
Cables with connectors preinstalled	If the cable length is accurately determined and the cable does not have to be pulled through a conduit, the cables can be ordered from a fiber optic cable assembly supplier with connectors installed on the ends, pre-tested, and with protective covers on the ends.

7.13 Signal Loss Budget

Introduction

The optic fiber cable/connector/splice signal loss budget is the maximum allowable optic signal loss between a transmitting UCNE and its receiving UCNE partner when interconnected by fiber optic cable that includes ST-type connectors and optional splices. It is the difference between the minimum transmit power and the minimum receiver sensitivity, minus the desired signal loss margin (3 dB) that is necessary to account for the decreasing transmit power characteristics of aging transmitters.

ATTENTION

In the calculations that follow, the loss in the ST-type connector at the cable end that attaches to the UCNE front panel XMTR connector is already included in the transmitter output specification and does not need to be included in the cable/connector/splice signal loss budget. However, any other connectors or splices must be included, including the ST-type connector at the far end of the fiber link that attaches to the UCNE partner's RCVR connector.

Signal loss budget calculation

The signal loss budget for the UCNE is calculated as follows. Table 7-8 provides a conversion table between dBm and microwatts.

Transmit Power - Receiver Sensitivity = Budgeted Signal Loss

<u>Component</u>	<u>dB Level</u>
Transmitter	-14.0
Receiver	(-) -27.0
	13.0
Less transmitter power loss over time	(-) 3.0
Signal loss budget	10.0

Summarizing the above calculation, the rated transmit power is -14 dB prior to transmitter aging. The rated receiver sensitivity is -27 dB. The transmitter aging loss is 3 dBm maximum. The worst case signal loss budget for the cable, splice, and connector loss is 10 dB.

Continued on next page

7.13 Signal Loss Budget, Continued

Power level conversion chart

Table 7-8 is useful for converting optical power levels between two common units of measurement, dBm and microwatts, for various types of optical power meters.

Table 7-8 dBm to Microwatts Conversion Table

dBm	Microwatts
+3	2000
0	1000
-3	500
-6	250
-9	125
-12	62
-14	40
-15	32
-17	20

Types of splices

Two types of splices are available: mechanical and fusion.

The mechanical-type of splice consists of a glass tube into which the optic fiber is secured by ultra violet (UV) curable epoxy.

The fusion-type splice is produced by carefully aligning the optic fibers and then fusing them together by heating the optic fibers.

In both cases, the signal loss in the resulting splice is very dependent upon the skill of the technician. The losses at all splices and cable installations need to be certified and recorded at the time of installation for future reference as mentioned in subsection 7.14.

Continued on next page

7.13 Signal Loss Budget, Continued

Cable span distance formula

As discussed previously in subsection 4.4, UCNE Configuration Rules, the maximum total length of UCN-A or UCN-B, including fiber optic links and coaxial trunk cables, must not exceed 2 kilometers which is based on cable delays. Therefore, because of cable/connector/splice attenuation of optical signals in a fiber optic link, it is possible that the fiber optic link distance may be less, but never more than 2 kilometers.

To calculate the maximum distance of the fiber optic link based on attenuation, use the following equation.

$$\frac{10.0 - (\text{losses in splices}) - (\text{losses in ST-type connectors})}{\text{Maximum cable loss in dB/km}} = D$$

Where D is the distance in kilometers.

The maximum distance for 62.5/125 micron cable with two mechanical splices and an ST-type connector at each end of the cable (one connector is already included in the loss signal budget previously discussed) can be calculated using Tables 7-9 and 7-10.

$$\frac{10.0 - (4.0 \times 0.5) - (1 \times 1.0)}{4} = 1.75 \text{ kilometers}$$

Table 7-9 Optical Power Loss

Type of Splice	Average Loss (dB)	Maximum Loss (dB)
Fusion	0.2	0.3
Mechanical	0.3	0.5
ST-Type Connector	0.5	1.0

Table 7-10 Optic Fiber Losses (@ 820 nm UCNE Wavelength)

Type of Fiber	Average Loss (dB/km)	Maximum Loss (dB/km)
62.5/125 Micron	3.75	4.0

7.14 Power Level Measurement

Record post installation measurements

After the cables have been connected to the fiber optic transmitters and power has been applied to the equipment, measurements of transmitter output, cable loss and power input to the fiber optic receivers should be recorded for future reference. This procedure is detailed in the *Universal Control Network Installation* manual.

Section 8 – Remote User LCN Access (RULA)

8.1 Overview

Section contents

The topics covered in this section are:

Topic		See Page
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Introduction

This section describes the purpose and features of the Remote User LCN Access (RULA) product and defines its hardware and software requirements.

8.2 Description

Introduction

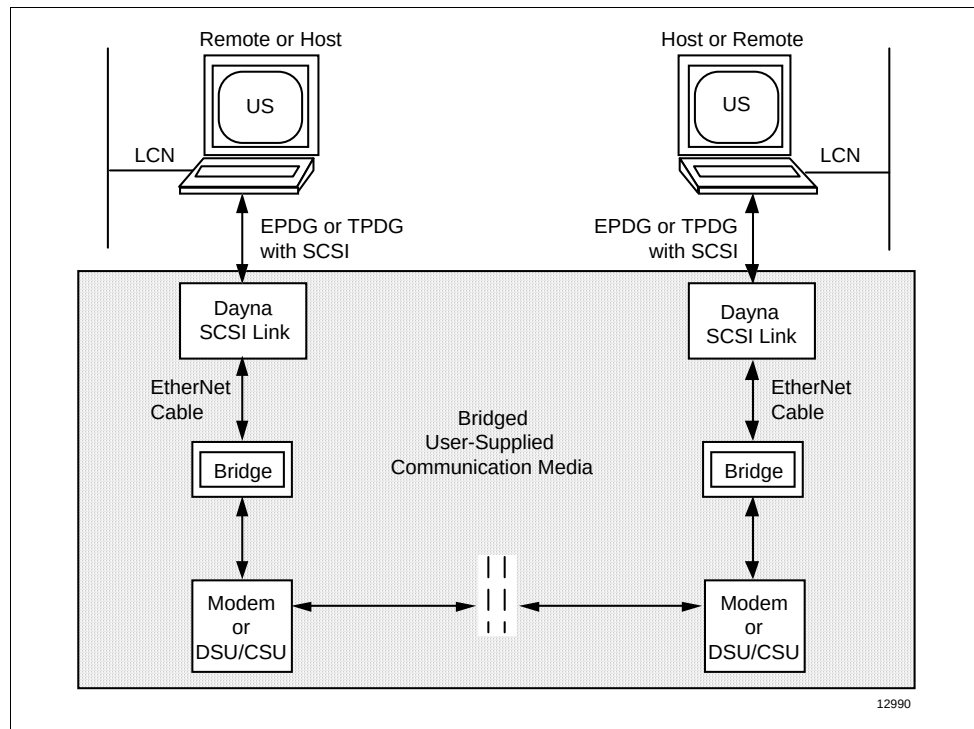
Remote User LCN Access (RULA) is Universal Station external load module software and Ethernet communication media that allows a station on one LCN to access and support a distant LCN by remoting display and keyboard information to one of its stations. RULA is intended to be used for:

- remote engineering,
- remote diagnostics,
- remote troubleshooting, and
- remote system administration.

The Universal Station connected to the LCN being viewed is called the “Host.” Either station can be the Host and as such provide a mirror image of its display and keyboard lamps to the Remote station. Remote keyboard and touchscreen input is sent to the Host station where it is processed as input to the Host. The Host’s screen is updated, then returned to the Remote screen.

Figures 8-1, 8-2, and 8-3 show the RULA hardware components in different operational scenarios. The LCN is optional at the Remote site — only a Universal Station is required to support a RULA session there.

Figure 8-1 RULA Overview Block Diagram



Continued on next page

8.2 Description, Continued

Introduction,
continued

Figure 8-2 RULA Block Diagram – Local Ethernet LAN

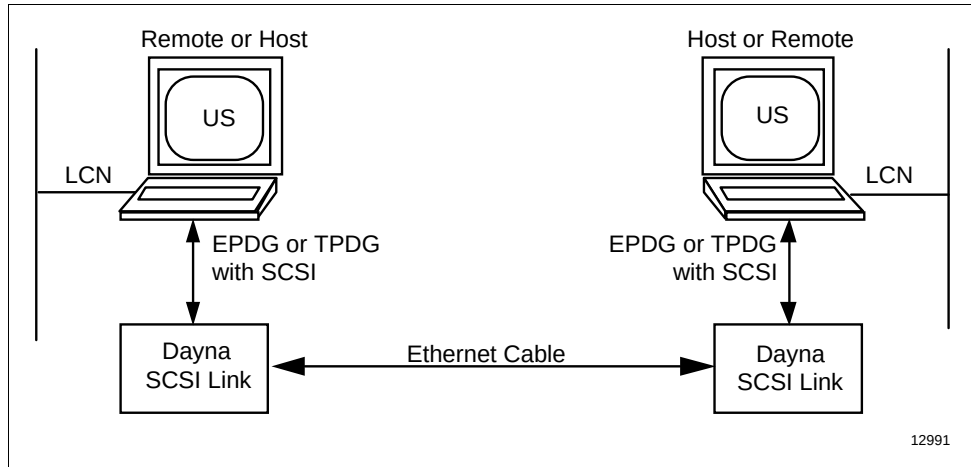
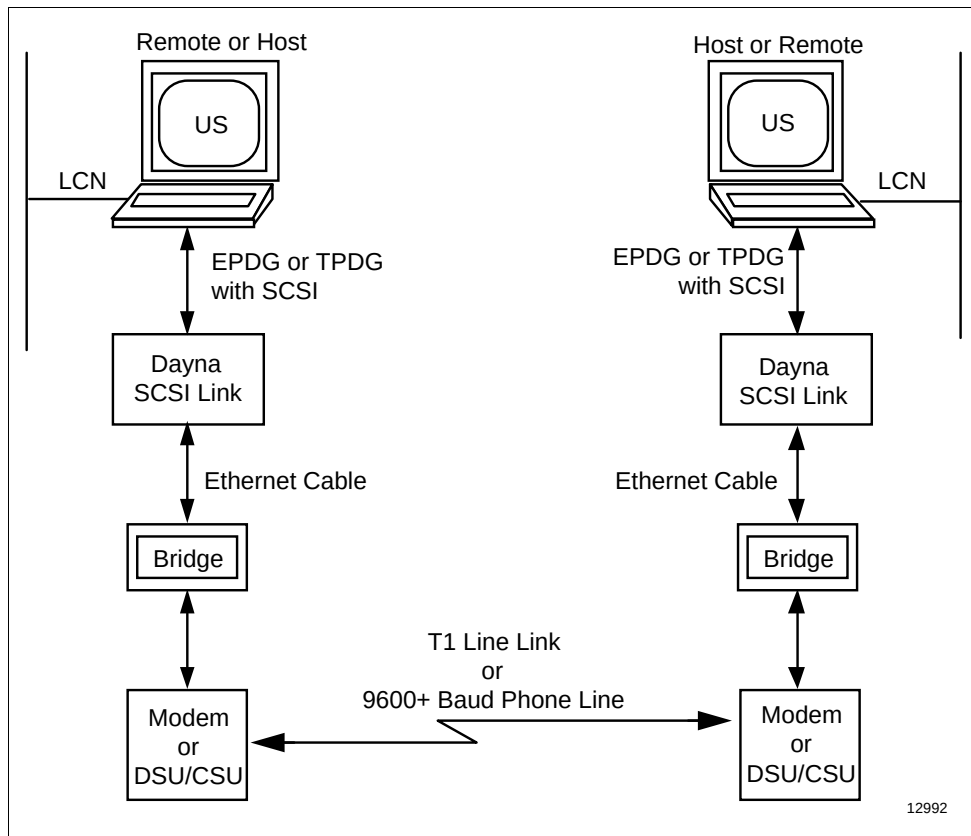


Figure 8-3 RULA Block Diagram – T1 High-Speed or 9600+ Baud Low-Speed Line



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8.2 Description, Continued

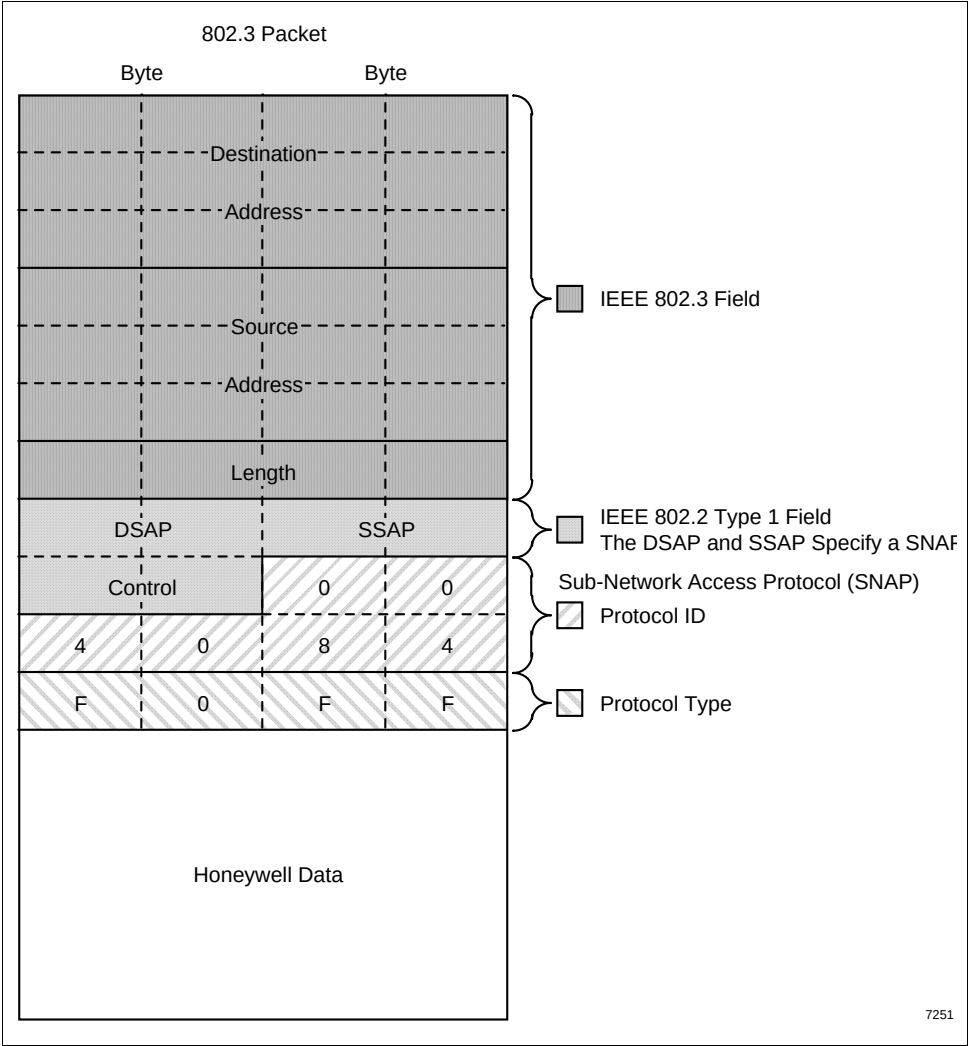
Station behavior	Generally, when a RULA work session is in progress, the information shown on the connected stations is identical, including: • display output, • keyboard, touchscreen, and cursor input (including mouse and trackball), • keylamp states and audible annunciator, • configurable buttons, and • audible keyboard responses occurring on the Host station (clicks when keys are pressed) are echoed on the Remote station.
Functions supported by Honeywell	The top half of Figure 8-1 indicates the extent of the RULA functions provided and supported by Honeywell: • the operations of the Host and Remote Universal Stations, and • the input and output of Ethernet packets directly to an Ethernet network through SCSI to Ethernet converters, which are DaynaPORT SCSI/Link-3 devices purchased from a third party.
Communication media	RULA communications may be accomplished using a wide range of interface hardware. The pair of stations may be physically connected by a communication media such as: • Ethernet cable, • dedicated line, • fiber optic, or • intercontinental phone line.
Routing	Currently, communications are not TCP/IP routed by RULA; instead they are directly addressed and are bridgeable.
Communication addressing scheme	RULA uses point-to-point Ethernet addressing to facilitate messaging. The two DaynaPORT SCSI/Link-3 devices involved in the RULA session must have a live Ethernet addressable connection. This may be as simple as an Ethernet cable connecting the two DaynaPORT SCSI/Link-3 devices. Alternatively, for long distance RULA sessions, the user may bridge the Ethernet line over a communication media.

8.3 RULA Protocol Definition

Protocol

RULA messages may be filtered in bridges using the Sub-Network Access Protocol (SNAP) values. The SNAP protocol is identified in the IEEE 802.2 DSAP (Destination Service Access Point) and SSAP (Source Service Access Point). Figure 8-4 illustrates the RULA protocol definition.

Figure 8-4 RULA Protocol Definition



WARNING

WARNING—RULA does not currently support the XID (Exchange Identification) and TEST commands of the IEEE 802.2 layer.

8.4 RULA Hardware Requirements

Major hardware components

Table 8-1 lists the major hardware components required to support RULA.

Table 8-1 RULA Major Hardware Components

Hardware	Required or Optional	Comment
US, U ^X S, UWS, or Micro TDC 3000	Required	RULA is only available for the Universal Station LCN node type. The memory requirement is 6 MB for Universal Personality.
Bernoulli Cartridge Drive or Zip Disk Drive	Required	For all types Universal Stations (US, U^XS, Micro TDC 3000, and UWS), a local cartridge drive is required in order to transfer files to/from an LCN running an incompatible software release. For the US and U^XS, the cartridge drive power supply provides DC power to the DaynaPORT SCSI/Link-3, located on the drive mounting tray. For a lone Remote station not connected to a local LCN, a local cartridge drive is required for station media.
Engineering Keyboard	Required	An engineering keyboard is necessary for an quick disconnect key sequence and to provide complete access to engineering functions during a RULA session. The engineering keyboard is not mandatory for the use of RULA, since RULA can be disconnected by target action on the RULA Control display.
EPDG2 or TPDG (U ^X S)	Required	Only these SCSI peripheral boards work with RULA, allowing operation of the DaynaPORT SCSI/Link-3 with the station.
SCSI Cable	Required	Connects a Universal Station's SCSI peripheral board to the DaynaPORT SCSI/Link-3.
DaynaPORT SCSI/Link-3	Required	A third party vendor supplied device that provides SCSI to Ethernet protocol translation, allowing the Universal Station access to Ethernet. Outputs onto a standard Ethernet cable and receives on the same cable. Output is guaranteed to be compliant with IEEE 802.3 Ethernet standards.
Ethernet Cable	Required	Provides the means for transmitting data using the TCP/IP protocol. May be connected to optional bridges, routers, or modems.
Bridge/Router	Optional	Any bridge or router used for RULA must allow a user to connect the Ethernet networks of the Host and Remote sites. The router must be IEEE 802.3 compliant and support ARP proxy.
Modem or DSU/CSU	Optional	Allows user to connect two bridge devices by way of a communication medium, such as a normal dial-up phone line. DSU/CSU adapter devices are needed instead of modems to connect devices through a T1 line, for digital (instead of analog) telephone communications.

Continued on next page

8.4 RULA Hardware Requirements, Continued

Memory requirement The minimum requirement for the station is 6 MB for the Universal Personality in Software Release 500.

If other external load module software is to run in the station, there may not be enough memory available to run RULA.

Hardware mounting kits Table 8-2 lists the four types of Honeywell supplied hardware mounting kits that are available to install a DaynaPORT SCSI/Link-3. The mounting kits include hardware installation instructions.

Table 8-2 RULA Hardware Mounting Kits

Model #	Description
MP-RLKT01	US or U ^X S in 1983 (Classic) Console Furniture
MP-RLKT02	Micro TDC 3000
MP-RLKT03	US or U ^X S in 1993 (Ergonomic) Console Furniture
MP-RLKT04	Universal Work Station (UWS)

CE Compliant hardware mounting kits Table 8-3 lists the four types of Honeywell supplied hardware mounting kits that are available to install a DaynaPORT SCSI/Link-3. The mounting kits include hardware installation instructions.

Table 8-3 CE Compliant RULA Hardware Mounting Kits

Model #	Description
MP-RLKT11	US or U ^X S in 1983 (Classic) Console Furniture
MP-RLKT12	Micro TDC 3000
MP-RLKT13	US or U ^X S in 1993 (Ergonomic) Console Furniture
MP-RLKT14	Universal Work Station (UWS)

HVTS A hardware verification test for testing the RULA SCSI interface is included with the standard HVTS software (R420 or later).

8.5 RULA Software Requirements

Software requirements

RULA requires the Universal Personality , with or without X windows. *The Universal Personality software loaded into the pair of RULA stations must be from the same software release series; for example, Release 500.*

When a RULA connection is requested, the Universal Station checks the software revision of both RULA nodes. If the first two digits of the software version do not match, a connection is not allowed. Additional software compatibility guidelines are described later in this manual.

8.6 Specifications

Universal Station hardware specifications

Table 8-4 lists the Universal Station hardware specifications.

Table 8-4 Universal Station Hardware Specifications

Description	Requirement
LCN Module Type	US, U ^X S
Minimum US/U ^X S Memory	6 MB for Universal Personality in Software Release R520
Engineering Keyboard	Required
SCSI Circuit Board	EPDG2 or TPDG
Cartridge Drive	At least one

DaynaPORT converter hardware mounting kits

Table 8-5 is a list of the hardware kits that are available for the installation of a DaynaPORT converter in existing Honeywell equipment.

Table 8-5 DaynaPORT Converter Hardware Mounting Kits

Equipment Description	Kit Model Number
Classic (1983) Console Furniture (US or U ^X S)	MP-RLKT01
Micro TDC 3000 Ergonomic	MP-RLKT02
(1993) Console Furniture (US or U ^X S)	MP-RLKT03
Universal Work Station (UWS)	MP-RLKT04

CE Compliant DaynaPORT converter hardware mounting kits

Table 8-6 is a list of the hardware kits that are available for the installation of a DaynaPORT converter in existing Honeywell equipment.

Table 8-6 DaynaPORT Converter Hardware Mounting Kits

Equipment Description	Kit Model Number
Classic (1983) Console Furniture (US or U ^X S)	MP-RLKT11
Ergonomic (1993) Console Furniture (US or U ^X S)	MP-RLKT12
Micro TDC 3000	MP-RLKT13
Universal Work Station (UWS)	MP-RLKT14

Continued on next page

8.6` Specifications, Continued

DaynaPORT converter specifications

Table 8-7 lists the specifications of the Model DP0802H (110 Vac) and Model DP0802AS (240 Vac) DaynaPORT SCSI/Link-3 converter .

Table 8-7 DaynaPORT Converter Specifications

Specification	Description
Vendor	Dayna Communications, Inc. Sorenson Research Park 849 W. Levoy Drive Salt Lake City, Utah 84123-2544 (801) 269-7200 Fax: (801) 269-7363
LAN Interfaces	10BASE-5, Transceiver (AUI) Connector - Honeywell certified 10BASE-2, BNC Connector (thinnet) (not Honeywell certified) 10BASE-T, Twisted Pair RJ-45 Connectors (not Honeywell certified) CE Compliant models: DP0802H 120 Vac DP0802HS 240 Vac
Power	External 12 Vdc 0.8 A adapter 100-130 or 210-250 Vac 50-60 Hz factory configurable 25 W maximum
AUI Power	250 mA maximum
Operating Environment	+10° to +40°C 5% to 95% RH, noncondensing
Altitude	0 to 12,000 feet
Certification	FCC Class A

Router (optional)

Table 8-8 lists the optional Router requirements.

Table 8-8 Optional Router Requirements

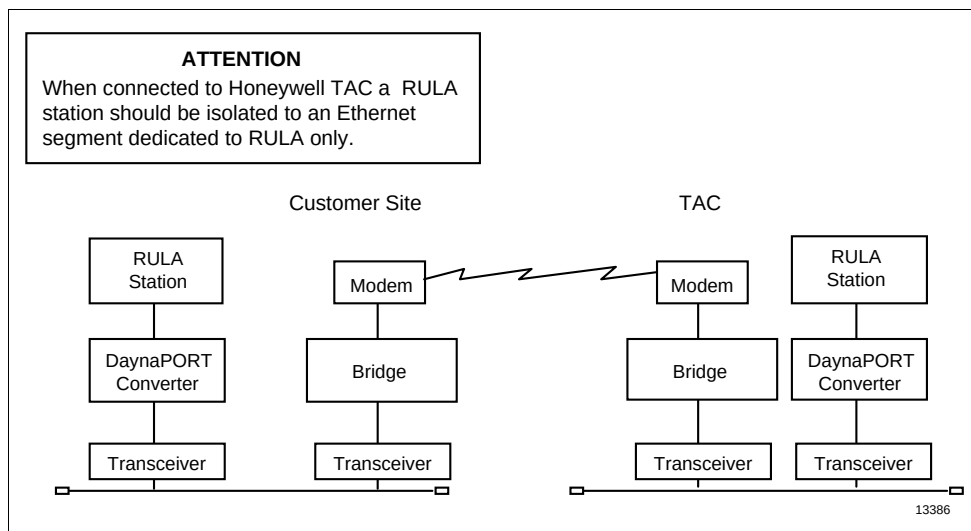
Protocol	Description
Routing	OSPF, RIP, or EGP Must be IEEE 802.3 compliant and support proxy ARP (Address Resolution Protocol)
Wide Area Network	PPP (Point-to-Point Protocol) required for TAC support, or Frame Relay (ANSI)

8.7 Honeywell Technical Assistance Center (TAC)

TAC support

TAC centers worldwide use RULA and other technologies to provide optional Solution (TPS) System support services. In order to provide a secure, high performance on-line TAC connection, dedicated communications equipment is required. A customer site must meet specific configuration requirements for an on-line TAC connection. A normal dial-up voice grade (analog) phone line must be available for an on-line TAC connection. Figure 8-5 illustrates typical TAC configuration requirements.

Figure 8-5 TAC Configuration Requirements



Nonstandard configurations

Nonstandard configuration must be certified on a project basis. This optional service is available from Honeywell at additional cost. If you have further questions about an on-line TAC connection, ask your Honeywell representative for assistance.

Section 9 – Cabinetry

9.1 Overview



SHOCK HAZARD

Troubleshooting, opening this equipment and removing any panels or covers will expose the **user** to the risk of a shock hazard. There are no **user** serviceable parts inside this equipment. Refer all servicing only to **qualified service personnel**.

Section contents

The topics covered in this section are:

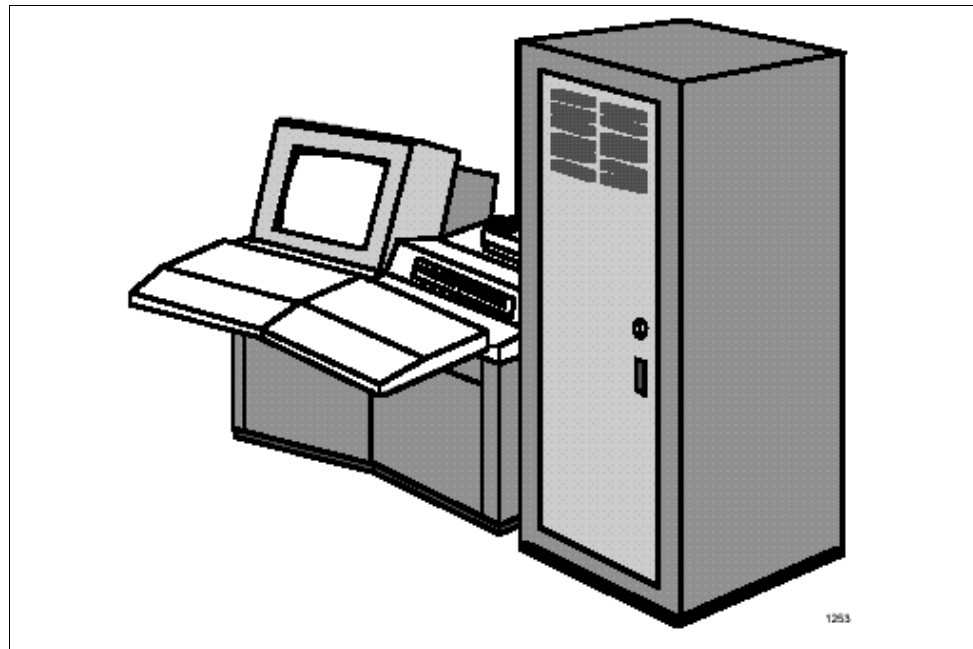
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9.1 Overview, Continued

Introduction

This section provides an overview of the consoles and cabinets used to house the Local Control Network hardware and supporting peripherals.

Figure 9-1 TPS System Console and Equipment Cabinet



Introduction continued

Of particular interest is the console furniture in which Universal Station, Global User Station and Universal Station^X are installed. Two styles of console furniture are available, the “Classic” style, introduced in 1983, and the “Ergonomic” style, introduced in 1993. Figure 9-1 is an illustration of a typical TPS system that includes a Classic style console with an LCN Universal Station and equipment cabinet.

9.2 Classic Furniture

Overview

The Universal Station and Universal Station^X have the same specifications and can occupy the same Classic furniture.

9.2.1 General Specifications

Electrical requirements

Voltage	120 Vac +10%, -15% Single Phase 2 wire (plus safety ground) 240 Vac +10%, -15% Single Phase 2 wire (plus safety ground)
Frequency	50 Hz or 60 Hz +3%, -6% with no more than 8% harmonic distortion
Power	1.4 kW maximum (per cabinet)
Grounding	Console must be grounded by a separate conductor (green or yellow/green wire) run with the power conductors.
Current	15 amperes maximum

Environmental requirements

Storage	Temperature: -35°C to 70°C Humidity: 5 to 95% Noncondensing Altitude: 10,668 meters (35,000 feet)
Operational	Temperature: 0°C to 50°C Humidity: 5 to 95% Noncondensing Altitude: 2,286 meters (7,500 feet)
Limitations	Certain peripheral devices have more critical requirements. The Universal Station will be limited by the requirements of those devices.

Continued on next page

9.2.1 General Specifications, Continued

Shipping requirements

Weight	Single Monitor: 204 kg (450 pounds) maximum Dual Monitor: 363 kg (800 pounds) maximum
Bearing Load	The unit requires a floor capable of supporting an average loading of 391 kg/m ² (80 pounds/square foot) for single and double bay consoles. Single bay unit requires point loading capability of 51 kg (113 pounds) and the double bay unit requires point loading capability of 91 kg (200 pounds) when either unit is resting on the leveling feet.
Moving	When moving, the lifting force must be applied by a method which will bridge both edges of the base. Any accessory items used in lifting or moving and any moving devices must be prevented from bearing against end panels, doors, covers, keyboards, or table tops as these are nonstructural parts and may be damaged.

Operating conditions

Weight	Single Bay: 204 kg (450 pounds) maximum Double Bay: 363 kg (800 pounds) maximum
Bearing Load	The unit requires a floor capable of supporting an average loading of 391 kg/m ² (80 pounds/square foot) for single and double bay consoles. Single bay unit requires point loading capability of 51 kg (113 pounds) and the double bay unit requires point loading capability of 91 kg (200 pounds) when either unit is resting on the leveling feet.
Heat:	1400 Watts (4778 BTU/hr.) maximum per bay. Exhaust temperature is 20°C maximum above inlet temperature. Required air flow is 95 liters/second (200 CFM) maximum per bay.

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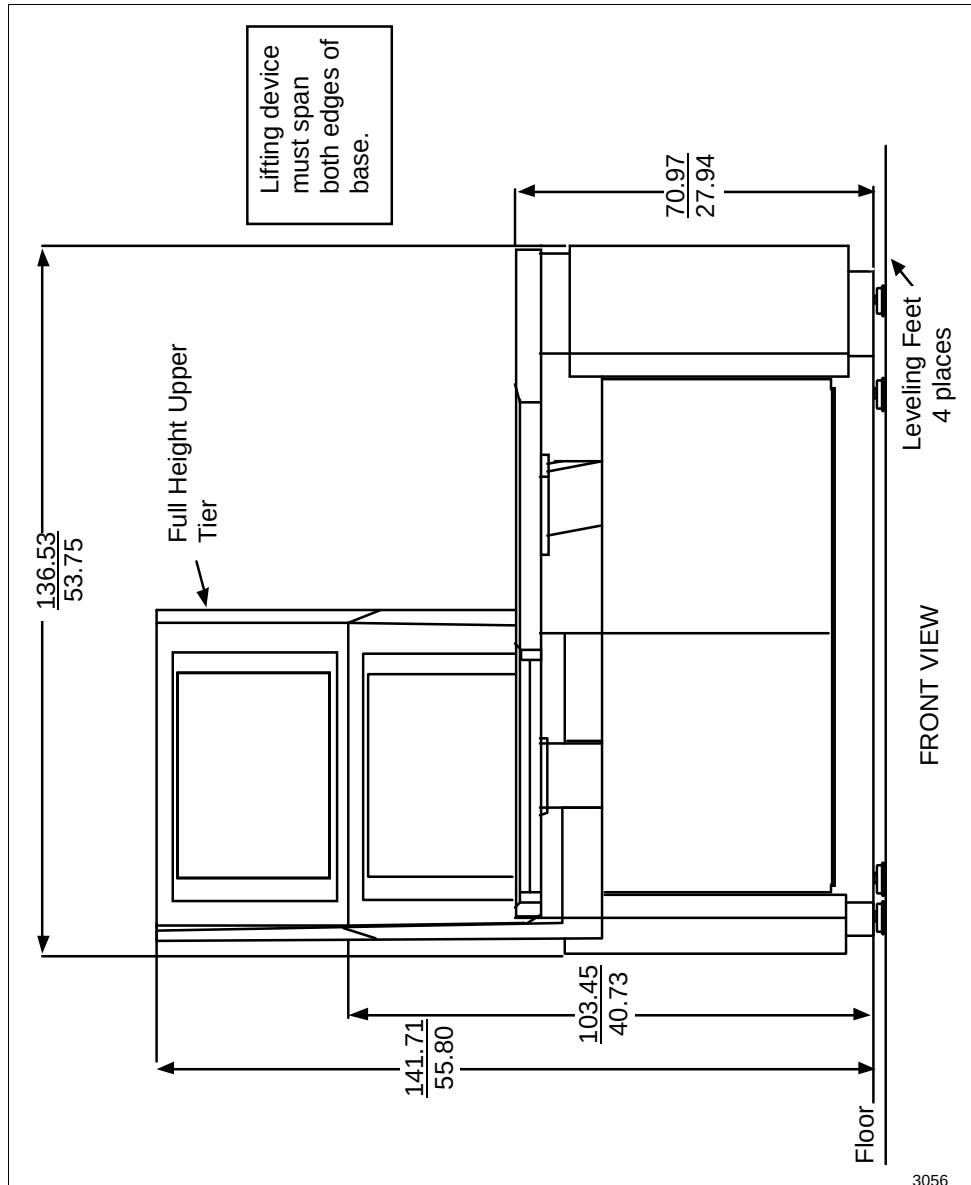
9.2.2 Classic Furniture Description

Overview

Figures 9-2 through 9-10 provide detailed illustrations of Classic furniture that houses both single or multiples of the Universal Station and Universal Station^X modules (nodes).

Front view

Figure 9-2 Classic Furniture Dimensions Front View

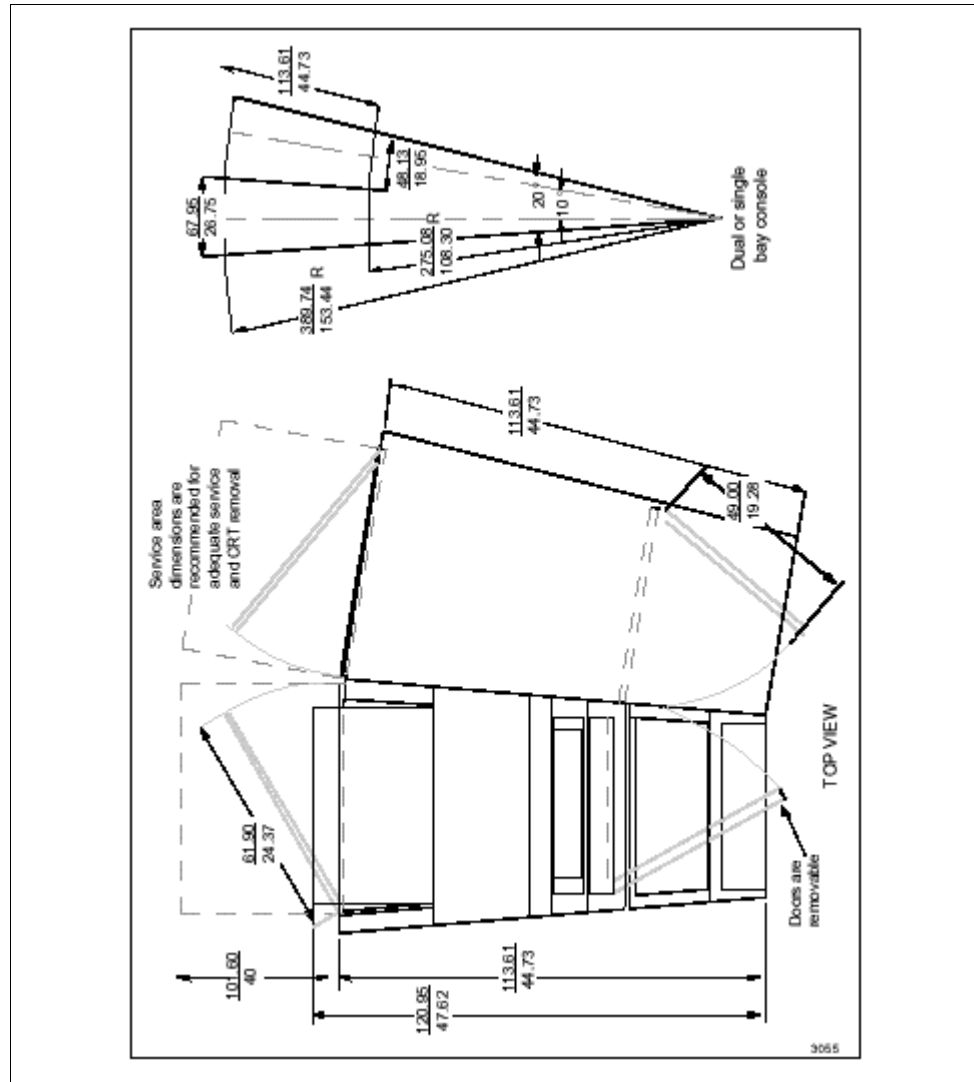


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9.2.2 Classic Furniture Description, Continued

Top view

Figure 9-3 Classic Furniture Dimensions Top View

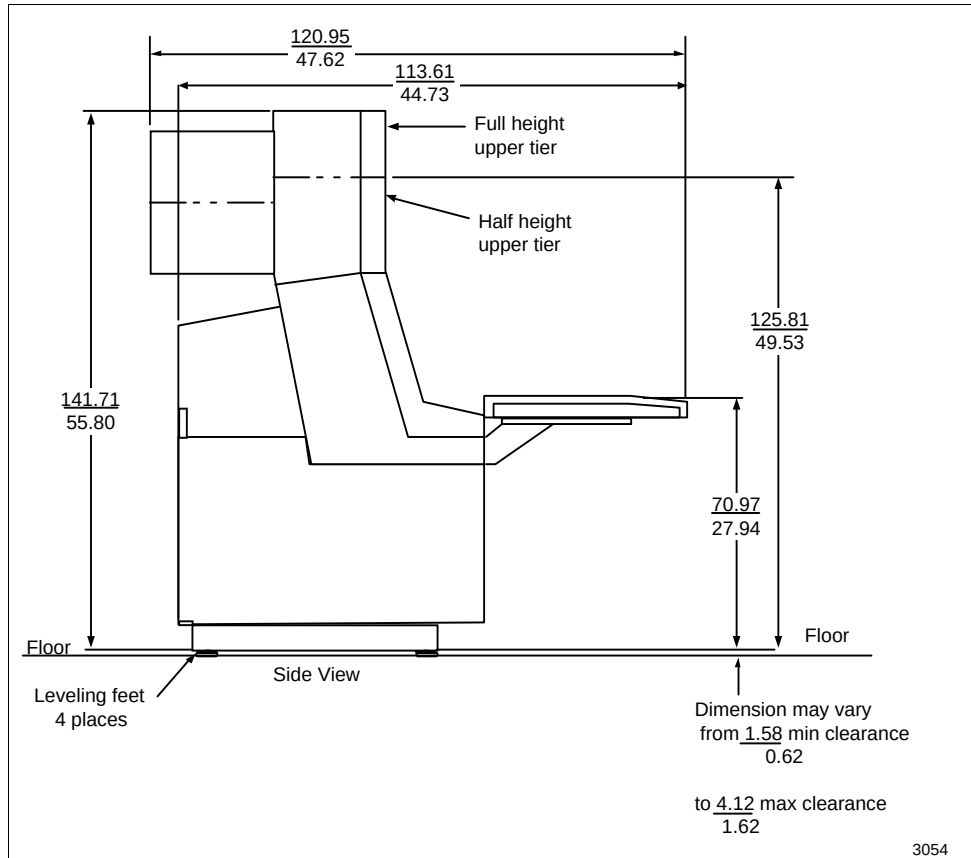


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9.2.2 Classic Furniture Description, Continued

Side view

Figure 9-4 Classic Furniture Dimensions Side View

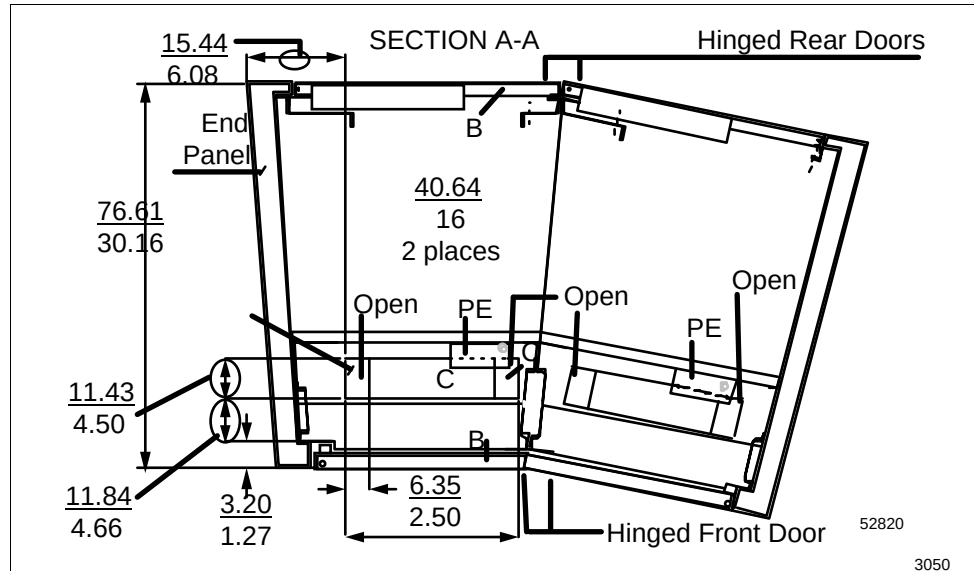


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9.2.2 Classic Furniture Description, Continued

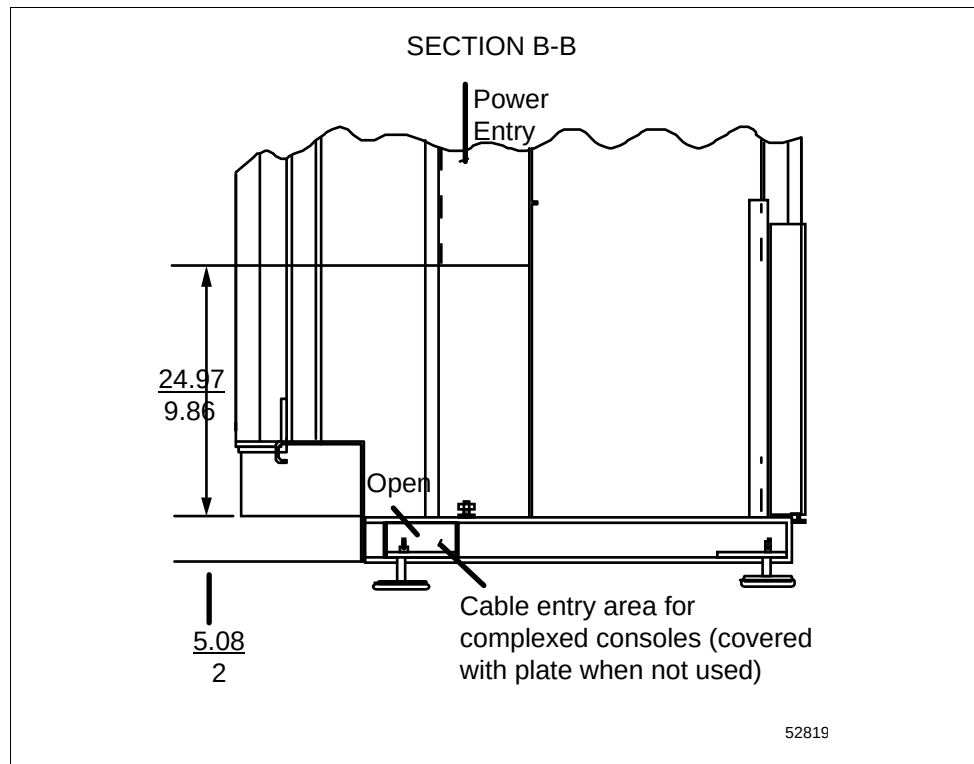
Wire and cable entry

Figure 9-5 Classic Furniture Wire and Cable Entry



Wire and Cable Entry

Figure 9-6 Classic Furniture Universal Station Cable Entry

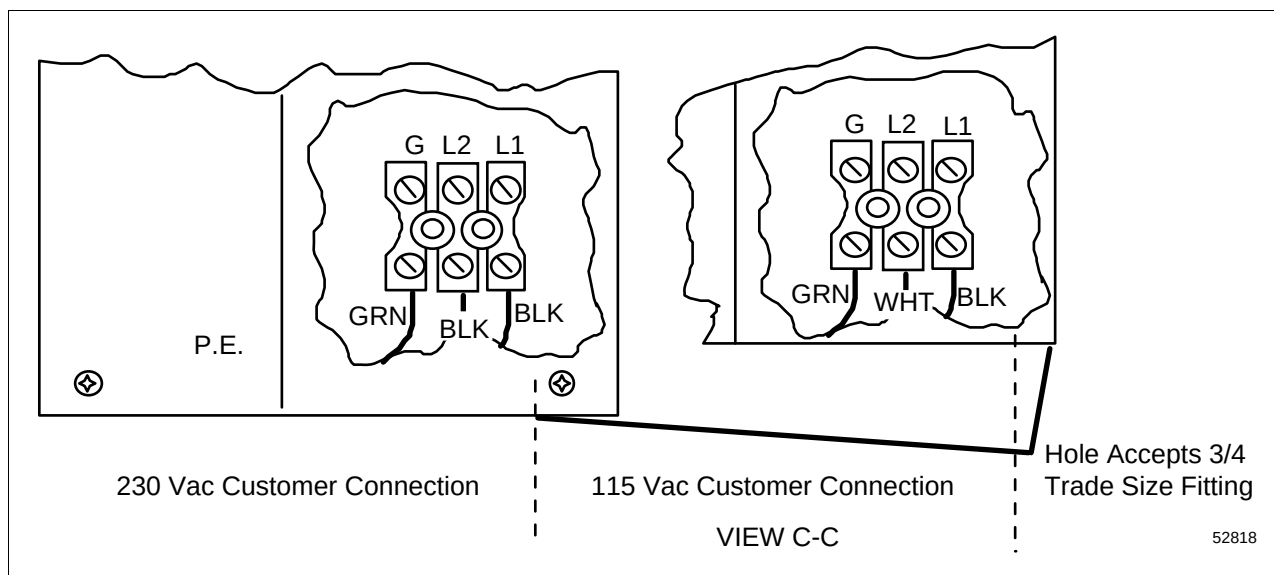


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9.2.2 Classic Furniture Description, Continued

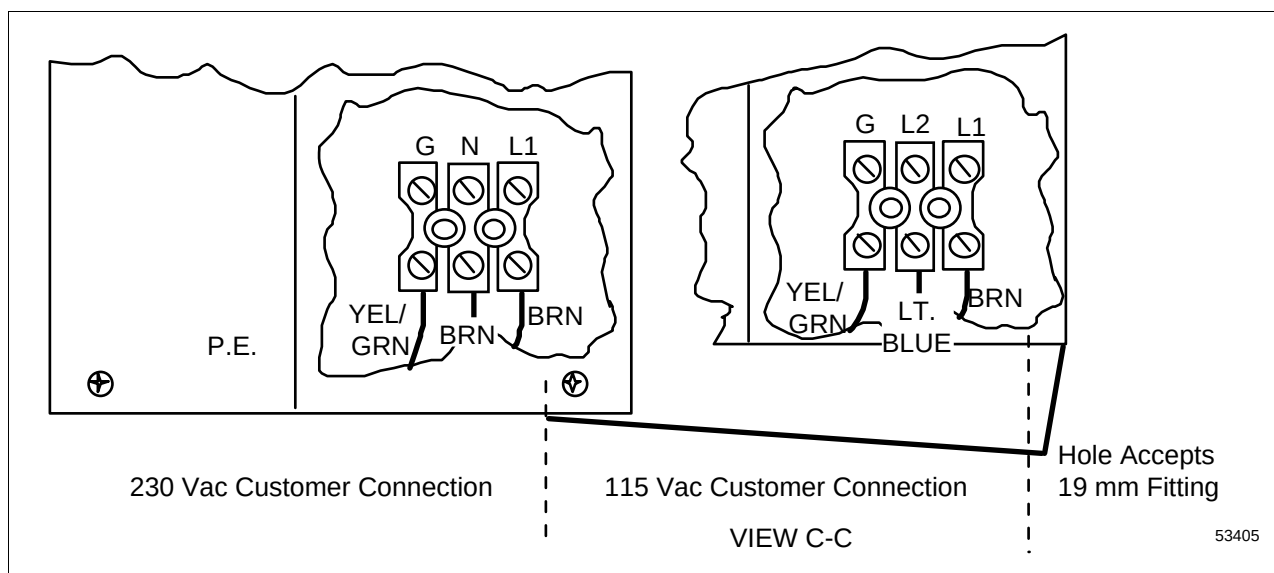
Power entry—US

Figure 9-7 Classic Furniture AC Power Wiring(U.S. Installations)



Power entry—Europe

Figure 9-8 Classic Furniture AC Power Wiring(European Installations)

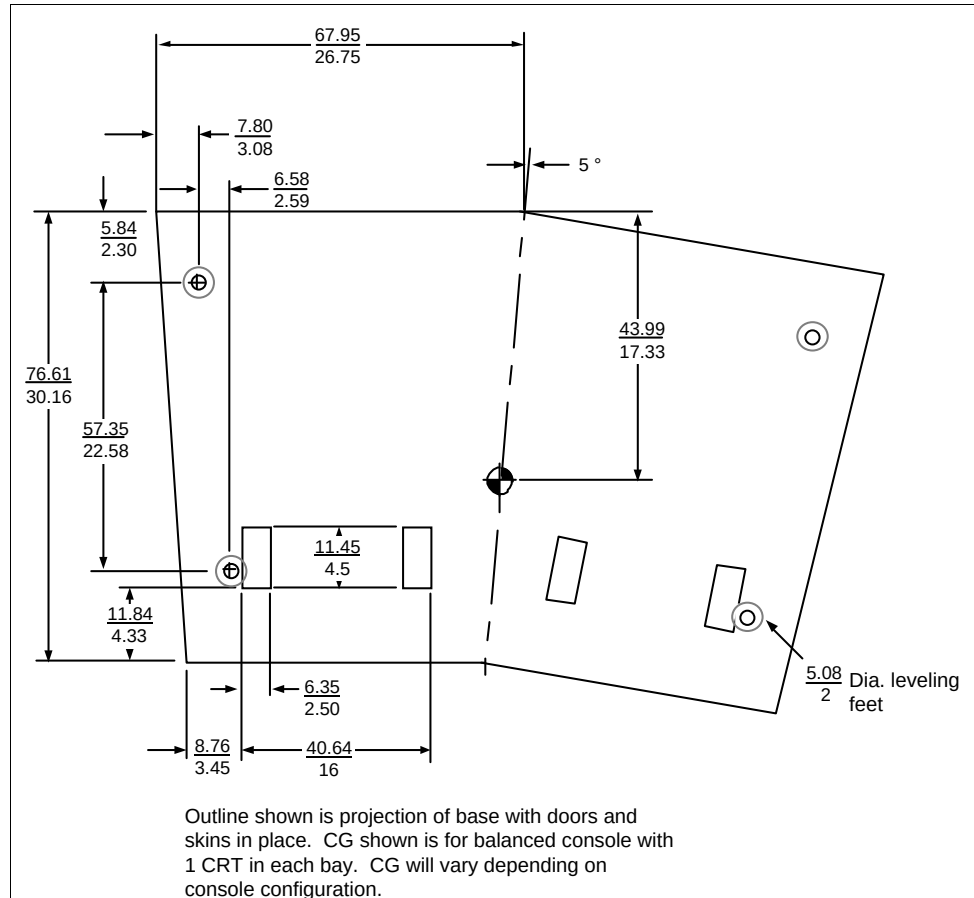


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9.2.2 Classic Furniture Description, Continued

Furniture base outline

Figure 9-9 Classic Furniture Base Outline

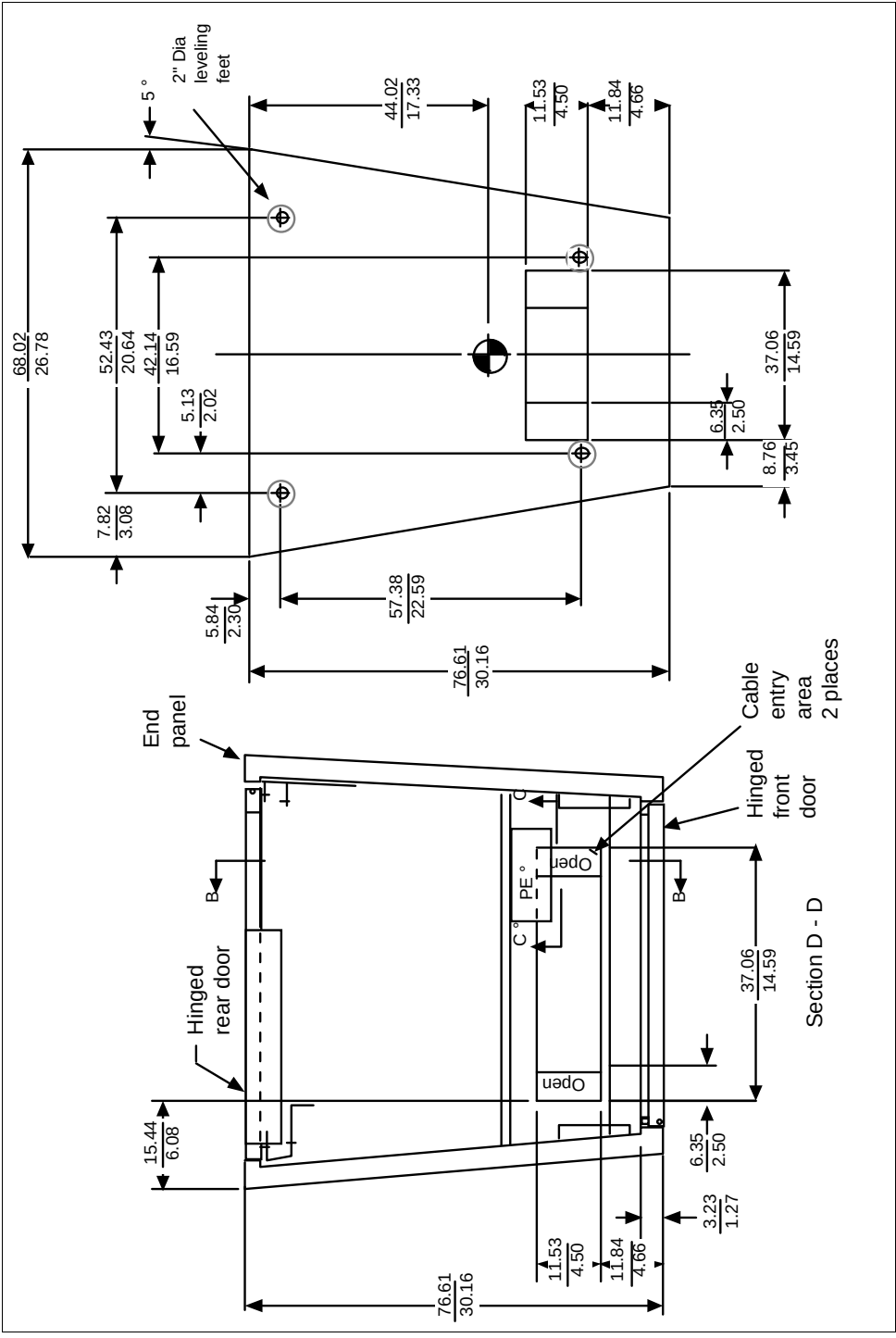


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9.2.2 Classic Furniture Description, Continued

Furniture base outline
with doors

Figure 9-10 Classic Furniture Base Outline with Doors

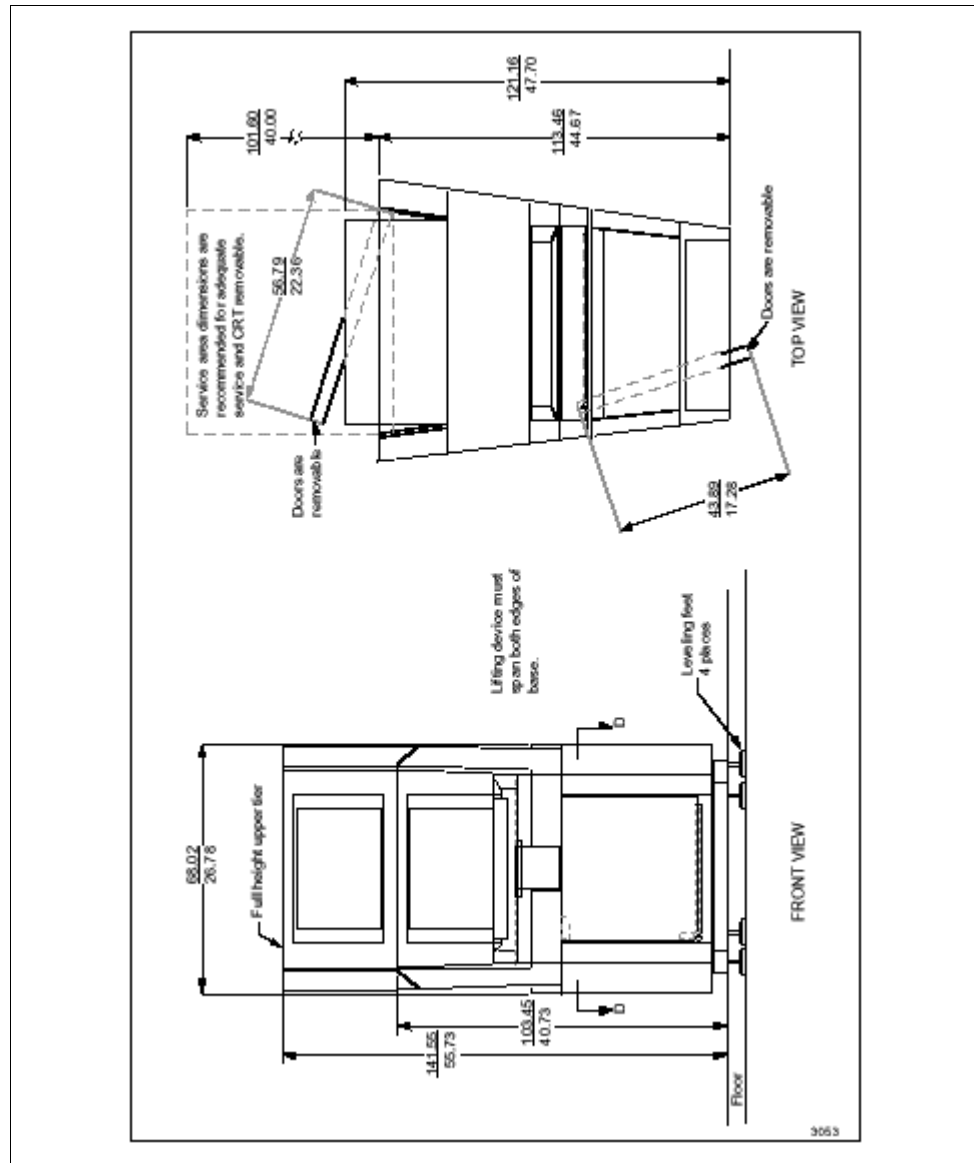


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9.2.2 Classic Furniture Description, Continued

Furniture front and top view

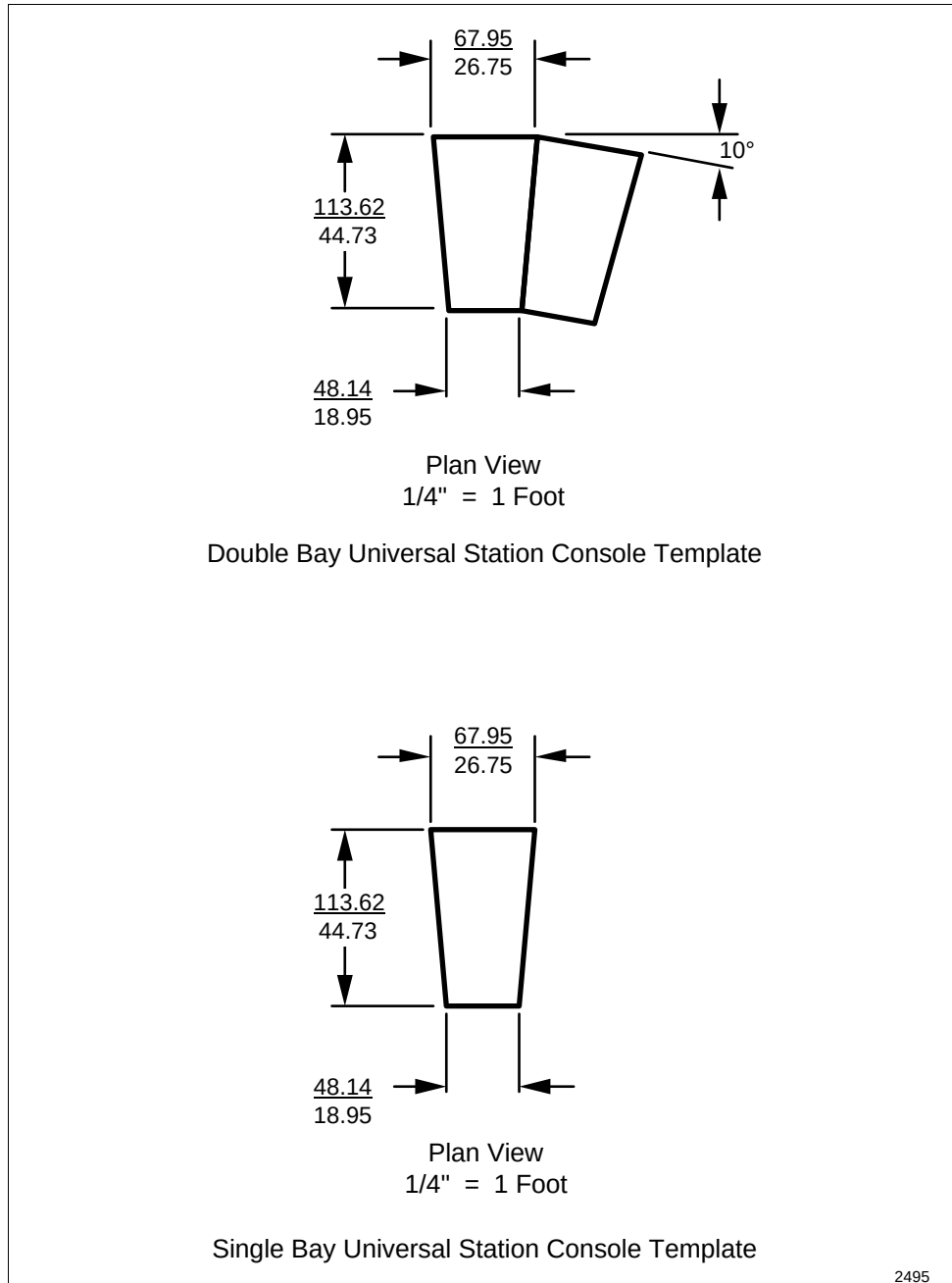
Figure 9-11 Classic Furniture Front and Top View



9.2.3 Classic Furniture Template

Console templates

Figure 9-12 Classic Furniture Console Templates (Footprints)



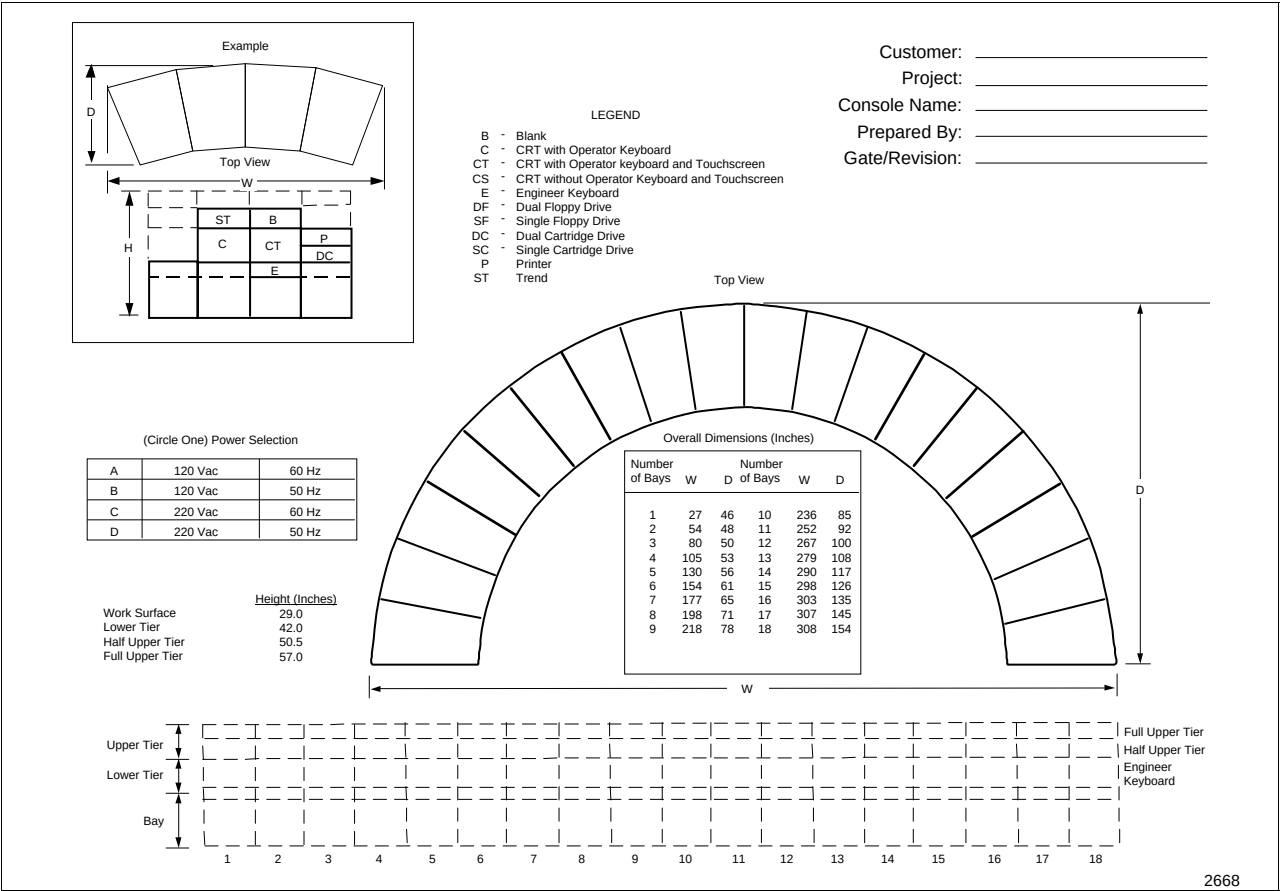
Continued on next page

9.2.3 Classic Furniture Template, Continued

**TotalPlant Solution
(TPS) System console
layout**

The console planning template shown here is available in printed pads of the template. Parts of the template are shown in the three figures that follow.

Figure 9-13 TPS Classic Furniture Console Layout



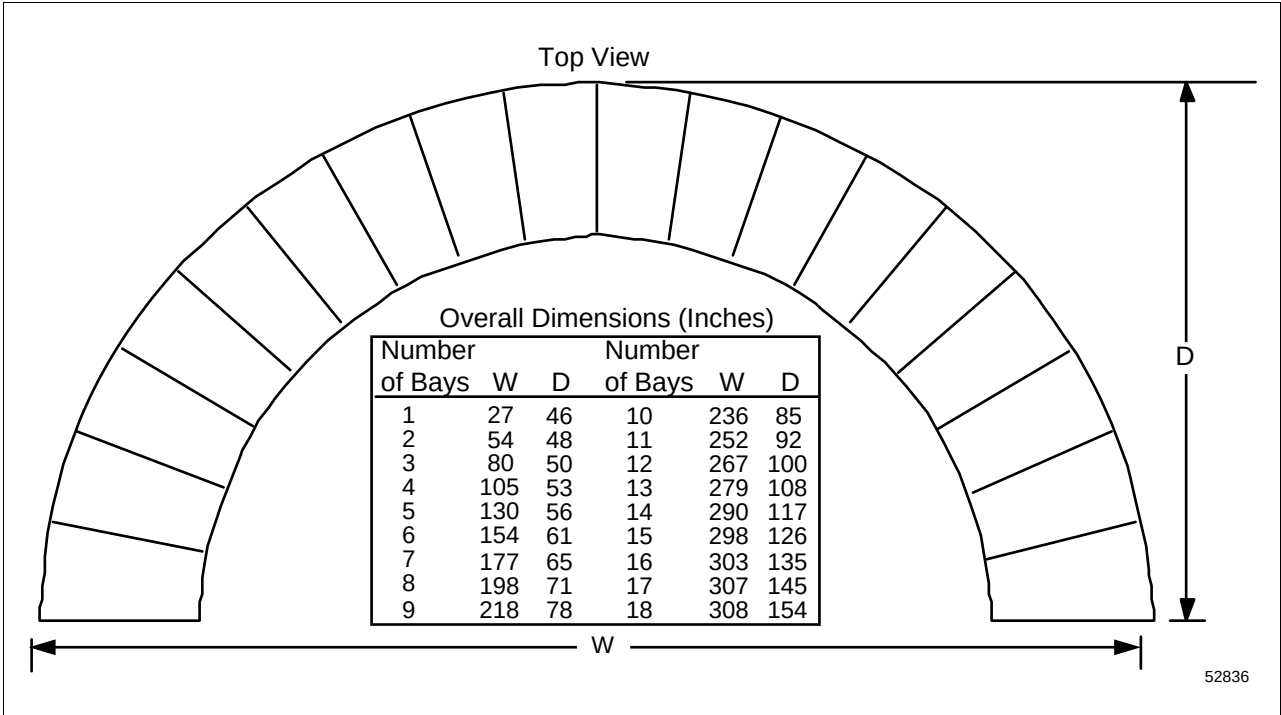
2668

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9.2.3 Classic Furniture Template, Continued

Console dimensions

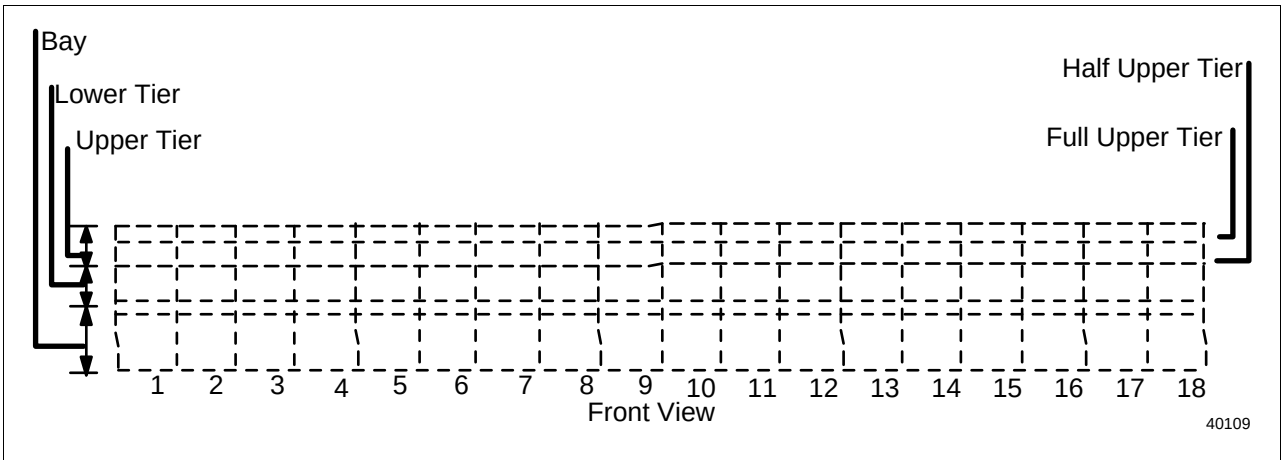
Figure 9-14 Console Dimensions



Template

This part of the template is used to note the location of peripherals to the console.

Figure 9-15 Peripheral Template

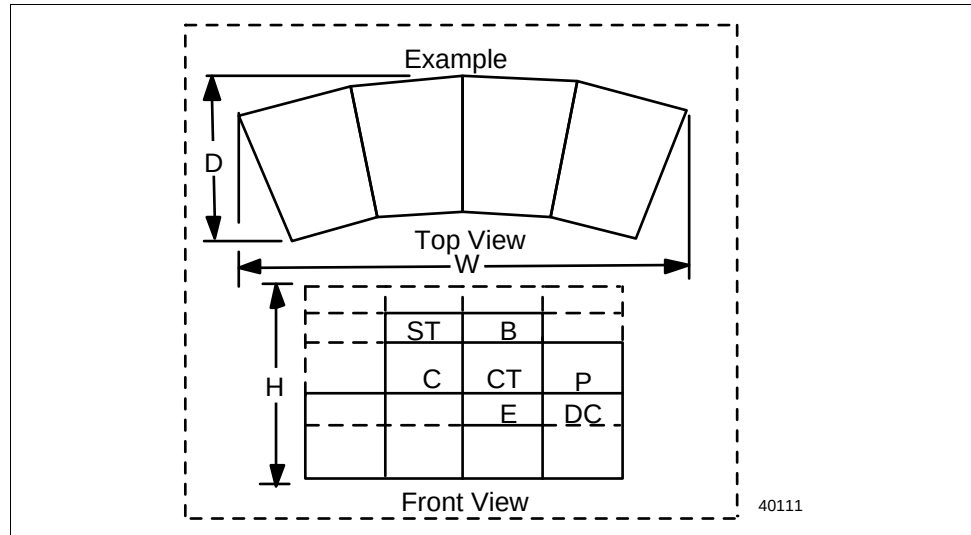


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9.2.3 Classic Furniture Template, Continued

Key to notation

Figure 9-16 Key to Notation for Peripherals—Classic Furniture



Legend table

Table 9-1 Legend to Peripheral Notation

Notation	Description
B	Blank Panel
C	CRT with Operator Keyboard
CD	CD ROM Drive
CT	CRT with Operator Keyboard and Touchscreen
CS	CRT without Operator Keyboard and Touchscreen
DC	Dual Cartridge Drive
E	Engineer Keyboard
P	Printer
SC	Single Cartridge Drive
ST	Trend Pen Recorder

9.3 Ergonomic Furniture

Overview

The Universal Station, Global User Station and Universal Station^X nodes with supporting hardware and peripherals can be installed in the Ergonomic furniture.

9.3.1 General Specifications

Electrical requirements

Voltage	120 Vac +10%, -15% Single Phase 2 wire (plus safety ground) 240 Vac +10%, -15% Single Phase 2 wire (plus safety ground)
Frequency	50 Hz or 60 Hz +3%, -6% with no more than 8% harmonic distortion
Power	1.4 kW maximum (per cabinet)
Grounding	Console must be grounded by a separate conductor (green or yellow green wire) run with the power conductors.
Current	15 amperes maximum

Environmental requirements

Storage	Temperature: -35°C to 70°C Humidity: 5 to 95% Noncondensing Altitude: 10,668 meters (35,000 feet)
Operational	Temperature: 0°C to 50°C Humidity: 5 to 95% Noncondensing Altitude: 2,286 meters (7,500 feet)
Limitations	Certain peripheral devices have more critical requirements. The Universal Station will be limited by the requirements of those devices.

Continued on next page

9.3.1 General Specifications, Continued

Shipping requirements

Weight	Single Monitor: 149.6 kg (330 pounds) maximum Dual Monitor: 181.4 kg (400 pounds) maximum
Bearing Load	A unit with a single monitor requires a floor capable of supporting an average loading of 32.2 kg/m ² (70.9 pounds/square foot) and point loading capability of 37.4 kg (82.5 pounds). A unit with double monitors requires a floor capable of supporting an average loading of 39 kg/m ² (86.0 pounds/square foot) and point loading capability of 45.4 kg (100 pounds)
Moving	When moving, the lifting force must be applied by a method which will bridge both edges of the base. Any accessory items used in lifting or moving and any moving devices must be prevented from bearing against end panels, doors, covers, keyboards, or table tops as these are nonstructural parts and may be damaged.

Operating conditions

Weight	Single Monitor: 149.6 kg (330 pounds) maximum Double Monitor: 181.4 kg (400 pounds) maximum
Bearing Load	A unit with single or double monitors requires a floor capable of supporting an average loading of 39 kg/m ² (86 pounds/square foot). A unit with a single monitor requires point loading capability of 37.4 kg (82.5 pounds) and a unit with double monitors requires point loading capability of 45.4 kg (100 pounds) when either unit is resting on the leveling feet.
Heat:	1400 Watts (4778 BTU/hr.) maximum per bay. Exhaust temperature is 20°C maximum above inlet temperature. Required air flow is 95 liters/second (200 CFM) maximum per bay.

9.3.2 Ergonomic Furniture Description

Overview

Figures 9-17 through 9-22 provide detailed illustrations of Ergonomic furniture that houses both single or multiple Universal Stations or a single Global User Station or Universal Station^X modules (nodes). A Global User Station or Universal Station^X module can have one other module mounted in the station.

The furniture is available in three heights referred to as 650 mm (small), 700 mm (medium), and 750 mm (tall). This terminology refers to the height of the keyboard above the floor. Refer to Figures 9-17, 9-18, and 9-19.

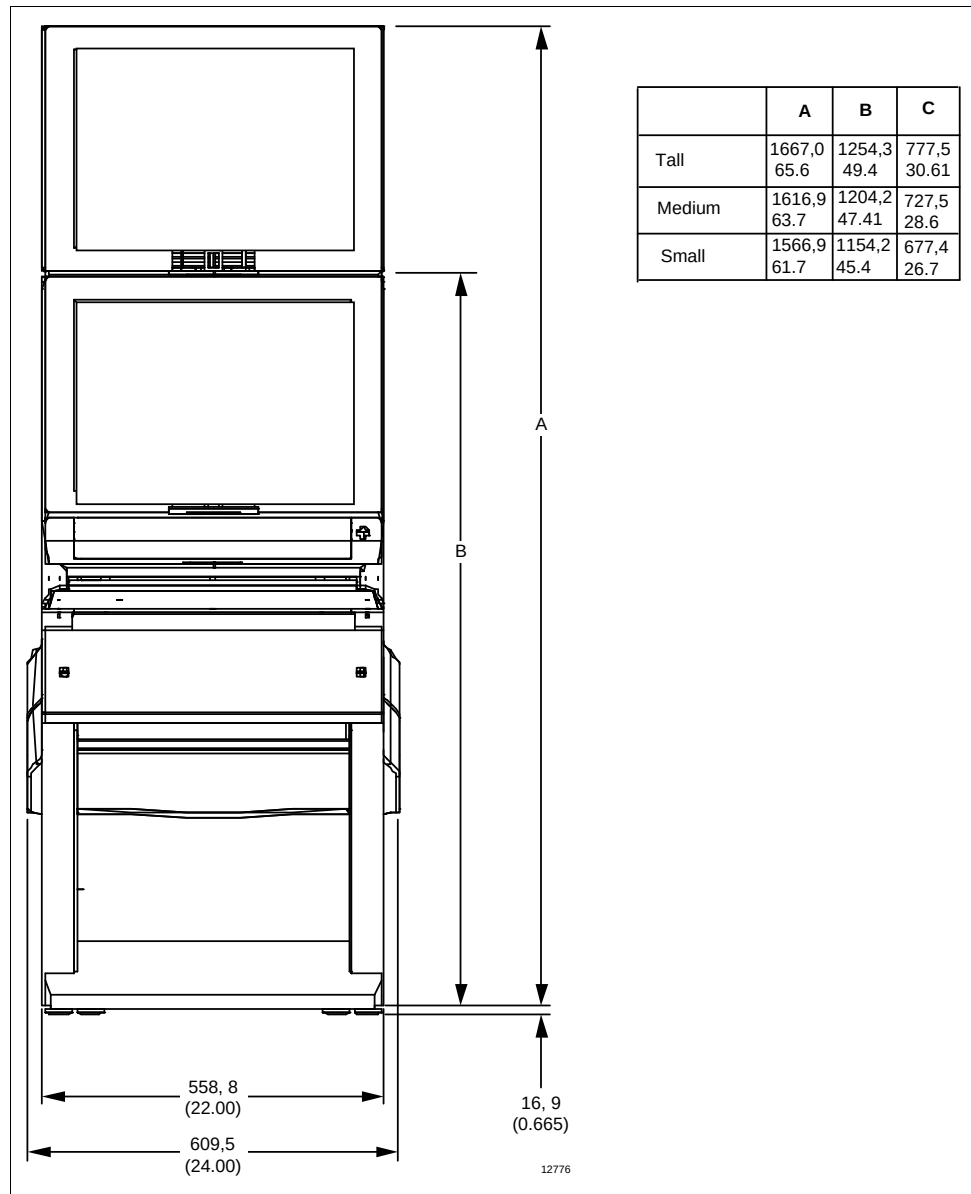
The work-surface furniture has the same footprint dimensions as the OIS.

Continued on next page

9.3.2 Ergonomic Furniture Description, Continued

Front view

Figure 9-17 Ergonomic Furniture Front View

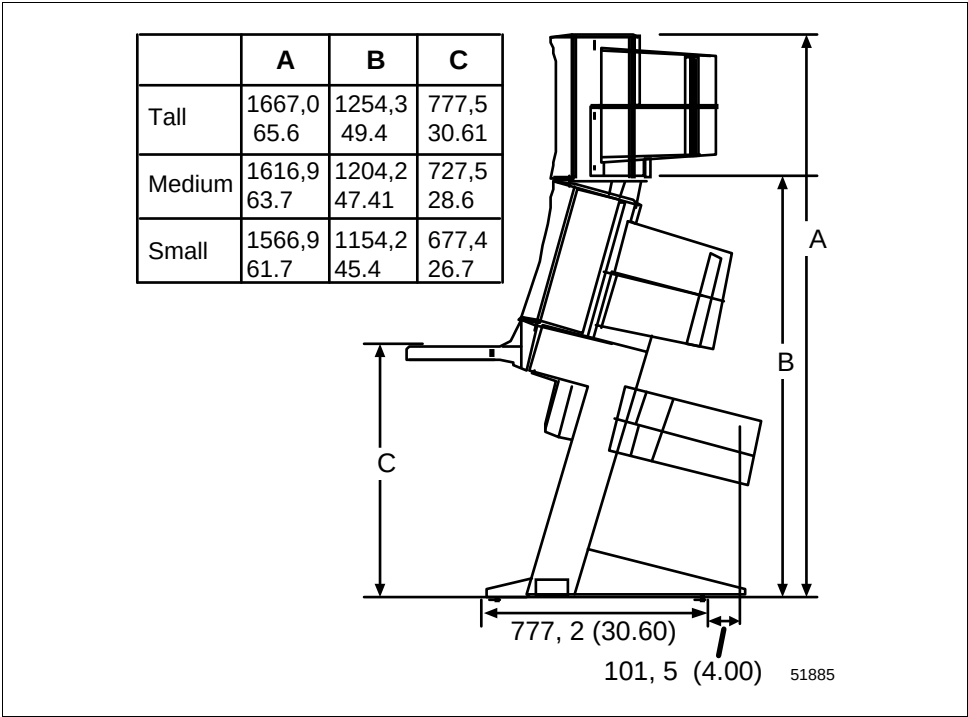


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9.3.2 Ergonomic Furniture Description, Continued

Side view

Figure 9-18 Ergonomic Furniture Side View

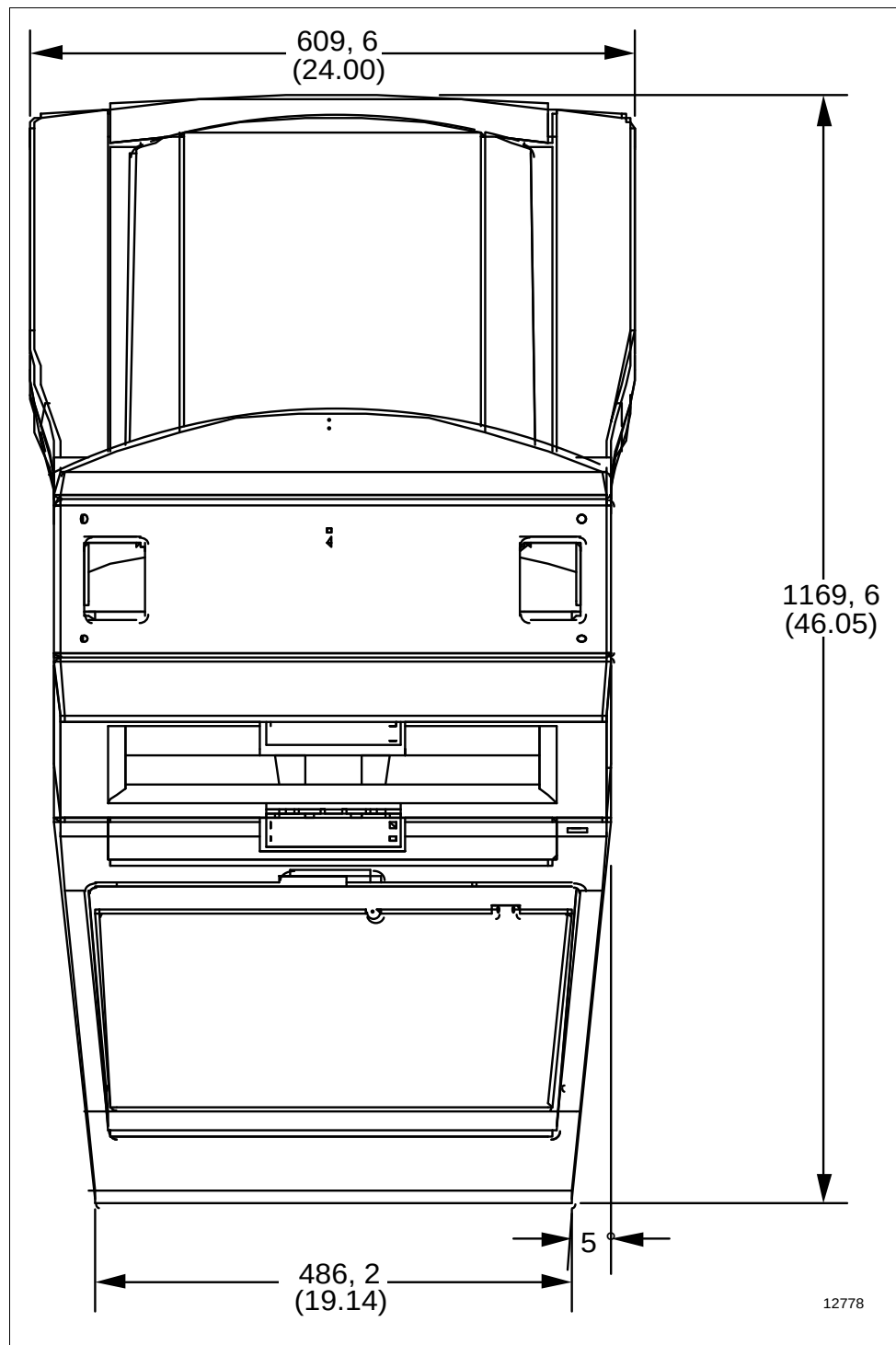


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9.3.2 Ergonomic Furniture Description, Continued

Top view

Figure 9-19 Ergonomic Furniture Top View

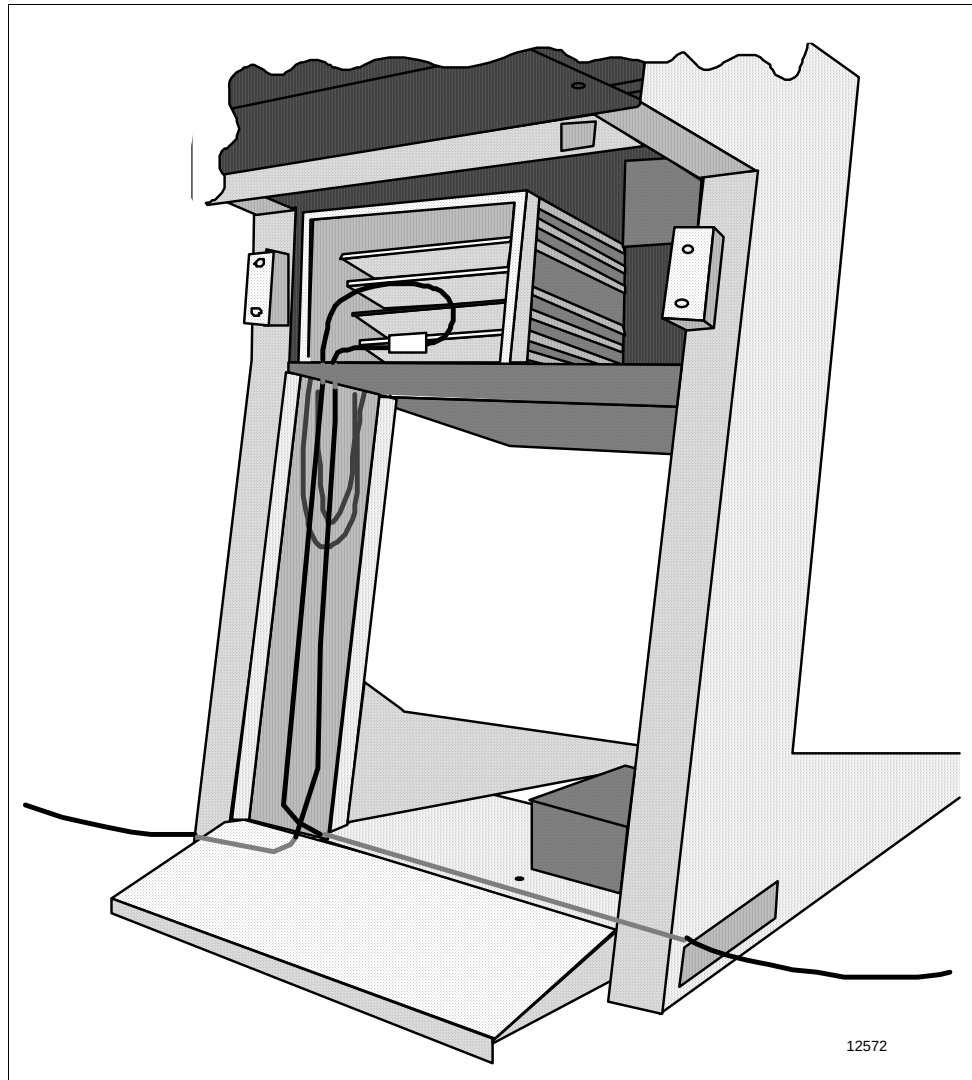


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9.3.2 Ergonomic Furniture Description, Continued

Cabling

Figure 9-20 Ergonomic Furniture Cabling



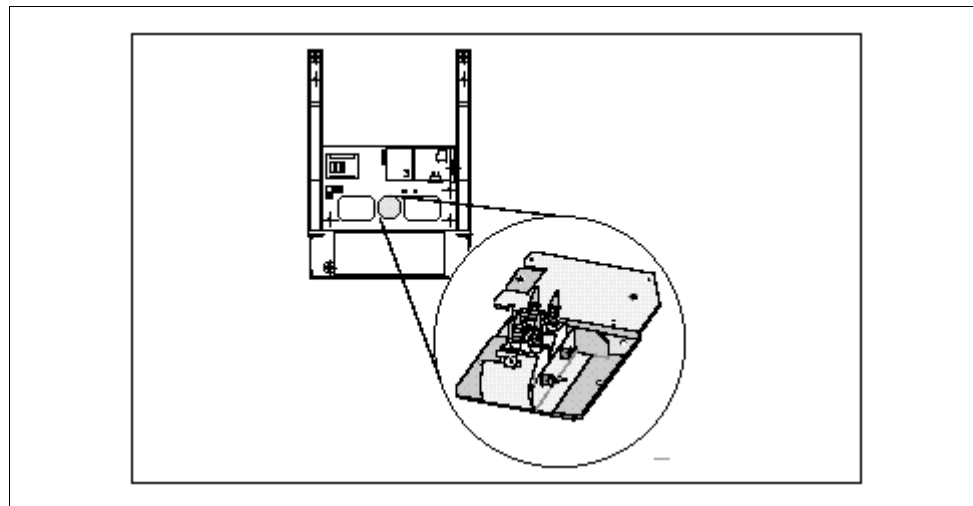
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9.3.2 Ergonomic Furniture Description, Continued

UCN tap location

UCN Trunk cable taps (2) can be mounted in the bottom of the footrest compartment.

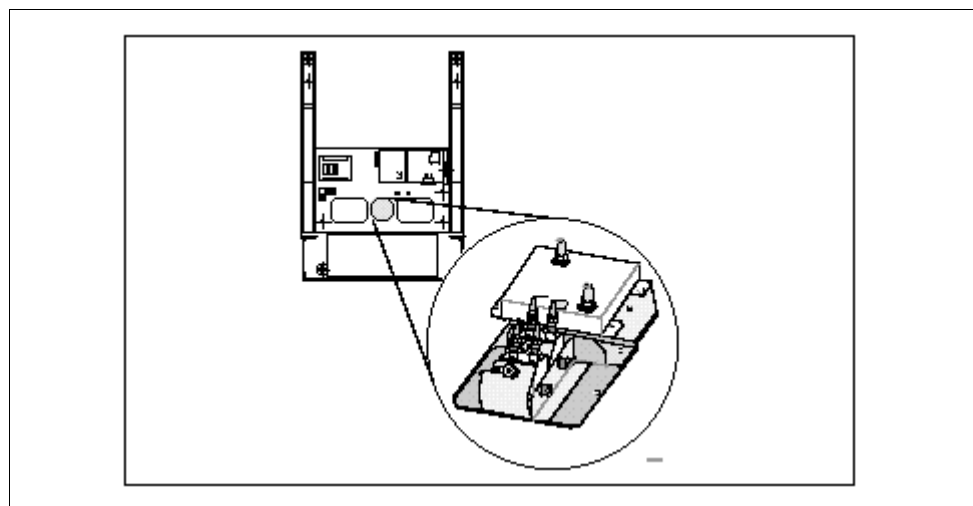
Figure 9-21 UCN Trunk Cable Mounting in New Furniture



MAU mounting

The Global User Station can have a LCN Media Adapter Unit (MAU) mounted in the bottom of the footrest compartment.

Figure 9-22 LCN MAU Mounting for Console GUS in New Furniture

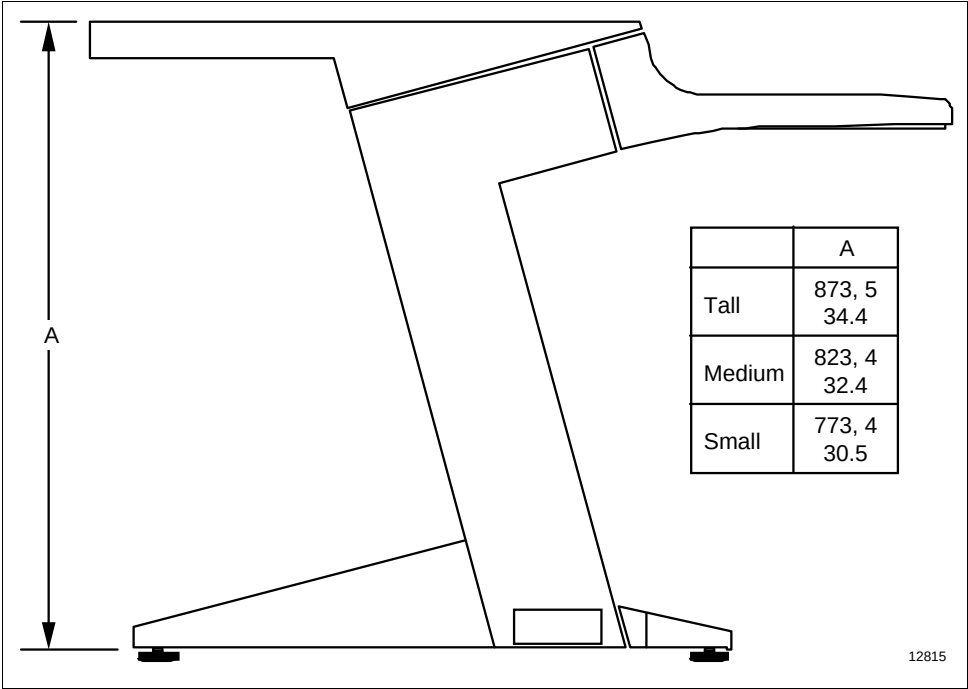


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9.3.2 Ergonomic Furniture Description, Continued

Work Surface side
view

Figure 9-23 Ergonomic Furniture Work Surface Side View

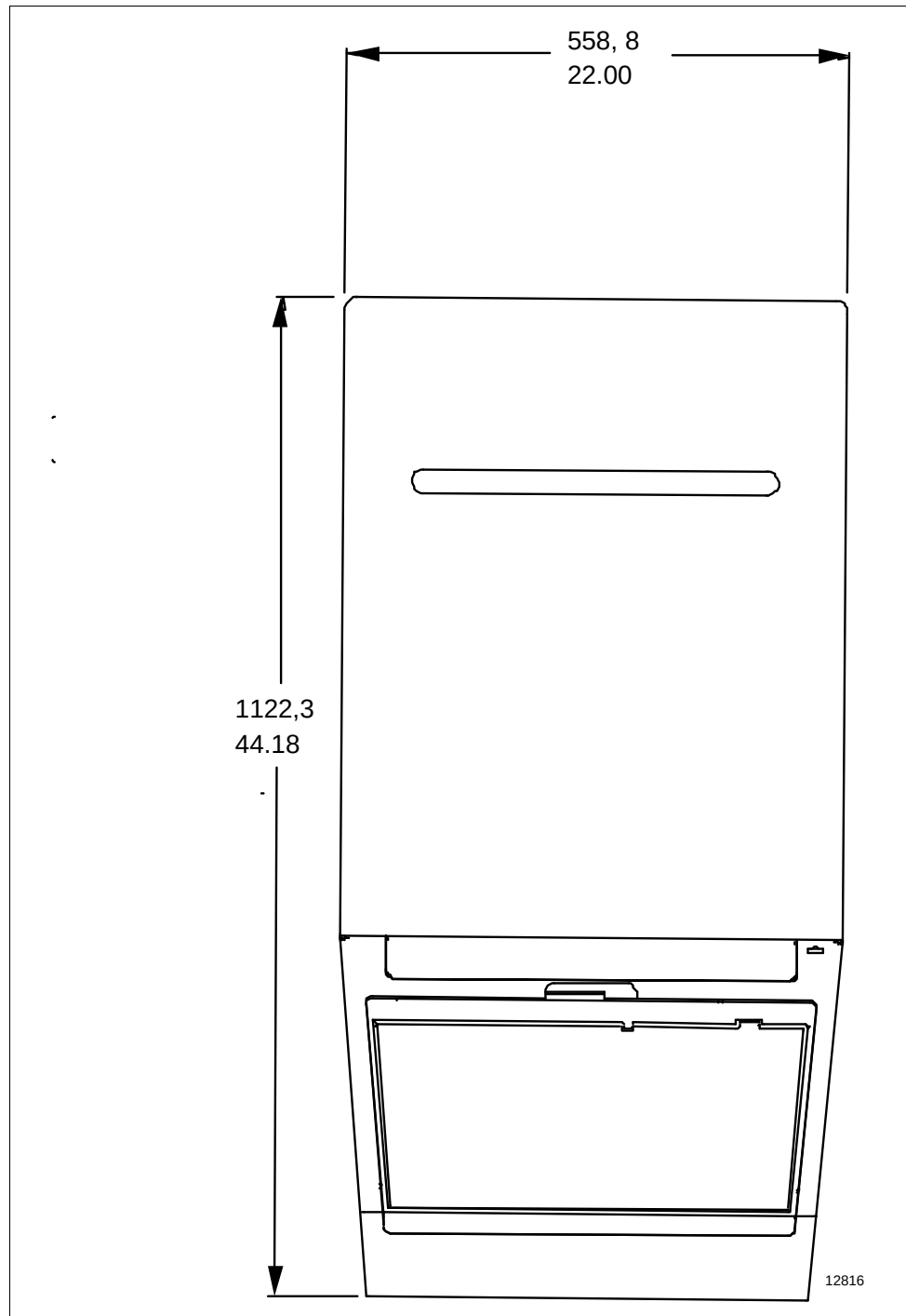


Continued on next page

9.3.2 Ergonomic Furniture Description, Continued

Work Surface top
view

Figure 9-24 Ergonomic Furniture Work Surface Top View

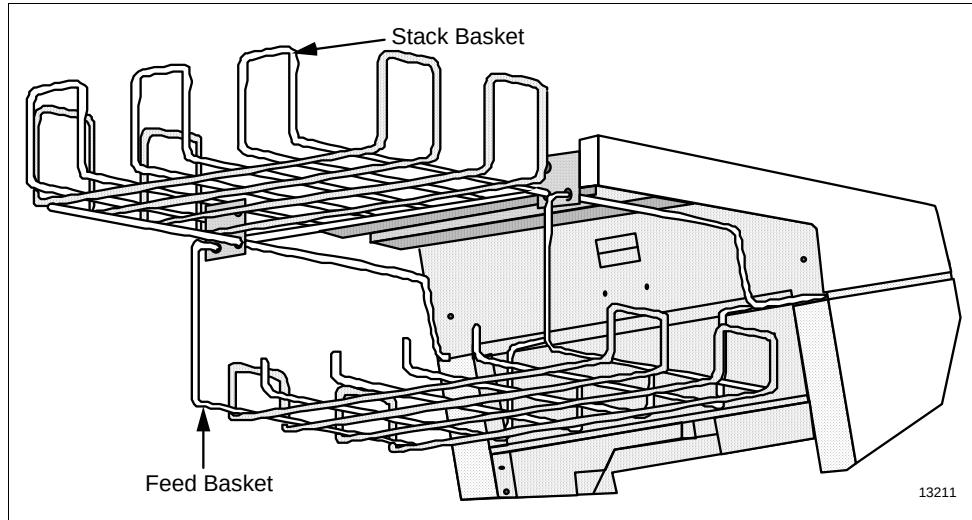


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9.3.2 Ergonomic Furniture Description, Continued

Printer paper basket accumulation kit

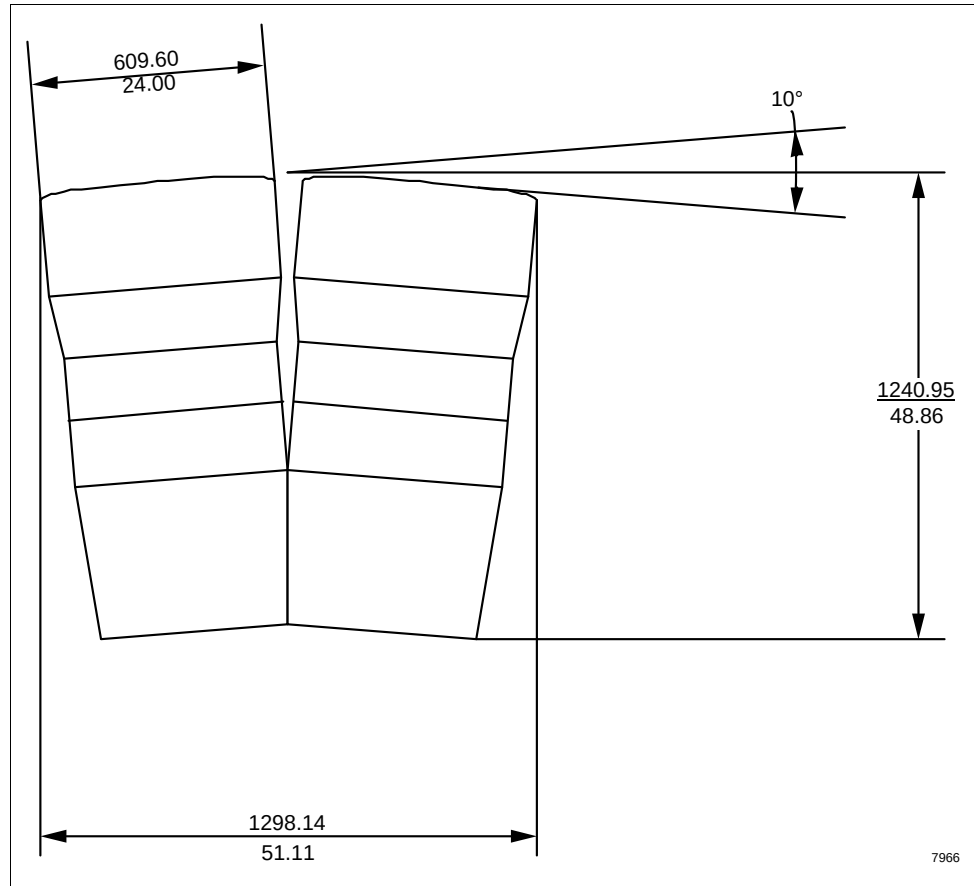
Figure 9-25 Printer Paper Basket Accumulation Kit



9.3.3 Ergonomic Furniture Template

Template

Figure 9-26 Ergonomic Furniture Template

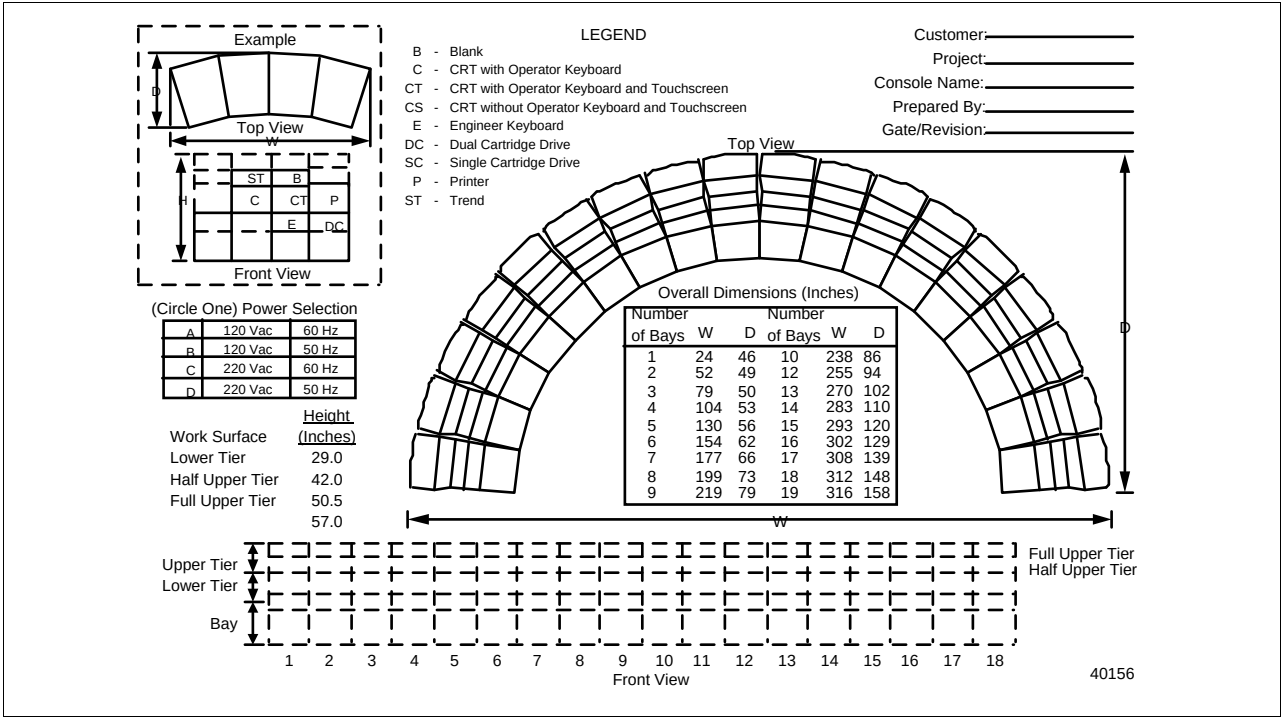


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9.3.3 Ergonomic Furniture Template, Continued

TPS console layout The console planning template shown here is available in printed pads of the template. Parts of the template are shown in the three figures that follow.

Figure 9-27 TPS Ergonomic Furniture Console Layout

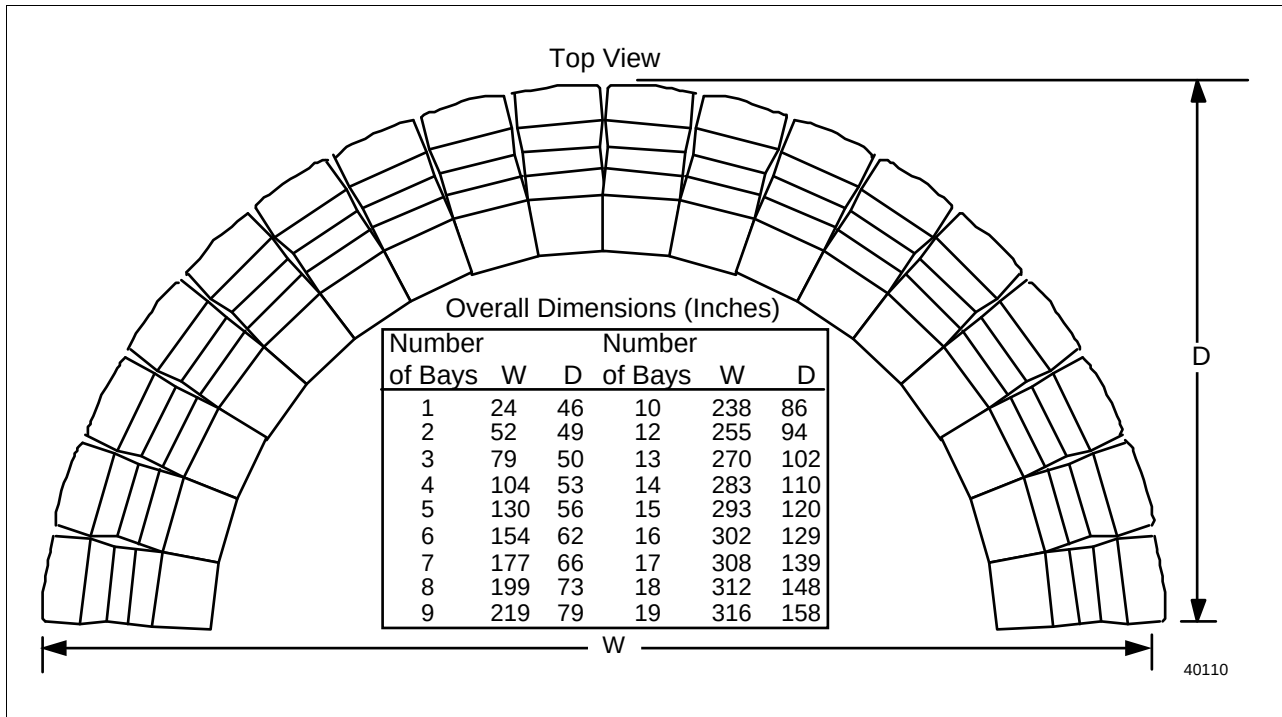


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9.3.3 Ergonomic Furniture Template, Continued

Console dimensions

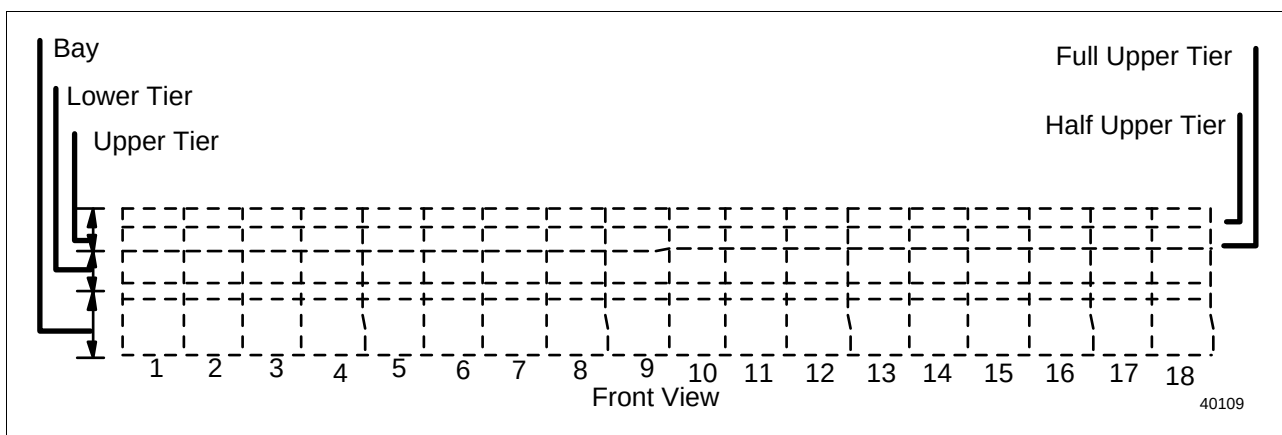
Figure 9-28 Console Dimensions



Template

This part of the template is used to note the location of peripherals to the console.

Figure 9-29 Peripheral Template

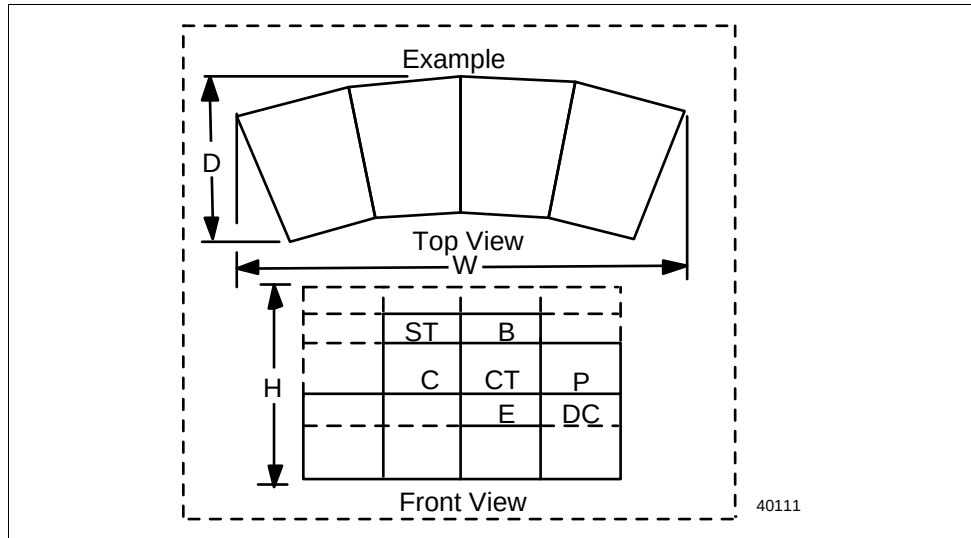


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9.3.3 Ergonomic Furniture Template, Continued

Key to notation

Figure 9-30 Key to Notation for Peripherals



Legend table

Table 9-2 Legend to Peripheral Notation—Ergonomic Furniture

Notation	Description
B	Blank Panel
C	CRT with Operator Keyboard
CD	CD ROM Drive
CT	CRT with Operator Keyboard and Touchscreen
CS	CRT without Operator Keyboard and Touchscreen
DC	Dual Cartridge Drive
E	Engineer Keyboard
P	Printer
SC	Single Cartridge Drive
ST	Trend Pen Recorder

9.4 Equipment Cabinet

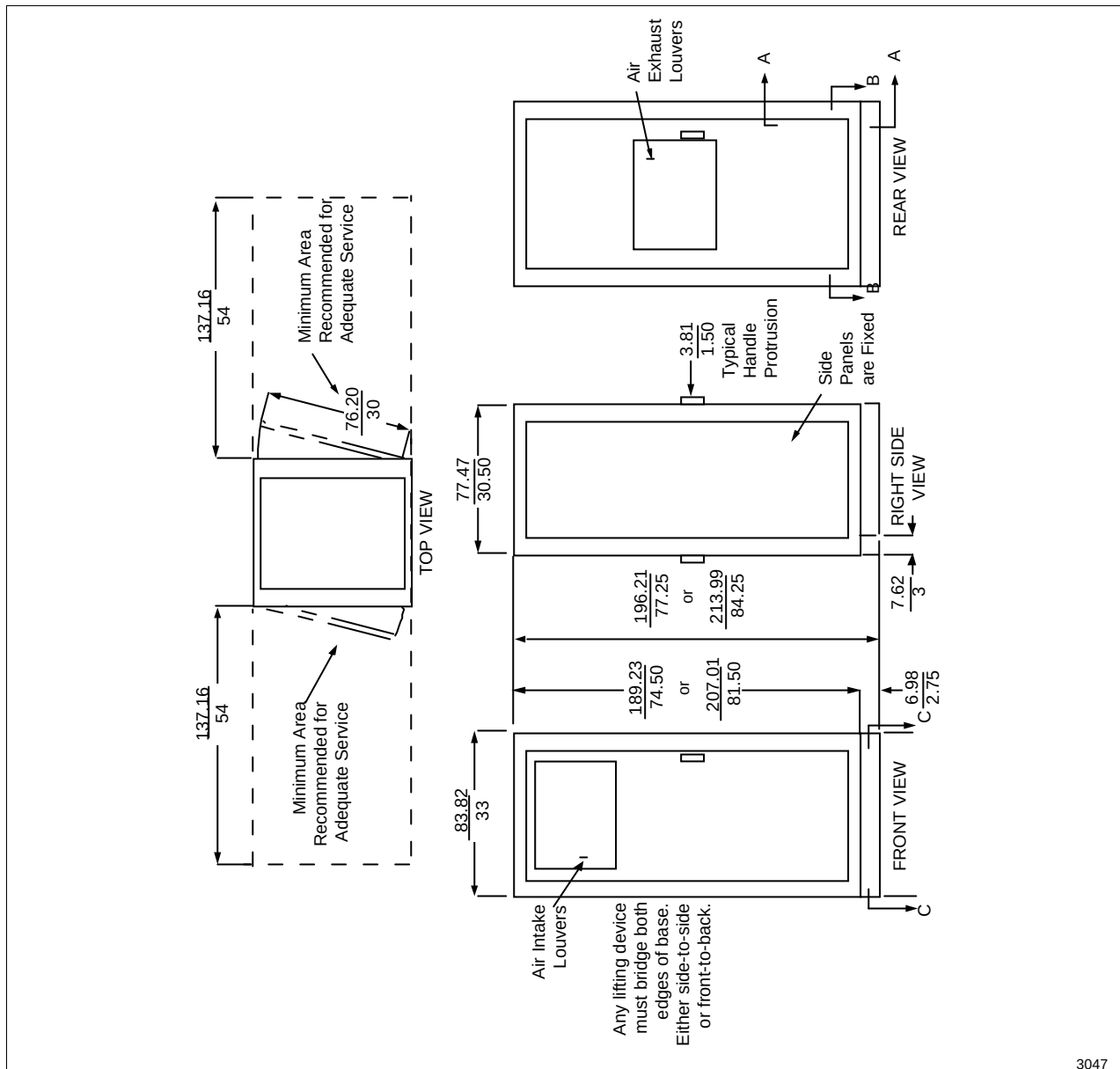
Overview

An LCN equipment cabinet is available to house the LCN modules. It will accommodate approximately 10 Five-Slot or Dual Node Module chassis. Figures 9-31 through 9-36 are related illustrations of the cabinet.

9.4.1 Equipment Cabinet Description

Cabinet dimensions

Figure 9-31 LCN Equipment Cabinet Dimensions



3047

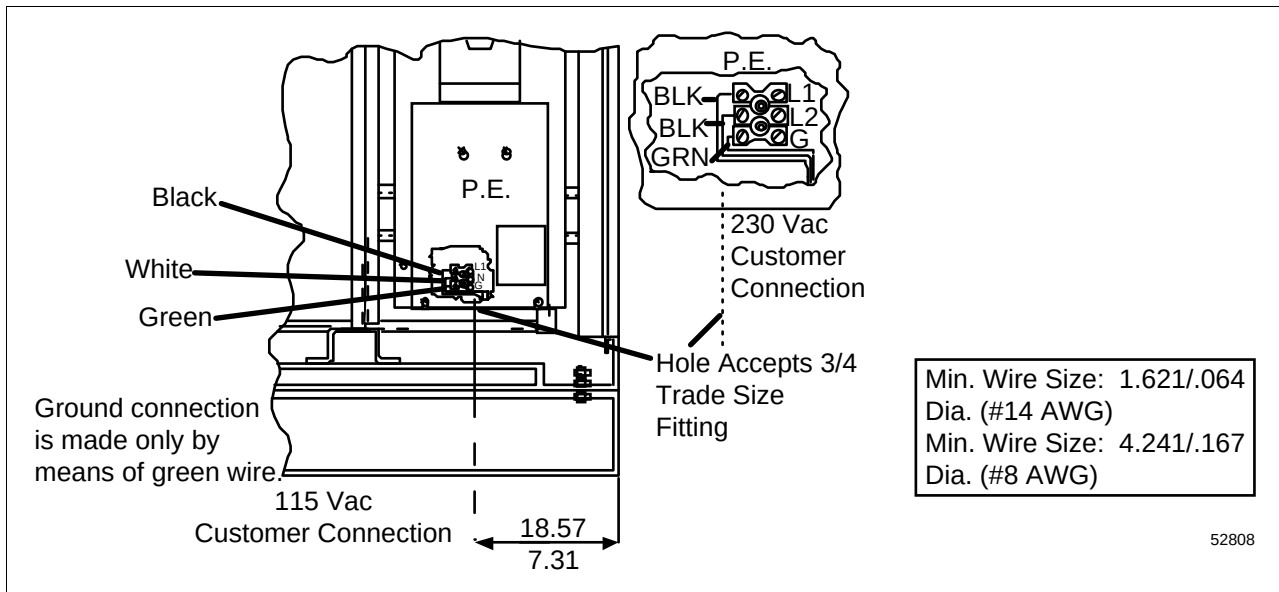
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9.4.1 Equipment Cabinet Description, Continued

AC power entry

The equipment cabinet is available with a second ac power entry distribution strip. Both are exactly alike and provide for distribution of ac power from an alternate power source. Redundantly paired modules are split between the two power entries.

Figure 9-32 LCN Equipment Cabinet AC Power Entry (US Compliant)

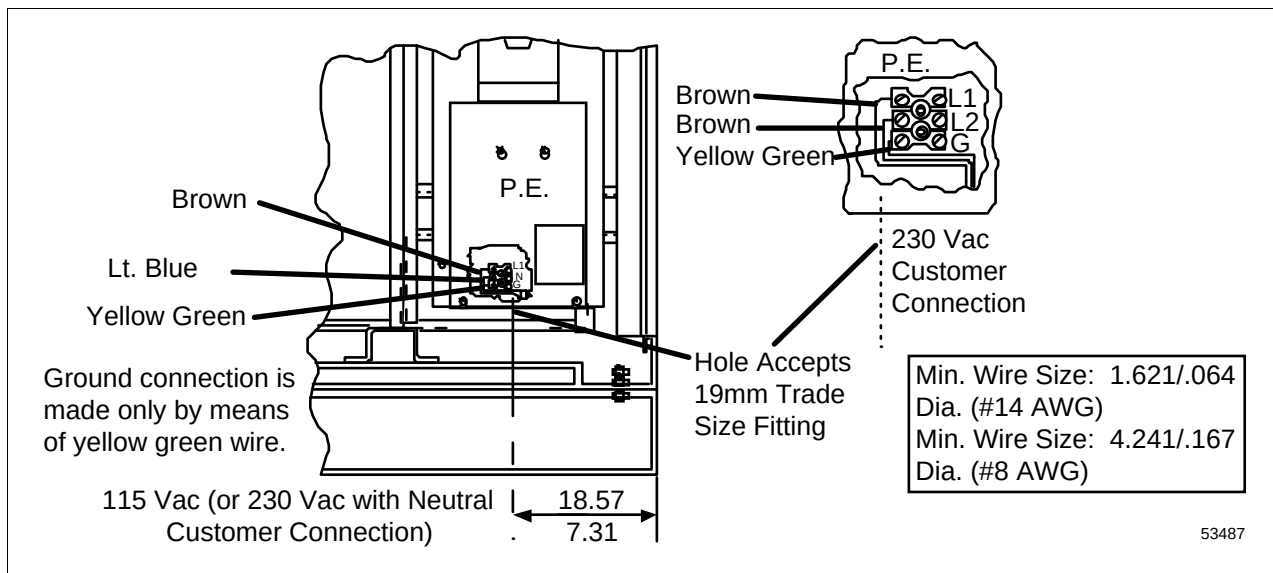


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9.4.1 Equipment Cabinet Description, Continued

AC power entry European ac power uses different wire color-coding as shown.

Figure 9-33 LCN Equipment Cabinet AC Power Entry (CE Compliant)

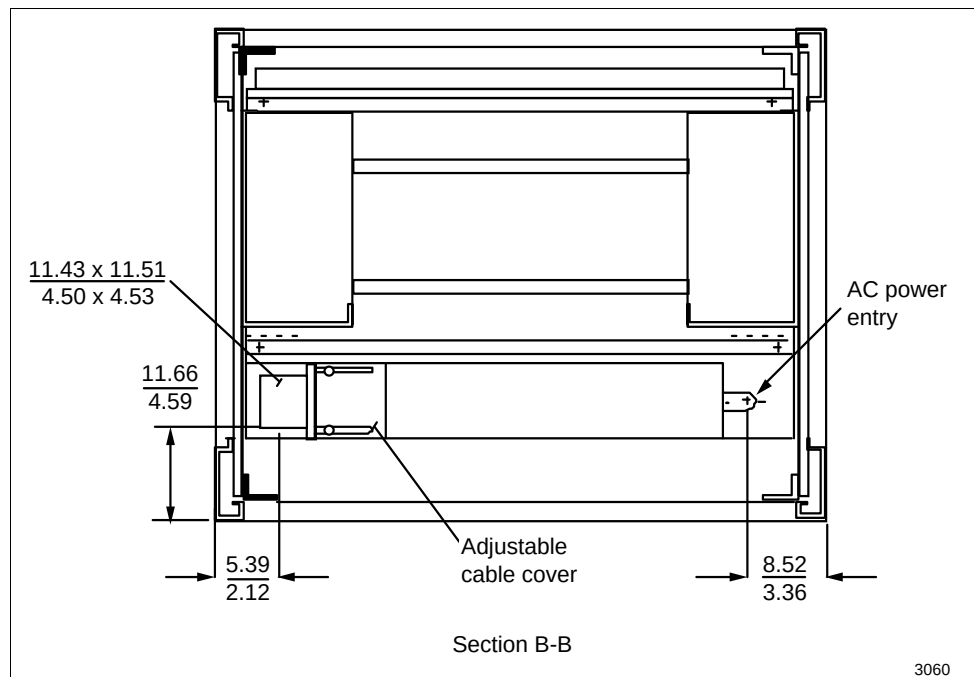


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9.4.1 Equipment Cabinet Description, Continued

AC power entry

Figure 9-34 LCN Equipment Cabinet AC Power Entry

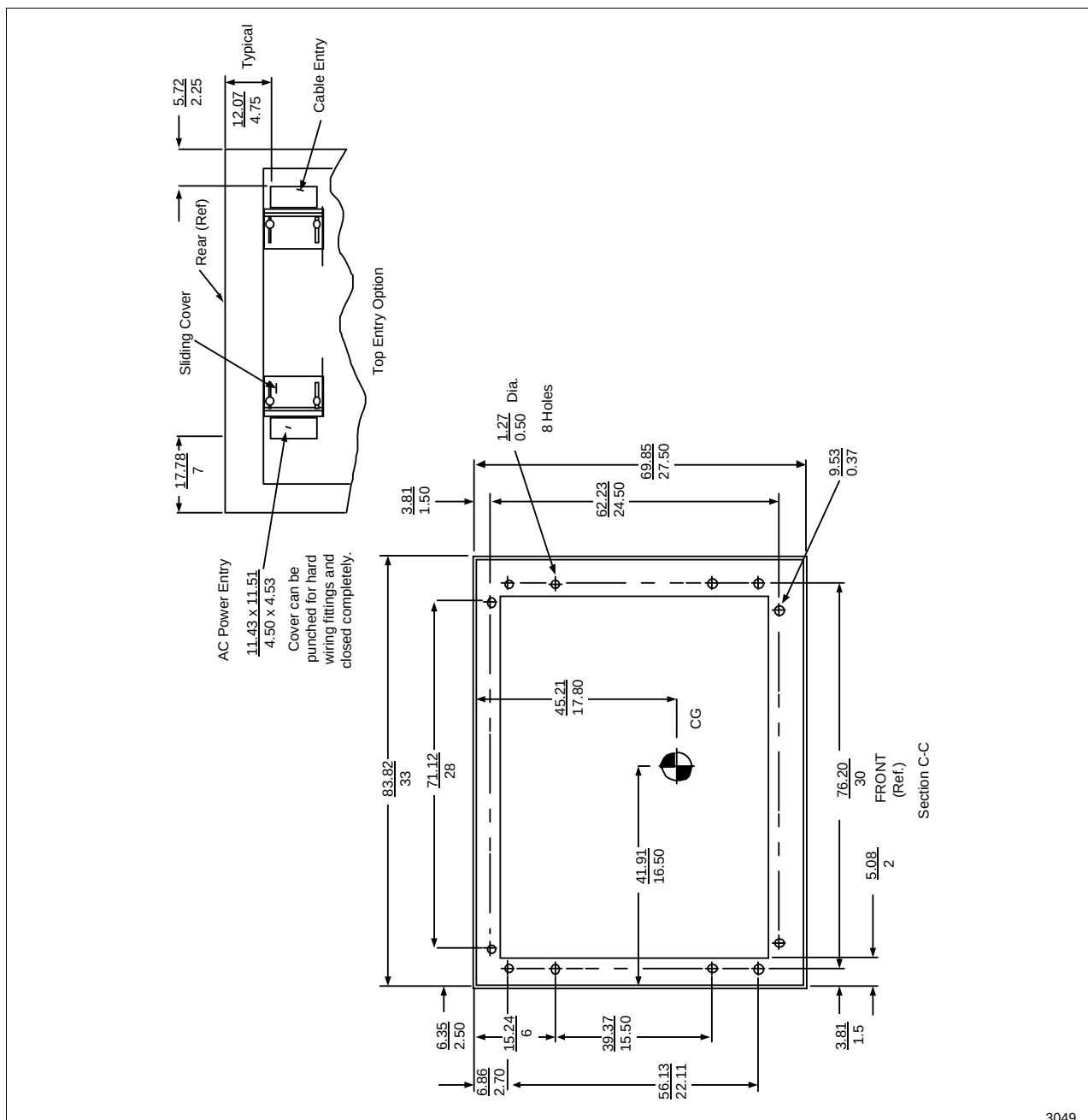


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9.4.1 Equipment Cabinet Description, Continued

Optional AC power entry

Figure 9-35 LCN Equipment Cabinet Optional AC Power Entry

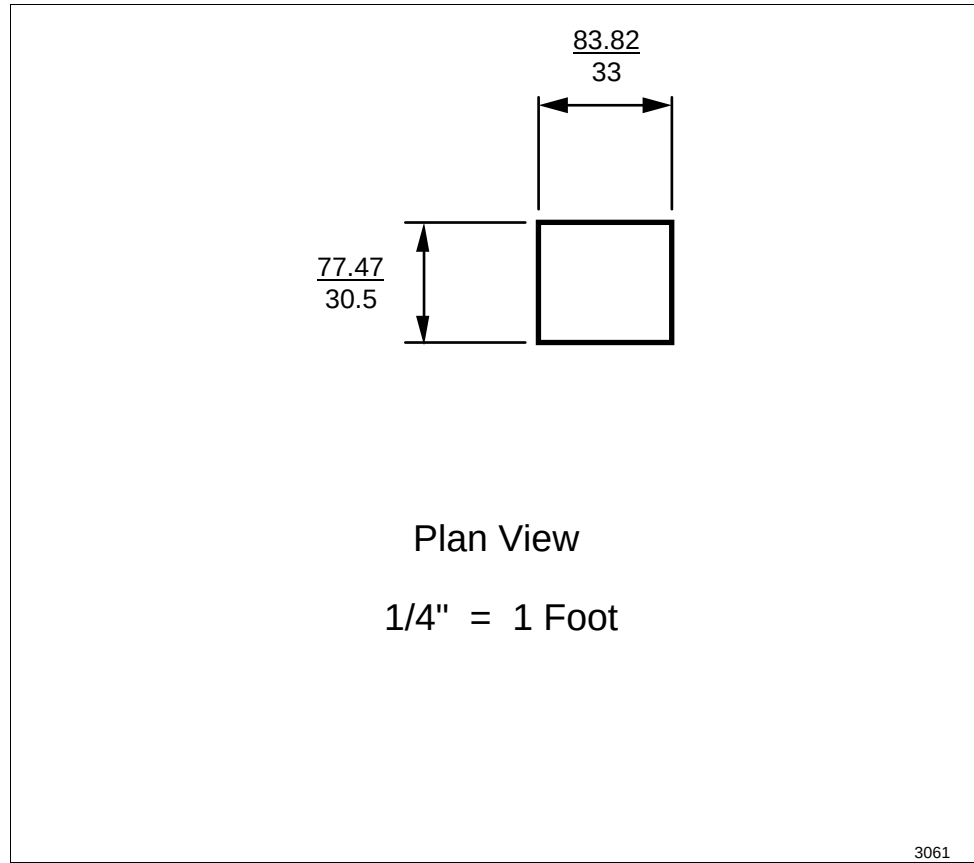


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9.4.2 Equipment Cabinet Template

Template (footprint)

Figure 9-36 LCN Equipment Cabinet Template (Footprint)



9.5 Global User Station PPC

Introduction

The Global User Station PPC is available in XC3000 console furniture, classic console furniture, and a deskside unit. Both console versions have the same physical attributes and the standard XC3000 and classic furniture.

The XC3000 and classic console furniture have the integrated keyboard (IKB) as standard equipment.

The deskside unit consists of

- MiniTower PPC
 - Monitor (Single or Dual)
 - Standard 17-inch
 - Optional 21-inch
 - Keyboard
 - Standard PC keyboard
 - Optional Integrated Keyboard (IKB)
-

9.5.1 XC3000 Global User Station Furniture

Scope

The XC3000 version is housed in the standard classic furniture. Refer to subsections 9.3 to 9.3.3 for measurements and templates.

9.5.2 Classic Global User Station Furniture

Scope

The classic version is housed in the standard XC3000 furniture. Refer to subsections 9.2 to 9.2.3 for measurements and templates.

9.5.3 Deskside Global User Station

Scope

The deskside MiniTower can be placed on or under a desk or work surface.

Dimensions

Table 9-3 Global User Station Deskside Dimensions

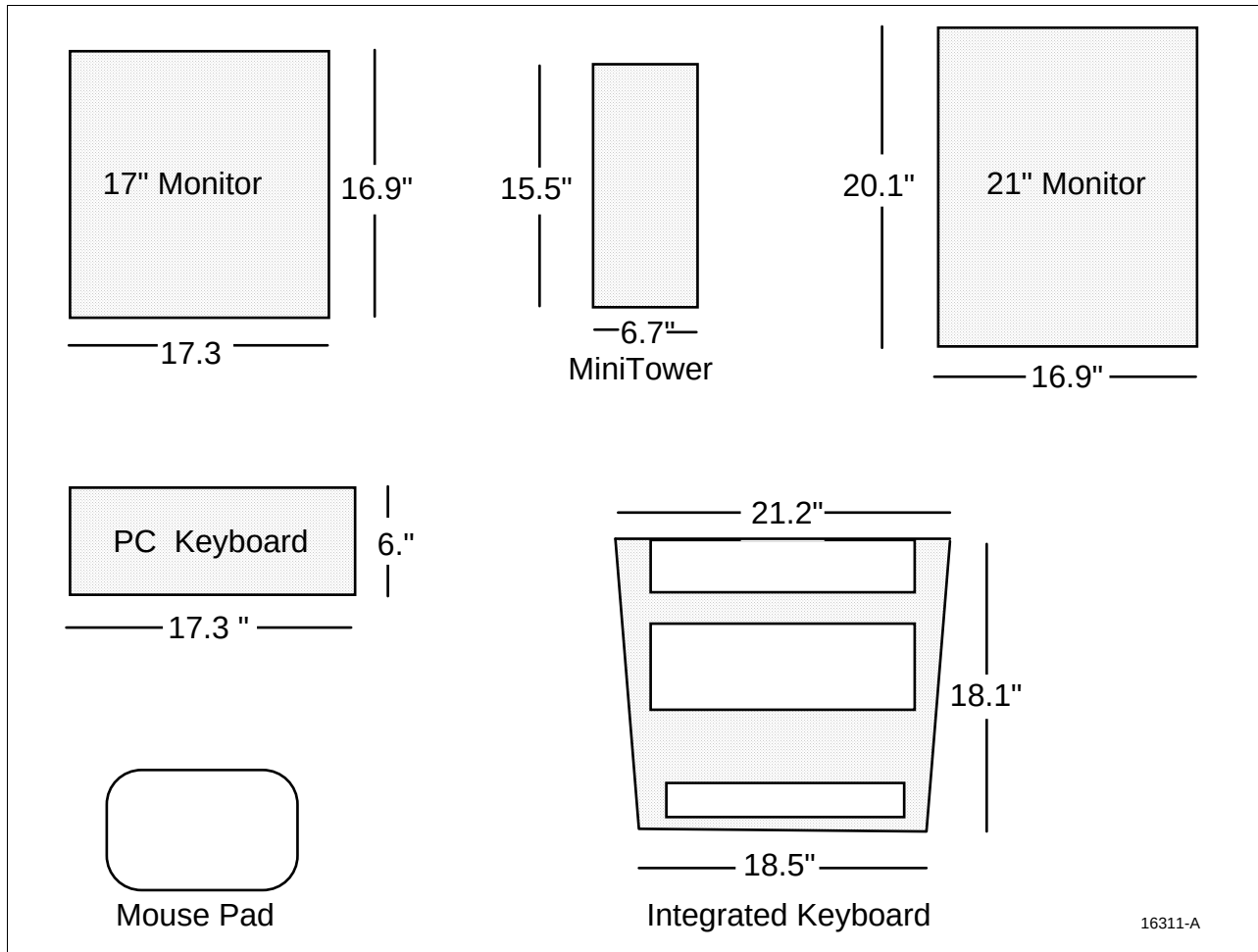
MiniTower Dimensions	Height: 404 mm (15.9 inches) Width: 170 mm (6.7 inches) Depth: 394 mm (15.5 inches) Typ. Weight: 7.7 kg (17 lb.) Input Power: 115 Vac/230 Vac selectable , 60/50 Hz
PC Keyboard	Width: 457 mm(18.0 inches) Depth: 165 mm (6.5 inches)
Desktop Integrated Keyboard(IKB)	Height: 88, 9 mm (3.5 inches) Max. Width: 539 mm (21.2 inches) Min Width: 469, 8 (18.5 inches) Depth: 459, 2 mm (18.08 inches)
17 Inch Monitor	H x W x D: 420 x 428 x 439 mm (16.5 x 16.9 x 17.3 inches) Weight 18 kg (39.7 lbs.) Input Power 90 to 264 Vac, 60 Hz/50 $\pm$ 3 Hz
21 Inch Monitor	H x W x D 470 x 500 x 510 mm (18.5 x 19.7 x 20.1 inches) Weight 34 kg (75 lb.) Input Power : 90 -127 Vac/180 - 254 Vac,47 - 63 Hz

9.6 Deskside Global User Station Templates

Model HI-D1000

The templates below are drawn on a scale of 1 inch to 1 foot. They can be used to plan the Global User Station Deskside unit installation.

Figure 9-37 Deskside Global User Station Templates



Section 10 – Hardware Specifications

10.1 Overview

Section contents

The topics covered in this section are:

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Introduction

The specifications in this section include environmental requirements, physical characteristics (size and weight), and electrical power requirements. The *TPS System Site Planning* manual contains additional information covering the measurement and use of the electrical values included in the following tables.

Implementation of different configurations

Many of the modules may be implemented with different configurations. For example, a Universal Station may be implemented in a 5-Slot Module chassis, in a 10-Slot Module chassis, or in a Dual Node Module chassis. In a Five-Slot or Ten-Slot Module chassis, either of two power supplies can be installed, the standard or the enhanced power supply. When installed in the Five-Slot Module chassis, the enhanced power supply eliminates the need for a separate power supply for floppy or cartridge drives. In cases where multiple implementation options exist, specifications are provided for each of the options when the specifications are significantly different.

Continued on next page

10.1 Overview, Continued

Measurement parameters

In the following tables, peak current was measured using a 1-ohm line resistance and 120 Vac RMS for the typical values, and a 1-ohm line resistance and 102 Vac RMS for the maximum values. Average RMS current and true RMS current were measured with 120 Vac RMS for the typical values, and with 102 Vac RMS for the maximum values.

Board current requirements

Current requirements in the tables assumes a typical board complement for each application. It is possible to derive hypothetical, worst-case ac current requirements by assuming that each power supply is loaded to its maximum output capability. These values are summarized in Table 10-17.

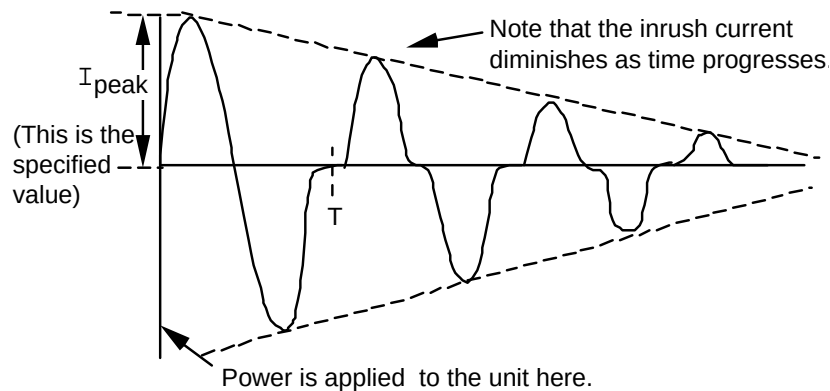
10.2 Definitions

AC voltage options

All voltages are in RMS with 6% or less harmonic distortion.

Inrush current

This is the peak current that occurs when power is first applied to a unit. It is measured in peak amperes as shown below.



6719

Power factor

This is the relationship of the actual power usage to the apparent power usage. In reality, a given unit may use only 100 watts of power when averaged over a period of time (the actual power), but if the RMS current and the RMS voltage are multiplied together (the apparent power), a different higher result is obtained. For this example (100 watts actual power), let us say that the RMS current is 1.28 A and the RMS voltage is 120 Vac. The apparent power would be $1.28 \text{ A} \times 120 \text{ Vac} = 153.6 \text{ VA}$. The power factor for this example would be $100 \text{ watts} / 153.6 \text{ VA} = .651$ or a power factor of 65.1%.

Average RMS current

As explained in the *TPS System Site Planning* manual, most current meters simply rectify and average the the ac current and have a scale that is calibrated to indicate the RMS value, assuming a sinusoidal input. The Average RMS values provided in the tables in this Section provide the customer with values that he would see displayed on such a meter.

Continued on next page

10.2 Definitions, Continued

True RMS current

The acronym RMS stands for Root Mean Square. The definition of Root Mean Square is the square root of the mean square of the variable values taken throughout one cycle. The formula for Root Mean Square current would be:

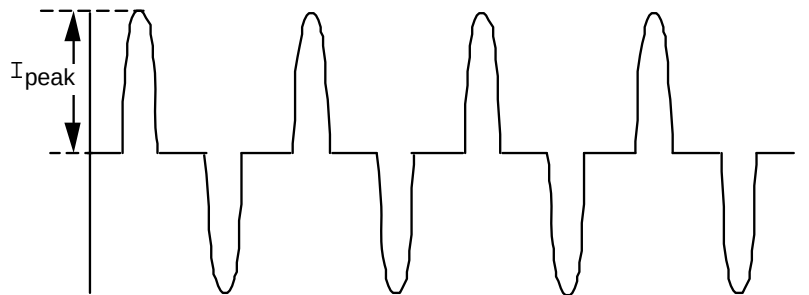
$$I_{\text{RMS}} = \sqrt{\frac{1}{T} \int_0^T i^2 dt}$$

T = the time period of one cycle

i = the instantaneous value of the current

Peak current

This is a measurement of the normal running current, but instead of an average or RMS value, the peak value is given. This is the highest value that the current reaches during a given half cycle. An illustration of the measurement of peak current is shown below. The waveform shown is typical for a load of the rectifier/capacitor-input type.



6720

Watts

This is the integral of the instantaneous voltage multiplied by the instantaneous current over a half cycle. It is the actual power usage referred to in the definition of power factor above.

BTU/hr

BTU/hr stands for British Thermal Units per hour. The British Thermal Unit is equivalent to the quantity of heat necessary to raise one pound of water one degree Fahrenheit (at 39.2°F). It provides a measurement of power converted to a form that is useful for sizing heating and cooling systems. The conversion from Watts to BTU/hr is:

$$\text{BTU/hr} = 3.41 \times \text{Watts}.$$

Continued on next page

10.2 Definitions, Continued

Typical values

The current values shown as typical (Typ.) in the tables in this appendix are values that you would measure from typical units running at 120 Vac RMS. The values given are, in fact, values measured from a sample of units.

Maximum value

The current values shown as maximum (Max.) in the tables were measured with a load approximately 30% higher than the load used for measurement of typical values to account for variations in components. Also, these measurements were taken with the minimum rated voltage, 102 Vac RMS, applied to the supply.

10.3 Specifications

Introduction

Specification for LCN components are listed in the following tables.

ATTENTION

ATTENTION—The weight figure for a Dual Node Module included in the tables which follow (18.2 kg or 40 lb) assumes a fully populated module (two nodes). The weight of a single node packaged in a Dual Node Module is 14.5 kg (32 lb).

10.3.1 Local Control Network Cabling Specifications

WARNING

WARNING—**TotalPlant** Solution (TPS) System LCN coaxial cables and associated components have been designed and tested by Honeywell IAC to meet our strict standards of performance. Other coaxial cables and components have not been tested and qualified for use in the LCN system. Therefore, no substitutions are approved by Honeywell IAC. Various lengths of coaxial cable are available as standard products and must be ordered from Honeywell IAC.

Minimum cable length	The minimum allowable LCN cable length between nodes is two meters. Use the cable pair, part number 51308111-002, which consists of two cables, one marked cable A and the other cable marked cable B. All LCN cables must be marked. It is essential that cable A is cable A to all nodes. Make certain that cables A and B are not crossed. Do not substitute unauthorized cables. These cables should not be daisy-chained together with T-connectors or with splices (barrel connectors) because this practice will add unacceptable impedance to the overall cable.
Indoor use only	The LCN cable physical and electrical characteristics are suitable for indoor use only. The LCN coaxial cable <u>must not</u> be routed outside the control room building.
Maximum number of nodes	The maximum allowable number of LCN nodes (Modules, Gateways, etc.) is 40 for an LCN coaxial bus segment up to 300 meters in length. Coaxial cable segments can be connected together with LCN Extenders so that up to 64 nodes, maximum, can be configured per system.
BNC connectors	Each LCN node must have two BNC T-connectors, part number 51190728-105. Note that these are not the same as those used for Data Hiway cables. The use of BNC splices (barrel connectors) or BNC T-connectors to join two short lengths of cable is not allowed because this introduces high impedance to the cable.
Terminator resistors	Terminator resistors, part number 30732052-001, are required at each end of both coaxial cables.
Cabinet cables	Coaxial cable assemblies (A & B), two meters (78.74 inches) in length, are used within cabinets and consoles to interconnect the LCN modules and gateways.

Continued on next page

10.3.1 Local Control Network Cabling Specifications, Continued

Factory cabling

The 2 meter interconnecting cables, along with the required T-connectors and terminators, are assembled and installed in each cabinet/console at the factory.

Available cable sets

System cables (LCN coaxial cable set), are available in standard and variable lengths. Each set consists of two cables, marked A and B. Table 10-1 lists the cable lengths that are available.

Table 10-1 LCN Cable Sets

Honeywell Part Number	Length (Meters)
51308112-003	3
51308112-008	8
51308112-015	15
51308112-030	30
51308112-050	50
51308112-100	100
51308112-200	200
51308112-xxx	Optional length – 300 meters maximum

Fiber optic cable

Fiber optic cable ordering information can be found in the *LCN Extenders and Fiber Optic Link Guidelines* manual.

10.3.2 US and UXS Environmental Specifications

Specifications

Environmental specifications for the Universal Station and Universal Station^X are listed in Table 10-2.

Table 10-2 Universal Station/Universal Station^X Environmental Specifications

Environmental Requirements				
Component	Temperature Range	Temperature Rate of Change	Relative Humidity	Environmental Class
Electronics Module Chassis, CRT, and Keyboard	0 to 50°C (32 to 122°F) Normal = 20°C	15°C (27°F) per hour	5 to 95%	A
Cartridge Disk Drive	10 to 46°C (50 to 115°F)	12°C (22°F) per hour	10 to 80%	A ₂
DAT Drive	5 to 40°C	<10°C per hour	15 to 95%	A
Floppy Disk Drive	10 to 46°C (50 to 115°F)	6°C (10.8°F) per hour	20 to 80%	A ₂
Floppy Diskettes	10 to 52°C (50 to 126°F)	N/A	20 to 80%	A ₂
Printer	10 to 38°C (50 to 100°F)	15°C (27°F) per hour	10 to 80%	A ₂
Printer Paper	16 to 24°C (61 to 75°F)	N/A	40 to 60%	A1
RULA (Dayna SCSI/Link-3)	10 to 40°C (50 to 104°F)		5 to 95%	A1
Trend Pen	10 to 50°C (50 to 122°F)	1°C (1.8°F) per min.	5 to 90%	A1
CAUTION				
This equipment is for use in a controlled environment. Although TPS will operate at 0-50°C (32-122°F), Honeywell recommends a normal environment of a nominal 25°C (77°F) with a relative humidity of 40-50% to realize maximum life and enhanced reliability.				
Physical Characteristics				
Approximate Dimensions		Approximate Weight		
Height	103 cm (40.5 in.)	295 kg (650 lb)		
Width	112 cm (44 in.)			
Depth	127 cm (50 in.)			
Applies to a 2-station console configuration of system cabinets and an in-cabinet, 10-card electronics module chassis, two keyboards, and two monitors, or one monitor with matrix printer and dual floppy disk drive. For Trend Pens (three Trend Pen recorders with power supplies) add 20 cm (8 in.) to the top of the monitor shroud and 24 kg (54 lb) to the weight of the console.				

10.3.3 US Electrical Specifications

5/10-Slot specifications The electrical specifications assume a power factor of approximately 0.65..
Table 10-3 Universal Station Electrical Specifications (Five/Ten-Slot)

Universal Station Electrical Specifications (5/10-Slot Module with Standard Power Supply or 10-Slot Module with Enhanced (Acme) Power Supply)						
AC Voltage Options		Frequency Options		Inrush Current		
120, 240 Vac +10%, -15% 220 Vac +10%, -10% (The 240 Vac option for the printer is +6%, -15%)		50 Hz or 60 Hz +3%, -6% (The printer is +0.5 Hz, -0.5 Hz)		Module Chassis	10 A @ 120 Vac 20 A @ 240 Vac	
				Peripheral	10 A @ 120 Vac 20 A @ 240 Vac	
				Monitor	15 A @ 120 Vac 30 A @ 240 Vac	
Component		Average RMS ¹	True RMS ¹	Peak ¹	Watts	BTUs/Hour
Cartridge Drive (Dual) ²	Typ.	0.66	1.47	3.92	107	365
	Max.	1.51	3.05	7.03	210	716
Monitor	Typ.	0.64	1.43	3.80	102	348
	Max.	0.91	1.98	4.85	128	437
Floppy Drive (Dual) ⁴	Typ.	0.26	0.62	1.75	41	140
	Max.	0.71	1.58	3.95	100	341
Keyboard ³	Typ.	0.17	0.46	1.28	28	96
	Max. ⁵	0.33	0.87	2.30	50	171
Printer (ASPI-32)	Typ.	0.15	0.30	0.70	36	123
	Max. ⁵	0.50	1.00	2.90	120	410
Printer (ASPI-46)	Typ.	1.10	1.75	9.10	210	716
	Max.	N/A	N/A	N/A	N/A	N/A
Trend Pen Recorder ⁴	Typ.	0.51	1.21	2.75	84	287
	Max. ⁶	0.73	1.64	4.12	100	341
Five-Slot Module ⁷	Typ.	1.32	2.82	7.52	210	716
	Max.	1.94	3.94	9.66	271	924
Ten-Slot Module ⁷	Typ.	2.24	4.66	12.0	371	1266
	Max.	3.30	6.40	15.9	482	1646
¹ Amperes. Divide the figures by 2 for 240 Vac operation or by 1.83 for 220 Vac operation. ² Uses Peripheral Power Supply 51107595. ³ Early versions of US use Keyboard Power Supply 51107593. Later versions get keyboard power from the module supply. ⁴ Uses Peripheral Power Supply 51108894. ⁵ Indicated maximum figures assumes that all power supplies are fully loaded. ⁶ 8 pen, 2-3 pen, 1-2 pen, 2 speed. ⁷ Includes an FDC or SIO board. Maximum figures measured with voltage at 102 Vac.						

Continued on next page

10.3.3 US Electrical Specifications, Continued

Five-Slot specifications

The electrical specifications assume a power factor of approximately 0.65.

Table 10-4 Universal Station Electrical Specifications (Five-Slot)

Universal Station Electrical Specifications (5-Slot Module with Enhanced (Acme) Power Supply)						
AC Voltage Options		Frequency Options		Inrush Current		
120, 240 Vac +10%, -15% 220 Vac +10%, -10% (The 240 Vac option for the printer is +6%, -15%)		50 Hz or 60 Hz +3%, -6% (The printer is +0.5 Hz, -0.5 Hz)		Module Chassis Vac	14 A @ 120	
					28 A @ 240	
				Vac Peripheral Vac	10 A @ 120	
					20 A @ 240	
				Vac Monitor Vac	15 A @ 120	
					30 A @ 240	
				Vac		
Component		Average RMS ¹	True RMS ¹	Peak ¹	Watts	BTUs/Hour
Cartridge Drive (Dual) ²	Typ.	0.39	0.95	2.60	68	232
	Max.	0.87	1.89	4.75	122	416
Monitor	Typ.	0.64	1.43	3.80	102	348
	Max.	0.91	1.98	4.85	128	437
Floppy Drive (Dual) ²	Typ.	0.44	1.03	2.70	69	235
	Max.	0.77	1.72	4.33	110	376
Printer (ASPI-32)	Typ.	0.15	0.30	0.70	36	123
	Max. ⁴	0.50	1.00	2.90	120	410
Printer (ASPI-46)	Typ.	1.10	1.75	9.10	210	716
	Max.	N/A	N/A	N/A	N/A	N/A
Trend Pen Recorder ³	Typ.	0.51	1.21	2.75	84	287
	Max. ⁵	0.73	1.64	4.12	100	341
Five-Slot Module ⁶	Typ.	0.98	2.17	6.10	156	532
	Max.	1.50	3.14	7.90	206	702
¹ Amperes. Divide the numbers by 2 for 240 Vac operation or by 1.83 for 220 Vac operation. ² Powered by module supply. This is the additional load on that power supply. ³ Uses Peripheral Power Supply 51108894. ⁴ Indicated maximum figures assume that all power supplies are fully loaded. ⁵ 8 pen, 2-3 pen, 1-2 pen, 2 speed. ⁶ Includes an FDC or SIO board and an Operator's and Engineer's keyboard. The maximum figures are measured with the voltage at 102 Vac.						

10.3.3 US Electrical Specifications, Continued

Dual Node specifications

The electrical specifications assume a power factor of approximately 0.65.

Table 10-5 Universal Station Electrical Specifications (Dual Node)

Universal Station Electrical Specifications (Dual Node Module Node)						
AC Voltage Options		Frequency Options		Inrush Current		
120, 240 Vac +10%, -15% 220 Vac +10%, -10% (The 240 Vac option for the printer is +6%, -15%)		50 Hz or 60 Hz +3%, -6% (The printer is +0.5 Hz, -0.5 Hz)		Module Chassis	10 A @ 120 Vac	
					20 A @ 240 Vac	
				Peripheral	10 A @ 120 Vac	
					20 A @ 240 Vac	
				Monitor	15 A @ 120 Vac	
					30 A @ 240 Vac	
Component		Average RMS ¹	True RMS ¹	Peak ¹	Watts	BTUs/Hour
Cartridge Drive ² (dual)	Typ.	0.66	1.48	3.92	107	365
	Max.	1.42	2.95	7.00	210	716
Monitor	Typ.	0.63	1.41	3.76	102	348
	Max.	0.91	1.98	4.85	128	437
Floppy Drive ^{3,8} (dual)	Typ.	0.25	0.62	1.76	41	140
	Max.	0.71	1.59	3.95	100	341
Printer (ASPI-32)	Typ.	0.15	0.30	0.70	36	123
	Max. ⁴	0.50	1.00	2.90	120	410
Printer (ASPI-46)	Typ.	1.10	1.75	9.10	210	716
	Max.	N/A	N/A	N/A	N/A	N/A
Trend Pen Recorder ^{3,5,8}	Typ.	0.51	1.21	3.26	84	287
	Max. ⁶	0.73	1.64	4.12	100	341
2 Board (Upper)Node ⁶	Typ.	0.53	1.40	4.05	98	336
	Max.	0.91	2.00	5.33	128	436
3 Board (Lower) Node ⁷	Typ.	0.79	1.83	5.10	132	451
	Max.	1.19	2.60	6.60	172	586

- 1 Amperes. Divide the numbers by 2 for 240 Vac operation or by 1.83 for 220 Vac operation.
- 2 Uses Peripheral Power Supply 51107595.
- 3 Uses Peripheral Power Supply 51108894.
- 4 Indicated maximum numbers assume that all power supplies are fully loaded.
- 5 8 pen, 2-3 pen, 1-2 pen, 2 speed.
- 6 Includes Operator's and Engineer's keyboards. Maximum figures measured with voltage at 102 Vac.
- 7 Includes an FDC or SIO board and Operator's and Engineer's keyboards. The maximum figures are measured with the voltage at 102 Vac.
- 8 Requires 3 board (lower) node.

10.3.4 UXS Electrical Specifications

5-Slot specifications (Classic furniture)

The electrical specifications assume a power factor of approximately 0.65.

Table 10-6 Universal Station^X Electrical Specifications (Classic Furniture – Five-Slot)

Universal Station ^X Electrical Specifications (Five-Slot Module with Enhanced (Acme) Power Supply)						
AC Voltage Options		Frequency Options		Inrush Current		
120, 240 Vac +10%, -15% 220 Vac +10%, -10% (The 240 Vac option for the printer is +6%, -15%) 90-132 Vac, 198-264 Vac Auto Select		50 Hz or 60 Hz +3%, -6% (The printer is +0.5 Hz, -0.5 Hz)		Module Chassis Vac	14 A @ 120	
					28 A @ 240	
				Peripheral Vac	10 A @ 120	
					20 A @ 240	
				Monitor Vac	15 A @ 120	
					30 A @ 240	
				Vac		
Component		Average RMS ¹	True RMS ¹	Peak ¹	Watts	BTUs/Hour
Cartridge Drive (Single) ²	Typ. Max.	0.18 0.45	0.31 0.76	0.73 1.34	30.92 71.23	105 242
Monitor	Typ. Max.				165	563
Hard Drive	Typ. Max.	0.10 0.28	0.18 0.47	0.41 0.83	17.62 44.23	60.07 150.83
DAT Drive	Typ. Max.	0.05 0.13	0.08 0.23	0.20 0.40	8.46 21.54	28.85 73.45
Printer (ASPI-32)	Typ. Max. ⁴	0.15 0.50	0.30 1.00	0.70 2.90	36 120	123 410
Printer (ASPI-46)	Typ. Max.	1.10 N/A	1.75 N/A	9.10 N/A	210 N/A	716 N/A
Trend Pen Recorder ³	Typ. Max. ⁵	0.51 0.73	1.21 1.64	2.75 4.12	84 100	287 341
Five-Slot Module ⁶	Typ. Max.	1.53 2.45	3.25 4.85	8.48 11.49	247.11 342.78	842 1168.86
¹ Amperes. Divide the figures by 2 for 240 Vac operation or by 1.83 for 220 Vac operation. ² Powered by the module power supply. This is the additional load on that power supply. ³ Uses Peripheral Power Supply 51108894. ⁴ Indicated maximum numbers assume that all power supplies are fully loaded. ⁵ 8 pen, 2-3 pen, 1-2 pen, 2 speed. ⁶ Includes an FDC or SIO board and an Operator's and Engineer's Keyboard. Maximum figures are measured with the voltage at 102 Vac.						

10.3.4 UXS Electrical Specifications, Continued

Dual Node specs (Ergonomic furniture)

The electrical specifications assume a power factor of approximately 0.65.

Table 10-7 Universal Station Electrical Specifications (Ergonomic Furniture – Dual Node)

Universal Station Electrical Specifications (Dual Node Module Node)						
AC Voltage Options		Frequency Options		Inrush Current		
120, 240 Vac +10%, -15% 220 Vac +10%, -10% (The 240 Vac option for the printer is +6%, -15%)		50 Hz or 60 Hz +3%, -6% (The printer is +0.5 Hz, -0.5 Hz)		Module Chassis	6 A @ 120 Vac 12 A @ 240 Vac	
				Peripheral	10 A @ 120 Vac 20 A @ 240 Vac	
				Monitor	15 A @ 120 Vac 30 A @ 240 Vac	
Component		Average RMS ¹	True RMS ¹	Peak ¹	Watts	BTUs/Hour
Cartridge Drive (Dual) ²	Typ.	0.66	1.48	3.92	107	365
	Max.	1.42	2.95	7.00	210	716
Monitor	Typ.	0.63	1.58	4.20	95	323
	Max.	1.20	2.40	6.60	160	545
Floppy Drive (Dual) ^{3,8}	Typ.	0.25	0.62	1.76	41	140
	Max.	0.71	1.59	3.95	100	341
Printer (ASPI-32)	Typ.	0.15	0.30	0.70	36	123
	Max. ⁴	0.50	1.00	2.90	120	410
Printer (ASPI-46)	Typ.	1.10	1.75	9.10	210	716
	Max.	N/A	N/A	N/A	N/A	N/A
Trend Pen Recorder ^{3,5,8}	Typ.	0.51	1.21	3.26	84	287
	Max.	0.73	1.64	4.12	100	341
2 Board (Upper) Node ⁶	Typ.	0.53	1.40	4.05	98	336
	Max.	0.91	2.00	5.33	128	436
3 Board (Lower) Node ⁷	Typ.	0.79	1.83	5.10	132	451
	Max.	1.19	2.60	6.60	172	586
¹ Amperes. Divide the numbers by 2 for 240 Vac operation or by 1.83 for 220 Vac operation. ² Uses Peripheral Power Supply 51107595. ³ Uses Peripheral Power Supply 51108894. ⁴ Indicated maximum numbers assume that all power supplies are fully loaded. ⁵ 8 pen, 2-3 pen, 1-2 pen, 2 speed. ⁶ Includes Operator's and Engineer's keyboards. Maximum figures measured with voltage at 102 Vac. ⁷ Includes an FDC or SIO board and Operator's and Engineer's Keyboards. The maximum figures are measured with the voltage at 102 Vac. ⁸ Requires 3 board (lower) node.						

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10.3.4 UXS Electrical Specifications, Continued

5-Slot specifications (Ergonomic furniture) The electrical specifications assume a power factor of approximately 0.65.

Table 10-8 Universal Station^X Electrical Specifications (Ergonomic Furniture – Five-Slot)

Universal Station ^X Electrical Specifications (Five-Slot Module with Enhanced (Acme) Power Supply)						
AC Voltage Options		Frequency Options		Inrush Current		
120, 240 Vac +10%, -15% 220 Vac +10%, -10% (The 240 Vac option for the printer is +6%, -15%) 90-132 Vac, 198-264 Vac Auto Select		50 Hz or 60 Hz +3%, -6% (The printer is +0.5 Hz, -0.5 Hz)		Module Chassis	14 A @ 120 Vac	
					28 A @ 240 Vac	
				Peripheral	10 A @ 120 Vac	
					20 A @ 240 Vac	
				Monitor	15 A @ 120 Vac	
					30 A @ 240 Vac	
Component		Average RMS ¹	True RMS ¹	Peak ¹	Watts	BTUs/Hour
Cartridge Drive (Single) ²	Typ.	0.18	0.31	0.73	30.92	105
	Max.	0.45	0.76	1.34	71.23	242
Monitor	Typ.	0.85	1.59	4.2	95	323
	Max.	1.20	2.40	6.6	160	545
Hard Drive	Typ.	0.10	0.18	0.41	17.62	60.07
	Max.	0.28	0.47	0.83	44.23	150.83
DAT Drive	Typ.	0.05	0.08	0.20	8.46	28.85
	Max.	0.13	0.23	0.40	21.54	73.45
Printer (ASPI-32)	Typ.	0.15	0.30	0.70	36	123
	Max. ⁴	0.50	1.00	2.90	120	410
Printer (ASPI-46)	Typ.	1.10	1.75	9.10	210	716
	Max.	N/A	N/A	N/A	N/A	N/A
Trend Pen Recorder ³	Typ.	0.51	1.21	2.75	84	
	Max. ⁵	0.73	1.64		100	
Five-Slot Module ⁶	Typ.	1.53	3.25	8.48	247.11	
	Max.	2.45	4.85		342.78	
1 Amperes. Divide the figures by 2 for 240 Vac operation or by 1.83 for 220 Vac operation. 2 Powered by the module power supply. This is the additional load on that power supply. 3 Uses Peripheral Power Supply 51108894. 4 Indicated maximum numbers assume that all power supplies are fully loaded. 5 8 pen, 2-3 pen, 1-2 pen, 2 speed. 6 Includes an FDC or SIO board and an Operator's and Engineer's Keyboard. Maximum figures are measured with the voltage at 102 Vac.						

10.3.5 Application Module Specifications

Specifications

The electrical specifications assume a power factor of approximately 0.65.

Table 10-9 Application Module Specifications

PHYSICAL CHARACTERISTICS						
Height Width Depth Weight	Ten-Slot Module		Five-Slot Module		Dual Node Module	
	31 cm (12.5 in.)		17.8 cm (7 in.)		17.8 cm (7 in.)	
	48 cm (19 in.)		48 cm (19 in.)		48 cm (19 in.)	
	61 cm (24 in.)		61 cm (24 in.)		61 cm (24 in.)	
	31.5 kg (69 lb)		31.5 kg (46 lb)		31.5 kg (40 lb)	
ELECTRICAL CHARACTERISTICS						
AC Voltage Options 120, 240 Vac +10%, -15% 220 Vac +10%, -10%			Frequency Options 50 of 60 Hz +3%, -6%			
Inrush Current						
Five-Slot Module (Standard Power Supply) 10 A @ 120 Vac 20 A @ 240 Vac		Five-Slot Module (Enhanced Power Supply) 14 A @ 120 Vac 28 A @ 240 Vac		Dual Node Module (Single Node) 6 A @ 120 Vac 12 A @ 240 Vac		
Component		Average RMS ¹	True RMS ¹	Peak ¹	Watts	BTUs/Hour
Five-Slot Node	Typ.	1.05	2.33	6.38	168	573
	Max. ⁴	1.60	3.31	8.20	219	747
Five-Slot Node (redundant)	Typ.	1.45	3.05	8.02	231	788
	Max. ⁴	2.91	5.62	13.20	411	1027
Dual Node Module ³	Typ.	0.40	1.04	3.12	67	228
	Max. ⁴	0.61	1.44	3.96	87	297
1 Amperes. Divide the figures by 2 for 240 Vac operation or by 1.83 for 220 Vac operation. 2 Amperage figures are valid for either the standard (51400712) or enhanced (51195066) power supply. Figures for redundant AMs are for one-half of a redundant pair. 3 The AM requires 2 boards and may be installed in a 2-slot node or in 3-slot node. 4 Maximum figures measured with the voltage at 102 Vac. The Application Module will operate properly during an interruption in the input ac voltage of up to 40 ms duration.						

10.3.6 Application Module^X Specifications

Specifications The electrical specifications assume a power factor of approximately 0.65.

Table 10-10 Application Module^X Specifications

PHYSICAL CHARACTERISTICS						
Height Width Depth Weight	Ten-Slot Module		Five-Slot Module			
	31 cm (12.5 in.)		17.8 cm (7 in.)			
	48 cm (19 in.)		48 cm (19 in.)			
	61 cm (24 in.)		61 cm (24 in.)			
	31.5 kg (69 lb)		31.5 kg (46 lb)			
ELECTRICAL CHARACTERISTICS						
AC Voltage Options			Frequency Options			
120, 240 Vac +10%, -15%			50 of 60 Hz +3%, -6%			
220 Vac +10%, -10%						
Inrush Current						
10-Slot Module (Enhanced Power Supply)		Five-Slot Module (Enhanced Power Supply)				
14 A @ 120 Vac		14 A @ 120 Vac				
28 A @ 240 Vac		28 A @ 240 Vac				
Component		Average RMS ¹	True RMS ¹	Peak ¹	Watts	BTUs/Hour
Five-Slot Node	Typ.	1.34	2.89	7.66	215	735
	Max. ³	2.36	4.69	11.2	329	1123
Ten-Slot Node	Typ.	1.82	3.87	10.6	295	1006
	Max. ³	2.97	5.73	14.5	420	1433
1 Amperes. Divide the figures by 2 for 240 Vac operation or by 1.83 for 220 Vac operation. 2 Amperage figures are valid for enhanced (51195066) power supply. 3 Maximum figures measured with the voltage at 102 Vac.						
The Application Module ^X will operate properly during an interruption in the input ac voltage of up to 40 ms duration.						

10.3.7 Gateway Module Specifications

Specifications

The electrical specifications assume a power factor of approximately 0.65.

Table 10-11 Gateway Module (PLCG, HG, NIM, CG, PLNM) Specifications

PHYSICAL CHARACTERISTICS						
Height Width Depth Weight	Five-Slot Module			Dual Node Module		
	17.8 cm (7 in.)			17.8 cm (7 in.)		
	48 cm (19 in.)			48 cm (19 in.)		
	61 cm (24 in.)			61 cm (24 in.)		
	21.5 kg (46 lb)			21.5 kg (40 lb)		
ELECTRICAL CHARACTERISTICS						
AC Voltage Options			Frequency Options			
120, 240 Vac +10%, -15% 220 Vac +10%, -10%			50 of 60 Hz +3%, -6%			
Inrush Current						
Five-Slot Module (Standard Power Supply)	Five-Slot Module (Enhanced Power Supply)		Dual Node Module (Single Node)			
10 A @ 120 Vac 20 A @ 240 Vac	14 A @ 120 Vac 28 A @ 240 Vac		6 A @ 120 Vac 12 A @ 240 Vac			
Component		Average RMS ¹	True RMS ¹	Peak ¹	Watts	BTUs/Hour
Five-Slot Node	Typ.	1.23	2.67	7.16	197	672
	Max. ⁴	1.84	3.75	9.20	256	873
Dual Node Module ³	Typ.	0.63	1.50	4.30	105	358
	Max. ⁴	0.95	2.12	5.48	136	465
1 Amperes. Divide the figures by 2 for 240 Vac operation or by 1.83 for 220 Vac operation. 2 Amperage figures are valid for either the standard (51400712) or enhanced (51195066) power supply. 3 The HG and PLNM modules must be installed in the 3-slot (lower) node. 4 Maximum figures measured with the voltage at 102 Vac. All Gateway Modules will operate properly during an interruption in the input ac voltage of up to 40 ms duration. The physical, electrical, and environmental specifications for the Gateways also apply to the Computer Interface Unit (CIU) portion of the Computing Module 60.						

10.3.8 Network Gateway Specifications

Specifications

The electrical specifications assume a power factor of approximately 0.65.

Table 10-12 Network Gateway Module Specifications

PHYSICAL CHARACTERISTICS						
Height Width Depth Weight	Five-Slot Module			Dual Node Module		
	17.8 cm (7 in.)			17.8 cm (7 in.)		
	48 cm (19 in.)			48 cm (19 in.)		
	61 cm (24 in.)			61 cm (24 in.)		
	21.5 kg (46 lb)			21.5 kg (40 lb)		
ELECTRICAL CHARACTERISTICS						
AC Voltage Options			Frequency Options			
120, 240 Vac +10%, -15% 220 Vac +10%, -10%			50 or 60 Hz +3%, -6%			
Inrush Current						
Five-Slot Module (Standard Power Supply)		Five-Slot Module (Enhanced Power Supply)		Dual Node Module (Single Node)		
10 A @ 120 Vac 20 A @ 240 Vac		14 A @ 120 Vac 28 A @ 240 Vac		6 A @ 120 Vac 12 A @ 240 Vac		
Component		Average RMS ¹	True RMS ¹	Peak ¹	Watts	BTUs/Hour
Five-Slot Node (single cable)	Typ.	0.87	2.21	5.91	152	518
	Max. ³	1.48	3.11	7.85	203	692
Dual Node Module ³ (single cable)	Typ.	0.77	1.72	4.70	108	368
	Max. ³	1.00	2.23	5.86	144	491
Five-Slot Node (redundant cable)	Typ.	1.21	2.61	7.01	193	658
	Max. ³	1.85	3.76	9.20	257	876
Dual Node Module ³ (redundant cable)	Typ.	0.90	2.04	5.55	150	512
	Max. ³	1.38	3.00	7.56	200	682
1 Amperes. Divide the figures by 2 for 240 Vac operation or by 1.83 for 220 Vac operation.						
2 Amperage figures are valid for either the standard (51400712) or enhanced (51195066) power supply.						
3 Maximum figures measured with the voltage at 102 Vac.						

10.3.9 History Module (WREN) Specifications

WREN History Module specifications The electrical specifications assume a power factor of approximately 0.65.

Table 10-13 History Module (WREN) Specifications

PHYSICAL CHARACTERISTICS						
Height Width Depth Weight	Winchester Drive Module (Dual)	Five-Slot Module		Dual Node Module		
	22.2 cm (8.75 in.)	17.8 cm (7 in.)		17.8 cm (7 in.)		
	48 cm (19 in.)	48 cm (19 in.)		48 cm (19 in.)		
	41 cm (16 in.)	61 cm (24 in.)		61 cm (24 in.)		
	20 kg (43 lb)	31.5 kg (46 lb)		18.2 kg (40 lb)		
ELECTRICAL CHARACTERISTICS						
AC Voltage Options 120, 240 Vac +10%, -15% 220 Vac +10%, -10%			Frequency Options 50 of 60 Hz +3%, -6%			
Inrush Current						
Five-Slot Module (Standard Power Supply) 10 A @ 120 Vac 20 A @ 240 Vac		Five-Slot Module (Enhanced Power Supply) 14 A @ 120 Vac 28 A @ 240 Vac		Dual Node Module (Single Node) 6 A @ 120 Vac 12 A @ 240 Vac		
Component		Average RMS ¹	True RMS ¹	Peak ¹	Watts	BTUs/Hour
Five-Slot Node	Typ.	1.18	2.56	6.85	189	645
	Max. ⁴	1.78	3.63	9.02	246	839
Dual Node Module ³	Typ.	0.70	1.58	4.30	98	336
	Max. ⁴	0.90	1.99	5.32	128	436
Winchester Drive PS (dual drive)	Typ.	0.77	1.75	4.57	127	451
	Max.	1.51	3.05	7.03	210	614
Winchester Drive PS (single drive)	Typ.	0.44	1.04	2.77	72	246
	Max.	0.82	1.79	4.50	115	392
1 Amperes. Divide the numbers by 2 for 240 Vac operation or by 1.83 for 220 Vac operation. 2 Amperage figures are valid for either a standard (51400712) or enhanced (51195066) power supply. 3 Requires 2 boards. They can be installed in either the 2-slot (upper) node or 3-slot (lower) node. 4 Maximum figures are measured with the voltage at 102 Vac.						

10.3.10 History Module (WDA) Specifications

WDA History Module specifications The electrical specifications assume a power factor of approximately 0.65.

Table 10-14 History Module (WDA) Specifications

PHYSICAL CHARACTERISTICS						
Height Width Depth Weight	Winchester Drive Module (Dual)	Description				
	22.2 cm (8.75 in.)	The Winchester Disk Assembly (WDA), which is installed in the top two slots of the Five-Slot Module chassis, contains the Winchester drives that are mounted in trays. An enhanced power supply provides power for the boards and drive(s).				
	48 cm (19 in.)					
	41 cm (16 in.)					
	20 kg (43 lb)					
ELECTRICAL CHARACTERISTICS						
AC Voltage Options			Frequency Options			
120, 240 Vac +10%, -15%			50 or 60 Hz +3%, -6%			
220 Vac +10%, -10%						
Inrush Current						
		Five-Slot Module (Enhanced Power Supply)				
		14 A @ 120 Vac 28 A @ 240 Vac				
Component		Average RMS ¹	True RMS ¹	Peak ¹	Watts	BTUs/Hour
Five-Slot Node (single drive)	Typ.	0.86	1.94	5.45	136.46	465.34
	Max. ⁴	1.43	3.02	7.66	195.94	668.16
Five-Slot Node (redundant single drives)	Typ.	0.92	2.07	5.81	147.16	501.81
	Max. ⁴	1.82	3.70	9.20	253.19	863.37
Five-Slot Node (dual drives)	Typ.	0.92	2.07	5.81	147.16	501.81
	Max. ⁴	1.82	3.70	9.20	253.19	863.37
Five-Slot Node (redundant dual drives)	Typ.	1.03	2.25	6.22	162.32	553.51
	Max. ⁴	2.55	5.25	12.38	376.14	1282.64
1 Amperes. Divide the numbers by 2 for 240 Vac operation or by 1.83 for 220 Vac operation. 2 Amperage figures are for the enhanced (51195066) power supply. 3 Requires 2 boards. They be installed in slots 1 and 2. The WDA is installed in slots 4 and 5. 4 Maximum figures are measured with the voltage at 102 Vac.						

10.3.11 Universal Work Station Specifications

Standard PS

The electrical specifications assume a power factor of approximately 0.65.

Table 10-15 Universal Work Station Specifications (Standard Power Supply)

PHYSICAL CHARACTERISTICS						
Height Width Depth Weight	Electronics Tower		Description			
	72.24 cm (28.44 in.) 18.4 cm (7.23 in.) 57.15 cm (22.5 in.) 34.02 kg (75 lb)		The electronics and disk drive(s) for the Universal Work Station reside in an electronics tower. The circuit boards are installed in a Five-Slot Module chassis with a standard power supply that provides power for the boards only. The drive(s) is powered by a peripheral power supply.			
ELECTRICAL CHARACTERISTICS						
AC Voltage Options 120, 240 Vac +10%, -15% 220 Vac +10%, -10%			Frequency Options 50 of 60 Hz +3%, -6%			
Inrush Current						
Five-Slot Module (Standard Power Supply) 10 A @ 120 Vac 20 A @ 240 Vac			Monitor 15 A @ 120 Vac 30 A @ 240 Vac			
Component		Average RMS ¹	True RMS ¹	Peak ¹	Watts	BTUs/Hour
Cartridge Disk ² (dual)	Typ. Max.	0.66 1.51	1.47 3.05	3.92 7.03	107 210	365 716
Monitor	Typ. Max.	0.64 0.91	1.43 1.98	3.80 4.85	102 128	348 437
Floppy Disk Drive ² (dual) ²	Typ. Max.	0.26 0.71	0.62 1.58	1.75 3.95	41 100	140 341
Printer (ASPI-32)	Typ. Max. ³	0.15 0.50	0.30 1.00	0.70 2.90	36 120	123 410
Printer (ASPI-46)	Typ. Max.	1.10 N/A	1.75 N/A	9.10 N/A	210 N/A	716 N/A
Five-Slot Module ⁴	Typ. Max.	1.32 1.94	2.82 3.94	7.52 9.66	210 271	717 925
1 Amperes. Divide the numbers by 2 for 240 Vac operation or by 1.83 for 220 Vac operation. 2 Uses a peripheral power supply (51107595). 3 The indicated maximum figures assumes that all power supplies are fully loaded. 4 Includes an FDC or SIO board and Engineer's and Operator's keyboards. The maximum figures are measured with voltage at 102 Vac.						

10.3.11 Universal Work Station Specifications, Continued

Enhanced PS

The electrical specifications assume a power factor of approximately 0.65.

Table 10-16 Universal Work Station Specifications (Enhanced Power Supply)

PHYSICAL CHARACTERISTICS						
Height Width Depth Weight	Electronics Tower		Description			
	72.24 cm (28.44 in.) 18.4 cm (7.23 in.) 57.15 cm (22.5 in.) 34.02 kg (75 lb)		The electronics and disk drive(s) for the Universal Work Station reside in an electronics tower. The circuit boards are installed in a Five-Slot Module chassis with an enhanced power supply that provides power for both the boards and drive(s).			
ELECTRICAL CHARACTERISTICS						
AC Voltage Options			Frequency Options			
120, 240 Vac +10%, -15% 220 Vac +10%, -10%			50 or 60 Hz +3%, -6%			
Inrush Current						
Five-Slot Module (Enhanced Power Supply)			Monitor			
14 A @ 120 Vac 28 A @ 240 Vac			15 A @ 120 Vac 30 A @ 240 Vac			
Component		Average RMS ¹	True RMS ¹	Peak ¹	Watts	BTUs/Hour
Cartridge Disk ² (dual)	Typ.	0.39	0.95	2.60	68	232
	Max.	0.87	1.89	4.75	122	416
Monitor	Typ.	0.64	1.43	3.80	102	348
	Max.	0.91	1.98	4.85	128	437
Floppy Disk Drive ² (dual) ²	Typ.	0.44	1.03	2.70	69	235
	Max.	0.77	1.72	4.33	110	376
Printer (ASPI-32)	Typ.	0.15	0.30	0.70	36	123
	Max. ⁴	0.50	1.00	2.90	120	410
Printer (ASPI-46)	Typ.	1.10	1.75	9.10	210	716
	Max.	N/A	N/A	N/A	N/A	N/A
Five-Slot Module ³	Typ.	0.98	2.17	6.10	156	532
	Max.	1.50	3.14	7.90	206	702
1 Amperes. Divide the numbers by 2 for 240 Vac operation or by 1.83 for 220 Vac operation.						
2 Powered by the module's enhanced power supply (51195066).						
3 Includes an FDC or SIO board and Engineer's and Operator's keyboards. The maximum figures are measured with voltage at 102 Vac.						
4 The indicated maximum figures assumes that the power supply is fully loaded.						

10.3.12 Module Worst Case Power Usage

Module worst case power usage

The amperage figures that are listed in Table 10-16 assume that the module's power supply is loaded to its maximum rated output current capacity with the input voltage at 102 Vac RMS. The power factor is approximately 0.65.

Table 10-17 Module Worst Case Power Usage

Module	Avg RMS ¹	True RMS ¹	Peak ¹	Watts	BTUs/hr
Five-Slot Module with Standard Power Supply (51400712)	2.91	5.62	13.2	411	1403
Five-Slot Module with Enhanced Power Supply (51195066)	3.47	6.35	15.1	492	1679
Ten-Slot Module with Standard Power Supply (51400712)	4.51	8.46	20.2	660	2253
Ten-Slot Module with Enhanced Power Supply (51195066)	5.05	9.06	21.3	712	2431
Dual Node Module (One Node)	1.22	2.64	6.73	175	598
1 Amperes. Divide the numbers by 2 for 240 Vac operation or by 1.83 for 220 Vac operation.					

10.3.13 Scanner Application Module Specifications

Specifications The electrical specifications assume a power factor of approximately 0.65.
 Table 10-18 Scanner Application Module (SAM) Specifications

PHYSICAL CHARACTERISTICS						
Height Width Depth Weight	Five-Slot Module					
	17.8 cm (7 in.)					
	48 cm (19 in.)					
	61 cm (24 in.)					
	31.5 kg (46 lb)					
ELECTRICAL CHARACTERISTICS						
AC Voltage Options			Frequency Options			
120, 240 Vac +10%, -15% 220 Vac +10%, -10%			50 or 60 Hz +3%, -6%			
Inrush Current						
	Five-Slot Module (Enhanced Power Supply)					
	14 A @ 120 Vac 28 A @ 240 Vac					
Component		Average RMS ¹	True RMS ¹	Peak ¹	Watts	BTUs/Hour
Five-Slot Module ²	Typ.	1.34	2.89	7.66	215	735
	Max. ³	2.36	4.69	11.2	329	1123
1 Amperes. Divide the figures by 2 for 240 Vac operation or by 1.83 for 220 Vac operation.						
2 Amperage figures are valid for enhanced (51195066) power supply.						
3 Maximum figures measured with the voltage at 102 Vac.						
The Scanner Application Module will operate properly during an interruption in the input ac voltage of up to 40 ms duration.						

10.3.14 Deskside GUS Specifications

Specifications

Environmental specification for the deskside GUS are listed below.

Table 10-19 Deskside GUS Environment Specifications

Environmental Requirements					
Component	Temperature Range	Temperature Rate of Change	Relative Humidity	Altitude Above Sea Level	Environmental Class
Electronics Module Operating	+5° to +32° C (+41° to +90° F)	5° per Hour	20 to 80% Non-Condensing	10,000 Ft.	A
Storage	-25° to +65° C (-13° to 145° F)			30,000 Ft.	
17" Monitor Operating	0° to 40°C 32° to 104° F		10 to 80%		
Storage	-20° to 45°C 5° to 113°F		5 to 95%		
21" Monitor Operating Storage	0 to 40° C 20 to 70°C		10 to 80% 5 to 95%	9,850 Ft. 39,370 Ft.	

Table 10-20 Deskside GUS Electrical Specification

Component	Voltage Range	Current	Watts	Weight
Electronics module (Maximum)	90 to 132 Vac 180 to 264 Vac 47 to 63 Hz	4.6 A @ 110 Vac 2.3 A @ 230 Vac	350 @ 110 Vac 350 @ 220 Vac	
17" Monitor	90 to 264 Vac 60 Hz/50 Hz ±3 Hz		100 W ac	21 kg 46.3 lb.
21" Monitor	90 to 132 Vac 198 to 264 Vac 47 to 63 Hz	1.75 A @ 110 Vac .9 A @ 230 Vac	135 @ 110 Vac 135 @ 220 Vac	

Continued on next page

10.3.14 Deskside GUS Specifications, Continued

ATTENTION

RFI specification—Deskside GUS stations are protected from RFI transmissions up 3 volts/meter in the 80 MHz to 1000 MHz frequency range. **Do not use walkie-talkies within 3 meters.**

EMC testing

The Deskside GUS has been tested to the following standards:

- Emission —complies to EN50081-1
 - Immunity—complies to EN50082-1
-

10.3.15 Console Mount GUS Specifications

Specifications

Environmental specification for the console mount GUS are listed below.

Table 10-21 Console Mount GUS Environment Specifications

Environmental Requirements					
Component	Temperature Range	Temperature Rate of Change	Relative Humidity	Altitude Above Sea Level	Environmental Class
Electronics Module Operating	+5° to +40° C (+41° to +104° F)	5° per Hour	20 to 80% Non-Condensing	10,000 Ft.	A
Storage	-25° to +65° C (-13° to 145° F)			35,000 Ft.	
21" Monitor Operating	0 to 50°C (32 to 122°F) Normal = 20°C	15°C (27°F) per hour	10 to 90%	9850 Ft.	A
Storage	-20 to 60°C		5 to 95%	39,370 Ft.	

Table 10-22 Console Mount GUS Electrical Specification

Component	Power	Current
Electronics Module	110 Vac @ 60 Hz 220 Vac @ 50 Hz	4.6 A @ 110 V 2.3 A @ 220 V
Primary Monitor	90 to 132 Vac 180 to 264 Vac	2.3 A @ 110 V 1.1 A @ 220 V
Second Monitor	90 to 132 Vac 180 to 264 Vac	2.3 A @ 110 V 1.1 A @ 220 V
ZIP, CD-ROM, Tape Drives are powered by the Electronics Module	_____	_____

ATTENTION

RFI specification—Console GUS stations are protected from RFI transmissions up to 10 volts/meter in the 80 MHz to 450 MHz frequency range and up to 3.0 volts/meter in the 450 MHz to 1 GHz frequency range. **Do not use Walkie-Talkies within .5 meters.**

Continued on next page

10.3.15 Console Mount GUS Specifications, Continued

EMC testing

Table 10-23 EMC Test Standards

Standard	Description	Level	Level Rating	Frequency Range
EN55011	Radiated Emission	Group 1 Class A		30 MHz to 1 GHz
EN55011	Conducted Emission	Group 1 Class A		150 kHz to 30 MHz
IEC801-3:1990	Radiated Immunity	2	3 V/m:80% modulation at 1 kHz	450 MHz to 1 GHz
IEC801-3:1990	Radiated Immunity	3	10 V/m:80% modulation at 1 kHz	80 MHz to 450 MHz
IEC801-2:1991	ESD: Contact Discharge	2	4 KV	
IEC801-2:1991	ESD:Air Discharge	3	8 KV	
IEC801-4:1988	Transients	2	0.5 KV: Data Line	
IEC801-4:1988	Transients	2	1 KV: Power Line	
IEC801-5:1993	Surge	2	0.5 KV:Line to Line	
IEC801-5:1993	Surge	2	1 KV:Line to Ground	
IEC801-4-11	Voltage Dips, Interruptions		230 V $\pm$ 10%	
IEC801-6	Conducted Radio Frequency Immunity	2	3 V/m:80% modulation at 1 kHz	150 kHz to 80 MHz

Appendix – UCLM: CCP AND UCIM Redesigned

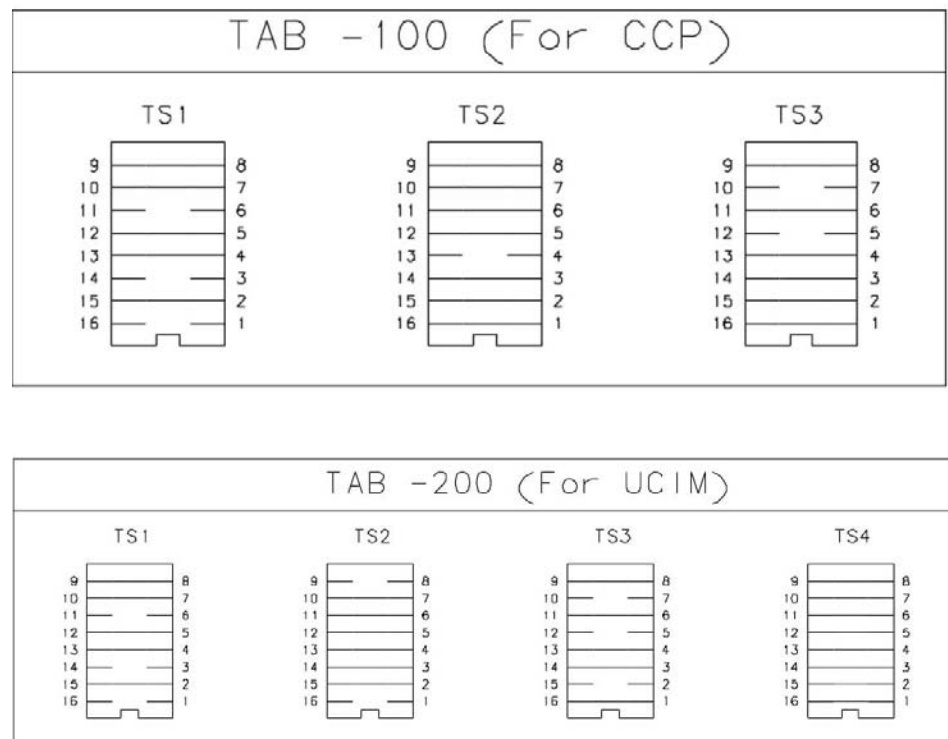
CCP (51402426) and UCIM (51402436) boards have been redesigned and are now known as UCLM-51306409. UCLM is a combination of CCP and UCIM.

Ordering Information To order the boards use the Honeywell part numbers as given below:

Part Number	Board Name
51306409-100	CCP
51306409-200	UCIM

Switch settings of UCLM board

Figure E-1 illustrates the redesigned CCP and UCIM settings



There is no change in the procedure to load AM personality on UCLM. Procedure remains the same as it was in CCP and UCIM.

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